EDIC / Emergency & Critical Care

HIGH-YIELD STUDY REFERENCE

17 clinically organized chapters

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EDIC / Emergency & Critical Care

01 — Resuscitation, Airway & Shock

Airway Assessment and Protection

Topic 1: Airway protection and the decision to intubate

- **Core concept:** The need for an artificial airway is determined by the patient's ability to maintain oxygenation, ventilation, and airway protection rather than by a single isolated reflex.
- **Key discriminator:** Absence of a gag reflex is unreliable because it may be absent in healthy adults and does not reliably measure airway protection.
- **Clinical trigger:** Depressed consciousness with sonorous respirations, progressive upper-airway obstruction, respiratory fatigue, or inability to handle secretions should prompt immediate airway reassessment.
- **Management priority:** Effective bag-mask ventilation remains an appropriate bridge when oxygenation and ventilation are adequate; an advanced airway should not replace a functioning basic intervention without a clinical reason.
- **Common mistake:** Treating a low Glasgow Coma Scale value or an absent gag reflex as an automatic substitute for an airway assessment.
- **Red flag:** Stridor, drooling, marked respiratory distress, agitation with progressive obstruction, or rapidly worsening hypoxemia indicates a threatened airway.
- **Clinical trigger:** In a critically ill child, unusual quietness or exhaustion can be a sign of impending respiratory failure even when overt agitation has ceased.

قرار intubation لا يعتمد على gag reflex أو رقم واحد فقط. المطلوب تقييم قدرة المريض على حماية airway والمحافظة على oxygenation وventilation، مع الانتباه لعلامات الانسداد المتفاقم.

Topic 2: Critical airway anatomy

- **High-yield fact:** The cricoid cartilage is the only complete cartilaginous airway ring and lies at the C6 vertebral level.
- **Key discriminator:** C6 marks the junction of the larynx with the trachea and the pharynx with the oesophagus.
- **High-yield fact:** The cricothyroid membrane lies between the thyroid and cricoid cartilages.
- **Clinical Pearl:** Point-of-care ultrasound can identify the cricothyroid membrane when surface palpation is difficult.
- **Exam trap:** A statement that ultrasound is ineffective for locating the cricothyroid membrane is false within the supplied material.

احفظ علاقة cricoid مع C6 وموقع cricothyroid membrane بين thyroid وcricoid. هذه landmarks مهمة خصوصًا عند التحضير لـfront-of-neck access.

Topic 3: Positioning for laryngoscopy

- **Core concept:** Airway positioning should align external landmarks while preserving oxygen reserve and laryngoscopic access.
- **Key discriminator:** In an infant, the external auditory canal and anterior shoulder should be horizontal and parallel to the bed; a shoulder roll may be required because of the large occiput.
- **Key discriminator:** In an obese patient, ramp the head and trunk until the external auditory meatus and sternal notch are horizontally aligned.
- **Clinical Pearl:** The ramped position improves preoxygenation and facilitates the laryngeal view in obesity.
- **Common mistake:** Placing a roll beneath the infant's occiput, which may worsen neck flexion.
- **Clinical Pearl:** The AQRAB pediatric airway panel emphasizes preparing suction, bag-mask ventilation, equipment sized for the child, and a primary as well as rescue airway plan before induction.

في infant يكون الرأس كبيرًا نسبيًا، لذلك غالبًا نرفع الكتفين لا مؤخرة الرأس. في obesity نستخدم ramped position حتى يتساوى external auditory meatus مع sternal notch.

Topic 4: Pediatric airway differences

- **High-yield fact:** The pediatric glottic opening is higher and more anterior, the tongue is relatively larger, and the epiglottis is proportionally larger and floppier.
- **Management priority:** A straight blade can directly lift the floppy epiglottis in young children.
- **Key discriminator:** The short pediatric trachea increases the risk of right mainstem intubation after apparently successful tube placement.
- **Clinical Pearl:** Video laryngoscopy reduces errors in small infants in the supplied material and is favored when a difficult airway is anticipated.
- **Common mistake:** Advancing the tube according to adult depth expectations.
- **Red flag:** Sudden unilateral ventilation or deterioration after pediatric intubation should prompt immediate assessment for a tube placed too deeply.
- **Management priority:** After pediatric tube placement, reconfirm position and ventilation following movement or clinical deterioration.

الـpediatric airway أعلى وأمامي أكثر، والـtrachea أقصر؛ لذلك visualization أصعب وmainstem intubation أسهل. افحص depth مباشرة عند أي تدهور بعد intubation.

Preoxygenation Rapid Sequence Intubation and Tube Placement

Topic 5: Preoxygenation and peri-intubation risk

- **Core concept:** Preoxygenation increases the oxygen reserve available during apnea and is especially important when rapid desaturation is expected.
- **High-yield fact:** Flush-rate oxygen through a nonrebreather mask at approximately 40-90 L/min produced a higher end-tidal oxygen concentration than a standard 15 L/min nonrebreather in the supplied material.
- **Clinical trigger:** Obesity, severe hypoxemia, and limited pulmonary reserve increase the risk of peri-intubation hypoxic arrest.
- **Management priority:** Optimize position and preoxygenation and maximize first-pass success before induction whenever time permits.
- **Common mistake:** Assuming a standard-flow nonrebreather provides the same nitrogen washout as flush-rate delivery.
- **Management priority:** In pediatric airway emergencies, optimize oxygen reserve and ensure an effective bag-mask fallback rather than allowing repeated laryngoscopy to interrupt oxygenation.

الهدف من preoxygenation هو إطالة safe apnea time. المريض البدين أو شديد hypoxemia قد ينهار سريعًا، لذلك positioning ورفع جودة first pass مهمان قبل إعطاء أدوية RSI.

Topic 6: Rapid sequence intubation drug selection

- **Core concept:** Induction-agent selection must account for hemodynamics and the underlying respiratory problem.
- **High-yield fact:** Propofol may produce hypotension through vasodilation, sympathetic suppression, and myocardial depression; the supplied source gives an RSI induction dose of 1.5 mg/kg and an anticipated mean arterial pressure reduction of about 10 mmHg.
- **Clinical Pearl:** Ketamine is favored in severe asthma within the supplied material because of bronchodilating and sympathomimetic effects; a cited dose is 1-2 mg/kg.
- **High-yield fact:** Etomidate is associated with myoclonus in procedural sedation and is described as useful when hemodynamic stability is important.
- **Red flag:** Succinylcholine may cause clinically important hyperkalemia in susceptible neuromuscular and myopathic conditions.
- **Exam trap:** Prolonged paralysis after suxamethonium should suggest pseudocholinesterase deficiency.

اختر induction agent حسب physiology: propofol قد يهبط الضغط، بينما ketamine قد يكون مفيدًا في severe asthma. لا تتعامل مع كل أدوية RSI كأنها متساوية hemodynamically.

Topic 7: Laryngoscope blades and laryngeal view

- **High-yield fact:** A Macintosh blade is placed in the vallecula to lift the epiglottis indirectly; a Miller blade lifts it directly.
- **Key discriminator:** Cormack-Lehane grade I shows the full glottic aperture; grade II shows a partial glottic view or arytenoids; grade III shows only the epiglottis; grade IV shows no glottic structures.
- **Clinical Pearl:** Hyperangulated video laryngoscopy requires less cervical movement and is useful when cervical spine movement should be minimized.
- **Common mistake:** Equating visualization of the epiglottis alone with a grade II view.

Macintosh يدخل في vallecula بينما Miller يرفع epiglottis مباشرة. في Cormack-Lehane، رؤية epiglottis فقط تعني Grade III، وهي نقطة امتحانية متكررة.

Topic 8: Confirmation of tracheal tube placement

- **Core concept:** Continuous quantitative waveform capnography is the definitive routine confirmation method for tracheal tube placement in the supplied material.
- **Key discriminator:** A sustained recognizable carbon dioxide waveform over repeated breaths supports tracheal placement; a transient reading alone is insufficient.
- **Clinical trigger:** During cardiac arrest, pulmonary blood flow is reduced, so the end-tidal carbon dioxide value may be low despite correct tracheal placement.
- **Management priority:** If tracheal placement cannot be confirmed during resuscitation, remove the tube and restore effective face-mask ventilation.
- **Common mistake:** Relying only on tube depth, chest movement, auscultation, or a single carbon dioxide detection.
- **Clinical Pearl:** A sudden sustained rise in end-tidal carbon dioxide during CPR suggests return of spontaneous circulation.

المهم هو sustained waveform وليس مجرد رقم عابر. في cardiac arrest قد يكون ETCO2 منخفضًا بسبب ضعف pulmonary blood flow، لكنه يبقى أداة مهمة لتأكيد tube ولمراقبة CPR.

Topic 9: Post-intubation deterioration

- **Core concept:** Sudden hypoxemia after intubation requires immediate restoration of oxygenation while the airway and equipment are assessed.
- **Management priority:** Disconnect the ventilator and manually ventilate with a bag-valve device to assess compliance and separate ventilator or circuit failure from airway or thoracic pathology.
- **Key discriminator:** Unilateral ventilation with an excessively deep tube suggests right mainstem intubation; withdraw incrementally and reassess.
- **Differential point:** Consider tube displacement or obstruction, tension pneumothorax, equipment disconnection, severe pulmonary pathology, and persistent shock.
- **Common mistake:** Waiting for chest radiography before correcting an obvious depth-related ventilation problem.

عند deterioration بعد intubation، ارجع سريعًا إلى manual ventilation. هذا يحافظ على الأكسجة ويساعدك على التفريق بين مشكلة ventilator ومشكلة tube أو الصدر.

Difficult Airway and Front of Neck Access

Topic 10: Failed intubation and rescue oxygenation

- **Core concept:** After failed intubation, the immediate objective is oxygenation rather than repeated traumatic attempts.
- **Management priority:** A second-generation supraglottic airway is a rescue option after failed intubation in the supplied difficult-airway pathway.
- **Red flag:** Inability to intubate and inability to oxygenate after induction is a front-of-neck access emergency.
- **Common mistake:** Continuing repeated laryngoscopy while oxygen saturation falls.
- **Exam trap:** The definitive step in a true cannot-intubate cannot-oxygenate state is emergency front-of-neck access, not another routine intubation attempt.

بعد failed intubation يصبح الهدف oxygenation. إذا نجح SAD حافظ على الأكسجة، أما CICO الحقيقي فيحتاج front-of-neck access دون إضاعة الوقت بمحاولات متكررة.

Topic 11: Emergency cricothyrotomy

- **Core concept:** Cricothyrotomy provides emergency airway access through the cricothyroid membrane when oral or nasal oxygenation and ventilation have failed.
- **Key discriminator:** A vertical midline skin incision improves exposure when landmarks are difficult; the supplied material describes a 3-5 cm vertical incision over the membrane.
- **Clinical Pearl:** The cricothyroid membrane lies immediately inferior to the thyroid cartilage and superior to the cricoid cartilage.
- **Common mistake:** Delaying the procedure in a fully developed cannot-intubate cannot-oxygenate state.
- **Pediatric discriminator:** In young children, the supplied material favors needle cricothyrotomy with jet ventilation or tracheotomy rather than adult-style surgical cricothyrotomy because the larynx is small and easily injured.

في CICO يكون cricothyrotomy إجراء إنقاذ حياة. الأطفال الصغار مختلفون تشريحيًا، لذلك لا تُطبق عليهم adult surgical technique بصورة تلقائية.

Topic 12: Upper airway obstruction syndromes

- **Clinical trigger:** Fever, drooling, dysphagia, muffled voice, stridor, and tripod or sniffing posture indicate a potentially unstable supraglottic airway.
- **Management priority:** Minimize agitation and obtain experienced airway support; unnecessary throat manipulation can precipitate complete obstruction.

- **Differential point:** Croup typically produces a barking cough and inspiratory stridor after a viral prodrome, whereas epiglottitis has a more abrupt toxic presentation with drooling and difficulty handling secretions.
- **Red flag:** Stridor, agitation, severe retractions, or rapidly progressive distress indicates impending complete obstruction.
- **Clinical Pearl:** A supraglottic airway may be ineffective when the obstruction is below the device.
- **Exam trap:** A normal-looking posterior oropharynx does not exclude a dangerous supraglottic process.
- **Imaging sign:** The pediatric airway infographic depicts the lateral-neck thumbprint sign as an epiglottitis recognition clue; imaging recognition must be integrated with airway risk rather than treated as sole confirmation.

المريض الذي لديه drooling وstridor ولا يتحمل الاستلقاء قد يفقد airway بسرعة. لا تزعجه بفحص غير ضروري؛ حافظ على الوضع المريح واستدعِ فريق airway الخبير.

Topic 13: Laryngotracheal injury and tracheostomy bleeding

- **Clinical trigger:** Dysphonia, neck soft-tissue emphysema, crepitus, laryngeal tenderness, deformity, or severe dyspnea after neck trauma suggests laryngotracheal injury.
- **Management priority:** Early specialist airway involvement is required because conventional intubation may worsen disruption.
- **Red flag:** Even a small sentinel bleed from a tracheostomy may precede major tracheo-innominate hemorrhage.
- **Clinical Pearl:** The innominate artery is the vessel of concern in tracheo-innominate fistula.
- **Common mistake:** Dismissing transient tracheostomy bleeding because it has stopped.

crepitus أو dysphonia بعد neck trauma قد يعني laryngotracheal injury. كذلك tracheostomy sentinel bleed ليس حدثًا بسيطًا لأنه قد يسبق نزفًا كارثيًا.

Procedural Sedation Airway Safety

Topic 14: Sedation depth

- **Key discriminator:** Moderate sedation preserves purposeful response to verbal or light tactile stimulation, spontaneous ventilation, airway patency, and cardiovascular function.
- **Key discriminator:** Deep sedation requires repeated or painful stimulation for purposeful response; airway patency and ventilation may be impaired.
- **Common mistake:** Labeling transient hypopnea and airway obstruction requiring jaw thrust or bag-mask support as moderate sedation.
- **Clinical Pearl:** Sedation depth is a continuum and can unintentionally deepen after additional medication.

الفرق المهم هو نوع stimulation المطلوب للاستجابة وتأثر airway. الحاجة إلى repeated painful stimulation مع airway compromise تعني deep sedation.

Topic 15: Sedation monitoring and fasting

- **Management priority:** Continuous clinical observation remains essential; capnography should be considered for moderate sedation and is recommended for prolonged deep sedation in the supplied material.
- **Clinical Pearl:** End-tidal carbon dioxide may detect ventilatory deterioration before oxygen saturation falls.
- **High-yield fact:** The supplied fasting intervals are 2 hours after clear liquids, 4 hours after breast milk, and 6 hours after other liquids or solids.
- **Exam trap:** Urgency may justify proceeding despite fasting status after assessing aspiration risk; fasting time is not an absolute substitute for clinical judgment.
- **Red flag:** Older patients have increased respiratory-depression risk because of reduced ventilatory reserve, airway vulnerability, comorbidity, and impaired drug clearance.

- **Management priority:** For pediatric procedural sedation, perform pre-sedation assessment of airway risk, comorbidities, medications, allergies, and prior sedation problems; have rescue equipment immediately available.
- **Clinical Pearl:** When feasible, assign a dedicated observer separate from the proceduralist and continue recovery observation until the child approaches baseline and meets discharge criteria.
- **Exam trap:** Supplemental oxygen may keep SpO2 normal while capnography already shows worsening ventilation.

يُقيّم ضمن المخاطر والضرورة ولا fasting. قد يتأخر SpO2 مبكرًا، بينما hypoventilation يكشف capnography ينبغي أن يؤخر إجراءً إسعافيًا بلا تفكير.

Pediatric sedation لا ينتهي بانتهاء الإجراء؛ يجب مراقبة ventilation وعودة الوعي والحالة الأساسية قبل الخروج.

Topic 16: Sedation-related laryngospasm

- **Clinical trigger:** Stridor, paradoxical chest movement, falling oxygen saturation, and loss of the capnographic plateau during ketamine sedation suggest laryngospasm.
- **Management priority:** Stop stimulation, reposition the airway, suction secretions, provide high-flow oxygen, and escalate to positive-pressure ventilation with 100% oxygen when obstruction or desaturation persists.
- **Clinical Pearl:** Pressure at the Larson point is described as an adjunct during positive-pressure ventilation.
- **Escalation:** Persistent laryngospasm may require deeper sedation, neuromuscular blockade, and intubation.
- **Risk discriminator:** Active respiratory infection or active asthma increases laryngospasm concern in the supplied pediatric sedation material.

laryngospasm أثناء sedation يُعالج تدريجيًا لكن بسرعة: airway maneuvers ثم suction وoxygen، ثم positive pressure، ومع الفشل قد نحتاج paralysis وintubation.

Mechanical Ventilation Fundamentals

Topic 17: Oxygenation versus ventilation

- **Core concept:** Oxygenation and carbon dioxide clearance are adjusted through different ventilator variables.
- **High-yield fact:** Increasing PEEP primarily improves oxygenation by maintaining alveolar patency and increasing functional residual capacity.
- **High-yield fact:** Increasing minute ventilation improves carbon dioxide clearance; in lung-protective ventilation, increasing respiratory rate may be preferred to increasing tidal volume.
- **Common mistake:** Increasing tidal volume to correct hypercapnic acidosis despite an established lung-protective target.
- **Exam trap:** FiO2 and PEEP address oxygenation; tidal volume and respiratory rate primarily determine ventilation.
- **Context-dependent target:** AQRAB cites 94–98% oxygen saturation in RCUK emergency care, with 88–92% considered when hypercapnic respiratory failure is a risk; these are context-specific targets rather than universal settings.

افصل دائمًا بين oxygenation وPEEP. ventilation وFiO2 للأكسجة، بينما minute ventilation يحدد CO2، مع المحافظة على lung-protective tidal volume.

Topic 18: Pressure support ventilation

- **Core concept:** Pressure-support ventilation is a spontaneous mode in which patient effort initiates each breath.
- **Key discriminator:** Trigger equals patient effort, limit equals pressure, and cycle equals flow.
- **Common mistake:** Describing pressure support as a mandatory time-triggered mode.

- **Clinical Pearl:** The patient must have an active respiratory drive for a purely spontaneous pressure-support breath.

في PSV المريض يبدأ النفس، ventilator يحدد مقدار pressure support، وينتهي النفس عندما ينخفض flow إلى معيار محدد.

Topic 19: Airway pressure release ventilation

- **Core concept:** APRV applies a prolonged high airway pressure with brief releases to a lower pressure for carbon dioxide clearance.
- **Key discriminator:** P-high and T-high maintain recruitment; P-low and a short T-low permit release ventilation while limiting derecruitment.
- **High-yield fact:** Spontaneous breathing is preserved throughout the respiratory cycle.
- **Common mistake:** Assuming APRV routinely requires deep sedation and neuromuscular blockade.
- **Clinical Pearl:** Eliminating spontaneous effort removes a defining physiological feature of APRV.

APRV يبقي الرئة مفتوحة معظم الوقت ويستخدم releases قصيرة لطرد CO2. الفخ هو اعتبار paralysis ضرورة، بينما preserved spontaneous breathing جزء أساسي من الفكرة.

Topic 20: Lung-protective ventilation and driving pressure

- **High-yield fact:** A supplied lung-protective strategy uses tidal volume 6 mL/kg with permissive hypercapnia.
- **Number/threshold:** Driving pressure equals plateau pressure minus PEEP; an example of plateau pressure 28 cm H2O and PEEP 8 cm H2O gives a driving pressure of 20 cm H2O.
- **Management priority:** When severe acidosis occurs while tidal volume is already limited, increase respiratory rate rather than abandoning the protective tidal volume; the supplied material accepts permissive hypercapnia until pH falls below 7.2 and keeps rate below 35/min in the cited example.
- **Clinical Pearl:** ECCO2R can reduce ventilatory demand and permit lower tidal volume, respiratory rate, and driving pressure.
- **Common mistake:** Using actual rather than protective tidal-volume principles simply to normalize carbon dioxide.

الهدف تقليل ventilator-induced injury حتى لو قبلنا بعض hypercapnia. Driving pressure هو Pplat - PEEP، وارتفاعه يعني strain أكبر على الرئة.

Topic 21: Pressure volume loops and stress index

- **Key discriminator:** Beaking or a duckbill appearance on the pressure-volume loop indicates alveolar overdistension.
- **Core concept:** During constant-flow volume-controlled ventilation, the stress index reflects the shape of the pressure-time curve.
- **Management target:** A decremental PEEP trial seeks a stress index near 1.0 in the supplied material.
- **Common mistake:** Interpreting terminal flattening of the inspiratory limb as improved compliance.

beaking يعني أن alveoli وصلت إلى overdistension. Stress index قريب من 1 يشير إلى أقرب pressure rise للخطية أثناء constant flow.

Topic 22: Obstructive physiology and auto PEEP

- **Clinical trigger:** Persistent expiratory flow when the next inspiration begins indicates incomplete exhalation and intrinsic PEEP.
- **Management priority:** Allow more expiratory time by reducing excessive respiratory rate and using an appropriate inspiratory-to-expiratory ratio; a ratio of 1:3 is cited for COPD NIV.
- **Red flag:** Dynamic hyperinflation may reduce venous return and cardiac output and can cause sudden hemodynamic collapse.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 9

1.5 mg/kg	High-yield fact: Propofol may produce hypotension through vasodilation, sympathetic suppression, and myocardial depression; the supplied source gives an RSI induction dose of 1.5 mg/kg and an anticipated mean arterial pressure reduction of about 10 mmHg.
1-2 mg/kg	Clinical Pearl: Ketamine is favored in severe asthma within the supplied material because of bronchodilating and sympathomimetic effects; a cited dose is 1-2 mg/kg. High-yield fact: Etomidate is associated with myoclonus in procedural sedation and is described as useful

- **Common mistake:** Increasing respiratory rate in severe airflow obstruction without checking whether expiration reaches zero flow.

في COPD أو asthma قد يبدأ النفس التالي قبل انتهاء الزفير، فينشأ auto-PEEP. العلامة المهمة هي استمرار expiratory flow، والعلاج يعتمد على إطالة وقت الزفير.

Topic 23: Patient ventilator asynchrony

- **Key discriminator:** Reverse triggering occurs when a ventilator-delivered breath induces a delayed diaphragmatic contraction rather than the patient triggering the ventilator.
- **Clinical trigger:** Vigorous spontaneous effort with large tidal volumes and large negative pleural-pressure swings may cause patient self-inflicted lung injury.
- **Core concept:** A low displayed airway pressure does not exclude injurious transpulmonary pressure when pleural pressure becomes markedly negative.
- **Common mistake:** Judging inspiratory injury solely from the ventilator pressure display.

المريض قد يؤذي رئته بجهد شهيق قوي حتى لو كان ventilator pressure منخفضًا، لأن transpulmonary pressure يعتمد أيضًا على pleural pressure.

Topic 24: Noninvasive ventilation

- **Core concept:** Noninvasive positive-pressure ventilation unloads respiratory muscles and can improve ventilation without immediate intubation in selected patients.
- **Key discriminator:** In bilevel support, the IPAP-EPAP difference determines ventilatory assistance and carbon dioxide clearance.
- **Number/threshold:** The supplied COPD example increases IPAP from 15 to 20 cm H2O while maintaining EPAP at 3 cm H2O when hypercapnic acidosis persists.
- **Number/threshold:** A maximum IPAP of 30 cm H2O is cited, with higher pressures reserved for expert guidance.
- **Clinical trigger:** Failure to improve gas exchange, worsening consciousness, inability to protect the airway, intolerance, or progressive fatigue should prompt escalation.
- **Common mistake:** Prophylactically intubating every patient who requires NIV.

في bilevel NIV، الفرق بين IPAP وEPAP هو driving support للتهوية. عدم التحسن أو تدهور الوعي يعني أن NIV قد يفشل ويجب التحضير للتصعيد.

Topic 25: Ventilator effects on circulation and fluid responsiveness

- **Core concept:** Positive-pressure inspiration increases intrathoracic pressure, reduces venous return, and may alter right-ventricular preload and afterload.
- **Key discriminator:** An end-expiratory occlusion interrupts the next positive-pressure breath, transiently increasing venous return; an increase in cardiac output greater than 10% after a 15-second hold indicates fluid responsiveness in the supplied material.
- **Clinical Pearl:** Pulse-pressure variation and stroke-volume variation are most useful under controlled mechanical ventilation and become unreliable when their assumptions are violated.
- **Common mistake:** Treating respiratory inferior-vena-cava variation as universally reliable regardless of breathing effort or ventilator conditions.
- **Red flag:** High PEEP with right-ventricular dilation and shock may require cautious PEEP reduction when overdistension is increasing pulmonary vascular resistance.

PPV يغير venous return، لذلك يمكن استغلال respiratory changes لتقدير fluid responsiveness، لكن هذه الاختبارات ليست موثوقة في كل مريض وخصوصًا مع spontaneous breathing.

Topic 26: Oxygenation indices

- **Number/formula:** Oxygenation Index equals mean airway pressure multiplied by FiO2 as a percentage, divided by PaO2 in mmHg.
- **Exam trap:** Entering FiO2 as 0.8 instead of 80 produces a result 100 times too low.
- **Differential point:** The P/F ratio is a simple PaO2-to-FiO2 ratio; the alveolar-arterial gradient also incorporates PaCO2 through the alveolar gas equation.
- **Clinical Pearl:** Oxygenation Index includes the intensity of ventilatory pressure support, whereas the P/F ratio does not.

OI لا يقيس الأكسجة فقط بل يضيف مقدار ventilatory pressure المستخدم. انتبه أن FiO2 تدخل كنسبة مئوية في المعادلة الواردة بالمصدر.

Comparison: oxygenation versus ventilation

Feature	Oxygenation	Ventilation	Exam point
Main variables	FiO2 and PEEP	Minute ventilation: rate and tidal volume	Do not raise tidal volume reflexively to fix CO2 under lung protection
Typical problem	Low arterial oxygen	Excess CO2 or hypercapnic acidosis	Mechanisms require different ventilator adjustments

Cardiac Arrest and Peri Arrest Care

Topic 27: High quality adult CPR

- **Number/threshold:** Before an advanced airway, the supplied adult compression-to-ventilation ratio is 30:2.
- **Number/threshold:** Compression rate is 100-120/min with a depth of 5-6 cm and full chest recoil.
- **Management priority:** Minimize interruptions and resume compressions promptly after rhythm assessment or shock delivery.
- **Exam trap:** A rhythm changing from nonshockable to ventricular fibrillation during a two-minute cycle does not justify unnecessary prolonged interruption of ongoing CPR.

جودة CPR تعتمد على rate وdepth وتقليل interruptions. وجود فريق متقدم لا يلغي قاعدة 30:2 قبل advanced airway.

Topic 28: Shockable and nonshockable rhythms

- **Core concept:** Ventricular fibrillation and pulseless ventricular tachycardia are shockable; asystole and pulseless electrical activity require CPR, drug therapy, and correction of reversible causes.
- **Management priority:** Defibrillation should be delivered with minimal interruption to compressions.
- **Clinical Pearl:** Confirm apparent asystole in more than one lead because an organized rhythm or ventricular fibrillation may appear as asystole in a single lead.
- **Common mistake:** Attempting synchronized cardioversion for a pulseless shockable rhythm.

VF وpulseless VT يحتاجان defibrillation، بينما PEA وasystole لا يُصدمان. الفكرة الأساسية هي rhythm recognition مع استمرار CPR.

Topic 29: Refractory ventricular fibrillation

- **Management option:** Changing defibrillator pad position to an anterior-posterior vector is described for refractory ventricular fibrillation.
- **Device discriminator:** When an implanted defibrillator is present, place the external pad at least 8 cm away; an anterior-posterior arrangement is acceptable.
- **Conflict note:** One supplied passage supports dual sequential defibrillation based on early studies, whereas another advises against its use outside research.

- **Exam trap:** Do not merge vector change and dual sequential defibrillation into the same intervention.

فالمصادر dual sequential defibrillation أما .anterior-posterior بوضع vector يمكن تغيير refractory VF في داخل الملف متعارضة، لذلك لا تُحسم هنا.

Topic 30: Resuscitation drugs

- **Number/threshold:** In nonshockable adult arrest, adrenaline 1 mg is given once IV or IO access is obtained and then every 3-5 minutes in the supplied algorithm.
- **Number/threshold:** Pediatric arrest adrenaline is 10 micrograms/kg, equivalent to 0.01 mg/kg IV or IO, repeated every 3-5 minutes in the cited pediatric source.
- **Number/threshold:** Pediatric amiodarone after the third shock for pulseless ventricular tachycardia is 5 mg/kg IV or IO.
- **Common mistake:** Confusing the 1 mg/mL intramuscular anaphylaxis preparation with the diluted intravenous cardiac-arrest preparation.
- **Medication safety:** AQRAB gives the pediatric IV/IO arrest adrenaline example as 0.01 mg/kg (maximum 1 mg) from 0.1 mg/mL; for 10 kg, 0.1 mg corresponds to 1 mL of that concentration. The IM anaphylaxis concentration is different.
- **Common mistake:** AQRAB states that routine calcium or bicarbonate is not recommended in pediatric arrest unless a specific indication exists.

أكبر فخ دوائي هو الخلط بين concentration وroute في adrenaline. جرعات cardiac arrest تختلف تمامًا عن IM anaphylaxis.

انتبه أن جرعة Pediatric arrest وتركيزها الوريدي يختلفان عن IM anaphylaxis. راجع mg وmL وconcentration معًا.

Topic 31: Capnography during CPR

- **Core concept:** With minute ventilation held relatively constant, end-tidal carbon dioxide reflects pulmonary blood flow and therefore the cardiac output generated by compressions.
- **Key discriminator:** Persistently very low ETCO2, such as 8 mmHg in the supplied example, suggests ineffective compressions or very low flow.
- **Key discriminator:** A sudden sustained increase in ETCO2 indicates return of spontaneous circulation; one supplied example rises from 2.0 to 5.3 kPa.
- **Exam trap:** Changes in ventilation can alter ETCO2 independently of cardiac output.

أثناء CPR يصبح ETCO2 مؤشرًا على perfusion بشرط ثبات ventilation. الارتفاع المفاجئ المستمر قد يعني ROSC، بينما القيمة المنخفضة جدًا توحي بأن flow غير كافٍ.

Topic 32: Hypothermic cardiac arrest

- **Core concept:** Severe hypothermia permits prolonged resuscitation and requires active rewarming before futility is judged.
- **Management priority:** Extracorporeal rewarming with cardiopulmonary bypass or ECMO provides rapid core rewarming in hypothermic arrest.
- **Number/threshold:** A cited cardiopulmonary-bypass rewarming rate is 6-9 degrees Celsius per hour.
- **Number/threshold:** Below 30 degrees Celsius, the supplied algorithm limits defibrillation attempts to three and withholds intravenous resuscitation drugs until temperature exceeds 30 degrees Celsius.
- **Red flag:** Serum potassium at or above 12 mEq/L is described as a very poor prognostic marker, but must be interpreted in clinical context.
- **Common mistake:** Terminating resuscitation solely because the initial arrest duration is prolonged while the patient remains severely hypothermic.

في hypothermic arrest لا تُحكم على futility قبل rewarming المناسب. ECMO أو bypass يقدمان أسرع rewarming، وتوجد تعديلات على shocks والأدوية تحت 30°C.

Topic 33: Pediatric and neonatal resuscitation

- **Core concept:** Pediatric arrest is commonly preceded by respiratory failure; effective bag-mask ventilation is therefore a critical early intervention.
- **Management priority:** Continue effective bag-mask ventilation during pediatric out-of-hospital arrest when it provides visible chest rise and adequate oxygenation.
- **Number/threshold:** Neonatal compressions use a 3:1 compression-to-ventilation ratio when the heart rate remains below 60/min after ventilation steps.
- **Number/threshold:** The supplied neonatal material gives five initial inflation breaths: 30 cm H2O for a term infant and 25 cm H2O below 32 weeks, with PEEP 5-6 cm H2O when possible.
- **Common mistake:** Advancing to an invasive airway merely because the patient is in arrest despite effective bag-mask ventilation.
- **Management priority:** Pediatric cardiac arrest following respiratory failure or shock calls for recognition of deterioration, effective CPR, prompt rhythm assessment and IV/IO access without interrupting compressions.
- **High-yield fact:** For an initial nonshockable pediatric arrest rhythm, AQRAB reports the 2025 AHA pathway as supporting early epinephrine; IV/IO drug delivery is preferred over endotracheal administration when feasible.
- **Number/threshold:** In the cited neonatal pathway, begin ventilation within 60 seconds after initial steps if apnea/gasping or heart rate below 100/min persists.
- **Number/threshold:** In AQRAB, neonatal compressions begin when heart rate remains below 60/min after 30 seconds of effective ventilation visibly inflating the lungs; use coordinated 3:1 compressions and ventilation.
- **Exam trap:** If a newborn remains bradycardic without visible chest rise, first correct mask seal, head position and obstruction rather than starting compressions before effective ventilation.
- **Number/threshold:** AQRAB lists a neonatal target temperature of 36.5–37.5 degrees Celsius and discourages routine suctioning unless obstruction is suspected.

في الأطفال يكون respiratory cause شائعًا، لذلك BMV الفعال قد يكون أهم من التسرع إلى advanced airway. في neonate النسبة 3:1 والتركيز أولًا على ventilation.

إذا لا توجد chest rise في حديث الولادة فالمشكلة قد تكون ventilation غير فعالة. صحح mask seal وposition وobstruction قبل الانتقال إلى compressions.

Topic 34: Maternal cardiac arrest

- **Clinical trigger:** Pregnancy beyond approximately 20 weeks creates clinically important aortocaval compression in the supine position.
- **Management priority:** Manually displace the uterus to the left while maintaining effective supine chest compressions.
- **Number/threshold:** Resuscitative hysterotomy is described when there is no return of spontaneous circulation within 4 minutes, with the aim of delivery by 5 minutes.
- **Common mistake:** Using lateral tilt in a way that compromises chest-compression quality.

في maternal arrest يجب إزالة aortocaval compression دون إضعاف chest compressions. manual left uterine displacement يسمح ببقاء المريضة supine للـCPR.

Topic 35: Pulmonary embolism cardiac arrest

- **Clinical trigger:** Pulmonary embolism-related arrest commonly presents as pulseless electrical activity with very low ETCO2 despite good compressions.

- **Management priority:** Treat the underlying embolic obstruction; the supplied material advises prolonged CPR for at least 60-90 minutes after thrombolysis in this setting.
- **Clinical Pearl:** Emergency echocardiography by a skilled operator can support the diagnosis, but a low ETCO2 is nonspecific.
- **Common mistake:** Stopping resuscitation immediately after thrombolytic administration before the drug has time to act.

PE arrest غالبًا يظهر كـPEA. إذا أعطي thrombolysis فقد يلزم CPR طويل لأن إزالة الانسداد ليست فورية.

Post Cardiac Arrest Care

Topic 36: Immediate post ROSC priorities

- **Management priority:** Support oxygenation and ventilation, avoid hyperoxia, maintain perfusion, identify the arrest cause, and prevent secondary brain injury.
- **Number/threshold:** A post-ROSC mean arterial pressure above 65 mmHg is cited in the supplied material.
- **Clinical Pearl:** Early neurologic examination can be unreliable after arrest because of sedation, temperature management, and evolving injury.
- **Common mistake:** Using a single early examination finding to make an irreversible prognosis.

بعد ROSC يبدأ ضرر reperfusion؛ لذلك المطلوب controlled oxygenation، ضغط كافٍ، ومعالجة السبب. لا تتسرع في neuroprognostication المبكر.

Topic 37: Targeted temperature management

- **Number/threshold:** A target temperature range of 32-36 degrees Celsius is repeatedly cited for comatose survivors in the supplied material.
- **Management priority:** Initiate temperature control as soon as possible after return of spontaneous circulation and do not delay it for routine diagnostic procedures.
- **Adverse effects:** Hyperglycemia, impaired coagulation, and increased infection risk are described complications.
- **Common mistake:** Allowing uncontrolled fever after cardiac arrest.

المادة تكرر target بين 32 و36°C، مع بدء مبكر ومراقبة complications مثل hyperglycemia وcoagulopathy وinfection.

Topic 38: Multimodal neuroprognostication

- **Core concept:** Neurologic prognosis after arrest should be multimodal and delayed until confounders have been addressed.
- **Red flag:** Status myoclonus within 48-72 hours combined with a malignant EEG pattern is described as a strong poor-outcome predictor.
- **Common mistake:** Using a single neuron-specific enolase value at 24 hours, an isolated early imaging result, or one examination sign as definitive.
- **Number/threshold:** One supplied source identifies 72 hours as the earliest point for reliable examination-based prognostication after relevant confounders are excluded.

التنبؤ العصبي لا يعتمد على test واحد. يجب دمج examination وEEG وimaging وbiomarkers بعد زوال sedation وtemperature-related confounders.

Shock Recognition and Hemodynamic Reasoning

Topic 39: Shock phenotypes

- **Core concept:** Shock is inadequate tissue perfusion and can be categorized by flow, vascular tone, and filling pressure rather than blood pressure alone.
- **Hypovolemic profile:** Low cardiac output, low mean arterial pressure, low pulmonary capillary wedge pressure, and low venous oxygen saturation indicate inadequate circulating volume.
- **Distributive profile:** Low systemic vascular resistance with preserved or increased cardiac output produces warm extremities and bounding pulses early in vasodilatory shock.
- **Cardiogenic profile:** Low cardiac output with high systemic vascular resistance and elevated filling pressures indicates pump failure with congestion.
- **Obstructive profile:** Impaired cardiac filling or outflow produces shock despite adequate or elevated venous pressure.
- **Common mistake:** Excluding shock because blood pressure is initially normal, especially in children or compensated hemorrhage.
- **Clinical Pearl:** Multiple shock mechanisms can coexist; a single warm, cold, or obstructive label may not capture the full clinical state.
- **Management priority:** Repeat assessment of mental status, pulses, capillary refill, skin temperature/mottling, heart and lung findings, urine output and the response after each intervention.
- **Number/threshold:** AQRAB identifies urine output under 0.5 mL/kg/hour as a concerning perfusion finding; interpret with context and trend.

افهم shock من CO وSVR وfilling pressures، لا من BP وحده. warm shock يعني vasodilation، بينما cardiogenic shock يجمع low flow مع congestion وhigh resistance.

Topic 40: Hemodynamic formulas

- **Number/formula:** Mean arterial pressure equals systolic pressure plus twice the diastolic pressure, divided by three.
- **Number/formula:** Systemic vascular resistance equals [(MAP minus CVP) divided by cardiac output] multiplied by 80.
- **Number/formula:** Cardiac Power Output in watts equals MAP multiplied by cardiac output, divided by 451.
- **Number/formula:** Cerebral perfusion pressure equals MAP minus intracranial pressure.
- **Number/formula:** Shock index equals heart rate divided by systolic blood pressure; a value above 0.7 is cited as suggesting occult shock and possible transfusion need.
- **Exam trap:** Calculate the pressure gradient using MAP minus CVP rather than MAP alone when deriving systemic vascular resistance.
- **Exam trap:** The abbreviation MAP means mean airway pressure in the oxygenation index but mean arterial pressure in cerebral perfusion pressure and systemic vascular resistance formulas.

هذه المعادلات تربط الضغط بالـflow. في SVR لا تنسَ طرح CVP، وshock index قد يكشف occult hemorrhage قبل حدوث hypotension الواضح.

اختصار MAP قد يعني mean airway pressure أو mean arterial pressure؛ حدد معنى المتغير قبل التعويض بالمعادلة.

Topic 41: Venous oxygen saturation and carbon dioxide gap

- **Key discriminator:** ScvO2 samples the upper-body venous return from the superior vena cava; SvO2 is true mixed venous blood sampled in the pulmonary artery.
- **Clinical Pearl:** The venous-to-arterial carbon dioxide gap is inversely related to cardiac output and reflects the adequacy of blood flow for carbon dioxide washout.
- **Number/threshold:** A normal venous-to-arterial carbon dioxide gap of 0.3-0.8 kPa, or 2-6 mmHg, is cited.
- **Common mistake:** Assuming ScvO2 and SvO2 are interchangeable under all shock states.

ScvO2 ليس mixed venous الحقيقي لأنه يمثل upper body أساسًا. CO2 gap يرتفع عندما يكون flow غير كافٍ لإزالة CO2 من الأنسجة.

Topic 42: Static preload versus fluid responsiveness

- **Core concept:** A static pressure such as CVP does not reliably establish whether cardiac output will increase after fluid administration.
- **High-yield fact:** The supplied normal CVP range is 8-12 mmHg, but its interpretation depends on ventricular compliance and intrathoracic pressure.
- **Clinical Pearl:** Dynamic tests such as passive leg raising, pulse-pressure variation, stroke-volume variation, and end-expiratory occlusion assess response rather than static filling.
- **Common mistake:** Giving repeated fluid solely because CVP is low or withholding fluid solely because CVP is high.
- **Key discriminator:** A fluid-responsiveness test requires a measurable increase in stroke volume or cardiac output, not just a blood-pressure change.
- **Common mistake:** Improving blood pressure or a single lactate measurement does not prove restoration of tissue perfusion and does not automatically justify repeating another fluid bolus.

CVP يصف pressure ولا يثبت fluid responsiveness. الاختبار الأفضل هو maneuver يغيّر preload مؤقتًا ثم نقيس هل CO أو stroke volume ارتفع فعلًا.

Topic 43: PiCCO and advanced monitoring

- **Core concept:** PiCCO combines transpulmonary thermodilution for calibration and volumetric measurements with pulse-contour analysis for continuous cardiac output.
- **Number/threshold:** A Global End-Diastolic Volume Index of approximately 680-800 mL/m2 is cited as normal; 750 mL/m2 represents appropriate preload in the supplied example.
- **Clinical Pearl:** Volumetric preload measures are less dependent on myocardial compliance and intrathoracic pressure than CVP or wedge pressure.
- **Common mistake:** Interpreting pulse-contour output without recognizing conditions that distort arterial waveform analysis.

PiCCO يجمع calibration بالـthermodilution مع GEDI. continuous pulse contour يقيس volume preload بدل الاعتماد على pressure فقط.

Topic 44: Focused POCUS during shock reassessment

- **Core concept:** POCUS answers a focused question and must be interpreted with symptoms, perfusion, examination and serial response.
- **Key discriminator:** B-lines indicate an alveolar-interstitial pattern, not a uniquely cardiogenic diagnosis; consider distribution and the broader lung/cardiac picture.
- **Exam trap:** Absent lung sliding is not specific for pneumothorax; apnea, mainstem intubation and pleural processes can mimic it, while an absent lung point does not exclude a large pneumothorax.
- **Management priority:** Persistent shock, worsening vasopressor requirements or organ dysfunction warrant higher-acuity monitored care and early specialist escalation.

POCUS أداة لفهم physiology وليست إجابة مستقلة؛ اربط الصورة بالتروية والسياق والتغير بعد العلاج.

Topic 45: Serial shock reassessment and escalation

- **Core concept:** A transient improvement in blood pressure is not proof of adequate tissue oxygen delivery or a safe endpoint.
- **Clinical trigger:** Persisting confusion, mottling, poor urine output, rising lactate, respiratory distress or increasing support suggest ongoing hypoperfusion despite an apparently satisfactory pressure.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 16

1 mg	Number/threshold: In nonshockable adult arrest, adrenaline 1 mg is given once IV or IO access is obtained and then every 3-5 minutes in the supplied algorithm.
0.01 mg/kg	and then every 3-5 minutes in the supplied algorithm. Number/threshold: Pediatric arrest adrenaline is 10 micrograms/kg, equivalent to 0.01 mg/kg IV or IO, repeated every 3-5 minutes in the cited pediatric source.
1 mg/mL	mg/kg IV or IO. Common mistake: Confusing the 1 mg/mL intramuscular anaphylaxis preparation with the diluted intravenous cardiac-arrest preparation.
0.01 mg/kg	intravenous cardiac-arrest preparation. Medication safety: AQRAB gives the pediatric IV/IO arrest adrenaline example as 0.01 mg/kg (maximum 1 mg) from 0.1 mg/mL; for 10 kg, 0.1 mg corresponds to 1 mL of that concentration. The IM anaphylaxis
0.5-1 mg	mL/kg/hour. Red flag: Large intravenous adrenaline boluses of 0.5-1 mg in a perfused patient can cause hypertensive crisis, myocardial ischemia, and lethal arrhythmia; those doses are reserved for cardiac arrest.

- **Management priority:** Document the suspected shock mechanism, treatment already given, measurable response, remaining uncertainty and the next action or escalation.

لا تحكم على نجاح الإنعاش من BP وحده؛ اعد فحص perfusion والوعي والبول والرئة بعد كل تدخل.

Comparison: hypovolemic versus cardiogenic shock

Feature	Hypovolemic shock	Cardiogenic shock	Exam point
Main mechanism	Inadequate circulating volume	Pump failure with congestion	Find and treat the source, not BP alone
Cardiac output	Low	Low	Both can have poor forward flow
Filling pressure	Low	Raised	Repeated blind fluid may harm pump failure
Management direction	Control bleeding / restore volume	Support perfusion and correct pump cause	Assess lung congestion before repeating boluses

Comparison: pediatric warm versus cold shock

Feature	Warm shock	Cold shock	Exam point
Hemodynamics	High output, low SVR	Low output, high SVR	Different physiologic deficits
Extremities	Warm, flushed	Cold	Look beyond blood pressure
Pulse and refill	Bounding; flash refill	Weak; delayed refill	Perfusion clues distinguish phenotypes
Treatment reasoning	Restore vascular tone	Address impaired cardiac output	Do not use one indiscriminate vasoactive strategy

Fluids Vasopressors and Inotropes

Topic 46: Initial vasopressor selection

- **Management priority:** Norepinephrine is repeatedly identified as the initial vasopressor for undifferentiated hypotensive shock and septic shock after inadequate response to fluid resuscitation.
- **Number/threshold:** A mean arterial pressure target of at least 65 mmHg is cited for septic shock and early post-ROSC care.
- **Clinical Pearl:** A short-acting peripheral vasopressor bolus may be used as a bridge while definitive infusion access is established in the supplied material.
- **Red flag:** Vasopressor extravasation can cause tissue necrosis; the supplied treatment is phentolamine 5-10 mg infiltrated subcutaneously or through the original catheter.
- **Common mistake:** Persisting with additional fluid despite pulmonary edema and ongoing vasodilatory hypotension.
- **Source-specific number:** AQRAB provides a reference norepinephrine starting example of 8–12 micrograms/min with titration, emphasizing that it is not an IV-push dose and depends on the formulation and local protocol.
- **Management priority:** If severe hypotension persists during individualized early septic shock resuscitation, the AQRAB text advises monitored vasopressor support alongside reassessment rather than waiting for a mandatory number of fluid bags.

بعد fluid غير الكافي يكون norepinephrine هو الاختيار المتكرر في الملف. لا تؤخر vasopressor عند ظهور fluid overload، وراقب extravasation باكرًا.

Topic 47: Inotropes and inodilators

- **High-yield fact:** Milrinone inhibits phosphodiesterase 3, increases cyclic AMP, improves contractility, and causes systemic and pulmonary vasodilation.
- **Key discriminator:** Milrinone acts independently of beta receptors and may retain effect when beta receptors are blocked or downregulated.

- **Differential point:** Dobutamine is a beta-agonist, whereas milrinone is a phosphodiesterase inhibitor.
- **Common mistake:** Using an inodilator without accounting for its potential to worsen hypotension.

Milrinone هو inodilator وليس beta-agonist، لذلك قد يعمل مع beta-blockade لكنه قد يزيد hypotension بسبب vasodilation.

Topic 48: Cardiogenic shock

- **Hemodynamic profile:** Low cardiac output, high systemic vascular resistance, and elevated wedge pressure indicate cardiogenic shock; simultaneous high CVP suggests right-sided or biventricular involvement.
- **Management priority:** Norepinephrine is preferred over epinephrine for hypotensive cardiogenic shock in the supplied material while definitive mechanical or surgical treatment is arranged.
- **Clinical Pearl:** Intra-aortic balloon counterpulsation inflates at the dicrotic notch and deflates just before systole, augmenting diastolic coronary perfusion and reducing afterload.
- **Number/formula:** Cardiac Power Output integrates MAP and cardiac output; a calculated value of 0.27 W is described as profoundly low.
- **Red flag:** Mechanical complications after myocardial infarction require urgent surgical management; vasopressors are temporizing, not definitive.

cardiogenic shock هو low flow مع norepinephrine. congestion يدعم perfusion مؤقتًا، لكن mechanical complication مثل papillary muscle rupture تحتاج definitive surgery.

Topic 49: Septic and vasoplegic shock

- **Core concept:** Septic shock commonly combines low systemic vascular resistance with myocardial dysfunction; the myocardial component may be reversible.
- **Management priority:** After initial crystalloid resuscitation, persistent hypotension or organ hypoperfusion requires norepinephrine rather than unlimited additional fluid.
- **Number/threshold:** One supplied definition uses persistent hypotension after 30 mL/kg crystalloid or lactate at least 4 mmol/L.
- **Rescue mechanism:** Methylene blue inhibits nitric oxide synthase and soluble guanylate cyclase, opposing nitric-oxide-mediated vasodilation in refractory vasoplegia.
- **Clinical Pearl:** Positive blood cultures occur in only a proportion of clinically septic patients; negative cultures do not exclude an infectious cause.
- **Clinical trigger:** Suspected infection with acute organ dysfunction should prompt assessment of consciousness, oxygenation, capillary refill, urine output, lactate, and anatomic source; fever and WBC alone cannot confirm or exclude sepsis.
- **Management priority:** Obtain cultures before antibiotics when this causes no clinically meaningful delay; treat shock or probable/definite sepsis promptly and look for a source requiring intervention.
- **Source-specific timing:** AQRAB quotes an SSC 2026 source-control goal ideally within 6 hours after diagnosis when early intervention is needed; it is a quoted source recommendation, not independently reverified.
- **Source-specific volume:** AQRAB quotes a conditional SSC 2026 crystalloid approach of 30 mL/kg over three hours for sepsis-associated hypoperfusion, individualized and repeatedly reassessed.
- **Exam trap:** AQRAB cautions that qSOFA alone is insufficient to screen out sepsis; lactate is a nonspecific risk/perfusion marker.

في septic shock المشكلة الأساسية غالبًا vasodilation مع myocardial depression. بعد fluid المناسب، متبدل hypotension المستمر يحتاج norepinephrine، وليس liters إضافية بلا نهاية.

ليس كل hypotension يحتاج سوائل بلا حد؛ حجم السوائل يقرأ مع سبب الصدمة واستجابة التروية والرئة.

Topic 50: Pediatric warm and cold shock

- **Warm shock:** High cardiac output, low systemic vascular resistance, warm flushed extremities, bounding pulses, and flash capillary refill favor norepinephrine to restore vascular tone.
- **Cold shock:** Low cardiac output with high systemic vascular resistance, cool extremities, weak pulses, and delayed capillary refill indicates a different hemodynamic deficit.
- **Clinical Pearl:** Children may maintain blood pressure despite significant perfusion failure; hypotension is a late finding.
- **Common mistake:** Applying a single vasopressor strategy without identifying the child's hemodynamic phenotype.
- **Clinical Pearl:** AQRAB emphasizes mental state, pulses, peripheral perfusion and urine output because pediatric shock may precede measurable hypotension.

warm shock هو vasodilated high-output state، بينما cold shock هو low-output vasoconstricted state.

الطفل قد يبقى normotensive رغم shock مهم.

Topic 51: Neurogenic shock

- **Clinical trigger:** Hypotension with relative bradycardia after spinal cord injury, especially with priapism or other cord signs, suggests neurogenic shock.
- **Core concept:** Loss of sympathetic tone produces vasodilation and impaired cardiac sympathetic stimulation.
- **Management priority:** Crystalloid may support preload, but norepinephrine is used to restore vascular tone and perfusion in the supplied material.
- **Differential point:** Spinal shock describes loss of neurologic reflex activity; neurogenic shock describes the hemodynamic state.
- **Common mistake:** In trauma with hypotension and suspected neurogenic shock, AQRAB advises continued search for hemorrhage instead of automatically attributing shock to spinal injury.

العلامة الفاصلة هي hypotension مع غياب tachycardia المناسبة بعد spinal injury. لا تخلط neurogenic shock مع hemodynamics neurologic findings spinal shock.

Topic 52: Obstructive shock and tamponade

- **Clinical trigger:** Hypotension with jugular venous distension, muffled heart sounds, and a compatible cause suggests cardiac tamponade.
- **Ultrasound discriminator:** Right-atrial collapse during ventricular systole is an early sign; right-ventricular diastolic collapse and a plethoric inferior vena cava support hemodynamic significance.
- **Management priority:** Intravenous fluid can temporarily augment preload while definitive pericardial drainage is arranged.
- **Common mistake:** Treating fluid as definitive therapy for tamponade.
- **Red flag:** Shock with acute chest or back pain and suspected proximal aortic dissection may reflect hemopericardium and tamponade.
- **Exam trap:** Pericardial fluid or a plethoric IVC alone does not establish tamponade; right-sided chamber collapse and the full circulation assessment provide context.
- **Differential point:** Right-ventricular strain may be acute or chronic; POCUS cannot by itself prove or exclude pulmonary embolism.
- **Management priority:** AQRAB distinguishes urgent drainage of unstable tamponade from a temporary cautious preload-supporting fluid bolus, with trauma or procedural bleeding requiring an appropriate surgical pathway.

tamponade يمنع filling، لذلك fluid قد يشتري وقتًا فقط. definitive treatment هو drainage أو surgery حسب السبب، والـecho يبحث عن chamber collapse وIVC plethoric.

Hemorrhagic Shock and Transfusion

Topic 53: Recognition of occult hemorrhage

- **Core concept:** Normal blood pressure does not exclude early hemorrhagic shock because tachycardia and vasoconstriction may preserve pressure initially.
- **Number/threshold:** Shock index equals heart rate divided by systolic pressure; a value above 0.7 is cited as a marker of occult shock and possible transfusion need.
- **Clinical trigger:** In an unstable trauma patient, assume hemorrhage until another cause is established.
- **Red flag:** Class III hemorrhage in the supplied material includes approximately 30-40% blood-volume loss, heart rate 120-139/min, hypotension, reduced pulse pressure, respiratory rate 30-40/min, and altered behavior.
- **Common mistake:** Waiting for a low hemoglobin concentration before initiating resuscitation during active major bleeding.

occult قد تكشف pulse pressureو mental statusو والنبض shock index. قد يكون BP طبيعيًا في بداية النزف shock قبل انخفاض hemoglobin أو الضغط.

Topic 54: Damage control resuscitation

- **Core concept:** Hemorrhage control and restoration of oxygen-carrying capacity must occur together; crystalloid alone does not correct ongoing blood loss.
- **High-yield fact:** The lethal triad is hypothermia, acidosis, and coagulopathy.
- **Management priority:** Activate a massive-transfusion pathway according to hemodynamics, bleeding pace, and inability to control hemorrhage rather than waiting for a laboratory threshold.
- **High-yield fact:** A balanced 1:1:1 ratio of red cells, plasma, and platelets is repeatedly cited.
- **Clinical Pearl:** Permissive hypotension is described before definitive hemorrhage control in selected patients; the supplied target range is systolic pressure 70-90 mmHg.
- **Common mistake:** Applying permissive hypotension without considering situations in which organ perfusion targets differ.
- **Source qualification:** For a cited ruptured AAA pathway, AQRAB describes permissive systolic pressure of approximately 70–90 mmHg while conscious, with limited evidence and the need for individualized perfusion assessment; do not generalize this target to every shock state.

أهم lethal triad. balanced blood productsو dilution وتجنب إيقاف النزف الهدف massive hemorrhage في من إعطاء crystalloid بكميات كبيرة.

Topic 55: Massive transfusion complications

- **Core concept:** Rapid transfusion of citrated blood products can produce ionized hypocalcemia because citrate binds calcium and magnesium.
- **Management priority:** Monitor and replace calcium during massive transfusion when clinically indicated.
- **Key discriminator:** TRALI causes acute permeability pulmonary edema and hypoxemia within 6 hours of transfusion without circulatory overload.
- **Key discriminator:** TACO reflects circulatory overload and is more likely when significant cardiac or renal comorbidity is present.
- **Management priority:** Stop the transfusion immediately when a serious transfusion reaction is suspected and begin supportive evaluation and treatment.
- **Clinical Pearl:** One adult red-cell unit raises hemoglobin by about 1 g/dL and hematocrit by about 3% in the supplied material.

TRALI فرّق بين transfusion بعد respiratory deterioration عند. citrate بسبب calcium راقب MTP في .TACO volume overloadو permeability edema

Topic 56: Tranexamic acid

- **Core concept:** Tranexamic acid inhibits fibrinolysis and is time-sensitive in major traumatic bleeding.
- **Number/threshold:** The supplied trauma material emphasizes administration within 3 hours.
- **Pediatric number:** A cited pediatric acute-bleeding dose is 15 mg/kg IV over 10 minutes.
- **Common mistake:** Treating tranexamic acid as a replacement for hemorrhage control or balanced resuscitation.

TXA يساعد على تثبيت clot لكنه لا يوقف مصدر النزف. الفائدة في trauma مرتبطة بالوقت، والمادة تؤكد إعطاءه خلال 3 ساعات.

Comparison: TRALI versus TACO

Feature	TRALI	TACO	Exam point
Mechanism	Permeability pulmonary edema	Circulatory overload	Pulmonary edema is not always fluid overload
Context	Acute hypoxemia after transfusion	Circulatory stress, often cardiac/renal risk	Stop transfusion and assess mechanism

Anaphylactic Shock

Topic 57: Recognition of anaphylaxis

- **Clinical trigger:** Acute skin or mucosal involvement plus respiratory compromise or hypotension strongly suggests anaphylaxis after a likely exposure.
- **Alternative trigger:** Hypotension after a known allergen, or involvement of two organ systems after a suspected allergen, can establish the clinical syndrome.
- **High-yield fact:** Skin manifestations are most common, but their absence does not safely exclude a severe reaction.
- **Core concept:** IgE cross-linking on mast cells releases histamine and other mediators, producing vasodilation, increased permeability, bronchospasm, and mucosal edema.
- **Common mistake:** Treating urticaria alone and anaphylaxis as equivalent conditions.
- **Differential point:** Histamine-mediated reactions and bradykinin-mediated angioedema can differ in response to allergy-directed treatment; both may threaten the airway.

anaphylaxis هو systemic reaction وليس rash فقط. وجود airway أو breathing أو circulation compromise بعد exposure يجب أن يدفع للعلاج الفوري.

Topic 58: Immediate anaphylaxis management

- **Management priority:** Lay the patient flat with or without leg elevation unless respiratory distress requires another position, then give intramuscular adrenaline promptly.
- **Number/threshold:** Adult IM adrenaline is 0.5 mg of 1 mg/mL solution in several supplied UK-oriented passages.
- **Pediatric number:** Weight-based pediatric IM adrenaline is 0.01 mg/kg of 1 mg/mL solution into the anterolateral thigh.
- **Number/threshold:** A repeat IM dose after 5 minutes without improvement is repeatedly cited, although another pediatric source gives 5-10 minutes.
- **Common mistake:** Delaying adrenaline while administering antihistamines or corticosteroids.
- **Clinical Pearl:** Routine corticosteroids and antihistamines are not recommended as emergency substitutes for adrenaline in the supplied 2021 guideline material.
- **Management priority:** Stop a suspected ongoing infusion or exposure when feasible, summon resuscitation help, and monitor response while giving indicated IM epinephrine.
- **Positioning:** Avoid standing or walking during an acute reaction. If breathing is difficult, allow a supported posture with legs extended; AQRAB cites a left-lateral position in pregnancy.

- **Medication safety:** AQRAB specifies 1 mg/mL epinephrine IM in the mid-anterolateral thigh at 0.01 mg/kg capped at 0.5 mg per dose in its ASCIA pathway; the example 20 kg = 0.20 mg = 0.20 mL. Autoinjector prescribing uses separate product rules.
- **Source-dependent fluid:** AQRAB cites adult RCUK initial crystalloid increments of 500–1,000 mL and pediatric increments of 10 mL/kg, whereas its ASCIA example uses 10–20 mL/kg; choose a source framework and reassess rather than mix regimens.
- **Exam trap:** Inhaled salbutamol may address persistent bronchospasm but does not replace IM epinephrine or reverse circulatory shock and laryngeal edema.

العلاج الأول هو IM adrenaline مبكرًا، وليس steroid أو antihistamine. استخدم 1 mg/mL IM في anterolateral thigh وانتبه لوجود اختلاف مصدري في بعض الجرعات والفواصل.

في Anaphylaxis لا تجعل المريض يمشي أو يقف فجأة؛ الوضعية جزء من دعم الدورة الدموية ويُراعى ضيق التنفس والحمل.

Topic 59: Refractory anaphylaxis

- **Definition:** Refractory anaphylaxis in the supplied material requires ongoing respiratory or cardiovascular support despite two IM adrenaline doses.
- **Management priority:** Start a titrated intravenous adrenaline infusion by an experienced clinician after repeated IM treatment and fluid resuscitation fail.
- **Number/threshold:** One supplied infusion preparation is adrenaline 1 mg in 100 mL saline at 0.5-1.0 mL/kg/hour.
- **Red flag:** Large intravenous adrenaline boluses of 0.5-1 mg in a perfused patient can cause hypertensive crisis, myocardial ischemia, and lethal arrhythmia; those doses are reserved for cardiac arrest.
- **Clinical Pearl:** Glucagon is described for refractory anaphylaxis in a patient taking beta blockers.
- **Source-specific number:** For beta-blocker-associated refractory anaphylaxis, AQRAB gives a cited RCUK glucagon example of 1 mg IV in adults and 20–30 micrograms/kg IV (maximum 1 mg) in children, with specialist prescribing and local confirmation.
- **Medication safety:** Do not administer undiluted 1 mg/mL epinephrine as a routine IV bolus to a patient with a pulse; confirm the concentration and route explicitly.

بعد فشل جرعتين IM مع fluid، الانتقال يكون إلى titrated IV infusion بخبرة ومراقبة، وليس large IV bolus.

وجود beta-blocker قد يجعل glucagon مهمًا.

Topic 60: Anaphylaxis investigation and observation

- **High-yield fact:** Mast-cell tryptase typically peaks 1-2 hours after symptom onset and may return toward baseline within 6-24 hours.
- **Common mistake:** Using a normal tryptase result to rule out anaphylaxis.
- **Risk discriminator:** Severe initial reaction, delayed adrenaline, or need for more than one adrenaline dose increases concern for a biphasic response.
- **Observation:** The supplied material contains risk-stratified observation periods of 2 hours after rapid complete response to one early dose, at least 6 hours after two doses, and more than 12 hours after more than two doses or severe respiratory risk.
- **Conflict note:** Another supplied passage recommends 6-12 hours for adults more generally; observation duration is therefore source-and risk-dependent within this corpus.
- **Source-specific timing:** AQRAB describes NICE tryptase sampling after treatment begins and again ideally 1–2 hours from onset (no later than 4 hours), with later baseline evaluation; a normal level does not exclude the clinical diagnosis.
- **Source-dependent observation:** AQRAB also quotes an ASCIA minimum observation period of four hours after the last epinephrine dose, with overnight care for higher-risk situations; its starting point differs from the UK risk-stratified framework.
- **Discharge safety:** Document recurrence advice, trigger avoidance, an action plan, practical device teaching, access to rescue medication, and allergy referral; AQRAB reports NICE two-autoinjector provisions with a stated exception for readily avoided drug triggers.

PASS 9 — Source-backed clinical recovery | Domain 01

Resuscitation, Airway & Shock

dosing or medical-correctness certification.

Unresponsive choking adult

Management priority: In the CLEAN V3 choking scenario, when an adult becomes unconscious, start chest compressions immediately.

إذا فقد المصاب بالاختناق وعيه، يذكر المصدر البدء الفوري بـ chest compressions.

Anaerobic threshold physiology

Core concept: The anaerobic threshold is the oxygen-consumption level beyond which anaerobic glycolysis supplements aerobic energy production and lactate rises.

بعد تجاوز anaerobic threshold لا تكفي الطاقة الهوائية وحدها، ويشارك المسار اللاهوائي ويرتفع lactate.

EDIC AUDIT PASS 11 | DOMAIN 01 | SOURCE-ONLY CONTEXT RESOLUTION

Anaphylaxis: source-dependent observation after two doses

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

Source-dependent threshold: CLEAN V3 cites a minimum 6-hour observation period after symptoms resolve when two IM adrenaline doses were required. Other source passages use risk-stratified observation durations. These are protocol contexts, not a universal discharge rule.

مدة 6 ساعات مذكورة في حالة احتاجت جرعتين IM Adrenaline؛ لا تُطبَّق تلقائيًا على كل حالات Anaphylaxis.

يدعم التشخيص tryptase لكنه لا ينفيه إذا كان طبيعيًا. مدة observation تختلف داخل المصادر، والأطول مطلوب عندما تكون reaction شديدة أو احتاجت جرعات متعددة.

Topic 61: Histamine-mediated versus bradykinin-mediated angioedema

- **Key discriminator:** Histamine-mediated swelling can be accompanied by hives or itching, whereas ACE-inhibitor-associated angioedema often lacks hives and can appear after longstanding exposure.
- **Clinical trigger:** Voice change, tongue swelling, drooling or stridor require early expert airway assessment regardless of cause.
- **Differential point:** Recurrent swelling or abdominal attacks can prompt consideration of hereditary angioedema and the C1-inhibitor/C4 evaluation described in AQRAB.
- **Management priority:** When clinical anaphylaxis remains plausible, AQRAB advises IM epinephrine without delaying care for complement tests; distinct bradykinin pathways require their own specialist context.

فرّق بين Histamine وBradykinin من القصة السريرية، لكن علامات تهديد Airway لا تنتظر استكمال الفحوص.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 23

0.1 mg/mL, 0.1 mg, 1 mL	Medication safety: AQRAB gives the pediatric IV/IO arrest adrenaline example as 0.01 mg/kg (maximum 1 mg) from 0.1 mg/mL; for 10 kg, 0.1 mg corresponds to 1 mL of that concentration. The IM anaphylaxis concentration is different.
15 mg/kg	Number/threshold: The supplied trauma material emphasizes administration within 3 hours. Pediatric number: A cited pediatric acute-bleeding dose is 15 mg/kg IV over 10 minutes. Common mistake: Treating tranexamic acid as a replacement for hemorrhage control or balanced
0.5 mg, 1 mg/mL	another position, then give intramuscular adrenaline promptly. Number/threshold: Adult IM adrenaline is 0.5 mg of 1 mg/mL solution in several supplied UK-oriented passages.
0.01 mg/kg, 1 mg/mL	passages. Pediatric number: Weight-based pediatric IM adrenaline is 0.01 mg/kg of 1 mg/mL solution into the anterolateral thigh.
1 mg/mL, 0.01 mg/kg	Medication safety: AQRAB specifies 1 mg/mL epinephrine IM in the mid-anterolateral thigh at 0.01 mg/kg capped at 0.5 mg per dose in its ASCIA pathway; the example 20 kg = 0.20 mg = 0.20 mL. Autoinjector
0.5 mg, 0.20 mg, 0.20 mL	Medication safety: AQRAB specifies 1 mg/mL epinephrine IM in the mid-anterolateral thigh at 0.01 mg/kg capped at 0.5 mg per dose in its ASCIA pathway; the example 20 kg = 0.20 mg = 0.20 mL. Autoinjector prescribing uses separate product rules.
1 mg, 100 mL	repeated IM treatment and fluid resuscitation fail. Number/threshold: One supplied infusion preparation is adrenaline 1 mg in 100 mL saline at 0.5-1.0 mL/kg/hour.
1 mg	Source-specific number: For beta-blocker-associated refractory anaphylaxis, AQRAB gives a cited RCUK glucagon example of 1 mg IV in adults and 20–30 micrograms/kg IV (maximum 1 mg) in children, with specialist prescribing and local confirmation.
1 mg/mL	specialist prescribing and local confirmation. Medication safety: Do not administer undiluted 1 mg/mL epinephrine as a routine IV bolus to a patient with a pulse; confirm the concentration and route explicitly.

Comparison: septic shock versus anaphylactic shock

Feature	Septic shock	Anaphylactic shock	Exam point
Typical trigger	Infection and acute organ dysfunction	Acute reaction after a likely exposure	Clinical context defines the first decision
Main physiology	Low SVR, sometimes myocardial dysfunction	Mast-cell mediators: vasodilation, leak and bronchospasm	Both can lower vascular tone
Recognition	Perfusion change, organ dysfunction and possible source	Airway, breathing or circulatory signs; rash may be absent	Absence of rash or fever does not safely exclude either
Immediate priority	Resuscitation, appropriate antibiotics, source control and vasopressor when needed	IM epinephrine without waiting for tests or antihistamines	First-line treatment differs

Comparison: histamine-related versus bradykinin-related angioedema

Feature	Histamine-mediated	Bradykinin-mediated	Exam point
Typical context	Often acute; itching or hives may occur	ACE inhibitor exposure or recurrent HAE-type swelling	Absence of hives does not eliminate airway risk
Management discriminator	Treat anaphylaxis with IM epinephrine when indicated	Specific cause-oriented pathway with expert airway care	Do not delay indicated emergency anaphylaxis treatment for C4 testing

Rapid Recall: Must-Know Facts

01. The cricoid cartilage lies at C6.
02. The cricothyroid membrane lies between the thyroid and cricoid cartilages.
03. An absent gag reflex does not reliably determine the need for intubation.
04. Infant positioning aligns the external auditory canal with the anterior shoulder.
05. Obese-airway ramping aligns the external auditory meatus with the sternal notch.
06. A straight blade directly lifts the pediatric epiglottis.
07. The short pediatric trachea increases right mainstem risk.
08. Waveform capnography is the definitive routine confirmation of tracheal placement.
09. A sustained waveform matters more than an isolated carbon dioxide reading.
10. Sudden post-intubation hypoxemia requires manual ventilation while troubleshooting.
11. Failed intubation shifts the priority to rescue oxygenation.
12. Cannot intubate and cannot oxygenate requires emergency front-of-neck access.
13. Moderate sedation preserves response to verbal or light tactile stimulation.
14. Deep sedation may impair airway patency and ventilation.

15. Capnography is recommended for prolonged deep sedation.
16. PEEP primarily improves oxygenation by maintaining alveolar patency and FRC.
17. Driving pressure equals plateau pressure minus PEEP.
18. Lung-protective tidal volume in the supplied material is 6 mL/kg.
19. Persistent expiratory flow at the next inspiration indicates auto-PEEP.
20. Pressure support is patient-triggered, pressure-limited, and flow-cycled.
21. APRV preserves spontaneous breathing.
22. Beaking of the pressure-volume loop indicates overdistension.
23. Reverse triggering begins with a ventilator breath followed by diaphragmatic effort.
24. Oxygenation Index uses FiO2 as a percentage.
25. Adult CPR before an advanced airway uses 30:2.
26. Adult compression rate is 100-120/min.
27. Ventricular fibrillation and pulseless ventricular tachycardia are shockable.
28. Asystole and PEA are nonshockable.
29. Sudden ETCO2 rise during CPR suggests ROSC.
30. Low ETCO2 with constant ventilation reflects low generated blood flow.
31. Hypothermic arrest requires rewarming before premature futility decisions.
32. Pediatric arrest adrenaline is 0.01 mg/kg IV or IO in the supplied material.
33. Neonatal resuscitation uses a 3:1 compression-to-ventilation ratio.
34. Maternal arrest requires left uterine displacement.
35. SVR equals [(MAP minus CVP) divided by CO] multiplied by 80.
36. Shock index equals heart rate divided by systolic blood pressure.
37. Cardiogenic shock is low output with high SVR and elevated filling pressures.
38. The lethal triad is hypothermia, acidosis, and coagulopathy.
39. Trauma tranexamic acid is time-sensitive within 3 hours.
40. Refractory anaphylaxis requires a titrated intravenous adrenaline infusion, not a large IV bolus.

All 40 recall facts appear in the integrated source-derived chapter above; no new medical facts are introduced.

Conflict Resolution Register — Domain 01

A. Existing source-dependent conflicts carried from CLEAN V3 (not new cross-file conflicts):

1. Adult IM adrenaline dose

- **Information A:** CLEAN V3: passages citing 0.5 mg IM, 1 mg/mL
- **Information B:** CLEAN V3: PEER-derived passages citing 0.3 mg IM, 1 mg/mL
- **Nature:** Different source conventions for initial adult IM dose
- **Classification:** Conflict — source-dependent

2. Pediatric IM adrenaline repeat interval

- **Information A:** CLEAN V3: 3–5 or 5 min
- **Information B:** CLEAN V3: one Rosen-derived passage cites 5–10 min
- **Nature:** Different repeat-dose intervals
- **Classification:** Conflict — source-dependent

3. Anaphylaxis observation time

- **Information A:** CLEAN V3: risk-stratified 2 / 6 / over 12 hours
- **Information B:** CLEAN V3: other source describes 6–12 hours more generally
- **Nature:** Different observation frameworks
- **Classification:** Conflict — source-dependent

4. Refractory VF dual sequential defibrillation

- **Information A:** CLEAN V3: early-study improvement reported
- **Information B:** CLEAN V3: avoid dual sequential shocks outside research, favor vector change
- **Nature:** Different recommendations
- **Classification:** Conflict — source-dependent

B. Between-file source-framework differences requiring context (not silently adjudicated):

1. Anaphylaxis observation clock

- **File 1:** CLEAN V3: early low-risk observation may be 2 hours after complete resolution, with longer periods for risk.
- **File 2:** AQRAB p.169: ASCIA approach cites at least 4 hours after last epinephrine, with overnight care for higher-risk presentations.
- **Nature:** Different policy and starting time; no single interval imposed.
- **Classification:** Source-framework difference — not adjudicated

2. Septic shock definitions and assessment

- **File 1:** CLEAN V3: one cited pathway describes persistent hypotension after 30 mL/kg or lactate ≥4 mmol/L.
- **File 2:** AQRAB p.203: Sepsis-3 shock uses vasopressors for MAP ≥65 mmHg and lactate >2 mmol/L despite adequate fluids.
- **Nature:** Different labels/criteria and clinical framework; do not interchange without naming definition.
- **Classification:** Source-framework difference — not adjudicated

3. Shock index thresholds

- **File 1:** CLEAN V3: >0.7 cited as warning for occult shock / possible transfusion.
- **File 2:** AQRAB p.213: around 0.9 or higher is a concerning contextual marker.
- **Nature:** Nonidentical risk-warning thresholds; both are presented as context-dependent, not definitive cutoffs.
- **Classification:** Source-framework difference — not adjudicated

Merge Quality Note — Domain 01

- **Final integrated topic headings:** 61
- **Original CLEAN V3 topic headings retained:** 58
- **Additional AQRAB-specific topic headings:** 3
- **CLEAN V3 source clinical bullets retained:** 279
- **AQRAB-added nonduplicative clinical bullets:** 57
- **Total integrated English clinical bullets:** 336
- **Arabic topic explanations retained:** 58
- **Confirmed shared concepts, conservative minimum:** 24
- **CLEAN points not separately re-proven as overlaps:** 255
- **Within-CLEAN unresolved conflict groups:** 4
- **Between-file framework differences explicitly recorded:** 3
- **Incomplete source fragments withheld from independent claim:** 2 explicitly recognized (AQRAB p.394 pediatric-airway negated option; p.394 trailing truncated claim)
- **Rapid Recall count:** 40

Counting method: Source-only means a newly retained clinical bullet derived from a named source; AQRAB additions are included only where not already stated at the same granularity. Confirmed shared concepts are a conservative verified minimum and should not be confused with a complete sentence-by-sentence corpus duplicate audit; existing CLEAN facts whose overlap has not been re-proven are reported separately, not falsely called “file 1 only.”

Incomplete in source — not included. Examples withheld: the negated-answer pediatric-airway phrase in AQRAB p.394, and a visibly truncated appended explanation on the same page. No missing wording was filled by inference.

No external sources were used. Medical source names appearing in the original files are retained only as descriptions of those files’ own cited pathways; no outside guideline or web verification was performed.

Resuscitation, Airway & Shock | Pass 8 | Explicit CLEAN V3 source locations

Hereditary angioedema — C1-inhibitor concentrate

• High-yield fact: The supplied CLEAN V3 passage identifies C1-esterase-inhibitor concentrate as a treatment for acute hereditary-angioedema attacks; apply only within the cited clinical/protocol context.

C1-inhibitor concentrate الـ مذكور لهجمات HAE الحادة؛ لا يُعامل كجرعة أو بروتوكول كامل دون سياقه.

Resuscitation, Airway & Shock. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 27

0.01 mg/kg	31. Hypothermic arrest requires rewarming before premature futility decisions. 32. Pediatric arrest adrenaline is 0.01 mg/kg IV or IO in the supplied material. 33. Neonatal resuscitation uses a 3:1 compression-to-ventilation ratio.
1 mg/mL	Information A: CLEAN V3: passages citing 0.5 mg IM, 1 mg/mL Information B: CLEAN V3: PEER-derived passages citing 0.3 mg IM, 1 mg/mL
0.3 mg, 1 mg/mL	Information A: CLEAN V3: passages citing 0.5 mg IM, 1 mg/mL Information B: CLEAN V3: PEER-derived passages citing 0.3 mg IM, 1 mg/mL Nature: Different source conventions for initial adult IM dose

EDIC / Emergency & Critical Care

02 — Cardiovascular Emergencies

Acute Coronary Syndromes and Ischemic Heart Disease

Topic 1: Recognition of acute coronary syndromes

- **Core concept:** ACS is a spectrum of suspected acute myocardial ischemia; symptoms, ECG and myocardial biomarkers must be interpreted together.
- **Clinical trigger:** Chest discomfort, dyspnea, diaphoresis, syncope, nausea, epigastric discomfort or functional decline may represent coronary ischemia.
- **Key discriminator:** Older adults and patients with diabetes may have atypical or less obvious chest pain.
- **Management priority:** Evaluate airway, breathing, circulation, stability and persistent ischemic symptoms; start cardiac monitoring and obtain a 12-lead ECG promptly.
- **High-yield fact:** The AQRAB-cited US ACS pathway specifies an initial ECG within 10 minutes of first medical contact.
- **Red flag:** Repeat ECGs when symptoms persist, recur or change; a nondiagnostic initial tracing does not exclude ACS.
- **Differential point:** Chest pain may also arise from aortic dissection, pulmonary embolism, pleurisy, musculoskeletal disease or other conditions.
- **Exam trap:** Nitrate-related symptom relief does not establish a coronary cause.

ACS ممكن يظهر بضيق نفس أو غثيان أو تعب بدل chest pain خصوصًا عند كبار السن ومرضى diabetes.

التخطيط الأول الطبيعي لا يلغي التشخيص.

Topic 2: Stable angina, unstable angina, NSTEMI and STEMI

- **Key discriminator:** Stable angina is predictable exertional ischemic discomfort, generally relieved by rest or an indicated nitrate, without acute necrosis.
- **Key discriminator:** Unstable angina is new, worsening or resting ischemic symptoms without demonstrated acute myocardial injury after completion of a suitably timed testing pathway.
- **Key discriminator:** NSTEMI has acute myocardial injury with ischemic evidence but lacks qualifying persistent ST elevation; ECG may show depression, inversion or no diagnostic early change.
- **Key discriminator:** STEMI or a recognized acute coronary occlusion pattern is an urgent reperfusion problem defined by clinical and ECG context.
- **Number/threshold:** The cited NSTEMI biomarker definition includes a troponin rise/fall above the assay-specific 99th percentile with evidence of ischemia.
- **Exam trap:** Unstable angina and NSTEMI cannot be separated using a single ECG alone.

- **Historical teaching:** Stable angina pain is often described as 2–10 minutes, unstable angina as 10–30 minutes and infarction as longer than 30 minutes; the source presents these as historical patterns, not a stand-alone diagnostic rule.
- **Management priority:** Do not delay indicated coronary reperfusion solely to await an early troponin value.

UA وNSTEMI ممكن يعطوا نفس ECG؛ الفرق يعتمد على serial troponin والسياق السريري. STEMI أو occlusion pattern قد يحتاج reperfusion قبل ظهور ارتفاع troponin.

Comparing coronary syndromes

Feature	Stable angina	UA / NSTEMI	STEMI / Occlusion
Presentation	Predictable exertional symptoms	New/worsening/rest ischemia	Acute occlusive ischemic emergency
Troponin	No acute MI rise/fall	UA: no injury; NSTEMI: injury with ischemia	May be negative very early
ECG	Often normal at rest	Variable ST/T findings or nondiagnostic	ST elevation or recognized equivalent
Exam point	Clinical course matters	Serial troponin distinguishes UA/NSTEMI	Do not defer indicated reperfusion for early troponin

Topic 3: High-risk ischemic ECG patterns

- **Key discriminator:** Wellens pattern includes deeply inverted or biphasic precordial T waves in a patient with recent ischemic symptoms; urgent cardiology assessment is needed.
- **High-yield fact:** De Winter morphology can include upsloping ST depression with prominent precordial T waves as an acute coronary occlusion clue.
- **Clinical Pearl:** Consider posterior and right-sided ECG leads when the initial distribution and presentation warrant them.
- **Key discriminator:** Concordant ST elevation or depression in LBBB can be important; use Sgarbossa/modified-Sgarbossa criteria within the full clinical picture.
- **Exam trap:** New LBBB alone is not automatically diagnostic of STEMI; nor can a normal first ECG exclude acute occlusion.
- **High-yield fact:** Capture the possibility of post-infarction LV aneurysm when persistent ST changes coexist with well-formed Q waves.
- **Common mistake:** Treating simplified educational ECG sketches as calibrated diagnostic tracings; AQRAB explicitly labels its four-panel morphology figure as illustrative.

• Imaging/ECG localization: The AQRAB coronary-territory figure maps anterior/septal leads V1-V4 to LAD, inferior leads II/III/aVF to RCA or LCx, and lateral leads I/aVL/V5-V6 to LCx or a LAD diagonal branch; it is a teaching map, not a stand-alone culprit-artery diagnosis. [AQRAB original PDF p. 27, visual panel]

احفظ توزيع الـleads حسب منطقة الـMI، لكن مصدر الـocclusion يتأثر بالـcoronary anatomy ولا يُستنتج من الخريطة وحدها.

ابحث عن Wellens وde Winter وposterior MI وmodified Sgarbossa حتى لو ما في STEMI كلاسيكي. تفسير LBBB يحتاج سياقًا ومعايير مناسبة، وليس مجرد وجوده.

Topic 4: Cardiac troponin, myocardial injury and risk pathways

- **Core concept:** Cardiac troponin is sensitive to myocardial injury, but elevation by itself does not prove plaque-related type 1 myocardial infarction.
- **Key discriminator:** Distinguish acute myocardial injury, type 1 MI and type 2 MI using rise/fall, evidence of ischemia and clinical context.
- **Clinical trigger:** Sepsis can create an oxygen supply-demand mismatch compatible with type 2 myocardial infarction in the supplied example.
- **Management priority:** Use the local validated high-sensitivity troponin algorithm with assay-specific values and sampling times.
- **Exam trap:** A single negative early troponin cannot exclude MI when presentation is early or risk remains high.

- **Clinical Pearl:** HEART or EDACS supports selected risk assessment pathways; it is not a substitute for assessment of PE or aortic dissection.
- **Management priority:** Selected low-risk patients may have outpatient follow-up only after completion of a validated pathway and appropriate assessment.

ارتفاع troponin يعني myocardial injury وليس بالضرورة type 1 MI. لازم تربط الارتفاع بالتغير الزمني ووجود ischemia والمرض المرافق.

Topic 5: Initial ACS medication and safety decisions

- **Management priority:** Aspirin is given promptly in suspected ACS when there is no contraindication, with local loading and maintenance protocol.
- **Key discriminator:** P2Y12 inhibitor type and timing depend on STEMI/NSTE-ACS strategy, PCI plan, bleeding risk and contraindications.
- **Source-specific recommendation:** CLEAN V3 contains a blanket statement to add clopidogrel to aspirin for NSTEMI in the ED; see the cross-file conflict register rather than adopting it universally.
- **Exam trap:** Prasugrel is contraindicated with a prior stroke or TIA in the cited source table.
- **Management priority:** Select anticoagulation using the indication and plan; initial medical UFH dosing is not interchangeable with procedural PCI dosing.
- **Red flag:** Before antithrombotics or fibrinolysis, reconsider aortic dissection when abrupt severe pain, neurologic deficit or pulse/BP asymmetry occurs.
- **Common mistake:** Using routine oxygen when saturation is at least 90% without hypoxemia; the cited ACS pathway directs oxygen to hypoxemia or respiratory distress.
- **Common mistake:** Using nitroglycerin despite hypotension, suspected right-ventricular preload dependence or recent PDE-5 inhibitor exposure.
- **Clinical Pearl:** Opioids may delay absorption of oral P2Y12 agents; assess need selectively and do not use pain relief as proof of reperfusion.

أعط aspirin عند الاستطباب وبعد فحص الموانع، لكن P2Y12 وanticoagulation يعتمدون على تشخيص ACS وخطة PCI والنزف. الاشتباه بـdissection يغير القرار.

Topic 6: Reperfusion and invasive strategy

- **Management priority:** Primary PCI is preferred for an appropriate STEMI/acute occlusion when available within the system target.
- **Management priority:** If timely PCI is unavailable, assess fibrinolysis only for an eligible STEMI patient, within the specified symptom window and after contraindication screening.
- **Exam trap:** Fibrinolysis is not used for unstable angina or NSTEMI in the supplied pathway.
- **Clinical trigger:** Hemodynamic instability, refractory/recurrent ischemia or malignant arrhythmia in NSTE-ACS warrants urgent specialist invasive assessment.
- **Management priority:** Transfer/reperfusion handover should include symptom onset, ECG evolution, biomarker timing, treatments, contraindications and bleeding risk.
- **Source qualification:** Exact PCI and fibrinolysis clock targets are service- and protocol-dependent; AQRAB avoids treating obsolete historical targets as universal current rules.

قرار reperfusion يتعلق بـECG والتوقيت وإمكانية PCI وموانع fibrinolysis. لا تعطي fibrinolysis في UA أو NSTEMI.

Topic 7: Right-ventricular infarction and preload dependence

- **Clinical trigger:** Inferior STEMI with hypotension, elevated JVP and clear lungs raises concern for right-ventricular infarction.
- **Key discriminator:** Right-sided ECG leads and bedside echocardiography support assessment when RV infarction is suspected.
- **High-yield fact:** ST elevation in V4R is a characteristic source-described clue to right-ventricular involvement.

- **Core concept:** RV failure reduces forward flow through the lungs and therefore reduces left-ventricular filling.
- **Management priority:** Avoid hypotension-provoking nitrate treatment in suspected preload-dependent RV infarction; cautiously reassess any considered fluid support.
- **Differential point:** Hypotension with JVD and clear lungs also occurs with pulmonary embolism or cardiac tamponade; context and imaging distinguish mechanisms.

Inferior MI مع hypotension وJVP مرتفع ورئتين صافيتين يثير RV infarction؛ نقص امتلاء LV يجعل nitrates أو خفض preload خطرًا في هذا السياق.

Topic 8: Mechanical complications and early post-MI deterioration

- **Red flag:** Papillary muscle rupture, interventricular septal rupture and free-wall rupture are catastrophic mechanical complications of acute MI.
- **Clinical trigger:** Sudden shock, new murmur or pulmonary edema after MI requires urgent echocardiography and specialist evaluation.
- **Key discriminator:** Papillary muscle rupture may cause acute severe mitral regurgitation; mechanical rupture can produce tamponade physiology.
- **High-yield fact:** LV aneurysm after MI may be associated with persistent ECG abnormalities and mature Q waves.
- **Common mistake:** Assuming improvement in symptoms excludes a mechanical complication or recurrent ischemia.
- **Differential point:** New pleuritic post-MI pain also requires consideration of recurrent ischemia, tamponade, pulmonary embolism and infection before labeling Dressler syndrome.

ظهور murmur جديد مع shock أو pulmonary edema بعد MI لازم يفتح احتمال papillary muscle rupture أو VSD أو free-wall rupture؛ لا تفترض أنه heart failure عادي.

Arrhythmias, Conduction and Syncope

Topic 9: Rhythm-first assessment and unstable tachycardia

- **Core concept:** Determine pulse, hemodynamic stability and whether the rhythm is the cause of deterioration before choosing drugs.
- **Clinical trigger:** Rhythm-related hypotension, confusion, ischemic chest discomfort, shock or acute heart failure calls for urgent escalation.
- **Management priority:** An unstable tachyarrhythmia with a pulse generally moves to synchronized electrical treatment in the cited adult algorithms.
- **Key discriminator:** A new regular wide-complex tachycardia is treated as possible ventricular tachycardia until adequately assessed.
- **Exam trap:** An irregular or polymorphic wide-complex rhythm is not a routine adenosine candidate.
- **High-yield fact:** Sudden cardiac death is commonly associated with ventricular fibrillation in CLEAN V3.
- **Common mistake:** Failing to recognize sinus tachycardia as compensatory for sepsis or hypovolemia; treat the underlying driver.

لا تبدأ باسم rhythm فقط. أول قرار: هل اضطراب النظم يسبب hypotension أو shock أو ischemia أو pulmonary edema؟ عدم الاستقرار يستلزم علاجًا كهربائيًا مناسبًا.

Topic 10: Regular narrow-complex supraventricular tachycardia

- **Clinical trigger:** Regular narrow-complex tachycardia with abrupt palpitations may represent AVNRT or AVRT; QRS width alone does not prove the mechanism.
- **Key discriminator:** P waves may be buried within or near the prior T wave in PSVT.
- **Management priority:** Stable eligible regular narrow SVT can start with vagal maneuvers; modified Valsalva may outperform standard Valsalva.

- **Number/threshold:** The AHA 2025 adult example uses adenosine 6 mg rapid IV push with flush, followed by 12 mg if indicated.
- **Source-specific number:** The RCUK UK pathway separately lists an adenosine sequence 6, 12 and 18 mg under suitable monitored conditions; it is not a universal substitute for the AHA sequence.
- **Management priority:** Instability attributable to SVT calls for synchronized cardioversion rather than repeated drug attempts.
- **Exam trap:** Adenosine is not automatically safe in irregular wide rhythms, pre-excitation or clinically important bronchospasm.
- **Clinical Pearl:** SVT may use an electrical re-entry mechanism and may be triggered by stimulants, ectopic beats or thyrotoxicosis.

في stable regular narrow SVT ابدأ بـvagal maneuver، ثم adenosine فقط ضمن المسار المناسب. إذا ظهرت instability الأولوية لـsynchronized cardioversion.

Topic 11: Atrial fibrillation and atrial flutter

- **Key discriminator:** Atrial fibrillation produces irregularly irregular RR intervals without distinct organized P waves.
- **Clinical Pearl:** The atrial JVP 'a' wave may be absent in atrial fibrillation because coordinated atrial contraction is absent.
- **Number/threshold:** Atrial flutter has an atrial rate commonly cited around 250–300/min; a 2:1 response often produces a ventricular rate near 150/min.
- **Management priority:** Unstable AF with poor perfusion, hypotension or altered consciousness may require synchronized cardioversion.
- **Management priority:** For stable AF lasting over 24 hours or of uncertain onset, consider thromboembolic protection before elective cardioversion.
- **Source-specific timing:** One AQRAB/UK pathway describes three weeks of anticoagulation before and four weeks after planned cardioversion, with specialist exceptions.
- **Clinical Pearl:** AF can produce systemic embolism; acute limb ischemia is an important possible manifestation.
- **Exam trap:** Rhythm rate control and stroke-prevention anticoagulation are separate decisions.

AF يتميز بـirregularly irregular rhythm بدون P waves واضحة. في المريض غير المستقر عالج الحالة كهربائيًا، أما المستقر فراجع مدة النظم وخطر الخثار.

Topic 12: WPW and pre-excited atrial fibrillation

- **Key discriminator:** Classic pre-excitation may show short PR interval, delta wave and mildly widened QRS in sinus rhythm.
- **Clinical trigger:** Very rapid or irregular wide-complex tachycardia suggests possible pre-excited AF.
- **Exam trap:** Adenosine, beta-blockers, non-dihydropyridine calcium-channel blockers and digoxin should not be used for AF with WPW in the supplied sources.
- **Management priority:** CLEAN V3 describes procainamide or electrical cardioversion for suitable WPW-associated tachyarrhythmia, depending on instability and context.
- **Management priority:** In an unstable patient, electrical treatment and expert-level rhythm assessment take precedence over uncertain drug treatment.
- **Common mistake:** Treating all wide-complex rhythms as ordinary SVT with aberrancy.

في pre-excited AF لا تستخدم isolated AV-nodal blockade؛ قد يمر التوصيل بسرعة عبر accessory pathway ويؤدي لتدهور خطير.

Ordinary AF versus pre-excited AF

Feature	Ordinary AF	Pre-excited AF	Exam point
Rhythm	Irregular RR, no organized P	May be very rapid, irregular and wide	Consider accessory-pathway conduction
AV-nodal drugs	May be used in selected usual	Avoid isolated AV-nodal	Do not give routine adenosine

	AF pathways	blockade	
Instability	Synchronized cardioversion	Urgent electrical/expert pathway	Perfusion comes first

Topic 13: Ventricular tachycardia, polymorphic rhythms and torsades

- **Key discriminator:** Capture beats or fusion beats in a wide-complex tachycardia support ventricular tachycardia.
- **Red flag:** Long QT increases the risk of torsades de pointes, a potentially life-threatening polymorphic ventricular arrhythmia.
- **Management priority:** For acquired torsades, stop the causative exposure and correct electrolytes according to the cited pathway.
- **High-yield fact:** Magnesium is the source-described intervention for torsades associated with QT prolongation.
- **Source-specific number:** CLEAN V3 lists 120–150 J biphasic synchronized cardioversion for some VT-with-pulse scenarios; use the rhythm/device-specific resuscitation pathway and do not interchange this figure with other rhythms.
- **Exam trap:** A polymorphic or irregular wide rhythm is not a default indication for adenosine.
- **Clinical Pearl:** Bidirectional VT can occur in digoxin toxicity and the source indicates digoxin-specific antibody fragments for severe toxicity.

Wide-complex tachycardia خصوصًا مع capture أو fusion beats قد تكون VT. وجود prolonged QT مع polymorphic VT يغير التشخيص والعلاج.

Topic 14: Mobitz I, Mobitz II, 2:1 and complete AV block

- **Key discriminator:** Mobitz I has progressively increasing PR interval before a nonconducted P wave.
- **Key discriminator:** Mobitz II has constant conducted PR intervals before intermittent dropped QRS complexes.
- **Key discriminator:** Complete AV block shows atrial and ventricular activity without a consistent association.
- **Exam trap:** A 2:1 conduction pattern alone cannot reliably be categorized as Mobitz I or II.
- **Red flag:** Mobitz II, a broad escape rhythm, syncope or poor perfusion demands monitoring and pacing readiness.
- **Number/threshold:** First-degree AV block is defined in CLEAN V3 by PR >0.20 seconds.
- **Management priority:** Assess perfusion, medications, ischemia, oxygenation and electrolytes; treat unstable conduction disease without waiting for a label alone.

في Wenckebach يطول PR تدريجيًا ثم تسقط P، أما Mobitz II فالـPR ثابت بين الضربات الموصلة. 2:1 وحده لا يكفي لتحديد النوع.

Mobitz I versus Mobitz II

Feature	Mobitz I	Mobitz II	Exam point
PR sequence	Progressively lengthens	Fixed for conducted beats	Dropped QRS alone is inadequate
Risk	Evaluate clinical perfusion	Infranodal disease/pacing risk	Monitor and prepare pacing
2:1 block	Cannot type from ratio alone	Cannot type from ratio alone	Do not overclaim block subtype

Topic 15: Symptomatic bradycardia and temporary pacing

- **Clinical trigger:** Persistent bradycardia with hypotension, shock, ischemic discomfort, acute heart failure or altered mental status warrants action.
- **Number/threshold:** The AHA 2025 adult symptomatic-bradycardia algorithm in AQRAB uses atropine 1 mg IV, repeated every 3–5 minutes to a maximum of 3 mg.
- **Source-specific number:** A separate RCUK pathway in AQRAB lists atropine 500 micrograms repeated every 3–5 minutes to 3 mg; do not merge these two protocol doses.

- **Management priority:** If atropine is ineffective or advanced block is unstable, prepare transcutaneous pacing and/or vasoactive infusion with expert consultation.
- **Management priority:** Transvenous pacing is described for persistent hemodynamic compromise from bradyarrhythmia.
- **Clinical Pearl:** Junctional bradycardia may be regular, narrow-complex, at 40–60/min without discernible sinus P waves.
- **Common mistake:** Assuming advanced infranodal conduction block will always respond to atropine.

الدواء وحده قد لا يكفي في high-grade AV block. قيّم perfusion واستعد لـtranscutaneous أو transvenous pacing إذا استمر عدم الاستقرار.

Topic 16: Syncope, inherited arrhythmia and ectopy

- **Red flag:** Exertional or unwarned syncope, structural heart disease and abnormal ECG make an apparently brief episode potentially high risk.
- **Key discriminator:** Brugada syndrome is linked in CLEAN V3 to a cardiac sodium-channel disorder and risk of ventricular fibrillation.
- **Clinical trigger:** A family history of unexplained sudden death under age 40 is a concerning transient-loss-of-consciousness feature in the cited source.
- **Clinical Pearl:** Premature atrial and ventricular contractions are not equivalent; structural disease or syncope changes the importance of ectopy.
- **High-yield fact:** Accelerated idioventricular rhythm can occur after reperfusion; the supplied rate description is approximately 60–120/min.
- **Exam trap:** Not every PVC needs amiodarone and not every PAC indicates anticoagulation.

الإغماء أثناء المجهود، أو دون إنذار، أو مع ECG غير طبيعي يحتاج تقييمًا قلبيًا حتى لو استعاد المريض وعيه بسرعة.

Topic 17: Pacemaker sensing and mechanical-support clues

- **Key discriminator:** Failure to sense may produce pacemaker spikes at inappropriate points rather than responding to the patient's intrinsic beats.
- **Clinical trigger:** Bradycardia with pacemaker malfunction and poor perfusion needs device assessment and pacing-capable escalation.
- **Key discriminator:** Continuous-flow LVAD support can produce low pulse pressure; a low cuff systolic reading must be interpreted with device and perfusion context.
- **Management priority:** Suspected LVAD complication warrants heart-failure/mechanical-support team contact while continuing clinical assessment.
- **Common mistake:** Treating a monitor or BP number without assessing the patient's actual perfusion and device status.

في pacemaker failure to sense قد ترى pacing spikes في أماكن غير مناسبة من الـECG. وفي LVAD المستمر قد يكون pulse pressure منخفضًا رغم وجود دوران.

Heart Failure, Cardiomyopathy and Structural Support

Topic 18: Acute decompensated heart failure and cardiogenic pulmonary edema

- **Core concept:** Heart failure may present as inadequate forward flow with raised filling pressures, pulmonary congestion and systemic edema.
- **Clinical trigger:** Dyspnea, tachycardia, crackles, peripheral edema, raised JVP and occasionally pink frothy sputum suggest acute cardiogenic pulmonary edema.
- **Key discriminator:** AQRAB recommends assessing perfusion and congestion together; crackles alone do not determine fluid responsiveness or the need for fluid.
- **High-yield fact:** S3 gallop is associated with heart failure; a source example describes three or more B lines in multiple lung zones during acute heart failure.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 36

6 mg	Number/threshold: The AHA 2025 adult example uses adenosine 6 mg rapid IV push with flush, followed by 12 mg if indicated.
12 mg	Number/threshold: The AHA 2025 adult example uses adenosine 6 mg rapid IV push with flush, followed by 12 mg if indicated. Source-specific number: The RCUK UK pathway separately lists an adenosine sequence 6, 12 and 18 mg
1 mg	altered mental status warrants action. Number/threshold: The AHA 2025 adult symptomatic-bradycardia algorithm in AQRAB uses atropine 1 mg IV, repeated every 3–5 minutes to a maximum of 3 mg.
3 mg	Number/threshold: The AHA 2025 adult symptomatic-bradycardia algorithm in AQRAB uses atropine 1 mg IV, repeated every 3–5 minutes to a maximum of 3 mg. Source-specific number: A separate RCUK pathway in AQRAB lists atropine 500 micrograms repeated
3 mg	Source-specific number: A separate RCUK pathway in AQRAB lists atropine 500 micrograms repeated every 3–5 minutes to 3 mg; do not merge these two protocol doses.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 36

Acute coronary syndromes - diffuse ST depression and aVR elevation [#010]

High-yield fact: Diffuse ST depression with ST elevation in aVR is a high-risk ischemic pattern that warrants urgent assessment.

هبوط ST المنتشر مع ارتفاعه في aVR نمط إقفاري عالي الخطورة يستلزم تقييمًا عاجلًا.

Congenital long-QT syndromes [#014]

High-yield fact: Congenital long-QT syndrome can be associated with hearing impairment in Jervell-Lange-Nielsen syndrome. .Jervell–Lange-Nielsen.

ارتباط QT الطويل بضعف السمع يذكّر بمتلازمة Jervell–Lange-Nielsen.

Posterior myocardial infarction - ECG recognition [#017]

Key discriminator: Posterior myocardial infarction can produce anterior ST depression and should prompt assessment with posterior leads.

هبوط ST في الاشتقاقات الأمامية قد يكون انعكاسًا لاحتشاء خلفي؛ تفيد الأقطاب الخلفية عند الشك.

- **Number/threshold:** CLEAN V3 cites BNP below 100 as a diagnostic aid against acute HF as the main cause of dyspnea; do not apply this number without clinical context.
- **Management priority:** Search for precipitating myocardial ischemia, arrhythmia, hypertensive crisis, infection, renal dysfunction or acute valve pathology.
- **Management priority:** Use diuresis, vasodilator or appropriate positive-pressure support according to congestion, blood pressure and clinical response.
- **Clinical Pearl:** NIV may improve clinical outcomes in selected acute heart-failure presentations in the supplied material.
- **Exam trap:** Initial response to diuretic or mask ventilation does not establish safe discharge; reassess oxygenation and organ perfusion.

اربط acute HF في dyspnea وcrackles وJVP وperfusion بالسونار والـECG، ولا تجعل تحسن التنفس بعد أول علاج سببًا آليًا للخروج.

Topic 19: Diuretics, nitrates and ventilatory support in heart failure

- **Source-specific dose:** AQRAB quotes an ESC 2026 example of IV furosemide 40 mg initially in a patient not taking chronic diuretics; titration depends on renal status and prior diuretic use.
- **Source-specific dose:** The source describes IV nitroglycerin initiation at 10 micrograms/min in an ACS-with-edema example, monitored and titrated; this is not mg/min.
- **Clinical Pearl:** Positive-pressure ventilation may improve pulmonary mechanics but can reduce venous return and blood pressure.
- **Common mistake:** Morphine is not a default routine treatment for pulmonary edema; assess respiratory depression and hypotension risks.
- **Clinical Pearl:** Prehospital morphine for presumed pulmonary edema was associated in a CLEAN V3 source passage with more intubation, especially if the diagnosis was mistaken.
- **Management priority:** Reassess breathing, BP, urine output and congestion after treatment rather than automatically escalating diuresis.

فكر بالهدف العلاجي: diuretic عند congestion وnitrate إذا الضغط يسمح، وNIV لتحسين التنفس. راقب الضغط والتروية ولا تجعل morphine روتينيًا.

Topic 20: Cardiogenic shock and ischemic pump failure

- **Core concept:** Cardiogenic shock is failure to maintain effective forward flow despite adequate or increased circulating volume.
- **Clinical trigger:** Hypotension with cool extremities, elevated JVP, pulmonary crackles or oliguria may indicate pump failure.
- **Differential point:** Hypotension and clear lungs with raised JVP should redirect assessment toward RV infarction, PE or tamponade.
- **Management priority:** Identify acute MI, myocarditis, severe dysrhythmia or acute valvular failure and seek definitive treatment.
- **Clinical Pearl:** Vasopressors can support pressure and inotropes selected output, but neither repairs an occluded artery or torn valve.
- **Red flag:** A new murmur or sudden deterioration after MI suggests papillary muscle or septal rupture and requires urgent echocardiography.

Cardiogenic shock هو مشكلة forward flow مع congestion، وقد يكون سببه MI أو acute valve lesion. vasopressor يدعم التروية لكنه لا يصلح سببًا ميكانيكيًا.

Topic 21: Dilated, stress-induced and high-output cardiomyopathy

- **Core concept:** Dilated cardiomyopathy has ventricular enlargement and impaired contractility with a broad etiologic differential.
- **Key discriminator:** Takotsubo stress cardiomyopathy often follows substantial emotional or physical stress; the source highlights presentations in postmenopausal women.

- **High-yield fact:** High-output heart failure features increased cardiac output despite reduced systemic vascular resistance.
- **Differential point:** An elevated cardiac output does not exclude a clinical syndrome of heart failure when vascular demand and resistance are abnormal.
- **Management priority:** Coronary ischemia and structural causes should still be assessed when chest pain, shock or acute failure are present.
- **Clinical Pearl:** HCM can produce characteristic LVH ECG changes and deep lateral Q waves; source recommends echocardiography for assessment.

Dilated cardiomyopathy يسبب توسع وضعف انقباض، وTakotsubo قد يتبع High-output HF stress. مختلف

لأنه يجمع CO مرتفعًا مع SVR منخفضة.

Topic 22: LVAD and device-dependent circulation

- **Core concept:** Continuous-flow LVADs may generate very low pulse pressure; conventional noninvasive blood pressure can be misleading.
- **Clinical trigger:** An LVAD patient with new dyspnea, shock, syncope or possible device alarm requires device-aware clinical assessment.
- **Management priority:** Contact the heart failure/LVAD team and continue evaluation rather than assuming low numerical systolic pressure necessarily equals absent perfusion.
- **High-yield fact:** CLEAN V3 describes LVAD implantation for selected end-stage heart failure populations.
- **Exam trap:** Device and circulation assessment must accompany, not replace, the evaluation of ischemia, bleeding and infection when suggested by symptoms.

مريض LVAD قد يملك pulse pressure ضعيفًا؛ لا تفسر الضغط المطبوع دون تقدير التروية وفحص الجهاز

واستدعاء فريق المساندة القلبية.

Aortic and Peripheral Vascular Emergencies

Topic 23: Acute aortic syndrome: recognition and imaging

- **Clinical trigger:** Sudden severe chest, back or abdominal pain, syncope, neurologic deficit, pulse difference or shock raises suspicion for acute aortic syndrome.
- **Core concept:** Acute aortic syndrome includes dissection, intramural hematoma and penetrating aortic ulcer.
- **Key discriminator:** Compare extremity pulse/BP and assess cerebral, coronary, renal, mesenteric and limb malperfusion.
- **Management priority:** CTA is the usual definitive rapid imaging when the patient can undergo CT; use TEE or MRI in appropriate contexts.
- **Exam trap:** Normal chest radiograph, limited TTE or isolated D-dimer does not exclude high-risk aortic dissection.
- **Red flag:** Aortic root involvement can cause acute regurgitation, tamponade and coronary compromise.
- **Management priority:** Engage an aortic/cardiothoracic team early when clinical suspicion is substantial.

لا تشترط وصف tearing pain. الألم المفاجئ أو syncope أو neurologic deficit أو pulse asymmetry مع خطر

مناسب يكفي لرفع الاشتباه بـAAS.

Topic 24: Dissection anti-impulse treatment and type A/B decisions

- **Management priority:** In appropriate hypertensive AAS, use monitored anti-impulse therapy to reduce aortic wall stress and obtain urgent specialist input.
- **Number/threshold:** AQRAB and CLEAN V3 cite a heart-rate target of 60–80/min for appropriate hypertensive dissection.
- **Number/threshold:** The cited ACC/AHA strategy targets SBP below 120 mmHg or the lowest pressure preserving organ perfusion, not a rigid pressure for shock.

- **Management priority:** IV beta blockade generally precedes addition of a titratable vasodilator when appropriate; also manage severe pain.
- **Exam trap:** A vasodilator started before rate control may cause reflex tachycardia and increased aortic wall stress.
- **Key discriminator:** Stanford type A involves the ascending aorta and usually needs emergency surgery; uncomplicated type B initially uses closely monitored medical management.
- **Red flag:** Complicated type B with rupture, malperfusion, progression, refractory pain or uncontrolled hypertension needs intervention assessment.
- **Clinical Pearl:** A new early diastolic aortic-regurgitation murmur can accompany proximal dissection.

في hypertensive dissection يبدأ ضبط HR/contractility بمراقبة ثم vasodilator إذا لزم، لكن لا تطبق SBP <120 على المريض المصاب بـshock.

Stanford type A versus type B dissection

Feature	Type A	Uncomplicated type B	Exam point
Anatomy	Ascending aorta involved	Descending aorta	Ascending involvement changes urgency
Usual strategy	Emergency surgical pathway	Medical therapy with monitoring	Complicated B also merits intervention assessment
Key danger	Tamponade, AR, coronary malperfusion	Malperfusion, rupture or progression	Assess target organs, not label alone

Topic 25: Ruptured abdominal aortic aneurysm

- **Clinical trigger:** Older or vascular-risk patient with abrupt abdominal, back, flank or groin pain, collapse or unexplained shock may have ruptured AAA.
- **Exam trap:** Absence of a palpable pulsatile mass or an initially normal hemoglobin does not rule out rupture.
- **Management priority:** Notify vascular surgery early, activate transfer and major hemorrhage resources, and establish appropriate access and laboratory preparation without delaying definitive care.
- **Key discriminator:** Bedside ultrasound can demonstrate aneurysmal dilation but cannot reliably confirm or exclude rupture.
- **Management priority:** Obtain urgent CTA for anatomy and repair planning when hemodynamically feasible; in a deteriorating patient prioritize vascular control.
- **Source-specific threshold:** AQRAB describes a permissive SBP of 70–90 mmHg while conscious in a cited low-certainty ESVS rAAA approach; it is not a universal target.
- **Management priority:** EVAR or open repair depends on anatomy, physiology, available expertise and vascular-team assessment.
- **Clinical Pearl:** Previous AAA repair and GI bleeding can indicate an aortoenteric fistula, a surgical vascular emergency.

AAA المتمزق قد لا يظهر بالـclassic triad. POCUS يكشف التوسع ولا يثبت التمزق؛ في shock لا تؤخر السيطرة على النزف بانتظار تصوير روتيني.

Topic 26: Acute limb ischemia and threatened viability

- **Clinical trigger:** Sudden cold painful limb, pallor, pulselessness, paresthesia or paralysis suggests acute arterial compromise.
- **Core concept:** In the cited definition, acute limb ischemia refers to abrupt perfusion loss with symptoms generally under two weeks.
- **Management priority:** Record bilateral color, temperature, pulses, capillary refill, sensory and motor examination and arterial/venous Doppler findings.
- **Red flag:** New sensory loss or motor weakness indicates potentially threatened limb viability and needs emergency vascular assessment.
- **Exam trap:** A palpable proximal pulse does not exclude distal arterial occlusion; do not rely on one ABI or pulse alone.
- **Management priority:** The source cites therapeutic IV unfractionated heparin at diagnosis when no contraindication exists while revascularization is evaluated.

- **Differential point:** Assess dissection, major bleeding, vascular trauma or HIT before blindly applying anticoagulation.
- **Clinical Pearl:** Following reperfusion, monitor compartment syndrome, hyperkalemia, rhabdomyolysis, acidosis and renal injury.
- **Exam trap:** Pulses may still be present in compartment syndrome after revascularization.

الأهم في ALI هو sensory loss وweakness وليس مجرد وجود نبض قريب. لا تؤخر vascular surgery بسبب فحص غير ضروري.

Topic 27: Deep vein thrombosis: assessment and testing

- **Clinical trigger:** Unilateral swelling, calf discomfort and tenderness raise suspicion for DVT but are not sufficiently specific for diagnosis.
- **Differential point:** Consider cellulitis, muscle injury, ruptured Baker cyst, venous insufficiency or proximal pelvic/iliac obstruction.
- **Core concept:** Use pretest probability to select D-dimer and venous ultrasound; examine for accompanying PE symptoms or threatened circulation.
- **Number/threshold:** The NICE two-level DVT Wells score cited in AQRAB considers 2 or greater likely and 1 or less unlikely.
- **Source-specific timing:** Under the cited NICE pathway, probable DVT merits proximal ultrasound within 4 hours when possible, or interim anticoagulation and ultrasound within 24 hours as appropriate.
- **Source-specific timing:** A negative proximal ultrasound with a positive D-dimer leads to repeat proximal ultrasound in 6–8 days in the cited NICE pathway.
- **Exam trap:** A negative limited proximal venous scan is not equivalent to a whole-leg scan, and a positive D-dimer does not diagnose DVT.
- **Red flag:** Marked painful cyanotic swelling, neurologic deficit or shock raises concern for phlegmasia or significant VTE complications.

لا يثبت DVT بالنظر إلى الساق. Wells يحدد مسار D-dimer وcompression ultrasound، وإذا كان الفحص proximal فقط قد تحتاج إعادة حسب المسار.

Topic 28: Proximal, distal DVT and anticoagulation decisions

- **Management priority:** Confirmed proximal DVT is generally anticoagulated unless contraindicated; AQRAB cites a minimum three-month treatment period in the NICE pathway.
- **Clinical Pearl:** Selected isolated distal calf DVT without severe symptoms or extension risk may have serial imaging for two weeks rather than immediate anticoagulation, according to the cited CHEST pathway.
- **Management priority:** Severe symptoms, high extension risk or progression of isolated distal DVT favor anticoagulation after evaluating bleeding and follow-up.
- **Source-specific dose:** For acute adult DVT/PE, the AQRAB-cited apixaban label uses 10 mg PO twice daily for seven days, then 5 mg twice daily.
- **Exam trap:** Acute VTE treatment is not the same dose framework as reduced-dose atrial-fibrillation prevention or extended VTE prevention.
- **Clinical Pearl:** Renal dysfunction, pregnancy, cancer, APS, drug interactions, bleeding and planned procedures may change drug selection.

لا تستخدم جرعة AF من apixaban لعلاج acute DVT. أما isolated distal calf DVT فقد يناسبها ultrasound في حالات مختارة بشرط متابعة موثوقة. surveillance

Proximal versus selected distal DVT

Feature	Proximal DVT	Isolated distal calf DVT	Exam point
Default strategy	Anticoagulate if eligible	Some cases eligible for 2-week ultrasound surveillance	Extension risk and follow-up decide
Escalation	Assess PE and bleeding risk	Severe symptoms/extension favor treatment	Imaging plan is part of safe care

Topic 29: Superficial vein thrombosis and deep junction risk

- **Key discriminator:** A palpable tender red superficial cord suggests SVT but does not exclude coexisting DVT.
- **Management priority:** Use ultrasound when needed to measure clot length, distance to the deep-vein junction and associated deep thrombosis.
- **Source-specific dose:** For suitable lower-limb SVT at least 5 cm long and at least 3 cm from a deep junction, AQRAB cites fondaparinux 2.5 mg SC daily for 45 days.
- **Source-specific threshold:** SVT within 3 cm of a deep-vein junction is treated as higher risk; the cited ESVS pathway describes therapeutic-dose anticoagulation generally for three months.
- **Exam trap:** A superficial-vein photograph cannot establish deep clot extent; anatomy and follow-up matter more than appearance alone.
- **Management priority:** Safe outpatient care requires clinical stability, access to therapy and reliable surveillance.

SVT بوجود cord مؤلم لا يعني عدم وجود DVT. المهم طول الخثرة وبعدها عن deep-vein junction؛ هذه التفاصيل تحدد مسار العلاج.

DVT versus superficial vein thrombosis

Feature	DVT	SVT	Exam point
Anatomy	Deep venous clot	Superficial thrombosis	Ultrasound defines extension
Test	Wells-guided D-dimer/ultrasound	Clot length and junction distance	A palpable cord does not exclude DVT
Management	Proximal often anticoagulated	Risk-dependent dose/duration	Do not substitute AF doses for VTE

Topic 30: Hypertensive emergency versus severe BP without organ injury

- **Core concept:** Hypertensive emergency is elevated BP with acute or worsening target-organ injury; the number alone is insufficient.
- **Clinical trigger:** Evaluate encephalopathy, intracranial injury, ACS, acute HF, aortic disease, renal injury, retinopathy and pregnancy-related injury.
- **Management priority:** Repeat BP with correct technique, review medications, pain, substances and baseline pressure, and examine for relevant organ injury.
- **Exam trap:** A reading around 180/120 mmHg or a mild isolated headache does not by itself establish an emergency.
- **Management priority:** Without acute injury, do not reflexively use rapid IV pressure lowering; address contributors and plan appropriate follow-up.
- **Management priority:** With target-organ injury, provide monitored, titratable, syndrome-specific treatment rather than a single universal BP reduction target.
- **Clinical Pearl:** Aortic dissection, stroke, ACS, pulmonary edema and obstetric hypertensive disease require different pathways.

Hypertensive emergency يعني acute target-organ injury، وليس مجرد BP مرتفع. Mild headache وحده أو رقم 180/120 لا يثبتان حالة إسعافية.

Severe BP versus hypertensive emergency

Feature	High BP without injury	Hypertensive emergency	Exam point
Defining factor	No acute target-organ injury	Acute or worsening organ injury	180/120 alone is not diagnostic
Immediate action	Confirm measurement, evaluate causes	Monitored syndrome-specific therapy	Avoid one universal rapid-lowering target
Follow-up	Outpatient plan and continuity	Admission/escalation as indicated	Pregnancy has separate severe-range pathway

Topic 31: Pregnancy-specific severe-range hypertension

- **Number/threshold:** The AQRAB-cited obstetric severe BP range is SBP at least 160 mmHg or DBP at least 110 mmHg, confirmed on repeat measurement within 15 minutes.
- **Source-specific timing:** The cited pathway aims for pressure below 160/110 within 30–60 minutes using appropriate obstetric management.
- **Management priority:** Labetalol IV, hydralazine IV/IM or immediate-release oral nifedipine are guideline-listed options with contraindication assessment and monitoring.
- **Exam trap:** Magnesium sulfate is used for eclampsia seizure prevention/treatment when indicated, not as a substitute antihypertensive.
- **Source qualification:** The exact pregnancy drug doses in AQRAB p.69 are protocol examples; do not extrapolate immediate-release nifedipine to unrelated adult hypertensive emergencies.

أرقام الحمل مختلفة: استمرار SBP ≥160 أو DBP ≥110 بعد إعادة القياس يحتاج مسار obstetric عاجل.

Magnesium للـseizure وليس بديلًا عن خفض الضغط.

Pericardial and Valvular Emergencies

Topic 32: Cardiac tamponade: integrated hemodynamic diagnosis

- **Core concept:** Pericardial pressure impairs filling and output; tamponade is not defined by a fixed effusion volume.
- **Clinical trigger:** Hypotension, tachycardia, dyspnea, raised JVP and suspected effusion may indicate tamponade, but the full Beck triad is often absent.
- **Key discriminator:** Echo may show right-atrial systolic collapse, right-ventricular early-diastolic collapse, plethoric IVC or respiratory Doppler variation; findings depend on physiology.
- **Exam trap:** Electrical alternans, low-voltage ECG or enlarged cardiac silhouette is neither necessary nor sufficient to prove tamponade.
- **Management priority:** Hemodynamically unstable suspected tamponade needs immediate specialist drainage planning, not repeated temporizing fluid boluses.
- **Clinical Pearl:** A cautious temporary preload-supporting fluid bolus may bridge urgent drainage; avoid diuresis and positive-pressure ventilation where feasible.
- **Red flag:** Traumatic or procedure-related hemopericardium may need surgical management rather than routine medical pericardiocentesis.

Tamponade تشخيص physiology: لا حجم السائل ولا Beck triad ولا electrical alternans لوحدها تحدد وجوده.

الأهم perfusion مع echo.

Topic 33: Acute pericarditis recognition and differential

- **Clinical trigger:** Sharp pleuritic pain worse supine and relieved by sitting forward with possible pericardial friction rub suggests pericarditis.
- **Key discriminator:** Widespread concave ST elevation and PR depression with PR elevation in aVR can support pericarditis, but patterns vary.
- **Exam trap:** The ECG morphology or absence of an effusion cannot alone diagnose or exclude pericarditis.
- **Differential point:** Consider ACS, myocarditis, aortic disease, PE, pneumothorax and pneumonia according to context.
- **Management priority:** Assess hemodynamics, troponin/myocardial involvement and echo when effusion or structural disease is suspected.
- **Red flag:** Fever above 38 C, subacute onset, effusion above 20 mm, tamponade or poor NSAID response after at least a week are source-cited high-risk markers.
- **Clinical Pearl:** Immunodepression, trauma, anticoagulant use and myocardial involvement are additional source-cited caution features.

Pleuritic positional pain وdiffuse ST elevation/PR depression تدعم pericarditis لكنها لا تستبعد ACS. افحص effusion وmyocardial injury وعلامات الخطورة.

Pericarditis versus ACS

Feature	Pericarditis	Acute coronary ischemia	Exam point
Clinical clue	Pleuritic/positional discomfort	Variable ischemic symptoms	Symptoms alone cannot close differential
ECG	May have diffuse ST/PR changes	May have focal/occlusion ECG patterns	A single pattern is not definitive
Urgent safety check	Effusion/tamponade/ myocardial injury	Serial ECG/troponin and reperfusion decisions	Avoid false reassurance from morphology

Topic 34: Pericarditis treatment and post-cardiac injury syndrome

- **Management priority:** For suitable presumed nonbacterial first-episode pericarditis, aspirin or an NSAID plus colchicine is the usual source-described initial approach.
- **Clinical Pearl:** Colchicine lowers recurrence risk; check renal, bleeding, anticoagulant, pregnancy and other medication contraindications.
- **Exam trap:** Systemic corticosteroids are not routine first-line therapy for an uncomplicated first episode in the cited material.
- **High-yield fact:** Dressler syndrome is a post-cardiac injury inflammatory pericarditis occurring weeks to months after MI or another cardiac injury.
- **Red flag:** After MI, recurrent ischemia, mechanical rupture/tamponade, PE and infection must be evaluated before diagnosing Dressler.
- **Management priority:** Hemodynamic compromise from tamponade requires urgent intervention; anti-inflammatory medication alone is insufficient.

uncomplicated pericarditis في يعتمد العلاج المذكور على NSAID أو aspirin مع colchicine إذا مناسب. بعد MI استبعد recurrent ischemia والتمزق قبل اعتبار الألم Dressler.

Topic 35: Acute mitral and aortic regurgitation

- **Core concept:** Mitral regurgitation is abnormal systolic backward flow from left ventricle to left atrium.
- **Clinical trigger:** Acute severe valve regurgitation may present with abrupt pulmonary edema, circulatory collapse or shock.
- **Key discriminator:** Pulmonary edema is a major source-described chest radiograph finding with acute mitral regurgitation.
- **Red flag:** Papillary muscle rupture after MI may cause acute severe MR.
- **Red flag:** Ascending aortic dissection can compromise the aortic root and produce acute aortic regurgitation.
- **Management priority:** Urgent echocardiography and surgical/cardiology expertise are required when severe acute valvular dysfunction is suspected.
- **Exam trap:** A quiet murmur or a nondilated heart does not exclude a dangerous acute valve lesion.

Acute MR أو AR قد يسبب shock وpulmonary edema حتى لو murmur ليس واضحًا. ظهور نفخة جديدة بعد MI أو dissection علامة تستوجب echo عاجل.

Topic 36: Aortic stenosis, mitral stenosis and prosthetic valves

- **Core concept:** Aortic stenosis restricts left-ventricular systolic ejection through a narrowed aortic valve.
- **Key discriminator:** Aortic stenosis commonly produces a crescendo-decrescendo systolic murmur radiating to the carotids.
- **Number/threshold:** CLEAN V3 describes symptomatic severe aortic stenosis around valve orifice area 1 cm2, as a source teaching point rather than a sole clinical test.
- **Key discriminator:** Mitral stenosis can produce loud S1, opening snap and low-pitched apical diastolic rumble.
- **Management priority:** In severe AS, avoid abrupt loss of preload or blood pressure; in mitral stenosis, tachycardia can shorten filling time and worsen physiology.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 43

40 mg	Source-specific dose: AQRAB quotes an ESC 2026 example of IV furosemide 40 mg initially in a patient not taking chronic diuretics; titration depends on renal status and prior diuretic use.
10 mg	anticoagulation after evaluating bleeding and follow-up. Source-specific dose: For acute adult DVT/PE, the AQRAB-cited apixaban label uses 10 mg PO twice daily for seven days, then 5 mg twice daily.
5 mg	Source-specific dose: For acute adult DVT/PE, the AQRAB-cited apixaban label uses 10 mg PO twice daily for seven days, then 5 mg twice daily. Exam trap: Acute VTE treatment is not the same dose framework as reduced-dose atrial-fibrillation
2.5 mg	Source-specific dose: For suitable lower-limb SVT at least 5 cm long and at least 3 cm from a deep junction, AQRAB cites fondaparinux 2.5 mg SC daily for 45 days. Source-specific threshold: SVT within 3 cm of a deep-vein junction is treated as higher risk; the cited ESVS

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 43

Aortic dissection - anti-impulse therapy [#016]

Key discriminator: In aortic dissection, reduce aortic impulse with heart-rate control before isolated vasodilation when possible.

في تسلّخ الأبهر، ضبط سرعة القلب يسبق التوسع الوعائي المنفرد في سياق علاج Anti-impulse المذكور.

- **Red flag:** Prosthetic valve with dyspnea, embolism, shock or interrupted anticoagulation should prompt evaluation for dysfunction, thrombosis or infection.
- **Clinical Pearl:** Mitral valve prolapse is described as a relatively common valvular condition in the first source.

في severe AS يجب الانتباه إلى preload والضغط؛ وفي mitral stenosis يزيد tachycardia سوء الامتلاء.

prosthetic valve مع shock أو embolism يحتاج تقييم وظيفة الصمام.

Infective Endocarditis and Cardiovascular Diagnostic Principles

Topic 37: Infective endocarditis: recognition and blood cultures

- **Clinical trigger:** Fever or systemic infection with prosthetic valve/device, prior IE, injection drug use, persistent bacteremia or embolic signs raises IE suspicion.
- **Management priority:** AQRAB cites three peripheral blood-culture sets before antibiotics when feasible, without delaying resuscitation or urgent treatment of sepsis.
- **Management priority:** Empiric antimicrobial therapy depends on native/prosthetic valve status, local resistance, allergy, renal function and specialist input.
- **Exam trap:** Do not wait for a fever spike to draw cultures and do not apply one historical empiric drug regimen universally.
- **Clinical Pearl:** Duke diagnostic frameworks integrate microbiology, echo and clinical criteria; versions should not be conflated.
- **Red flag:** New regurgitation, heart failure, arrhythmia, neurologic deficits or shock may be early complications.

فكر بـIE إذا حمى أو bacteremia مع صمام صناعي أو جهاز أو IV drug use أو emboli. خذ blood cultures قبل المضاد إن أمكن دون تأخير علاج sepsis.

Topic 38: Endocarditis imaging, Duke frameworks and follow-up

- **Key discriminator:** TTE is usually initial cardiac imaging for suspected IE; TEE provides better resolution when suspicion remains high or prosthetic material complicates imaging.
- **Exam trap:** A negative initial blood culture or negative TTE does not by itself exclude infective endocarditis.
- **Source-specific timing:** The cited pathway suggests repeat echocardiography after 5–7 days, or sooner if deterioration occurs, when initial studies are nondiagnostic and suspicion remains.
- **Management priority:** Integrate microbiology, echocardiography, and the clinical diagnostic framework; advanced imaging may be appropriate in selected prosthetic/device cases.
- **Clinical Pearl:** Persistent bacteremia, abscess concern or new conduction disturbance increases the urgency of multidisciplinary review.
- **Management priority:** Coordinate infectious disease, cardiology, microbiology and cardiac surgery teams when IE is complicated or likely.

TTE أول فحص، لكن TTE سلبي لا ينفي IE مع شك قوي. استخدم TEE أو repeat imaging، خصوصًا مع prosthetic valve أو persistent bacteremia.

Topic 39: Embolic and structural complications of endocarditis

- **Key discriminator:** Right-sided IE more often causes pulmonary septic emboli, whereas left-sided IE more often causes systemic embolism or valvular failure.
- **Red flag:** Stroke, limb ischemia, splenic/renal infarct or new heart failure in the context of IE requires urgent complication assessment.
- **Clinical trigger:** New AV conduction abnormality may raise concern for perivalvular extension or abscess.
- **Exam trap:** Right/left embolic patterns are tendencies, not absolute rules; clinical findings and imaging remain necessary.

PASS 9 — Source-backed clinical recovery | Domain 02

Cardiovascular Emergencies

dosing or medical-correctness certification.

Coronary CT angiography: examination limitations

Exam limitation: CLEAN V3 notes that coronary CT angiography can be nondiagnostic and that image quality improves when heart rate is below 65/min; the cited ~20% rate belongs to its original context.

قد تكون صور CCTA غير تشخيصية؛ والرقم 20% وارد في مثال المصدر ولا يعمَّم على كل الأجهزة أو المرضى.

- **Management priority:** Prosthetic material, persistent infection or uncontrolled structural disease warrants endocarditis-team and surgical evaluation.

Right-sided IE غالبًا يعطي septic pulmonary emboli، أما left-sided IE غالبًا يسبب emboli جهازية؛ هذا نمط شائع وليس قاعدة مطلقة.

Topic 40: Bedside echocardiography and cardiovascular discriminators

- **Core concept:** Bedside echo complements hemodynamic examination by assessing ventricular function, pericardial fluid and selected structural abnormalities.
- **Exam trap:** Normal limited cardiac POCUS alone does not exclude high-risk aortic dissection.
- **Exam trap:** An effusion by itself does not prove tamponade; hemodynamic consequences and chamber findings matter.
- **Key discriminator:** JVP, lung findings, extremity perfusion, ECG and focused echo should be interpreted together in shock.
- **Clinical Pearl:** A suspected structural cardiomyopathy or prosthetic valve abnormality often requires formal echocardiography and specialist evaluation.

POCUS يساعد في تشخيص نوع shock وتقييم القلب لكنه ليس اختبارًا شاملًا لنفي dissection أو PE أو tamponade من علامة واحدة.

Rapid Recall: Must-Know Facts

01. Atypical ACS may present without classic chest pain.
02. Repeat an initially nondiagnostic ECG when ischemic symptoms persist.
03. UA and NSTEMI may look alike on ECG; serial troponin matters.
04. A negative early troponin does not exclude MI.
05. Troponin elevation alone does not prove type 1 MI.
06. Recognize Wellens, de Winter and posterior occlusion patterns.
07. New LBBB alone does not establish STEMI.
08. Do not postpone indicated STEMI reperfusion for biomarkers.
09. Do not fibrinolyse UA or NSTEMI in the cited pathway.
10. Before antithrombotics, reconsider dissection when its clues appear.
11. Hypotension + high JVP + clear lungs with inferior MI suggests RV infarction.
12. New murmur and shock after MI suggest mechanical rupture.
13. Assess rhythm-related instability before naming the rhythm.
14. Unstable tachyarrhythmia with pulse needs electrical-pathway assessment.
15. Stable regular SVT: vagal maneuvers first when appropriate.
16. Never assume irregular wide-complex tachycardia is ordinary SVT.
17. Atrial fibrillation is irregularly irregular without organized P waves.
18. Avoid isolated AV-nodal blockade in pre-excited AF.
19. Capture or fusion beats support VT in a wide rhythm.
20. Acquired torsades calls for trigger and electrolyte correction.
21. Mobitz I lengthens PR before a dropped QRS.
22. Mobitz II maintains conducted PR before a dropped QRS.
23. A 2:1 AV block cannot be subclassified from its ratio alone.
24. Unstable high-grade AV block may require pacing.
25. Exertional syncope and structural heart disease are cardiac red flags.
26. Pulmonary crackles alone cannot determine fluid therapy.
27. Acute valve regurgitation may cause abrupt pulmonary edema and shock.
28. Continuous-flow LVADs can produce very low pulse pressure.
29. Dissection can present without a classic tearing-pain description.
30. CTA commonly defines AAS when the patient is stable enough.
31. Hypertensive AAS: control impulse before adding a vasodilator when suitable.
32. Type A dissection usually needs emergency surgery.

33. POCUS aneurysm dilation does not prove AAA rupture.
34. Acute limb weakness or sensory loss threatens viability.
35. A proximal pulse does not exclude distal limb ischemia.
36. NICE DVT testing begins with pretest probability.
37. A superficial venous cord does not exclude coexisting DVT.
38. Hypertensive emergency requires acute organ injury.
39. Pericardial effusion size alone cannot diagnose tamponade.
40. Negative TTE does not exclude IE when suspicion remains high.

Recall statements are condensed from the documented points in this Domain; they do not create extra clinical source facts.

Conflict Resolution Register — Domain 02

A. Direct between-file recommendation conflict, not adjudicated:

1. Routine P2Y12 pretreatment for NSTEMI

- **File 1:** CLEAN V3: In addition to aspirin, clopidogrel should be given to NSTEMI patients in the ED.
- **File 2:** AQRAB PDF pp. 30–35: P2Y12 agent and timing depend on invasive strategy, PCI plan, bleeding risk and contraindications; do not apply one pretreatment policy to everyone.
- **Nature of conflict:** The first file gives a generalized ED instruction, while the second explicitly rejects universal pretreatment and requires stratified selection.
- **Classification:** Conflict — source-dependent

B. Source-specific protocol differences and internal timing variants (not misclassified as new cross-file conflicts):

1. Adult symptomatic-bradycardia atropine protocols

- **Source statements:** AQRAB PDF pp.17–19: RCUK 500 micrograms IV versus AHA 2025 1 mg IV, both with repeats to a 3 mg maximum.
- **Disposition:** Protocol-dependent difference within AQRAB; not invented as a new between-file conflict.

2. Adenosine and synchronized-cardioversion regimes

- **Source statements:** AQRAB PDF pp.12 and 17: AHA adenosine 6 then 12 mg versus RCUK 6, 12 then 18 mg; CLEAN V3 also quotes different device/rhythm-specific initial cardioversion energies.
- **Disposition:** Separate algorithm, indication and device before selecting an energy or dose.

3. Acute-aortic-syndrome heart-rate target

- **Source statements:** AQRAB PDF p.74 notes an older HR <=60/min reference and gives a 60–80/min ACC/AHA target; CLEAN V3 cites 60–80/min.
- **Disposition:** Historical update described within AQRAB; do not treat as two simultaneous universal rules.

4. Dressler syndrome time descriptions

- **Source statements:** CLEAN V3 cites one week to months and two weeks to months after MI; AQRAB characterizes inflammation weeks after cardiac injury.
- **Disposition:** Approximate descriptions overlap; no precise onset threshold imposed.

All cases above are recorded without outside guideline selection or independent medical adjudication.

Merge Quality Note — Domain 02

Unit of count = one independently rendered English clinical bullet under a numbered Topic. Bilingual explanation, comparison cells and recall repetitions are not counted as new facts. Provenance tags are supported by the cited passages. “Exclusive” is limited to the audited cardiovascular mappings, not a proof of absence from every page of both large sources.

- **Final integrated topics:** 40
- **Integrated primary clinical English points:** 270
- **Identified first-file-only retained points:** 57
- **Identified second-file-only retained points:** 186
- **Directly corroborated shared integrated points:** 27
- **Between-file clinical recommendation conflicts registered:** 1
- **Intra-source framework/protocol variations noted:** 4
- **Clearly unusable fragments explicitly withheld:** 7

- **Rapid Recall facts (re-expression only):** 40
- **Comparison tables:** 8

Incomplete in source — not included

- **CLEAN V3:** "1.2." under Cardiovascular threshold — Isolated value without variable or indication
- **CLEAN V3:** "Age × 10 ng/mL." — Fragmentary threshold without defined assay/test context
- **CLEAN V3:** "16 vials" linked to a VT vignette — Cannot infer specific antidote, indication or dose from fragment
- **CLEAN V3:** "Anesthesia of the affected limb." — Fragment with no reliable management intent
- **CLEAN V3:** "20 breaths/min." — Unattributed numerical claim
- **CLEAN V3:** "Nitroprusside: His blood pressure is 250/170." — No clear therapeutic recommendation or dosing context
- **AQRAB PDF p.395 and other digest passages:** Some isolated questions/answer fragments without negation or context — Require full clinical question context, so no independent fact constructed

were left for their clinically appropriate Domains, and no unintelligible case fragments were completed by guessing.

because that name or recommendation was already described by one of the two supplied documents. No independent source checking was performed.

addendum count is not part of the original Merge Quality Note editorial tally. No external clinical source was consulted.

Cardiovascular Emergencies. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

3 mg	Source statements: AQRAB PDF pp.17–19: RCUK 500 micrograms IV versus AHA 2025 1 mg IV, both with repeats to a 3 mg maximum. Disposition: Protocol-dependent difference within AQRAB; not invented as a new between-file conflict.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 48

ECG interpretation - limb-lead reversal [#008]

High-yield fact: An apparently inverted ECG axis can result from limb-lead reversal and should be checked before diagnosing disease.

انعكاس محور ECG قد يحدث بسبب تبديل أقطاب الأطراف؛ تحقق من وضعها قبل تشخيص مشكلة قلبية.

LVAD complications - pump thrombosis [#012]

Key discriminator: Marked LDH elevation in an LVAD patient can indicate hemolysis from pump thrombosis.

ارتفاع LDH عند مستخدم LVAD قد يشير إلى انحلال الدم بسبب خثار المضخة؛ اربط النتيجة بحالة الجهاز.

Pacemaker malfunction - oversensing [#013]

High-yield fact: Pacemaker oversensing can suppress pacing; return of appropriate pacing under magnet response supports a sensing problem.

قد يثبط Pacemaker oversensing التحفيز؛ استجابة الجهاز للمغناطيس تحتاج تفسيرًا بحسب نوعه.

Brugada syndrome - type 1 ECG pattern [#015]

Key discriminator: A coved ST-elevation pattern in right precordial leads suggests type 1 Brugada and increased arrhythmia risk.

ارتفاع ST المقبب في الاشتقاقات الصدرية اليمنى يدعم نمط Brugada من النوع الأول.

EDIC / Emergency & Critical Care

03 — Respiratory & Thoracic Emergencies

Pulmonary Embolism: Probability, Physiology and Treatment

Topic 1: Pulmonary embolism: clinical recognition and obstructive physiology

- **Core concept:** Pulmonary arterial obstruction increases right-ventricular afterload and may impair left-ventricular filling and systemic cardiac output.
- **Clinical trigger:** Sudden unexplained dyspnea, pleuritic pain, tachycardia, syncope, hemoptysis, or unilateral leg swelling suggests PE when the pretest context is compatible.
- **High-yield fact:** V/Q mismatch may produce hypoxemia; peripheral pulmonary infarction can cause pleuritic pain and hemoptysis.
- **Red flag:** Hypotension, shock or arrest with suspected PE indicates life-threatening cardiopulmonary compromise.
- **Differential point:** Consider pneumothorax, ACS, aortic dissection, pneumonia, heart failure and pericarditis.
- **Exam trap:** Normal SpO2, ECG or chest radiography and absence of leg symptoms do not exclude PE.

الـPE قد يظهر بضيق نفس مفاجئ، ألم pleuritic أو syncope؛ الصدمة تنتج من زيادة حمل RV ونقص امتلاء LV. تشبع طبيعي لا ينفي PE.

Topic 2: Pretest probability: Wells/Geneva, PERC and D-dimer

- **Management priority:** Establish clinical pretest probability using structured clinical assessment or a validated Wells/Geneva pathway before selecting testing.
- **Key discriminator:** PERC is a no-testing rule only for an appropriately selected, already very-low-probability patient; all items must be negative.
- **Exam trap:** A single positive PERC item does not diagnose PE; it means PERC cannot rule it out.
- **Key discriminator:** Appropriate negative sensitive D-dimer may exclude PE in a validated non-high-probability pathway.
- **Exam trap:** Positive D-dimer is nonspecific and does not diagnose PE.
- **Management priority:** Proceed to imaging when clinical probability is high or the selected rule-out strategy is positive.

ابدأ بـpretest probability: PERC فقط لمن كان احتمالهم منخفضًا جدًا مسبقًا، وD-dimer للاشتباه المنخفض/المتوسط. النتيجة الإيجابية ليست تشخيصًا.

Topic 3: D-dimer units, age adjustment and YEARS thresholds

- **Number/threshold:** The AQRAB-cited adult age-adjusted FEU D-dimer limit after age 50 is age × 10 micrograms/L in suitable low/intermediate-probability patients.
- **Key discriminator:** FEU and DDU reporting units are not interchangeable; apply the assay-validated threshold and pathway.
- **Number/threshold:** In the source-described YEARS pathway, zero criteria with D-dimer <1000 ng/mL, or one or more criteria with <500 ng/mL or the applicable age-adjusted threshold, may exclude PE according to that specific algorithm.
- **Common mistake:** Applying age-adjusted or YEARS cutoffs to high-probability patients or converting FEU limits mentally to DDU.
- **Source qualification:** Pregnancy requires an adapted diagnostic strategy, such as pregnancy-adapted YEARS where appropriate; the adult pathway is not a universal pediatric or pregnancy rule.

لا تخلط FEU مع DDU؛ إن نقلت حد D-dimer من وحدة لأخرى دون التحقق من الاختبار قد تغيّر قرار التصوير خطأً.

Topic 4: PE imaging and bedside echocardiography

- **Management priority:** CTPA is a usual rapid diagnostic study when PE imaging is indicated and feasible.
- **Differential point:** V/Q imaging may be considered when CTPA is unsuitable, accounting for renal disease, contrast exposure or pregnancy.
- **Clinical trigger:** An unstable patient may not tolerate transfer for CT; resuscitate and obtain urgent specialist input.
- **Key discriminator:** Bedside echo may show right-ventricular strain or another cause of shock, but RV dilation is not specific for acute PE.
- **Exam trap:** A normal limited echo does not rule out PE in a stable patient.
- **Clinical Pearl:** On CTPA, diagnose pulmonary arterial filling defects rather than labeling every dark region as thrombus.

CTPA يفيد لتشخيص الصمة في المسار المناسب؛ أما echo في المريض غير المستقر فيساعد بفهم RV لكنه لا يؤكد PE وحده.

Topic 5: Confirmed PE: severity and disposition

- **Core concept:** A confirmed PE requires a separate physiologic severity assessment beyond clot location and image appearance.
- **High-yield fact:** The AQRAB-cited 2026 framework describes A as asymptomatic/incidental, B as low-severity symptomatic, C as elevated severity, D as incipient cardiopulmonary failure, and E as established failure.
- **Management priority:** Reassess BP trend, perfusion, respiratory support, RV findings, cardiac biomarkers and comorbidity before disposition.
- **Management priority:** Selected A/B patients may be suitable for outpatient care only after validated risk assessment, medication access and dependable follow-up.
- **Red flag:** D/E physiology requires urgent monitored multidisciplinary PE assessment and readiness for advanced therapy.
- **Exam trap:** Stable RV strain by itself is not an automatic indication for systemic thrombolysis.

افصل التشخيص عن الخطورة: شكل الخثرة ومكانها وحدهما لا يكفيان؛ BP وperfusion وRV واحتياج الأكسجين والـbiomarkers هي التي تغيّر القرار.

Topic 6: Anticoagulation for pulmonary embolism: source-specific pathways

- **Management priority:** Assess active bleeding, recent surgery, neurologic history, renal/hepatic function, pregnancy and planned reperfusion before treatment.
- **Source-specific recommendation:** The AQRAB-cited 2026 adult pathway prefers LMWH over UFH when parenteral anticoagulation is needed, with specialist/protocol-dependent exceptions.

- **Source-specific recommendation:** For eligible adults, the same passage favors a DOAC over a vitamin K antagonist unless contraindicated.
- **Number/dose:** AQRAB gives the labeled adult VTE apixaban regimen as 10 mg PO twice daily for 7 days, then 5 mg PO twice daily; this is not an AF loading regimen.
- **Clinical Pearl:** Pregnancy, antiphospholipid syndrome, renal/hepatic disease, bleeding and interacting drugs may alter eligibility and drug selection.
- **Exam trap:** Do not delay escalation in shock while applying an algorithm intended for stable low-risk PE.

قبل مضاد التخثر قيّم النزف والكلى والكبد والحمل والتداخلات. جرعة apixaban لعلاج PE تختلف عن جرعات AF أو extended prevention.

Topic 7: High-risk PE, RV failure and reperfusion

- **Core concept:** Acute pulmonary vascular obstruction can cause RV dilation, reduced forward flow and obstructive shock.
- **Management priority:** Stabilize oxygenation, ventilation and circulation while activating an urgent PE response and bleeding-risk review.
- **Red flag:** Large indiscriminate fluid loads may worsen severe RV failure, while positive-pressure ventilation may precipitate collapse.
- **Management priority:** The source describes systemic thrombolysis, catheter-based treatment, mechanical thrombectomy and surgical embolectomy as options for selected severe/deteriorating PE.
- **Source qualification:** The AQRAB-cited pathway considers advanced therapy reasonable for specified established-failure cases and selectively for incipient failure, after weighing bleeding risk.
- **High-yield fact:** The CLEAN V3 respiratory cross-topic extract names embolectomy as a possible treatment when embolic obstruction requires intervention.
- **Exam trap:** An isolated image of large clot burden or RV strain does not by itself justify routine thrombolysis in a stable patient.

في PE مع shock يتأذى RV من الحمل المرتفع. إعطاء سوائل كبيرة أو تنبيب دون تخطيط قد يزيد انهيار الدورة؛ اطلب فريقًا خبيرًا لعلاج الانسداد.

Topic 8: PE treatment duration and follow-up

- **Number/threshold:** The AQRAB-cited adult acute PE guideline describes an initial anticoagulation phase of 3–6 months.
- **Management priority:** Assess whether the index event had a major reversible cause or persistent/no major reversible risk when considering longer treatment.
- **Clinical Pearl:** Continued anticoagulation beyond the initial phase may benefit selected first-PE patients with persistent risk factors after bleeding-risk assessment.
- **Source-specific follow-up:** Review PE-related symptoms and functional limitation for at least one year after the acute episode in the cited framework.
- **Common mistake:** Automatically extending or stopping anticoagulation without reassessing provoking factors and hemorrhage risk.

بعد العلاج الأولي أعد تقييم السبب والنزف والعودة المحتملة للخثار؛ استمرار الأعراض التنفسية يحتاج متابعة، لا مجرد إنهاء العلاج عند تاريخ ثابت.

Asthma: Recognition, Acute Treatment and Ventilatory Risk

Topic 9: Asthma: variable expiratory obstruction and objective diagnosis

- **Core concept:** Asthma is a heterogeneous inflammatory airway disease with variable respiratory symptoms and expiratory airflow limitation.
- **Clinical trigger:** Episodic wheeze, dyspnea, chest tightness or cough, often varying with night, exercise, allergens or viral illness.

- **Management priority:** Seek objective variable airflow limitation when feasible, including spirometry and bronchodilator response.
- **Number/threshold:** AQRAB cites a commonly used adult bronchodilator response of FEV1 increase at least 12% and 200 mL; it cautions that age-specific standards matter.
- **Key discriminator:** Obstruction on spirometry is considered relative to the appropriate FEV1/FVC lower limit of normal.
- **Clinical Pearl:** Atopy, eczema or allergic rhinitis may accompany asthma but are not obligatory for diagnosis.

الربو أعراضه متغيرة مع الوقت، والتشخيص يعتمد على قصة مناسبة مع دليل موضوعي على تغير airflow، وليس وجود wheeze في لحظة واحدة.

Asthma versus COPD — diagnostic distinction

Feature	Asthma	COPD	Exam point
Airflow limitation	Usually variable	Usually persistent post-BD	Objective testing and clinical context
Spirometry	Variable / reversibility supportive	Post-BD FEV1/FVC <0.70 cited	Overlap is possible
Core controller	ICS-containing	Long-acting bronchodilators; selected ICS	Do not use SABA-only adult asthma strategy

Topic 10: Asthma exacerbation: severity and danger signs

- **Red flag:** Silent chest, exhaustion, inability to speak, confusion, cyanosis or falling air entry indicates dangerous airflow limitation.
- **Number/threshold:** AQRAB lists room-air SpO2 below 92% and PEF below 50% predicted/personal best among important acute severity markers.
- **Number/threshold:** CLEAN V3 gives a severe-attack example with RR at least 25/min, HR at least 110/min, and markedly reduced PEF, in conjunction with limited speech.
- **Clinical Pearl:** Severe asthma generally causes low PaCO2 early; a normal or rising PaCO2 in ongoing marked distress may signal muscle fatigue.
- **Red flag:** Previous intubation for asthma identifies increased risk in future exacerbations.
- **Common mistake:** Assuming reduced wheeze means improvement without rechecking work of breathing and air entry.
- **Management priority:** Reassess mental status, circulation, respiratory effort, SpO2 and PEF when feasible, rather than a single observation.

الصدر الصامت مع fatigue أو confusion ليس تحسنًا. وفي الشديد يكون PaCO2 الطبيعي أو الصاعد علامة إنذار لأن المريض قد يكون فقد قدرة hyperventilation.

Topic 11: Acute asthma bronchodilator and antimuscarinic treatment

- **Management priority:** Rapid inhaled short-acting beta-agonist therapy is a principal early treatment in acute exacerbation.
- **Number/dose:** AQRAB gives adult moderate examples of salbutamol 4–6 puffs of 100 micrograms by pMDI/spacer or 2.5 mg nebulized, repeated as needed.
- **Number/dose:** AQRAB gives adult severe examples of 6–10 puffs by pMDI/spacer or 5 mg nebulized, with repeat treatment about every 20–30 minutes when indicated.
- **Number/dose:** Nebulized ipratropium 0.5 mg is a cited adjunct for severe attacks; source examples repeat with SABA during the initial treatment period.
- **Clinical Pearl:** A spacer is an effective delivery approach in a cooperative patient; use one puff at a time when applying the cited technique.
- **Common mistake:** High-dose albuterol can cause tremor, tachycardia, hypokalemia and hyperglycemia; monitor clinically relevant adverse effects.
- **Exam trap:** Do not equate nebulizer milligrams directly with inhaler microgram-puffs.

ابدأ SABA سريعًا وبحسب شدة النوبة، مع ipratropium عند الشدة أو ضعف الاستجابة. لا تجمع أرقام البخاخ والرذاذ كأنها نفس الجرعة.

Topic 12: Asthma systemic corticosteroids and magnesium

- **Management priority:** Give systemic corticosteroids early in moderate/severe attacks while continuing inhaled bronchodilator therapy.
- **Number/dose:** AQRAB cites adult oral prednisolone/prednisone 40–50 mg once daily for 5–7 days in its GINA 2026 discussion.
- **Key discriminator:** Oral corticosteroids are generally preferred if swallowing and absorption are reliable; IV methylprednisolone is for patients unable to take oral therapy or critically ill under an appropriate protocol.
- **Number/dose:** Selected poorly responsive severe attacks may receive magnesium sulfate 2 g IV over approximately 20 minutes in the supplied pathways.
- **Source-specific dose:** CLEAN V3 cites IV salbutamol 250 micrograms over 5–10 minutes for a selected acute severe asthma scenario; this is not a routine recommendation in AQRAB.
- **Common mistake:** Giving routine antibiotics, sedatives, aminophylline or IV beta-agonists in standard uncomplicated acute asthma management.
- **Exam trap:** Corticosteroids do not provide the immediate bronchodilation expected from SABA.

الـsteroid يقلل relapse لكنه لا يفتح القصبات فورًا. IV ليس أفضل تلقائيًا من PO إذا الامتصاص جيد؛ وmagnesium خيار لحالات شديدة مختارة.

Topic 13: Controlled oxygen and reassessment in acute asthma

- **Number/threshold:** The AQRAB GINA 2026 adult acute-asthma oxygen target is SpO2 92–95%, with ongoing titration and reassessment.
- **Management priority:** Review symptoms, speech, respiratory effort, air entry, oxygenation and airflow after initial treatment and at about one hour, earlier if worsening.
- **Red flag:** Poor effort, exhaustion, drowsiness, persistent desaturation or hemodynamic deterioration warrants immediate senior/critical-care escalation.
- **Exam trap:** Pulse oximetry alone does not establish adequate ventilation or exclude progressing fatigue.
- **Common mistake:** Ordering routine chest radiography, blood gases or antibiotics for every uncomplicated asthma exacerbation rather than investigating when risk/poor response requires it.
- **Management priority:** Evaluate alternate diagnoses or complications when clinical response is atypical.

هدف الأكسجين المستشهد به في AQRAB للبالغين 92–95%، لكن الاستجابة تقاس أيضًا بالوعي وeffort وair entry، وليس SpO2 فقط.

Topic 14: Near-fatal asthma and ventilatory escalation

- **Clinical trigger:** Drowsiness, exhaustion, silent chest and rising PaCO2 with severe hypoxemia suggest impending ventilatory failure.
- **Management priority:** Seek experienced airway/critical-care support without waiting for additional routine tests when ventilation is failing.
- **Red flag:** Sedatives may worsen airway/ventilatory failure; the AQRAB acute-care pathway advises avoiding them.
- **Key discriminator:** NIV is not a substitute for timely airway control in a tiring or deteriorating patient; if used in selected cases, continuous monitoring and a failure plan are required.
- **Clinical Pearl:** Intrinsic PEEP and dynamic hyperinflation can markedly impair venous return and cause hypotension after intubation.
- **Management priority:** Preserve expiratory time; increasing respiratory rate in severe obstruction may worsen trapping.
- **Common mistake:** Treating a normalizing PaCO2 in a severely distressed patient as recovery.

عند hypercapnia مع fatigue أو drowsiness في الربو الشديد حضّر فريق airway مبكرًا. أهم خطر بعد التهوية هو air trapping وانخفاض venous return.

Topic 15: Asthma differential diagnosis, aftercare and controller strategy

- **Differential point:** Consider anaphylaxis, pneumonia, PE, pneumothorax, heart failure and inducible laryngeal obstruction when presentation or treatment response is atypical.
- **Management priority:** If anaphylaxis is suspected, give indicated IM epinephrine first; inhaled bronchodilator is an adjunct and must not delay it.
- **Management priority:** Before discharge, arrange an ICS-containing controller strategy, inhaler technique review, action plan and prompt follow-up.
- **Source-specific recommendation:** AQRAB cites GINA 2026 preference for an ICS-containing strategy rather than SABA-only use for adult/adolescent asthma.
- **Clinical Pearl:** Frequent SABA use, previous ICU/intubation, recent severe attack and low lung function increase future risk.
- **Common mistake:** Escalating drug therapy without checking inhaler adherence and technique.

لا تُنهي الزيارة بعد تحسن wheeze فقط؛ اضبط ICS وتقنية البخاخ وخطة المتابعة. وأي wheeze مع anaphylaxis يحتاج epinephrine أولًا.

COPD: Obstructive Physiology and Acute Exacerbation

Topic 16: COPD definition, diagnostic spirometry and exacerbation triggers

- **Core concept:** COPD produces persistent airflow limitation associated with airway and/or alveolar abnormalities and chronic respiratory symptoms.
- **Number/threshold:** The cited GOLD spirometric criterion is post-bronchodilator FEV1/FVC below 0.70, interpreted with age and clinical context.
- **Clinical trigger:** Acute increase in dyspnea, cough and/or sputum over baseline suggests an exacerbation.
- **Differential point:** Triggers include infection, pollution or missed therapy; also reassess for heart failure, PE and pneumothorax.
- **High-yield fact:** Smoking, biomass exposure, occupational fumes and alpha-1 antitrypsin deficiency are source-listed risk factors.
- **Clinical Pearl:** COPD is commonly associated with secondary spontaneous pneumothorax.

COPD يعني انسدادًا مستمرًا غالبًا مع دخان أو تعرض مزمن. لكن التفاقم لا يُفسر تلقائيًا كعدوى؛ ابحث أيضًا عن PE وHF وpneumothorax.

Topic 17: Acute COPD: bronchodilator and steroid examples

- **Management priority:** Use short-acting inhaled bronchodilators, systemic corticosteroids when indicated, controlled oxygen and repeated severity assessment.
- **Number/dose:** AQRAB gives nebulized adult salbutamol 2.5–5 mg repeated according to clinical response and the local regimen.
- **Number/dose:** Nebulized ipratropium 0.5 mg is a source example, often combined with SABA.
- **Number/dose:** AQRAB and CLEAN V3 describe prednisone-equivalent 40 mg PO daily for five days in a cited GOLD-style COPD exacerbation pathway.
- **Source-specific difference:** AQRAB separately notes NICE prednisolone 30 mg for five days, illustrating protocol dependence rather than a universal single dose.
- **Management priority:** Assess baseline oxygen, previous hypercapnia/NIV, sputum purulence, comorbidity and response before disposition.
- **Common mistake:** Applying acute-asthma steroid or oxygen targets without accounting for COPD and CO2-retention risk.

علاج التفاقم يبدأ SABA مع أو بدون ipratropium وsteroid قصير عند الاستطباب؛ لا تجعل المضادات الحيوية تلقائية في كل حالة.

Topic 18: COPD oxygen titration and hypercapnia risk

- **Number/threshold:** In COPD at risk for hypercapnic failure, the cited initial oxygen target is SpO2 88–92% pending blood-gas assessment.
- **Core concept:** Excess oxygen may worsen CO2 retention through increased V/Q mismatch, the Haldane effect and a lesser reduction in ventilatory drive.
- **Management priority:** Use controlled oxygen delivery, such as a Venturi mask when a defined FiO2 is needed.
- **Source-specific timing:** AQRAB cites repeat ABG in 30–60 minutes after oxygen adjustment, earlier with clinical decline.
- **Exam trap:** Controlled oxygen is not withholding rescue oxygen from a hypoxemic patient.
- **Clinical Pearl:** SpO2 improvement without improvement in awareness or work of breathing may coexist with worsening ventilation.

أعط الأكسجين للمريض ناقص الأكسجة لكن بشكل مضبوط. عند خطر احتباس CO2 استهدف مبدئيًا 88–92%، وأعد ABG بعد تعديل الأكسجين.

Initial oxygen targets by clinical syndrome

Feature	Acute adult asthma	COPD with CO2-retention risk	Exam point
Source target	SpO2 92–95%	SpO2 88–92% initially	Adjust to physiology and protocol
Monitoring	Air entry, speech, effort, saturation	Controlled oxygen and repeat ABG	SpO2 alone misses hypoventilation
Danger	Silent chest, rising/normal PaCO2	Worsening pH with CO2 rise	Escalate despite apparent oxygenation

Topic 19: NIV for acute hypercapnic COPD exacerbation

- **Management priority:** Consider bilevel noninvasive ventilation for acute hypercapnic respiratory acidosis persisting despite initial therapy in a suitable monitored patient.
- **Number/threshold:** The cited pathway uses pH at or below 7.35 with PaCO2 above 45 mmHg; another AQRAB BTS pathway uses pH below 7.35 and PaCO2 above 6.5 kPa.
- **Number/setting:** CLEAN V3 cites a bilevel starting example of IPAP 8–12 cm H2O and EPAP 3–5 cm H2O, individually titrated rather than used as fixed targets.
- **Clinical trigger:** Increased effort with declining pH or rising PaCO2 despite bronchodilator and oxygen management warrants escalation consideration.
- **Red flag:** Worsening consciousness, inability to protect airway, persistent vomiting, intolerance or hemodynamic instability undermines NIV suitability.
- **Exam trap:** NIV is not an unmonitored discharge-level treatment; define the escalation and intubation plan.

الـNIV bilevel مناسب عند acute hypercapnic respiratory acidosis في مريض مختار بعد العلاج الأولي؛ أهم شيء مراقبة الفشل وعدم تأخير intubation.

Topic 20: COPD air trapping, auto-PEEP and ventilator troubleshooting

- **Core concept:** Rapid shallow breathing and insufficient expiratory time cause dynamic hyperinflation, intrinsic PEEP and respiratory muscle fatigue.
- **Key discriminator:** Expiratory flow failing to reach baseline before the next breath suggests incomplete expiration.
- **Management priority:** Reduce excessive respiratory rate and allow longer exhalation in ventilated obstructive disease rather than reflexively increasing ventilation rate.
- **Red flag:** High intrathoracic pressure from trapping can reduce venous return and precipitate hemodynamic collapse.
- **Management priority:** During abrupt deterioration, assess patient, tube, circuit, lungs and circulation together; review pressures and expiratory flow.
- **Exam trap:** Rapid large bagged breaths or a faster respiratory rate may worsen intrinsic PEEP in severe obstruction.

في COPD المُتَبَّب إذا expiratory flow لم يرجع للصفر قبل النفس التالي، فكر في auto-PEEP. رفع RR قد يزيد trapping ويخفض cardiac output.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 57

10 mg, 5 mg	antagonist unless contraindicated. Number/dose: AQRAB gives the labeled adult VTE apixaban regimen as 10 mg PO twice daily for 7 days, then 5 mg PO twice daily; this is not an AF loading regimen.
2.5 mg	Number/dose: AQRAB gives adult moderate examples of salbutamol 4–6 puffs of 100 micrograms by pMDI/spacer or 2.5 mg nebulized, repeated as needed. Number/dose: AQRAB gives adult severe examples of 6–10 puffs by pMDI/spacer or 5 mg nebulized, with repeat
5 mg	or 2.5 mg nebulized, repeated as needed. Number/dose: AQRAB gives adult severe examples of 6–10 puffs by pMDI/spacer or 5 mg nebulized, with repeat treatment about every 20–30 minutes when indicated.
0.5 mg	treatment about every 20–30 minutes when indicated. Number/dose: Nebulized ipratropium 0.5 mg is a cited adjunct for severe attacks; source examples repeat with SABA during the initial treatment period.
40–50 mg	bronchodilator therapy. Number/dose: AQRAB cites adult oral prednisolone/prednisone 40–50 mg once daily for 5–7 days in its GINA 2026 discussion.
2 g	methylprednisolone is for patients unable to take oral therapy or critically ill under an appropriate protocol. Number/dose: Selected poorly responsive severe attacks may receive magnesium sulfate 2 g IV over approximately 20 minutes in the supplied pathways.
2.5–5 mg	controlled oxygen and repeated severity assessment. Number/dose: AQRAB gives nebulized adult salbutamol 2.5–5 mg repeated according to clinical response and the local regimen.
30 mg	style COPD exacerbation pathway. Source-specific difference: AQRAB separately notes NICE prednisolone 30 mg for five days, illustrating protocol dependence rather than a universal single dose.
40–50 mg	18. SABA is early bronchodilator therapy; severe cases may add ipratropium. 19. Adult asthma prednisolone example: 40–50 mg daily for 5–7 days. 20. Selected refractory severe asthma: IV magnesium 2 g over ~20 min.
2 g	19. Adult asthma prednisolone example: 40–50 mg daily for 5–7 days. 20. Selected refractory severe asthma: IV magnesium 2 g over ~20 min. 21. Asthma acute oxygen target cited: SpO2 92–95%.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 57

Pulmonary embolism - ECG patterns [#018]

High-yield fact: In pulmonary embolism, sinus tachycardia and nonspecific ST-T changes are more frequent than the classic S1Q3T3 pattern.

لا تنتظر نمط S1Q3T3 في PE؛ تسرّع القلب الجيبي وتبدلات ST-T غير النوعية أكثر شيوعًا.

Topic 21: COPD ABG interpretation and post-hypercapnic alkalosis

- **Key discriminator:** Chronic hypercapnia may produce a high plasma bicarbonate as renal compensation for respiratory acidosis.
- **Clinical Pearl:** Rising PaCO2 together with falling pH in a COPD exacerbation suggests acute-on-chronic ventilatory failure.
- **Key discriminator:** Post-hypercapnic metabolic alkalosis can follow a reduction in previously elevated PaCO2 while bicarbonate remains high.
- **Exam trap:** A single elevated bicarbonate concentration does not prove a primary metabolic alkalosis.
- **Management priority:** Use serial blood gases and previous baseline to distinguish chronic retention from new ventilatory deterioration.

الـHCO3 المرتفع في COPD قد يعني تعويضًا كلويًا مزمنًا، وليس metabolic alkalosis أوليًّا؛ قارنه بالـpH وغازات سابقة و PaCO2.

Topic 22: Antibiotic selection and long-term COPD prevention

- **Management priority:** Consider antibiotics in exacerbation with increased sputum purulence plus increased volume and/or dyspnea, or when ventilation support is required.
- **Common mistake:** Antibiotics are not automatic for all exacerbations; viral triggers are common and pneumonia is separately assessed.
- **Source-specific regimen:** CLEAN V3 contains an amoxicillin-clavulanate five-day outpatient example for selected moderate exacerbations meeting antibiotic criteria.
- **Management priority:** Chronic care emphasizes smoking cessation, vaccination, pulmonary rehabilitation, inhaler technique and long-acting bronchodilators.
- **Clinical Pearl:** ICS in COPD is selected based on exacerbations, eosinophils and convincing asthma features; it is not given automatically to everyone.
- **Exam trap:** Asthma and COPD may coexist; clear asthma features require an ICS-containing approach rather than bronchodilator-only COPD management.

وجود purulent sputum مع زيادة الكمية أو dyspnea، أو الحاجة لدعم التهوية، يغير قرار antibiotic. في الوقاية راجع التدخين والبخاخات واللقاحات.

Acute asthma versus COPD antibiotic decisions

Feature	Acute asthma	Acute COPD exacerbation	Exam point
Antibiotics	Not routine	Selected purulence/cardinal symptoms or ventilation	Avoid automatic prescribing
Systemic steroid example	40–50 mg for 5–7 days, adult source	40 mg for 5 days, GOLD source	Different syndromes and protocols
Controller	ICS-containing	Long-acting bronchodilators; selective ICS	Review inhaler technique

Pneumonia, Respiratory Infection and Complications

Topic 23: Pneumonia recognition and respiratory red flags

- **Clinical trigger:** Fever, cough, sputum, pleuritic pain, focal crackles or bronchial breathing with a new infiltrate supports pneumonia.
- **Key discriminator:** Older adults may present with altered mental status rather than prominent respiratory symptoms.
- **Red flag:** Hypoxemia, tachypnea, hypotension, multilobar involvement, rising lactate or organ dysfunction suggests a severe course.
- **Management priority:** Assess oxygenation, work of breathing, perfusion, mental status, oral intake, immune status and clinical trajectory together.
- **Differential point:** Consider sepsis, acute heart failure, PE, pleural effusion, aspiration and other respiratory conditions.
- **High-yield fact:** The supplied CLEAN V3 review notes atypical pathogens including Mycoplasma, Chlamydia species and Legionella, with variable patterns.

pneumonia في cough لا يكفي والحرارة؛ قيّم oxygenation وperfusion والوعي وكمية السوائل التي يستطيع المريض شربها، خصوصًا كبار السن والأطفال.

Pneumonia versus PE versus pulmonary edema

Feature	Pneumonia	Pulmonary embolism	Exam point
Presentation	Fever/cough, focal signs	Sudden dyspnea, VTE risks, pleuritic pain	Symptoms overlap
Crackles	Often focal	May be absent/nonspecific	Do not exclude PE by chest exam
Priority	Severity, oxygenation, infection assessment	Pretest probability, PE testing pathway	Physiology takes priority

Topic 24: CRB-65/CURB-65 and severity-based disposition

- **Core concept:** CRB-65/CURB-65 assists community-acquired pneumonia severity stratification but does not override clinical instability.
- **Number/threshold:** CRB-65 gives one point each for confusion, respiratory rate at least 30/min, low blood pressure and age at least 65 years.
- **Number/threshold:** CURB-65 adds elevated urea above 7 mmol/L to CRB-65 features in the source summary.
- **Red flag:** Shock, hypoxemia, organ dysfunction, frailty, worsening trajectory or unsafe follow-up may require admission despite a low score.
- **Clinical Pearl:** Poor intake or dehydration is an important disposition factor in pediatric pneumonia, independently of respiratory signs.
- **Exam trap:** A score describes risk, not an absolute treatment or discharge command.

CURB-65 يساعد لكنه لا يعطي تصريح خروج آلي؛ hypoxemia أو shock أو frailty أو عدم القدرة على المتابعة أهم من score منخفض.

Topic 25: CAP imaging, pathogen testing and alternative diagnoses

- **Management priority:** Chest radiography is a standard imaging option; AQRAB cites trained lung ultrasound as a conditional alternative for selected adult CAP assessment.
- **Clinical trigger:** Test for specific pathogens when severity, immune status, exposure or potential treatment impact justifies the study.
- **Key discriminator:** Legionella urinary antigen principally detects L. pneumophila serogroup 1; a negative result cannot exclude every Legionella infection.
- **High-yield fact:** CLEAN V3 describes subacute nonproductive cough and interstitial findings as clues to certain atypical pneumonias.
- **Exam trap:** Sputum color or a single biomarker cannot identify a pathogen with certainty.

- **Differential point:** Unresolved hypoxemia or persistent symptoms should prompt reconsideration of empyema, abscess, PE or heart failure.

الصورة والـlung ultrasound يدعمان التشخيص، لكن لا تجعل pathogen test واحد يختار المضاد أو ينفي bacterial coinfection دون سياق.

Topic 26: Viral-positive CAP and source-dependent antibiotic recommendations

- **Source-specific recommendation:** The AQRAB ATS 2025 summary suggests no empiric antibiotics for selected viral-positive immunocompetent outpatients without comorbidities.
- **Source-dependent difference:** For viral-positive outpatients with comorbidities and hospitalized adults, ATS suggests antibiotics despite very-low-certainty evidence, whereas IDSA did not endorse those two recommendations.
- **Management priority:** Individualize the decision using severity, trajectory, imaging, inflammation, coinfection probability and antibiotic harm.
- **Exam trap:** A positive viral result does not automatically eliminate the possibility of bacterial coinfection.
- **Source qualification:** This is a documented disagreement described inside AQRAB, not an independent new conflict between the two uploaded files.

عند viral PCR إيجابي لا يوجد اتفاق مطلق في AQRAB عن antibiotics لكل الفئات؛ شدة الحالة وcomorbidities واحتمال coinfection تغير القرار.

Topic 27: CAP antibiotic duration, steroids and selected source examples

- **Source-specific duration:** AQRAB quotes ATS 2025 as suggesting at least 3 but fewer than 5 days for clinically stable adult outpatients/nonsevere inpatients.
- **Source-specific duration:** The same AQRAB passage cites at least 5 days for severe hospitalized adult CAP.
- **Key discriminator:** Organism, complications, instability and local pathways can lengthen therapy beyond the minimum source examples.
- **Source-specific steroid position:** The AQRAB-cited ATS pathway advises against routine systemic corticosteroids in nonsevere hospitalized CAP and conditionally supports them in selected severe CAP, excluding influenza pneumonia.
- **Source-specific regimen:** AQRAB gives adult low-severity CAP example of amoxicillin 500 mg PO three times daily for five days, or a selected doxycycline course; patient-specific factors require assessment.
- **Exam trap:** A positive test or short historical duration example is not sufficient to prescribe in an unstable or complicated case.

مدة المضاد تعتمد على severity والاستقرار والمضاعفات. لا تطبق مدة واحدة على كل CAP ولا تعط systemic steroids روتينيًا لكل مريض.

Topic 28: Atypical pneumonia and infection-specific clues

- **High-yield fact:** CLEAN V3 lists Mycoplasma pneumoniae, Chlamydia species and Legionella pneumophila among atypical pneumonia pathogens.
- **Clinical trigger:** Subacute nonproductive cough, hoarseness and interstitial infiltrates are a source-associated clue for Chlamydia pneumoniae.
- **Clinical Pearl:** Radiographic resolution of Legionella pneumonia may be slower than typical patterns in the CLEAN V3 examples.
- **Exam trap:** Legionella urinary antigen negativity does not exclude all serogroups; interpret with clinical findings and other tests.
- **Differential point:** Pontiac fever is a nonpneumonic illness and should not be equated automatically with Legionella pneumonia.
- **Number/threshold:** CLEAN V3 describes hospital-acquired pneumonia as an infection starting more than 48 hours after hospital admission.

في atypical pneumonia قد تتداخل Mycoplasma وChlamydia وLegionella. لا تربط organism بصورة واحد شعاعية واحدة كأنها تشخيص قاطع.

Upper Airway Infection and Acute Thoracic Differentials

Topic 29: Sore throat versus deep-neck and supraglottic infection

- **Core concept:** Most uncomplicated acute sore throats are self-limiting; use age-appropriate clinical rules and reassessment rather than antibiotics for everyone.
- **Clinical trigger:** Drooling, inability to swallow, stridor, muffled voice, trismus, asymmetric palatal bulge or floor-of-mouth swelling indicates possible deep-space or supraglottic disease.
- **Management priority:** Keep an airway-threatened patient calm and upright; involve experienced airway/anesthesia and ENT teams early.
- **Red flag:** Avoid forced oral examination, supine positioning or transfer to imaging when the patient is unstable.
- **Key discriminator:** Unilateral palatal bulge with trismus and uvular deviation supports peritonsillar abscess; neck stiffness and floor-of-mouth swelling may signal other deep-neck disease.
- **Management priority:** Stable patients may undergo contrast CT or selected ultrasound; suspected collections need source-control evaluation and indication-specific antimicrobials.

الـsore throat العادي لا يهدد airway. لكن drooling أو stridor أو trismus أو neck swelling يعني احتمال epiglottitis أو deep-neck infection.

Topic 30: Clinical chest sounds and respiratory differentials

- **Key discriminator:** Wheeze suggests airflow limitation but may also occur in pulmonary edema; a silent chest can accompany severe obstruction.
- **Red flag:** Stridor points toward upper airway compromise rather than ordinary lower-airway wheeze.
- **Clinical Pearl:** Fine crackles can occur with interstitial edema, fibrosis or pneumonia and are not specific for heart failure.
- **Key discriminator:** Absent unilateral breath sounds suggest pneumothorax, major pleural effusion or mainstem obstruction; correlate with clinical stability.
- **Key discriminator:** Dullness suggests effusion/consolidation, whereas hyperresonance suggests air or hyperinflation; neither is diagnostic alone.
- **Exam trap:** Bilateral basal crackles do not distinguish cardiogenic edema from ARDS or other diffuse lung disease without clinical context.

الـcrackles أو wheeze علامة وليست diagnosis. اربط مكان الصوت والفحص مع vitals وPOCUS و chest عند الحاجة imaging.

Topic 31: Pleurisy, costochondritis and the dangerous pleuritic-pain differential

- **Core concept:** Costochondritis is a costosternal pain syndrome; the CLEAN V3 material distinguishes it from swelling-associated costal cartilage disorders.
- **Key discriminator:** The source states costochondritis itself is not typically accompanied by costal cartilage swelling.
- **Management priority:** Symptomatic NSAID therapy is listed in CLEAN V3 for appropriate benign chest-wall pain presentations.
- **Differential point:** Acute pleuritic pain may arise from PE, pneumothorax, pneumonia or pericardial disease; risk context determines testing.
- **Red flag:** Sudden pleuritic pain with dyspnea, tachycardia, syncope or hypotension deserves urgent reassessment for a life-threatening cause.

الألم الجنبي قد يكون benign لكن لا تنس PE أو pneumothorax أو pneumonia. Costochondritis غالبًا ألم موضعي بلا تورم غضروفي.

Pleural Disease, Pneumothorax and Thoracic Procedures

Topic 32: Tension pneumothorax: life-threatening clinical diagnosis

- **Core concept:** Pressurized pleural air may impair venous return and produce respiratory and circulatory collapse.
- **Clinical trigger:** Sudden respiratory distress, unilateral markedly reduced breath sounds, hypotension and compatible risk suggest tension physiology.
- **Management priority:** An unstable patient with clinically suspected tension pneumothorax requires urgent decompression by trained clinicians without imaging delay.
- **Exam trap:** Tracheal deviation can be absent or late and is not a prerequisite to treatment.
- **Source-specific technique:** CLEAN V3 contains needle/finger thoracostomy descriptions with second intercostal midclavicular and fourth/fifth anterior-axillary approaches; procedural selection remains pathway- and expertise-dependent.
- **Red flag:** Positive-pressure ventilation can worsen tension physiology and associated shock.

عندما يوجد shock أو respiratory collapse مع unilateral absent breath sounds، لا تنتظر CXR أو tracheal deviation؛ اضغط على علاج tension فورًا بفريق مؤهل.

Tension pneumothorax versus non-tension stable PSP

Feature	Tension pneumothorax	Selected stable primary spontaneous PTX	Exam point
Circulation	May cause hypotension/shock	No physiologic compromise	Treat tension immediately
Imaging	Must not delay decompression	Imaging supports diagnosis/selection	Tracheal shift may be late
Management	Immediate trained decompression	Possible observation/ambulatory path	Do not generalize stable PSP care

Comparison synthesized only from the cited source topics immediately above.

Topic 33: Primary versus secondary spontaneous pneumothorax

- **Key discriminator:** Primary spontaneous pneumothorax occurs without recognized underlying chronic lung disease; secondary pneumothorax accompanies conditions such as COPD.
- **Source-specific recommendation:** AQRAB cites BTS 2023 allowance for conservative care in selected minimally symptomatic/asymptomatic adult primary spontaneous cases without compromise, irrespective of radiographic size.
- **Management priority:** Stable symptomatic cases require selection among observation, ambulatory treatment, aspiration or drainage according to symptoms, physiology, lung reserve and local capability.
- **Source-specific pathway:** CLEAN V3 includes size-specific observation, aspiration and tube-based follow-up examples; these are not automatically interchangeable with the selected BTS pathway.
- **Red flag:** Do not apply stable-primary observation criteria to tension pneumothorax, secondary lung disease, marked hypoxia or hemodynamic instability.
- **Exam trap:** Size alone does not set the intervention threshold across all modern context-specific pathways.

PSP المستقر قليل الأعراض قد يُراقَب في مسار مختار، بينما SSP أو tension لا تطبق عليهما القاعدة تلقائيًا.

Topic 34: Pneumothorax ultrasound and chest-drain monitoring

- **Key discriminator:** Absent lung sliding on focused thoracic ultrasound can support pneumothorax in the right clinical context.
- **Clinical Pearl:** B-lines and preserved lung sliding argue against free pleural air at the assessed point in suitable circumstances.
- **Management priority:** After chest-drain insertion, check respiratory status, vitals, drainage and tube function repeatedly using the local protocol.
- **Clinical Pearl:** Water-seal chamber bubbling with cough may help evaluate an ongoing air leak in the supplied chest-drain description.

- **Common mistake:** Treating a single ultrasound sign or chest-drain chamber observation as conclusive without assessing patient and equipment.
- **Source-specific follow-up:** CLEAN V3 includes chest-clinic follow-up in 2–4 weeks after selected aspiration pathways.

الـlung sliding يساعد في تقييم pneumothorax لكنه ليس وحده تشخيصًا قاطعًا. بعد chest tube افحص العلامات الحيوية وعمل الدرنقة باستمرار.

Topic 35: Pleural effusion assessment and Light criteria

- **Core concept:** Pleural effusion is fluid accumulation between visceral and parietal pleura and can result from heart failure, infection, trauma or PE.
- **Clinical trigger:** Dyspnea, reduced hemithorax expansion, stony dullness, diminished breath sounds and reduced tactile fremitus support effusion.
- **Number/formula:** Light criterion 1: pleural fluid protein/serum protein >0.5.
- **Number/formula:** Light criterion 2: pleural fluid LDH/serum LDH >0.6.
- **Number/formula:** Light criterion 3: pleural fluid LDH >two-thirds of the laboratory serum LDH upper limit; any one criterion classifies an exudate.
- **Exam trap:** The exudate classification does not specify its cause, and a diuresed heart-failure effusion may appear exudative.
- **Management priority:** Use thoracic ultrasound to characterize effusion and guide any aspiration in appropriate circumstances.

Light criteria للتفريق بين transudate/exudate وليس لتحديد السبب. أي معيار واحد يكفي، لكن تفسير diuretic-treated HF قد يكون مضللًا.

Pleural fluid interpretation — paired criteria

Feature	Exudative criterion	Common interpretation trap	Exam point
Protein ratio	Pleural/serum >0.5	Ratio is not a diagnosis	Any one criterion sufficient
LDH ratio	Pleural/serum >0.6	Do not forget lab values	Any one criterion sufficient
Pleural LDH	>2/3 serum upper normal	Diuresed HF may appear exudative	Integrate disease context

Comparison synthesized only from the cited source topics immediately above.

Topic 36: Parapneumonic effusion, empyema and pH-based drainage

- **Core concept:** Empyema is infected pleural fluid containing frank pus and generally requires antibiotics plus effective pleural drainage assessment.
- **Number/threshold:** Pleural pH at or below 7.2 supports intercostal drainage for suspected pleural infection when accessible fluid can be drained safely.
- **Number/threshold:** pH above 7.2 but below 7.4 is intermediate; LDH above 900 IU/L plus other findings may favor drainage.
- **Clinical Pearl:** Supporting features include fever, significant fluid volume, pleural glucose at or below 4.0 mmol/L, CT pleural enhancement or ultrasound septation.
- **Number/threshold:** Pleural pH at or above 7.4 does not require immediate drainage on pH criteria alone; when pH is unavailable, glucose below 3.3 mmol/L may raise concern in context.
- **Exam trap:** Local anesthetic, excess heparin and delayed sample handling may distort pH; sample quality changes interpretation.

في pleural infection وجود pus يعني source control. وإذا لم يوجد pus، فإن pH يوجّه قرار drain بشرط سحب وقياس عينة صحيحة.

Pleural infection: source pH decisions

Feature	pH ≤7.2	pH >7.2 and <7.4	Exam point
Risk	High risk, drain assessment	Intermediate zone	Sample quality matters
Further features	Accessible fluid and safety	LDH >900 with fever/glucose/US features	Do not interpret one number alone
pH ≥7.4	No drain from pH alone	Not applicable	Pus still requires source control

Topic 37: Chest drainage: procedural and traumatic distinctions

- **Source-specific tube example:** CLEAN V3 mentions small-bore 10–14 Fr drainage in selected primary spontaneous pneumothorax and 14 Fr in an empyema example.
- **Management priority:** Select drainage approach, tube caliber, image guidance and monitoring according to indication, local practice and procedural expertise.
- **Number/threshold:** CLEAN V3 describes more than 1500 mL immediate traumatic chest drainage as an urgent thoracic surgical trigger.
- **Number/threshold:** The source also describes continued traumatic blood output about 200 mL/h for three hours as a thoracotomy trigger; note an internally inconsistent 250 mL/h answer fragment.
- **Exam trap:** Pneumothorax and pleural-infection algorithms are not interchangeable with traumatic hemothorax management.
- **Common mistake:** Delaying decompression of unstable tension physiology for radiography or a procedural measurement.

لا تنقل قياس chest tube من empyema إلى trauma بشكل آلي. نزف hemothorax المستمر يحتاج تقييمًا جراحيًا وليس متابعة الجهاز فقط.

Topic 38: Hemoptysis: physiology before absolute volume

- **Core concept:** Clinically dangerous hemoptysis is defined chiefly by threat to airway patency, oxygenation and perfusion rather than one universal volume cutoff.
- **Number/threshold:** CLEAN V3 notes that massive-hemoptysis definitions vary from at least 100 to 1000 mL over 24 hours, preventing a single universal volume rule.
- **Management priority:** The source includes urgent interventional-radiology assessment for embolization in selected major bleeding cases.
- **Red flag:** A patient unable to maintain ventilation, oxygenation or circulation requires immediate prioritized resuscitation and source-control planning.
- **Exam trap:** A small measured amount may still be dangerous when airway reserve is limited; do not focus solely on volume terminology.

في hemoptysis، الاختناق وفشل oxygenation أو shock أهم من الحد الحجمي. الرقم في المصادر متغير، لذلك لا تجعل حجم الدم وحده يحكم.

Additional Pulmonary Emergencies and Clinical Principles

Topic 39: Acute respiratory distress syndrome (ARDS)

- **Core concept:** Inflammatory alveolar-capillary injury produces permeability edema with acute respiratory compromise.
- **Clinical trigger:** Sepsis, pneumonia or aspiration followed by rapidly worsening oxygen requirement and bilateral lung opacities suggests ARDS.
- **Key discriminator:** Assess timing, bilateral opacities, oxygenation and whether cardiac failure/fluid overload fully explains the process.
- **Clinical Pearl:** AQRAB distinguishes Berlin ARDS classification from an expanded 2024 global definition that includes certain nonintubated patients.
- **Management priority:** Treat the trigger and use critical-care lung-protective ventilation according to the established ventilation chapter.

- **Number/threshold:** The cited source approach uses tidal volume 4–8 mL/kg predicted body weight and plateau pressure under 30 cm H2O; avoid actual-weight tidal-volume calculations.
- **Exam trap:** Diffuse crackles are not specific for ARDS; oxygenation, context and named diagnostic criteria matter.

ARDS مشكلة نفاذية الحويصلات مع hypoxemia وbilateral opacities، ولا يكفي تشخيصها من crackles أو صورة وحدها. استخدم تعريفًا محددًا.

Topic 40: Foreign-body aspiration and occult obstruction

- **Clinical trigger:** Sudden choking, stridor or focal wheeze suggests airway foreign-body aspiration.
- **Red flag:** Ineffective cough, inability to speak or cry, cyanosis and collapse indicate severe obstruction requiring immediate age-appropriate rescue.
- **Exam trap:** A normal radiograph does not exclude a radiolucent aspirated foreign body.
- **Management priority:** Stabilize airway; persistent suspicion or retained objects require airway-specialist assessment and controlled removal planning.
- **Common mistake:** Performing blind finger sweeps during suspected obstruction.
- **Clinical Pearl:** Predominance in the right main bronchus is not an absolute exclusion rule for other locations, especially in children.

الـCXR الطبيعي لا ينفي جسمًا غريبًا غير ظاهر بالأشعة. شدة الاختناق أولًا، وبعدها bronchoscopy أو إزالة مدروسة دون blind finger sweep.

Topic 41: Pulmonary hypertension and right-heart consequences

- **Clinical trigger:** Exertional dyspnea is noted in CLEAN V3 as a common early presentation of pulmonary hypertension.
- **High-yield fact:** The supplied material distinguishes pre-capillary pulmonary hypertension by elevated pulmonary vascular resistance in a compatible hemodynamic profile.
- **Key discriminator:** Pulmonary artery occlusion pressure estimates left-sided filling pressure and informs interpretation alongside PVR.
- **Clinical Pearl:** Acute RV strain and failure from high-risk PE are important differentials but are not synonymous with chronic pulmonary arterial hypertension.
- **Common mistake:** Assigning an entire pulmonary-hypertension clinical group from a single pressure without the full hemodynamic context.

عند pulmonary hypertension قد يبدأ العرض بـexertional dyspnea؛ وبعد زيادة RV afterload يحدث انخفاض LV filling والدورة.

Topic 42: Inhalational lung injury and exposure-linked disease

- **Clinical trigger:** Exposure to fermenting silage-associated nitrogen dioxide can cause toxic pneumonitis termed silo filler's disease.
- **High-yield fact:** The CLEAN V3 summary names asbestos exposure as a cause of pneumoconiosis.
- **Clinical Pearl:** An occupational/exposure history can direct the differential in pulmonary illness unexplained by typical infection.
- **Differential point:** Acute dyspnea and hypoxemia still require simultaneous consideration of severe infection, PE and noninfectious pulmonary disorders.

بعض الإصابات الرئوية ترتبط بتعرض واضح مثل silage أو ألياف asbestos. التاريخ المهني مهم لتفسير pneumonitis أو ضيق النفس.

PASS 9 — Source-backed clinical recovery | Domain 03

Respiratory & Thoracic Emergencies

dosing or medical-correctness certification.

Oxygen-cylinder duration calculation

Number/formula: Cylinder duration (minutes) = available cylinder volume (L) / oxygen flow (L/min). The CLEAN V3 example is 460 L / 15 L/min = about 30.6 min.

احسب وقت الأسطوانة من الحجم المتاح مقسومًا على معدل الجريان؛ المثال يعطي نحو 31 دقيقة.

Flying after pneumothorax

Source-specific timing: A CLEAN V3 case states to wait 7 days after radiographic resolution of pneumothorax before air travel; preserve this as a cited source-specific interval rather than a universal rule.

فترة السبعة أيام في المصدر مشروطة بإثبات زوال pneumothorax بالأشعة؛ لا تُعرض كقاعدة لكل الحالات.

Topic 43: Acute lung findings in cardiogenic pulmonary edema

- **Clinical trigger:** Dyspnea, orthopnea, hypoxemia, basal/diffuse crackles and sometimes pink frothy sputum raise concern for cardiogenic pulmonary edema.
- **Differential point:** Evaluate cardiac ischemia, arrhythmia, valvular catastrophe, and alternate causes of diffuse infiltrates.
- **Management priority:** Choose oxygen when hypoxemic, selected NIV, nitrates when appropriate for hypertensive physiology and diuretics for congestion; reassess blood pressure.
- **Common mistake:** Giving reflex fluid boluses to a congested patient without identifying the hemodynamic problem.
- **Exam trap:** Crackles alone do not distinguish edema from pneumonia, ARDS and pulmonary fibrosis.

الـorthopnea وcrackles وJVP مرتفع يدعمون pulmonary edema القلبية، لكن يجب التفكير بـACS أو arrhythmia أو valve problem.

Rapid Recall: Must-Know Facts

01. Use structured pretest probability before any PE rule-out test.
02. PERC works only in an already very-low-risk patient.
03. Positive D-dimer is not a diagnosis of PE.
04. D-dimer FEU and DDU thresholds are not interchangeable.
05. High-probability PE usually proceeds to diagnostic imaging.
06. Normal SpO2, ECG or chest radiograph does not exclude PE.
07. RV dilation on echo is not specific for acute PE.
08. PE severity depends on physiology, RV, biomarkers and respiratory need.
09. Stable RV strain is not an automatic thrombolysis indication.
10. Large fluid loads may worsen severe PE-related RV failure.
11. Adult VTE apixaban source example: 10 mg BID ×7 days, then 5 mg BID.
12. An adult PE initial anticoagulation phase of 3–6 months is cited.
13. Asthma causes variable expiratory airflow limitation.
14. Prior asthma intubation raises future exacerbation risk.
15. Silent chest with fatigue is a life-threatening finding.
16. Normal/rising PaCO2 in severe asthma distress may mean fatigue.
17. SpO2 below 92% is a cited acute-asthma danger marker.
18. SABA is early bronchodilator therapy; severe cases may add ipratropium.
19. Adult asthma prednisolone example: 40–50 mg daily for 5–7 days.
20. Selected refractory severe asthma: IV magnesium 2 g over ~20 min.
21. Asthma acute oxygen target cited: SpO2 92–95%.
22. Steroids do not replace immediate inhaled bronchodilation.
23. If wheeze is anaphylaxis, IM epinephrine takes priority.
24. Before asthma discharge, ensure an ICS-containing strategy.
25. COPD: post-BD FEV1/FVC <0.70 is a cited diagnostic criterion.
26. COPD exacerbation requires reconsideration of PE, HF and PTX.
27. COPD-at-risk initial SpO2 target is 88–92%.
28. COPD oxygen changes require blood-gas reassessment.
29. COPD bilevel NIV addresses acute hypercapnic respiratory acidosis.
30. Worsening consciousness can indicate NIV failure.
31. Auto-PEEP: expiratory flow does not reach baseline.
32. Increasing respiratory rate can worsen obstructive air trapping.
33. High bicarbonate in COPD may be chronic compensation.
34. COPD antibiotics are selective, not automatic.
35. CAP severity scores cannot overrule hypoxemia or shock.
36. CURB-65 adds urea >7 mmol/L to CRB-65.
37. Viral-positive CAP does not automatically exclude coinfection.
38. Unstable suspected tension pneumothorax: decompress before imaging.
39. Light exudate: protein >0.5, LDH ratio >0.6, or fluid LDH >2/3 ULN.

40. Pleural infection with pH ≤7.2 merits drainage assessment if safe.

All recall statements restate facts already present in the integrated topics; they add no source data.

Conflict Resolution Register — Domain 03

Between-file conflicts: no directly opposed recommendations for the same clearly established patient population were verified within this Domain. This zero is a classification result, not a claim that the original files are clinically concordant in all circumstances.

Source-dependent pathway differences and internal tensions — retained without adjudication:

1. Selected stable primary spontaneous pneumothorax

- **Source statements:** CLEAN V3 includes size-linked observation/intervention examples and a >20%-volume intervention fragment; AQRAB cites BTS 2023 conservative care irrespective of size for selected stable, minimally symptomatic adult PSP.
- **Classification / disposition:** Different population detail and pathway framing; cannot claim a direct clinical contradiction without the complete CLEAN V3 case context.

2. COPD oral steroid course

- **Source statements:** CLEAN V3 and AQRAB cite prednisone-equivalent 40 mg daily for 5 days; AQRAB also names NICE prednisolone 30 mg for 5 days.
- **Classification / disposition:** Different published pathway conventions. Do not choose a universally correct regimen from these files.

3. Asthma intravenous beta-agonist use

- **Source statements:** CLEAN V3 contains IV salbutamol 250 micrograms over 5–10 min in a selected severe case; AQRAB says IV beta-agonists are not routine in standard acute-asthma management.
- **Classification / disposition:** Selected rescue intervention versus routine baseline strategy. Kept separate; no automatic conflict.

4. Viral-positive adult CAP and antibiotics

- **Source statements:** Within AQRAB, ATS 2025 recommends antibiotics for specified viral-positive comorbid/hospitalized populations while IDSA did not endorse those recommendations.
- **Classification / disposition:** Disagreement described inside the second source only. Source-dependent; no external adjudication.

5. Chest trauma drainage threshold

- **Source statements:** CLEAN V3 contains a traumatic hemothorax output example 200 mL/hour for 3 hours alongside an answer fragment referring to 250 mL/hour for 3 hours.
- **Classification / disposition:** Internal inconsistency in first-file wording; not a measured cross-file conflict.

6. Airway/respiratory domain boundary

- **Source statements:** Both sources contain mechanical-ventilation fundamentals under resuscitation and respiratory sections.
- **Classification / disposition:** General airway/RSI duplicated in Domain 01; this Domain retains only condition-specific ARDS and obstructive physiology.

Where evidence or original case context was insufficient, comparisons were not falsely counted as confirmed clinical conflicts.

Merge Quality Note — Domain 03

Counting unit: one authored English clinical bullet within the numbered Topic list. Arabic explanations, comparison-table cells and Rapid Recall restatements do not count as independent new clinical points. Provenance A = identifiable only in second-file mapping; C = identifiable in first-file mapping; B = corroborated by both in the audited source portions. These are traceable editorial classifications, not proof of global absence from every other page.

- **Final integrated clinical topics:** 43
- **Integrated English clinical points:** 256
- **Retained points mapped to CLEAN V3 only:** 39
- **Retained points mapped to AQRAB only:** 182
- **Corroborated points merged from both files:** 35
- **Confirmed direct between-file conflicts:** 0

- **Source-dependent contextual / intra-source differences:** 6
- **Explicitly withheld malformed or misleading fragments:** 7
- **Rapid Recall (no new facts):** 40
- **Comparison tables:** 7

Incomplete in source — not included

- **CLEAN V3:** “D 25%.” — Isolated number without condition, endpoint or unit context.
- **CLEAN V3:** “The respiratory quotient for pure lipid metabolism is 1.0.” — Ambiguous source claim without supporting explanation; not independently corrected.
- **CLEAN V3:** “A decrease in lung compliance leads to a reduced work of breathing. resistance of the lungs. rates.” — Corrupted, contradictory fragment lacks usable grammatical context.
- **CLEAN V3:** “Lung bases: Lung apices are favored rather than the bases.” — Missing clinical question and interpretation.
- **CLEAN V3:** “In severe pneumonia: D 25%.” — Isolated disconnected exam-answer element.
- **CLEAN V3:** “PE: Interventional radiology consultation for emergent embolization. Massive hemoptysis ...” — Two distinct topics collided; relevant hemoptysis facts recovered under Hemoptysis rather than mislabelled PE.
- **CLEAN V3:** “This rhythm shows a patient with narrow-complex PEA” under COPD. — ECG and respiratory diagnosis collision; no source basis for a causal statement.

Unrelated clinical statements from the source respiratory catch-all (e.g. toxin antidotes, unrelated procedures or cardiac arrhythmias) were deferred to their proper Domains rather than duplicated here. General airway/ventilator principles already reviewed in Domain 01 were also not recounted. Source duplicates were collapsed and case-question fragments without usable context withheld.

No external information was added. This source-only educational consolidation does not establish current guideline correctness or replace patient-specific clinical care and local policy.

Respiratory & Thoracic Emergencies | Pass 8 | Explicit CLEAN V3 source locations

> **Acute bronchitis — viral pattern**
>
> • High-yield fact: CLEAN V3 describes bronchitis as inflammation of the lower airways and states that viral pathogens are the typical cause.
>
> الـBronchitis هو التهاب بالمجاري التنفسية السفلية، والمصدر يذكر أن السبب المعتاد فيروسي.

Respiratory & Thoracic Emergencies. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 69

40 mg	Source statements: CLEAN V3 and AQRAB cite prednisone-equivalent 40 mg daily for 5 days; AQRAB also names NICE prednisolone 30 mg for 5 days.
30 mg	Source statements: CLEAN V3 and AQRAB cite prednisone-equivalent 40 mg daily for 5 days; AQRAB also names NICE prednisolone 30 mg for 5 days. Classification / disposition: Different published pathway conventions. Do not choose a universally correct regimen

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 69

Superior vena cava syndrome - clinical recognition [#022]

High-yield fact: Superior vena cava syndrome may arise from thoracic malignancy or intrathoracic obstruction, producing facial or upper-limb congestion.

احتقان الوجه أو الطرفين العلويين قد ينتج عن انسداد الوريد الأجوف العلوي بسبب مرض داخل الصدر.

EDIC / Emergency & Critical Care

04 — Gastrointestinal & Abdominopelvic Emergencies

Acute Abdominal Assessment and Surgical Red Flags

Topic 1: The undifferentiated acute abdomen

- **Core concept:** Stabilize threatened vital functions, assess perfusion, examine the abdomen, and seek the cause while treating pain.
- **Red flag:** Vascular hemorrhage or rupture, bowel ischemia, perforation and strangulated obstruction are high-risk abdominal alternatives.
- **Management priority:** Use resuscitation, analgesia, serial abdominal examinations, targeted imaging, and early surgical consultation when findings suggest urgent pathology.
- **Differential point:** Extra-abdominal causes may mimic abdominal disease; possible pregnancy also changes the dangerous differential.
- **Common mistake:** A urine test, clinical score, plain film or response to analgesia must not close an uncertain diagnosis prematurely.
- **Clinical Pearl:** Discharge requires reassessment, appropriate follow-up and explicit return precautions.

أعد الفحص بعد obstruction. أو perforation أو ischemia استبعد نزفًا أو IBS، أو Gastritis قبل تسمية الألم المسكنات ولا تجعل تحسن الألم وحده سببًا للخروج.

Topic 2: Mesenteric ischemia and pain out of proportion

- **Clinical trigger:** Severe abdominal pain out of proportion to examination should prompt assessment for bowel ischemia.
- **Red flag:** Early normal lactate or absent rigidity does not exclude evolving ischemia.
- **Management priority:** Resuscitate and involve surgery urgently when ischemia, peritonitis, shock or deteriorating physiology is suspected.
- **High-yield fact:** The CLEAN V3 material describes difficulty judging bowel viability after mesenteric revascularization because of vasospasm and reperfusion effects.
- **Number/threshold:** A planned second-look laparotomy after 24–48 hours is described in the CLEAN V3 acute mesenteric ischemia example, not as a universal rule.
- **Exam trap:** Do not wait for a rise in lactate before escalating a concerning ischemic abdomen.

طبيعيًا في البداية. lactate ؛ قد يكونmesenteric ischemiaالألم الشديد غير المتناسب مع الفحص علامة إنذار لـ surgical escalation. يحتاج peritonitis تدهور التروية أو ظهور

Topic 3: Perforation, peritonitis and urgent surgical escalation

- **Red flag:** Guarding, worsening constant pain, fever, shock, acidosis or peritoneal signs may signal bowel compromise.
- **Management priority:** Maintain resuscitation and urgent surgical review for suspected perforation or peritonitis.
- **High-yield fact:** Hemodynamic instability with a surgical abdominal process is a trigger for emergent surgical consultation and possible laparotomy in the source cases.
- **Common mistake:** A normal early examination does not override a worsening clinical trajectory.
- **Differential point:** Consider perforation, ischemia and strangulated obstruction alongside inflammatory abdominal conditions.

يغيّر المسار من المراقبة إلى التدخل العاجل. ثبّت المريض واستدعِ الجراحة perforation أو peritonitis وجود ولا تؤخر الإنقاذ بانتظار فحص واحد.

Topic 4: Pelvic and non-gastrointestinal mimics of abdominal pain

- **Clinical trigger:** Include ectopic pregnancy when pregnancy is possible in an acute abdominal-pain presentation.
- **Differential point:** Uterine, adnexal and cervical-motion tenderness are source-described clues to pelvic inflammatory disease.
- **Red flag:** Infertility treatment is associated with source-described risks including ectopic/heterotopic pregnancy, ovarian torsion and ovarian hyperstimulation syndrome.
- **Exam trap:** A gastrointestinal-looking pain pattern does not exclude a gynecologic or vascular emergency.
- **Management priority:** Use patient-specific pregnancy assessment, reassessment and appropriate specialty consultation; detailed gynecologic therapy is reserved for Domain 10.

الحوض قد يسبب ألمًا يبدو هضميًا. في المريضة المناسبة لا تنسَ الحمل وأسبابه الطارئة، وميّز الألم البطني عن torsion أو PID عند اختيار الفحوص.

Gastrointestinal Bleeding and Esophageal Emergencies

Topic 5: Initial gastrointestinal hemorrhage assessment

- **Core concept:** Assess hemodynamics, resuscitate and identify bleeding severity before definitive endoscopic or interventional control.
- **Clinical trigger:** Hematemesis, melena or hematochezia warrants recognition of the bleeding phenotype and the patient’s circulation.
- **Management priority:** Obtain source-directed endoscopy or intervention after stabilization as indicated; transfusion decisions depend on shock and clinical context.
- **Number/threshold:** In a stable adult without cardiovascular disease, the cited transfusion trigger is hemoglobin <7 g/dL, aiming for approximately 7–9 g/dL; this does not apply as a waiting rule in shock.
- **Clinical Pearl:** The Glasgow-Blatchford score is described in CLEAN V3 as a tool to estimate need for intervention and risk in GI hemorrhage.
- **Common mistake:** Do not assume that one hemoglobin result captures the severity of active hemorrhage.

إلى حد نقل الدم في مريض Hb قبل التنظير. لا تنتظر وصول resuscitationو perfusion هي أولوية GI bleeding مصدوم؛ وحدد هل النزف variceal أم nonvariceal.

Topic 6: Airway and procedural safety during massive hematemesis

- **Clinical trigger:** Heavy hematemesis during resuscitation or a procedure raises aspiration and airway-management concerns.
- **Management priority:** Have effective suction, airway tools and preoxygenation ready when preparing for airway intervention.
- **High-yield fact:** Resuscitation and bleeding-source control remain linked priorities.
- **Common mistake:** Undertaking an airway procedure without appropriate equipment and oxygenation preparation.

ضمن تجهيز preoxygenationوالأدوات و suction ؛ حضّرairway المريض الذي ينزف بغزارة قد يحتاج حماية الإنقاذ قبل الإجراء.

Topic 7: Nonvariceal upper-GI bleeding and high-risk ulcers

- **Management priority:** For nonvariceal upper-GI bleeding, endoscopy is generally arranged within 24 hours after resuscitation in the cited pathway.
- **Number/threshold:** Following endoscopic hemostasis of a high-risk ulcer, the source describes PPI 80 mg IV then 8 mg/hour for 72 hours, or an accepted high-dose alternative.
- **High-yield fact:** Peptic ulcer disease is described in CLEAN V3 as a common cause of nonvariceal upper-GI bleeding.
- **Exam trap:** A high-dose post-hemostasis regimen should not be detached from its indication.
- **Common mistake:** Delaying stabilization in order to rush directly to endoscopy.

خلال 24 ساعة حسب مسار الملف. endoscopy عادةً إلى nonvariceal بعد استقرار المريض، يوجه النزف المكثف هنا مرتبط بقرحة عالية الخطورة بعد hemostasis وليس لكل ألم معدة PPIعلاج الـ.

Topic 8: Suspected esophageal variceal hemorrhage

- **Clinical trigger:** Chronic alcohol exposure, ascites and caput medusae with upper-GI bleeding support a portal-hypertension/variceal source.
- **Management priority:** Suspected variceal bleeding follows a source-described endoscopy pathway within 12 hours after stabilization, with vasoactive treatment and appropriate antibiotics.
- **High-yield fact:** Octreotide is a somatostatin analogue that decreases portal venous pressure; CLEAN V3 describes it as a variceal vasoactive treatment.
- **Number/threshold:** CLEAN V3 separately cites terlipressin 2 mg IV initially, followed by 1–2 mg every 4–6 hours, under a British guideline pathway; this protocol-specific example is not reconciled here with other agent-selection statements.
- **Number/threshold:** The AQRAB-cited prophylaxis example is ceftriaxone 1 g IV daily for up to seven days, adjusted for patient factors and local policy.
- **Exam trap:** Balloon tamponade is described as temporary rescue in difficult variceal bleeding, not a substitute for definitive endoscopic treatment.
- **Clinical Pearl:** The source describes reducing esophageal balloon pressure by 5 mmHg every three hours in a Sengstaken-Blakemore example; procedural details require specialist context.

vasoactive therapyو resuscitation يستدعي ascites أو cirrhosis عند مريض لديه اشتباه variceal bleed مبكرًا. اختيار العامل الدوائي المختلف في المصدر لا يُحسم تلقائيًا endoscopy وتقييمantibiotics.

Topic 9: Mallory-Weiss tear and vomiting-associated hematemesis

- **Clinical trigger:** Hematemesis developing after repeated vomiting or retching suggests Mallory-Weiss tear.
- **Key discriminator:** CLEAN V3 uses recurrent vomiting in a young patient with bulimia as a clue to Mallory-Weiss bleeding.
- **Differential point:** Severe chest or neck pain, fever, crepitus or toxicity after forceful vomiting raises concern for esophageal perforation instead.
- **Management priority:** Address the patient's circulation and pursue appropriate endoscopic/surgical evaluation for concerning bleeding or perforation signs.
- **Exam trap:** Do not treat every post-vomiting presentation as a simple mucosal tear.

لكن الألم الصدري الشديد أو علامات Mallory-Weiss tear، قد يشير إلى repeated retching بعد Hematemesis بعد القيء تغيّر التشخيص تمامًا perforation.

Topic 10: Aortoenteric fistula as an uncommon bleeding source

- **Red flag:** Aortoenteric fistula may cause massive unexplained hematemesis or hematochezia.
- **High-yield fact:** The duodenum is cited as an important site in the CLEAN V3 aortoenteric fistula material.
- **Differential point:** Assess for vascular bleeding or rupture among dangerous acute-abdominal alternatives.
- **Exam trap:** A major bleed without clear liver disease is not automatically benign or nonvascular.

يجب ألا يُفسّر تلقائيًا بـ vascular catastrophe، نزف هضمي غزير غير مفسّر، خصوصًا مع علامات ulcer أو varices.

Topic 11: Diverticular lower-GI bleeding

- **Clinical trigger:** Painless bright red blood per rectum can be a presentation of diverticular bleeding.
- **Management priority:** First assess hemodynamic stability and resuscitate significant ongoing bleeding.
- **Key discriminator:** Diverticular bleeding and diverticulitis are distinct presentations: bleeding may be painless, whereas diverticulitis causes localized inflammatory pain.
- **Exam trap:** Do not assume the presence of diverticula automatically means infection.
- **Clinical Pearl:** In a stable source case, serial blood counts and GI consultation were part of the observation plan.

Diverticular bleeding قد يكون painless hematochezia، بينما diverticulitis استقرار. يسبب ألمًا والتهابًا. الضغط والفحص المتكرر يحددان أولوية التدخل.

Topic 12: Esophageal food bolus and foreign-body obstruction

- **Clinical trigger:** Inability to swallow saliva or manage secretions signals complete esophageal obstruction and aspiration risk.
- **Management priority:** Complete obstruction, esophageal disk battery or sharp-pointed esophageal object requires emergent endoscopic/specialist action.
- **Number/threshold:** Food bolus or other esophageal objects without complete obstruction generally need urgent removal within 24 hours in the cited pathway.
- **Key discriminator:** Identify the object, location, time since ingestion, number of objects, and airway/aspiration risk.
- **Common mistake:** Do not induce vomiting or force an impacted food bolus distally.
- **Exam trap:** A negative plain radiograph does not exclude a radiolucent foreign body.
- **Source limitation:** CLEAN V3 contains medication suggestions such as IV glucagon for impaction without a complete endoscopic-delay context; these are not promoted ahead of indicated endoscopy.

أهم سؤال: هل يستطيع المريض بلع اللعاب؟ عدم القدرة على التعامل مع secretions يعني انسدادًا كاملًا وخطر aspiration ويحتاج endoscopy طارئًا.

Topic 13: Esophageal perforation and post-instrumentation injury

- **Clinical trigger:** Forceful vomiting, recent endoscopy/instrumentation, foreign-body injury or caustic exposure followed by severe chest/neck pain should raise concern for perforation.
- **Red flag:** Fever, crepitus, toxicity or mediastinal/pleural signs after these exposures strengthen concern.
- **Management priority:** Maintain NPO, stabilize and obtain urgent GI and surgical input; use CT for suspected perforation/complications when feasible.
- **Exam trap:** Do not delay specialist escalation for imaging in a deteriorating patient.
- **Clinical Pearl:** Iatrogenic perforation can occur without a classic spontaneous-rupture history.

يستدعي التفكير بـ crepitus أو fever ألم صدر شديد مع instrumentation، عنيف أو vomiting بعد perforation. مبكرًا أهم من انتظار صورة واحدة GI/Surgery واستشارة NPO.

Topic 14: Reflux, pill esophagitis and progressive dysphagia

- **High-yield fact:** CLEAN V3 describes reflux as a common cause of esophagitis and GERD as a cause of heartburn, regurgitation and noncardiac chest discomfort.
- **Clinical trigger:** Recent doxycycline treatment with acute chest pain and odynophagia can suggest pill esophagitis.
- **Clinical trigger:** Progressive solid-food dysphagia may indicate esophageal stricture and predispose to impaction.
- **Exam trap:** Response to antacids does not exclude myocardial infarction.
- **Management priority:** Do not label dysphagia as uncomplicated GERD before considering obstructive or perforation danger.
- **Clinical Pearl:** The source mentions head-of-bed elevation and weight loss in symptomatic GERD management.

Heartburn قد ينتج عن GERD، لكن response للـantacid لا ينفي ACS. وإذا وُجد dysphagia متدرج أو odynophagia تحت كل الحالات بعد دواء، لا تضع GERD.

Topic 15: Achalasia, motility disturbance and delayed emptying

- **Key discriminator:** Achalasia is described by increased resting lower-esophageal sphincter tone and absent esophageal peristalsis.
- **High-yield fact:** The source differentiates diffuse esophageal spasm (uncoordinated contractions) from hypercontractile patterns (high-amplitude coordinated contractions).
- **Core concept:** Gastroparesis is delayed gastric emptying without a mechanical obstruction.
- **Management priority:** Metoclopramide is identified as a prokinetic treatment option for gastroparesis in CLEAN V3.
- **Exam trap:** Progressive dysphagia is not automatically uncomplicated acid reflux.

Achalasia مشكلة في LES relaxation وperistalsis، أما gastroparesis فهو delayed gastric emptying دون انسداد ميكانيكي. لا تخلط بين dysphagia وvomiting بسبب هذه الآليات.

Hepatobiliary, Liver and Pancreatic Emergencies

Topic 16: Biliary colic versus acute cholecystitis

- **Clinical trigger:** Biliary colic may cause steady right-upper-quadrant, epigastric or substernal pain with nausea/vomiting.
- **Key discriminator:** Acute cholecystitis usually produces persistent right-upper-quadrant pain accompanied by localized or systemic inflammation.
- **Management priority:** Ultrasound is the initial imaging study described for suspected cholecystitis.
- **Management priority:** Assess complications and the need for early surgical consultation/cholecystectomy.
- **Exam trap:** The term "colic" does not guarantee short, intermittent pain.

اسم biliary colic استمرار مضلل أحيانًا لأن الألم قد يكون ثابتًا. يرجح inflammatory findings مع RUQ pain وبدأ التقييم غالبًا بـ ultrasound، cholecystitis.

Topic 17: Ascending cholangitis: infection plus obstruction

- **Core concept:** Ascending cholangitis is infection involving an obstructed biliary tree, requiring sepsis recognition and obstruction assessment.
- **Clinical trigger:** Fever, jaundice and right-upper-quadrant pain form Charcot triad, but the triad may be incomplete.
- **Key discriminator:** Assess systemic inflammation, cholestasis, imaging evidence of obstruction and organ dysfunction rather than using the triad alone.
- **Clinical Pearl:** An older patient may present mainly with hypotension and lack prominent fever or jaundice.
- **Management priority:** Begin resuscitation and appropriate antimicrobials and arrange biliary drainage/source control.

- **Number/threshold:** The cited ASGE-derived pathway suggests decompression within 48 hours, with greater urgency in shock or deterioration.
- **Exam trap:** Antibiotics alone do not decompress an infected obstructed biliary system.

Cholangitis تعني infection مع biliary obstruction. غياب Charcot triad؛ لا يستبعدها، خصوصًا عند كبار السن. وحده لا يعالج الانسداد المستمر antibiotic.

Topic 18: Biliary drainage, ERCP and gallstone complications

- **Management priority:** When feasible, the source favors endoscopic over percutaneous biliary drainage for cholangitis.
- **Clinical trigger:** Pancreatitis complicated by cholangitis is a cited indication for urgent ERCP.
- **Common mistake:** Do not schedule urgent ERCP for every uncomplicated episode of biliary pancreatitis.
- **High-yield fact:** Ultrasound can demonstrate gallstones and may assess common-bile-duct dilation or obstruction.
- **Clinical Pearl:** Mild gallstone pancreatitis may warrant cholecystectomy during the same admission when clinically suitable.

acute biliary لكنه ليس إجراءً روتينيًا لكل biliary obstruction/cholangitis، يُستخدم للتعامل مع ERCP أو انسداد مناسب cholangitis اربط الحاجة للتصريف بوجود pancreatitis.

Topic 19: Acute pancreatitis: diagnostic criteria and causes

- **Core concept:** Acute pancreatitis involves pancreatic inflammatory injury; the source names gallstones and several other precipitating causes.
- **Number/threshold:** Diagnosis requires at least two of: characteristic pain, lipase/amylase >3 times upper limit of normal, or characteristic imaging.
- **Key discriminator:** Enzyme magnitude alone does not establish disease severity.
- **Clinical Pearl:** Hypertriglyceridemia is cited as a likely etiology when triglycerides exceed 1,000 mg/dL and another cause is not clear.
- **High-yield fact:** Azathioprine is described as a possible drug cause of pancreatitis.
- **Management priority:** Assess for a gallstone-related cause with transabdominal ultrasound.

تشخيص acute pancreatitis يتطلب lipase مثل 2 من 3، ولا يجوز تقييم الشدة من وحده. حدّد السبب عند وجود سياق drugs أو hypertriglyceridemia أو gallstones.

Topic 20: Acute pancreatitis: resuscitation, nutrition and imaging

- **Management priority:** Use moderately aggressive isotonic fluid resuscitation, preferentially lactated Ringer’s in the cited pathway, and reassess frequently.
- **High-yield fact:** Analgesia and antiemetics are supportive priorities; mild disease can receive early oral low-fat solid food as tolerated.
- **Key discriminator:** CT/MRI is reserved for diagnostic uncertainty or failure to improve after 48–72 hours; routine early CT solely to grade severity is discouraged.
- **Common mistake:** Avoid excessive fluid without repeated assessment of perfusion, lung findings, volume status, BUN and urine output.
- **Exam trap:** Do not delay feeding until pancreatic enzyme levels normalize.
- **Clinical Pearl:** Severe pancreatitis can cause multisystem organ failure; source cases mention abnormal glucose and systemic inflammatory findings as associated features.

والبول والرئتين. لا تؤخر BUNللتروية و reassessment ليست بلا حدود؛ أعطها مع pancreatitis السوائل في طبيعية enzymes التغذية حتى تصير.

Topic 21: Necrotizing pancreatitis and pancreatic collections

- **Key discriminator:** The CLEAN V3 review describes contrast-enhanced CT or MRI as imaging methods for necrotizing pancreatitis.
- **Management priority:** Do not use prophylactic antibiotics for sterile pancreatic necrosis or uncomplicated sterile inflammation.
- **High-yield fact:** Chronic pancreatitis can follow repeated acute episodes and is associated with pancreatic pseudocysts.
- **Clinical Pearl:** A pseudocyst is described as a well-circumscribed peripancreatic fluid collection.
- **Exam trap:** A new diagnosis of pancreatitis does not by itself require immediate CT or urgent ERCP.

حسب الحاجة، necrosis يساعدان بتقييم MRI أو contrast مع CT. وقائية antibiotics لا يعني اشتباه necrosis بينما pseudocyst من مضاعفات chronic pancreatitis.

Topic 22: Acute viral hepatitis and serologic discriminators

- **Key discriminator:** Acute hepatitis A is identified with IgM anti-HAV in the source material and spreads by the fecal-oral route.
- **Key discriminator:** HBsAg positivity with negative anti-HBs indicates ongoing hepatitis B infection in the cited case; anti-HBc IgM supports an acute HBV pattern.
- **Clinical trigger:** Hepatitis A may present with fever, abdominal pain, jaundice and fatigue, including in children.
- **High-yield fact:** Acute HBV may include arthralgias or rash and marked AST/ALT elevation.
- **Exam trap:** A falling transaminase level alongside worsening encephalopathy and coagulation can signal loss of functioning hepatocytes rather than recovery.
- **Clinical Pearl:** CLEAN V3 links chronic hepatitis C infection with hepatocellular carcinoma risk.

افصل HAV عن HBV عبر serology: IgM anti-HAV مرتبط بالتهاب A، و HBsAgالحاد مع anti-HBc IgM يدعم العدوى الحادة B أو التخثر انخفاض الإنزيمات وحده كتحسن عند تدهور الوعي. لا تفسر بحسب المصدر.

Topic 23: Cirrhosis and portal hypertension: pattern recognition

- **Core concept:** Cirrhosis reflects hepatocyte injury with replacement by fibrosis and regenerative nodules in the supplied teaching material.
- **Clinical trigger:** Longstanding alcohol misuse, ascites, caput medusae and splenomegaly may support a cirrhosis/portal-hypertension phenotype.
- **High-yield fact:** CLEAN V3 describes an AST:ALT ratio >2 as a possible feature of alcohol-associated hepatitis in a source example.
- **Management priority:** Assess for acute bleeding, infection, encephalopathy and deterioration in kidney or circulatory function.
- **Exam trap:** A cirrhotic patient with upper-GI bleeding still needs source assessment; not all bleeding is variceal.

Cirrhosis قد يظهر مع ascitesو splenomegalyو varices. هذه العلامات تساعد على تقدير portal hypertension لكنها لا تُغني عن تقييم نزف أو infection أو organ dysfunction.

Topic 24: Ascites and spontaneous bacterial peritonitis

- **Clinical trigger:** New confusion, worsening renal function or suspected infection in a patient with ascites should raise concern for spontaneous bacterial peritonitis.
- **Number/threshold:** Ascitic-fluid neutrophils >=250 cells/mm³ support treatment for spontaneous bacterial peritonitis in the supplied review.
- **Management priority:** Link diagnostic paracentesis and treatment decisions to the clinical state and ascitic-fluid findings.
- **Exam trap:** Absence of striking abdominal pain does not exclude an ascitic infection in a deteriorating patient.

حتى بدون صورة ألم SBP قد يكون علامة renal function أو تدهور confusion، ascitesو cirrhosis في مريض ascitic neutrophils ≥250/mm³ بطن واضحة. الـ معلومة امتحانية مهمة.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 78

80 mg	resuscitation in the cited pathway. Number/threshold: Following endoscopic hemostasis of a high-risk ulcer, the source describes PPI 80 mg IV then 8 mg/hour for 72 hours, or an accepted high-dose alternative.
2 mg, 1–2 mg	as a variceal vasoactive treatment. Number/threshold: CLEAN V3 separately cites terlipressin 2 mg IV initially, followed by 1–2 mg every 4–6 hours, under a British guideline pathway; this protocol-specific example is not reconciled here with other agent-selection
1 g	statements. Number/threshold: The AQRAB-cited prophylaxis example is ceftriaxone 1 g IV daily for up to seven days, adjusted for patient factors and local policy.

Domain 04 | Gastrointestinal & Abdominopelvic Emergencies

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

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Esophageal perforation - Boerhaave syndrome [#023]

Key discriminator: A mediastinal crunch (Hamman sign) following forceful vomiting should prompt consideration of esophageal perforation.

صوت فرقعة في المنصف بعد قيء عنيف قد يشير إلى انثقاب المريء (Boerhaave).

Upper gastrointestinal bleeding - initial stabilization [#024]

Key discriminator: Upper gastrointestinal bleeding originates proximal to the ligament of Treitz; resuscitation and airway assessment precede definitive endoscopy.

النزف الهضمي العلوي يقع قبل رباط Treitz؛ الإنعاش وتقييم الهواء يسبقان التنظير النهائي.

Topic 25: Acute liver failure and encephalopathy

- **Core concept:** Acute liver failure is described as encephalopathy and coagulopathy with acute hepatic injury in a patient without prior cirrhosis.
- **Number/threshold:** CLEAN V3 cites INR >1.5 or prolonged PT as part of a fulminant-hepatic-failure concern when altered sensorium is present.
- **Management priority:** Monitor glucose and contact critical care and liver/transplant services early in the source pathway.
- **Exam trap:** Do not interpret falling liver enzymes as reassurance if consciousness and coagulation worsen.
- **Source limitation:** The CLEAN V3 steroid statement for severe alcohol-associated hepatitis contains contradictory clauses; it is not converted into a treatment instruction.

الـAcute liver failure ليس مجرد encephalopathy مع coagulopathy في مرض حاد مرتفعة: وجود AST/ALT
دون cirrhosis سابقة علامة خطورة وتتطلب مراقبة glucose وcritical care.

Obstruction, Appendix and Colonic Emergencies

Topic 26: Small-bowel obstruction: initial assessment

- **Core concept:** Small-bowel obstruction may arise from adhesions, hernias, tumors or other mechanical causes.
- **High-yield fact:** Prior postoperative adhesions are described as a major cause of SBO, about 60–70% in CLEAN V3 source examples.
- **Clinical trigger:** Distension, vomiting, pain, obstipation or a prior abdominal operation may suggest SBO.
- **Management priority:** If uncomplicated adhesive SBO is likely, use NPO, IV fluid/electrolyte correction, decompression when appropriate and surgical oversight.
- **Key discriminator:** CT helps define the obstruction, transition point, cause and complications when clinically appropriate.
- **Exam trap:** Recent stool passage or a transition point alone does not prove complete obstruction or strangulation.

في SBO وعلامات السبب قيّم peritonitis/ischemia أولًا. History of surgery يرجح adhesions لكنه لا يثبت
السبب. المراقبة المحافظة تتطلب serial examination لا قرارًا نهائيًا.

Topic 27: Strangulated obstruction and threatened bowel

- **Red flag:** Continuous worsening pain, guarding, fever, tachycardia, shock, acidosis or concerning CT features suggest possible strangulation/ischemia.
- **Management priority:** Peritonitis, suspected strangulation, ischemia or deterioration requires urgent surgical assessment and possible operative management.
- **Common mistake:** A normal early lactate cannot reliably exclude evolving bowel ischemia.
- **Key discriminator:** A CT transition point alone is not proof of vascular compromise.
- **Clinical Pearl:** A strangulated hernia in the source material similarly requires immediate surgical consultation.

إذا أصبح الألم continuous/worsening أو ظهر guarding أو shock أو acidosis فلا تستمر بخطة conservative
أولي طبيعي lactate تحتاج مراجعة جراحية عاجلة حتى لو .SBO

Topic 28: Large-bowel obstruction: tumor, stricture and other causes

- **Core concept:** Large-bowel obstruction may arise from malignancy, stricture, volvulus or pseudo-obstruction.

- **Clinical trigger:** Marked distension, vomiting, failure to pass stool/flatus or worsening abdominal pain should prompt obstruction assessment.
- **Management priority:** Resuscitate, maintain NPO, correct deficits and involve surgery early; CT usually defines site, cause and complications if stable.
- **Red flag:** Cecal dilation, ischemia, perforation, peritonitis, sepsis or deterioration changes the urgency.
- **High-yield fact:** CLEAN V3 warns that flexible sigmoidoscopy cannot exclude malignancy in unexamined proximal colon.
- **Exam trap:** Do not confuse mechanical LBO with acute colonic pseudo-obstruction.

Large bowel obstruction قد ينجم عن malignancy أو volvulus أو stricture أو pseudo-obstruction. يجب تقييم perforation وischemia قبل اختيار والتروية decompression.

Topic 29: Sigmoid versus cecal volvulus

- **Core concept:** Colonic volvulus occurs when a bowel loop twists, producing luminal obstruction.
- **Key discriminator:** CLEAN V3 describes the classic "coffee-bean" pattern on abdominal radiography for sigmoid volvulus.
- **Management priority:** Eligible sigmoid volvulus without ischemia or perforation may receive endoscopic detorsion/decompression.
- **Management priority:** Cecal volvulus generally requires surgery; ischemia, perforation, peritonitis, failed decompression or deterioration requires urgent operation.
- **Exam trap:** Do not use the same initial treatment for sigmoid and cecal volvulus.
- **Differential point:** Distinguish volvulus from cancer, stricture and colonic pseudo-obstruction.

الفرق الامتحاني: uncomplicated sigmoid volvulus قد يبدأ بـ endoscopic detorsion، بينما cecal volvulus يحتاج غالبًا surgery. وجود ischemia أو perforation يمنع المسار البسيط.

Topic 30: Gallstone ileus and internal/ventral hernia

- **Core concept:** Gallstone ileus occurs after a large gallstone erodes from the gallbladder into bowel and causes mechanical obstruction.
- **Clinical Pearl:** CLEAN V3 associates internal hernias with mesenteric spaces opened after substantial weight loss in a source case.
- **High-yield fact:** Prior surgery is cited as a common antecedent of ventral hernia.
- **Red flag:** A hernia with obstruction, worsening constant pain, peritonitis or shock requires urgent surgical assessment.
- **Exam trap:** A prior surgical history does not establish the obstruction cause without evaluation.

Gallstone ileus هو mechanical obstruction بعد مرور حصاة كبيرة من المرارة إلى الأمعاء. أما internal أو ventral hernia فيجب تقييم قابليتها للانحشار والاختناق.

Topic 31: Appendicitis: evolution and diagnostic reasoning

- **Clinical trigger:** Periumbilical pain migrating to the right lower quadrant is a classic appendicitis clue.
- **Key discriminator:** Pain before vomiting is described as a useful historical discriminator, though no isolated finding establishes appendicitis.
- **High-yield fact:** Presentation varies with age, pregnancy, appendix position and disease stage.
- **Management priority:** Use a validated clinical score as an adjunct rather than a replacement for examination and judgment.
- **Exam trap:** Absence of fever or classic migration does not rule out appendicitis.

لا ينفيه fever أو migration لكن عدم وجود appendicitis، قد يدعم RLQ ثم ينتقل إلى periumbilical ألم يبدأ استخدم القصة مع الفحص والاحتمال والتصوير المناسب.

Topic 32: Appendicitis imaging and treatment pathway

- **Management priority:** Select ultrasound, CT or MRI according to age, pregnancy status, probability, body habitus and local availability.
- **Key discriminator:** A nondiagnostic ultrasound is not a negative exclusion test if clinical suspicion persists.
- **Management priority:** High probability, peritonitis or suspected complication warrants surgical consultation.
- **High-yield fact:** Selected uncomplicated appendicitis may be managed using an antibiotics-first shared plan with follow-up in the cited pathway.
- **Source limitation:** A CLEAN V3 appendix-diameter sentence is extraction-corrupted; the unqualified 6–7 mm statement is not used as a stand-alone diagnostic rule.
- **Exam trap:** One reassuring test must not override persistent high clinical concern.

اختيار US أو CT أو MRI يعتمد على العمر والحمل والـpretest probability. US غير الحاسم ليس سلبيًا عندما يستمر الاشتباه.

Topic 33: Named abdominal physical examination signs

- **High-yield fact:** Murphy sign: inspiratory arrest during right-upper-quadrant palpation may indicate cholecystitis.
- **High-yield fact:** Rovsing sign: left-lower-quadrant palpation provokes right-lower-quadrant pain.
- **Key discriminator:** Psoas sign: pain with hip extension or resisted flexion can support retrocecal appendicitis or psoas irritation.
- **Key discriminator:** Obturator sign: pain with internal rotation of a flexed hip supports pelvic appendiceal irritation.
- **Clinical Pearl:** Cullen sign and Grey Turner sign are bruising clues to retroperitoneal/hemorrhagic processes; they do not replace resuscitation and evaluation.
- **Exam trap:** Absence of any one named physical sign does not rule out appendicitis.

تزيد signs حسب موقعها. هذه تدعم appendicitis Rovsing/Psoas/Obturator، وMurphy للـcholecystitis الاشتباه ولا تُستخدم منفردة لنفي التشخيص.

Topic 34: Other gastrointestinal causes resembling appendicitis

- **Differential point:** Yersinia enterocolitica ileocolitis can mimic acute appendicitis.
- **High-yield fact:** A source pediatric intussusception presentation includes colicky pain, vomiting and currant-jelly stool.
- **Clinical Pearl:** The CLEAN V3 material describes intussusception as an important cause of SBO in children younger than two years.
- **High-yield fact:** Failure to pass meconium is a clue to neonatal Hirschsprung disease in the supplied material.

these findings remain GI differential clues here.

ومشكلات الحوض قد تُشابهها بحسب intussusception وileocolitis دائمًا؛ لا تجعل RLQ pain يعني appendicitis العمر والسياق.

Diarrhea, Inflammatory Bowel and Anorectal Emergencies

Topic 35: Acute gastroenteritis and dehydration assessment

- **Core concept:** Most uncomplicated acute infectious diarrhea is managed with supportive assessment and replacement of losses.
- **Clinical trigger:** Evaluate mental state, urine output, perfusion, dehydration, electrolyte loss, sepsis and exposure history.
- **Management priority:** Reduced-osmolarity oral rehydration solution is first-line for mild to moderate dehydration when oral intake is possible.
- **Management priority:** Use isotonic IV fluids for shock, severe dehydration, altered consciousness, ileus or unsuccessful ORS.
- **Number/threshold:** CLEAN V3 gives 245 mOsm/kg as an example of reduced-osmolarity oral rehydration solution.
- **Common mistake:** An antiemetic does not replace fluid and electrolyte replacement.

الـpriority في diarrhea هو hydration وperfusion وليس معرفة pathogen من أول دقيقة. ORS للجفاف الخفيف/المتوسط و IV isotonic عند shock أو فشل الشرب.

Topic 36: Watery versus inflammatory diarrhea and testing

- **Key discriminator:** Stool testing is selective for fever, bloody or mucoid stool, severe pain, sepsis, immunocompromise, outbreaks or results that would change management.
- **Management priority:** Empiric antibiotics are not routine for uncomplicated watery diarrhea.
- **Red flag:** Bloody stool can suggest Shiga-toxin-producing E. coli; avoid reflex antibiotic prescribing before assessment.
- **Common mistake:** Avoid antimotility drugs in children with acute diarrhea and in febrile or inflammatory diarrhea in the source guidance.
- **High-yield fact:** CLEAN V3 associates severe gastroenteritis with particular exposures and notes that norovirus may cause fever.

يحوّل التقييم إلى احتمال inflammatory infection. لا تُعطِ antibiotics أو antimotility therapy تلقائيًا قبل تقدير السبب والمخاطر. وجود fever أو severe pain أو blood

Topic 37: Diverticulitis: uncomplicated versus complicated

- **Core concept:** Diverticulitis causes localized inflammatory abdominal pain rather than the typical painless diverticular hemorrhage presentation.
- **Management priority:** CT helps confirm a first or uncertain diagnosis, severe disease, failed improvement or complications.
- **High-yield fact:** Antibiotics may be selective in mild uncomplicated disease in suitable immunocompetent patients.
- **Red flag:** Frailty, comorbidity, immunocompromise, vomiting, refractory symptoms, significant inflammation, abscess, peritonitis or obstruction favors escalation.
- **Exam trap:** Do not prescribe antibiotics solely because diverticula are present.

Diverticulitis يختلف عن diverticular bleeding: الألم والالتهاب هنا هما الأساس. Antibiotics قد تكون انتقائية في الحالة البسيطة المناسبة، لكن sepsis أو abscess أو immunocompromise تغيّر القرار.

Topic 38: Crohn disease: transmural injury and complications

- **Core concept:** Crohn disease can involve the GI tract from mouth to anus with transmural inflammation.
- **Red flag:** Consider bowel obstruction, fistula or abscess in a Crohn patient with new severe abdominal symptoms.
- **Management priority:** Cross-sectional imaging and surgical input may be needed when a structural complication is suspected.

- **High-yield fact:** CLEAN V3 links Crohn disease with small-bowel adenocarcinoma risk and pyoderma gangrenosum among associated complications.
- **Exam trap:** Do not assume worsening Crohn symptoms are uncomplicated inflammation without investigating infection or obstruction.

Crohn يمكن أن يصيب من mouth إلى anusوبـ transmural inflammationلذلك ؛ obstruction وfistula وabscess بسيطًا flare أهم من اعتبار كل ألم.

Topic 39: Ulcerative colitis and acute severe flare

- **Clinical trigger:** Recurrent abdominal pain with bloody diarrhea and possible arthritis can suggest ulcerative colitis.
- **Management priority:** Acute severe UC warrants hospital assessment for bleeding, systemic toxicity, hydration, infection and colonic dilation.
- **Management priority:** Test for C. difficile in acute severe UC and monitor for toxic megacolon and clinical decline.
- **High-yield fact:** Hospitalized acute severe UC patients should receive appropriate VTE prophylaxis unless contraindicated in the cited pathway.
- **Red flag:** Severe bleeding, perforation, toxic megacolon or failure to respond warrants urgent surgical assessment.
- **Common mistake:** Avoid opioids and anticholinergics in acute severe UC because of ileus/dilation concern.

في acute severe UC ابحث عن infection وخصوصًا C. difficile وراقب toxic megacolon. دخول المستشفى جزء من المسار المذكور في المصدر وVTE prophylaxis.

Topic 40: Toxic megacolon and severe inflammatory colitis

- **Clinical trigger:** Marked abdominal distension, tenderness and systemic instability in a colitis patient suggests toxic megacolon.
- **Management priority:** Reassess for perforation, infection, dehydration and organ dysfunction, with early surgical input for deterioration.
- **Differential point:** Persistent symptoms after exclusion of infective causes warrant evaluation for inflammatory bowel disease.
- **Exam trap:** Do not manage toxic dilation as routine outpatient colitis.
- **Clinical Pearl:** Medication choices that impair motility may worsen dilation/ileus in acute severe disease.

IBD مع severe distension وsystemic instability ليس flare بسيطًا. يحتاج تقييم perforation وsepsis ومراجعة جراحية عاجلة عند التدهور.

Topic 41: IBS and chronic non-alarm abdominal symptoms

- **Core concept:** Irritable bowel syndrome is described as chronic crampy pain with altered bowel habit, bloating, diarrhea or constipation and no established structural bowel disease.
- **Clinical Pearl:** Pain may improve after defecation in the source description.
- **Differential point:** Fecal calprotectin and further testing appear in the source discussion of differentiating IBS from IBD when concern persists.
- **Exam trap:** Do not apply a benign functional diagnosis to new severe pain, instability, bleeding or other acute-abdominal warning signs.

يحوّل red flags لكن ظهور defecation، مع تغير التبرز وقد يتحسن بعد chronic crampy painيرتبط بـ IBS البحث إلى organic disease.

Topic 42: Colorectal malignancy and high-risk bowel symptoms

- **Red flag:** Colorectal malignancy may be a cause of large-bowel obstruction and requires appropriate evaluation.

- **Exam trap:** Flexible sigmoidoscopy does not exclude disease in colon segments it does not visualize.
- **Differential point:** Hemorrhoids can coexist with another source of rectal bleeding; consider malignancy or IBD when appropriate.
- **Management priority:** Stabilize bleeding or obstruction first, then choose definitive diagnostic and specialist follow-up according to severity.

نزف شرجي أو تغير عادات التبرز لا يعني تلقائيًا hemorrhoids أو IBS. في الاشتباه بـcolorectal cancer، فحص جزء من القولون لا يستبعد مرضًا في الجزء غير المرئي.

Topic 43: Anal fissure: pain with defecation

- **Clinical trigger:** Severe pain during and after defecation, sometimes lasting hours, supports an anal fissure pattern.
- **Management priority:** The source describes conservative initial treatment with warm sitz baths, laxatives, dietary fiber and topical symptom-directed therapy.
- **High-yield fact:** Topical nifedipine/lidocaine or nitroglycerin appear among treatment approaches in CLEAN V3.
- **Red flag:** A lateral anal fissure is described as concerning for possible underlying systemic illness.
- **Exam trap:** Do not label all severe anorectal pain as a hemorrhoid.

Anal fissure يسبب pain وبعد أثناء defecation وقد يستمر بسبب spasm. Fissure جانبية تستدعي التفكير بسبب غير اعتيادي بدل وصفها بسيطًا دون تقييم.

Topic 44: Hemorrhoids: internal, external and thrombosed

- **Key discriminator:** Internal hemorrhoids occur above the dentate line, external hemorrhoids below it.
- **Clinical trigger:** A tense, bluish, highly tender perianal lump suggests thrombosed external hemorrhoid in source cases.
- **Management priority:** CLEAN V3 describes clot excision/elliptical excision as a selected treatment for a painful thrombosed external hemorrhoid.
- **Differential point:** Consider fissure, abscess, proctitis, IBD and malignancy when symptoms do not fit a simple hemorrhoid.
- **Exam trap:** Hemorrhoids do not automatically explain all rectal bleeding.

له internal hemorrhoid شديد الألم، بينما قد يكون bluish tender mass الـthrombosed external hemorrhoid موقع مختلف بالنسبة للـdentate line. النزف لا ينفي malignancy.

Topic 45: Perianal abscess and occult deep anorectal infection

- **Clinical trigger:** Perineal, rectal, back or buttock pain with little external change may represent a deep perianal/perirectal abscess.
- **Red flag:** Systemic illness, cellulitis, diabetes, immunocompromise or Crohn disease increase concern.
- **Management priority:** Drainage is central for a drainable perianal abscess; selected uncomplicated superficial abscesses may undergo I&D in the ED.
- **High-yield fact:** Antibiotics are adjunctive in selected high-risk/systemic presentations rather than a replacement for drainage.
- **Exam trap:** A normal external inspection does not exclude a deep anorectal abscess in a toxic patient.

هو العلاج الأساسي عند وجود Drainage. Abscess عميق قد يسبب ألمًا شديدًا مع فحص خارجي شبه طبيعي؛ وحدها ليست antibiotics تجمع قابل للتصريف source control.

Topic 46: Fistula, proctitis and pilonidal disease

- **Core concept:** Proctitis is inflammation of distal rectal mucosa and can be infectious, inflammatory, medication-related or radiation-associated.
- **Clinical trigger:** Pilonidal disease is centered near the superior natal/gluteal cleft; an abscess can present with pain and swelling.
- **Management priority:** Incision and drainage is described for a pilonidal abscess in the source material.
- **Differential point:** Consider fissure, fistula, abscess, proctitis, malignancy or IBD in atypical anorectal presentations.
- **Clinical Pearl:** A sensitive history about possible rectal foreign-body insertion can be important when examination and symptoms suggest it.

فلها سياق مختلف. اسأل عن fistula أو proctitis أما، natal cleft قرب أعلى pilonidal disease: موقع الألم مهم.
بصياغة غير حكمية إذا كان الألم الشرجي غير مفسر foreign body.

Topic 47: Rectal prolapse and other local emergencies

- **Key discriminator:** The supplied source identifies rectal procidentia as rectal prolapse.
- **Clinical trigger:** A protruding anorectal mass requires examination to distinguish prolapse from thrombosed hemorrhoid or another cause.
- **Source limitation:** CLEAN V3 includes disconnected reduction instructions without a clearly recoverable diagnosis or full technique; they are excluded.
- **Exam trap:** Do not assume all protruding perianal masses have the same treatment.

فرّق بحسب شكل الكتلة وموقعها قبل محاولة أي تدخل.thrombosed hemorrhoid ليس Rectal prolapse؛
مبتورة فلا تُستخدم كتعليمات تقنية Word في reduction بعض تفاصيل.

Comparison Tables

Tables are included only where the conditions share a useful exam discriminator. They contain no facts beyond the two supplied source documents.

Bleeding phenotypes

Feature	Nonvariceal upper-GI bleed	Suspected variceal upper-GI bleed	Exam point
Source clues	Peptic ulcer or another nonvariceal lesion	Cirrhosis / portal hypertension context	Identify source, not just color of blood
Endoscopy timing	Generally within 24 h after resuscitation	Cited pathway within 12 h	Stabilization first
Additional therapy	High-dose PPI after high-risk ulcer hemostasis	Vasoactive therapy plus appropriate antibiotics	Treat different pathophysiology

Biliary presentations

Feature	Biliary colic / cholecystitis	Ascending cholangitis	Exam point
Pattern	Biliary pain; cholecystitis adds persistent inflammation	Infected obstructed biliary tree	Fever/jaundice may be absent
Initial investigation	Ultrasound	Inflammation + cholestasis + obstruction evaluation	No single negative test closes concern
Key action	Assess for surgical management of cholecystitis	Resuscitation, antibiotics and drainage	Obstruction needs source control

Obstruction physiology

Feature	Uncomplicated SBO	Strangulated / ischemic SBO	Exam point
Pain	May be colicky or intermittent	Continuous or worsening	Clinical trajectory changes care

PASS 9 — Source-backed clinical recovery | Domain 04

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 85

80 mg, 8 mg/h	09. Variceal UGIB: the cited pathway targets endoscopy within 12 hours. 10. High-risk ulcer after hemostasis: cited IV PPI 80 mg, then 8 mg/h for 72 hours.
1–2 mg	source — not included. CLEAN V3 1569: Glucagon 1–2 mg IV for food impaction lacks complete context for failure, contraindications and urgent endoscopy — Incomplete in source — not included as a management recommendation.
125 mg	urgent endoscopy — Incomplete in source — not included as a management recommendation. CLEAN V3 1567: Oral vancomycin 125 mg QID is detached from complete CDI severity/treatment context — Incomplete in source — not included as a dosing instruction.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 85

Ileus versus mechanical bowel obstruction [#026]

High-yield fact: Paralytic ileus is functional impairment of peristalsis, not a fixed mechanical bowel blockage.

العلوص الشللي خلل وظيفي في حركة الأمعاء، وليس انسدادًا ميكانيكيًا ثابتًا.

Supplement IDs 026 | ProTest Emergency Medicine 5e (2021) | PDF p. 132 | Question 102

Ileus versus mechanical bowel obstruction [#027]

High-yield fact: Postoperative ileus is common after surgery, but persistent symptoms require exclusion of mechanical obstruction and metabolic causes.

استمرار العلوص بعد الجراحة يستلزم التفكير بانسداد ميكانيكي أو اضطراب استقلابي.

Supplement IDs 027 | 1200 by Subject | PDF p. 156 | Question 40

Post-bariatric surgery - internal hernia [#030]

Key discriminator: Internal hernia is an important cause of bowel obstruction after Roux-en-Y gastric bypass.

بعد Roux-en-Y gastric bypass، تذكّر الفتق الداخلي كسبب محتمل لألم البطن أو الانسداد.

Supplement IDs 030 | 1200 by Subject | PDF p. 134 | Question 4

Acute liver failure - etiologies [#031]

Key discriminator: Acute liver failure has diverse causes, including viral hepatitis and acetaminophen toxicity.

الفشل الكبدي الحاد له أسباب متعددة؛ من الأمثلة المذكورة التهاب الكبد الفيروسي وسمية Acetaminophen.

Supplement IDs 031 | 1200 by Subject | PDF p. 173 | Question 70

Gastrointestinal & Abdominopelvic Emergencies

dosing or medical-correctness certification.

Bowel obstruction and nitrous oxide

Exam trap: CLEAN V3 warns against Entonox (nitrous oxide/oxygen mixture) in suspected bowel obstruction because of the risk of gas expansion.

يحذر المصدر من Entonox في suspected bowel obstruction لاحتمال توسع الغاز.

Feature	Uncomplicated SBO	Strangulated / ischemic SBO	Exam point
Physiology	Without peritonitis or shock	May show guarding, fever, acidosis or shock	Normal early lactate does not exclude
Management	NPO, IV fluid, serial exam with surgery oversight	Urgent surgical evaluation	Do not persist with conservative plan

Volvulus treatment

Feature	Sigmoid volvulus	Cecal volvulus	Exam point
Initial option	Endoscopic detorsion if eligible and bowel viable	Usually surgical management	Not interchangeable
When surgery urgent	Ischemia, perforation or failed decompression	Usually indicated	Peritonitis overrides detorsion
Imaging	Classic coffee-bean appearance described	CT/source assessment for cause and compromise	Do not rely on X-ray alone

Inflammatory bowel disease

Feature	Crohn disease	Ulcerative colitis	Exam point
Pattern	Transmural, mouth to anus	Colitis with bloody diarrhea	Recognize different complication profiles
Major complication	Obstruction, abscess, fistula	Acute severe colitis, toxic megacolon	Image or escalate based on findings
Admission concern	Complication or sepsis	ASUC needs inpatient evaluation in source pathway	Exclude infection and protect against VTE

Diverticular presentations

Feature	Diverticular bleeding	Diverticulitis	Exam point
Dominant feature	Often painless hematochezia	Localized pain and inflammation	Do not treat all diverticula as infection
Initial priority	Resuscitate significant bleeding	CT when first/uncertain or complicated	Assessment differs
Antibiotics	Not automatic for bleeding	Selective in suitable mild uncomplicated disease	Host risk changes decisions

Anorectal conditions

Feature	Anal fissure	Perianal abscess	Exam point
Clue	Pain at/after defecation	Persistent focal or deep pain; may lack external signs	Normal external exam does not exclude deep abscess
Management	Sitz baths, stool softening and topical therapy	Drainage for drainable collection	Antibiotics are adjunctive for selected abscess cases
Exam pitfall	A lateral fissure suggests atypical cause	Toxic patient may have deep occult infection	Avoid assuming hemorrhoids

Rapid Recall: Must-Know Facts

Exactly 40 short facts; each restates a concept already present in this Domain. No new knowledge is introduced here.

01. Acute abdomen: resuscitate, reassess and identify surgical red flags.
02. Pain out of proportion raises concern for mesenteric ischemia.
03. Normal early lactate does not exclude bowel ischemia.
04. Peritonitis or perforation requires urgent surgical assessment.
05. Pregnancy-related emergencies can mimic GI abdominal pain.
06. Active GI bleeding requires perfusion assessment before endoscopy.
07. Stable, noncardiac adult UGIB: the source cites transfusion at Hb <7 g/dL.
08. Nonvariceal UGIB: endoscopy generally within 24 hours after resuscitation.
09. Variceal UGIB: the cited pathway targets endoscopy within 12 hours.
10. High-risk ulcer after hemostasis: cited IV PPI 80 mg, then 8 mg/h for 72 hours.

11. Variceal bleeding requires vasoactive treatment and appropriate antibiotics.
12. Repeated retching followed by hematemesis suggests Mallory-Weiss tear.
13. Massive unexplained GI bleeding can reflect aortoenteric fistula.
14. Inability to swallow saliva suggests complete esophageal obstruction.
15. An esophageal battery or sharp object requires emergent specialist removal.
16. Severe post-vomiting chest pain and crepitus: consider esophageal perforation.
17. Cholangitis is biliary infection plus obstruction.
18. Charcot triad is not sufficiently sensitive to exclude cholangitis.
19. Cholangitis requires antibiotics and timely biliary source control.
20. Pancreatitis diagnosis: two of pain, >3x enzymes, characteristic imaging.
21. Lipase magnitude does not determine pancreatitis severity.
22. Pancreatitis: reassess fluid therapy and begin oral feeding when tolerated.
23. Routine prophylactic antibiotics are not indicated for sterile pancreatitis.
24. Urgent ERCP is not routine for uncomplicated biliary pancreatitis.
25. Ascitic neutrophils >=250/mm3 support treatment for SBP.
26. Acute liver failure: acute injury with encephalopathy and coagulopathy.
27. SBO with ischemia or peritonitis requires surgery, not routine observation.
28. Prior surgery suggests adhesions but does not prove the SBO cause.
29. A transition point alone does not establish strangulation.
30. Sigmoid volvulus may be endoscopically detorsed if no ischemia/perforation.
31. Cecal volvulus generally requires surgical management.
32. A nondiagnostic ultrasound does not exclude appendicitis.
33. Rovsing: LLQ palpation reproduces RLQ pain.
34. Obturator: internal hip rotation may support pelvic appendicitis.
35. Reduced-osmolarity ORS is preferred for mild/moderate dehydration.
36. Bloody diarrhea requires cause-specific assessment before antibiotics.
37. Diverticular bleeding may be painless; diverticulitis causes inflammation.
38. Crohn disease may cause obstruction, fistula or abscess.
39. Acute severe UC: evaluate C. difficile and toxic megacolon.
40. Perianal abscess needs drainage; antibiotics alone do not replace source control.

Conflict Resolution Register — Domain 04

Confirmed direct contradictions between the two supplied files: 0 in the retained, source-matched statements. This is not a claim that the sources have been externally medically validated.

Source-dependent / protocol-context differences

- **Issue:** 1. Variceal vasoactive agent selection

CLEAN V3: CLEAN V3 describes octreotide as a treatment of choice and separately cites BSG-recommended terlipressin as first-line.

AQRAB: AQRAB p. 116 requires an appropriate vasoactive drug but does not name a preferred agent.

Nature: Internal wording/protocol emphasis; no confirmed direct AQRAB-vs-CLEAN conflict.

- **Issue:** 2. SBO water-soluble contrast

CLEAN V3: CLEAN V3 includes the isolated statement “Oral Gastrografin is not recommended” with image-dependent case context.

AQRAB: AQRAB pp. 120–121 presents supportive care and surgical oversight without a blanket oral-contrast rule.

Nature: Incomplete context, not evidence of contradictory protocols.

- **Issue:** 3. Variceal and nonvariceal bleeding protocols

CLEAN V3: CLEAN V3 includes case-specific medication examples.

AQRAB: AQRAB p. 116 separates the 12-hour variceal and 24-hour nonvariceal pathways.

Nature: Different indications and timing, not a direct conflict.

Internal source inconsistency retained without resolution

CLEAN V3 Domain 4 contains both an affirmative corticosteroid-treatment statement and an active-infection contraindication in the same severe alcohol-associated hepatitis bullet (para 1301). The context is internally irreconcilable as extracted. Classification: Conflict — source-dependent (within CLEAN V3; not an established between-file conflict). No steroid instruction is carried into the study content.

CLEAN V3 includes octreotide and terlipressin first-choice wording in separate variceal-treatment bullets. Both are recorded with their source pathway; no preferred agent is chosen by external inference.

Merge Quality Note — Domain 04

Counting unit: one authored English clinical point under the 47 numbered Topic headings. Arabic explanations, comparison-table cells, index/header material and Rapid Recall statements do not inflate the point count. Source categorization is conservative: “both” denotes a directly identifiable corroborating concept, not all plausible conceptual overlap.

- **Final integrated clinical topics:** 47
- **Integrated English clinical points:** 249
- **Authored points mapped to CLEAN V3 only:** 89
- **Authored points mapped to AQRAB only:** 152
- **Directly corroborated, integrated concepts from both files:** 8
- **Confirmed direct between-file conflicts:** 0
- **Protocol/context differences logged:** 3
- **Internal source contradictions explicitly withheld:** 1
- **Incomplete/malformed source fragment groups explicitly withheld:** 8
- **Comparison tables:** 7
- **Rapid Recall points (restatements):** 40

Incomplete in source — not included

- **CLEAN V3 1282:** Truncated hepatitis statement mixing unrelated comparison and missing terms — Incomplete in source — not included.
- **CLEAN V3 1301:** Mutually inconsistent corticosteroid treatment and contraindication clauses in a single bullet — Internal source inconsistency — not converted to an instruction.
- **CLEAN V3 1359:** Corrupted appendix-diameter text lacking clean imaging/measurement context — Incomplete in source — not included.
- **CLEAN V3 1381, 1382, 1388, 1428, 1472:** Orphan procedural responses without reconstructable clinical setting — Incomplete in source — not included.
- **CLEAN V3 1473:** STEC and ETEC terminology conflated within a single source item — Internal source inconsistency — not used as pathogen identification.
- **CLEAN V3 1474–1475, 1480, 1568:** Isolated numeric fragments without an interpretable indication — Incomplete in source — not included.
- **CLEAN V3 1569:** Glucagon 1–2 mg IV for food impaction lacks complete context for failure, contraindications and urgent endoscopy — Incomplete in source — not included as a management recommendation.
- **CLEAN V3 1567:** Oral vancomycin 125 mg QID is detached from complete CDI severity/treatment context — Incomplete in source — not included as a dosing instruction.

Cross-domain content allocation

- **CLEAN V3 para 1288:** Sexual-assault prophylaxis doses/drugs. Reserved for: Emergency systems, ethics or Obstetrics & Gynecology.
- **CLEAN V3 paras 1374, 1376, 1395, 1396, 1401–1407, 1413, 1417, 1424:** Toxicology, psychiatry, neurologic, trauma and interpreter topics. Reserved for: Respective later domains.
- **CLEAN V3 paras 1460–1463, 1468, 1479, 1481, 1493, 1495–1496:** Urologic, toxicologic, neurologic and trauma treatment examples. Reserved for: Respective later domains.
- **CLEAN V3 paras 1361–1362, 1416, 1489, 1539, 1552–1555:** Pediatric obstruction and airway-related GI anomalies. Reserved for: Pediatrics & Neonatology; brief GI differential retained.

Quality limitation: the classification counts describe traceable authored integrated points; they are not a machine-verified census of every source phrase, and no medical guideline was checked outside the two supplied files. This Domain adds no external medical information or outside-sourced clinical corrections.

Source integrity: the original two study documents remain unchanged. The output consolidates usable source content, flags missing context and preserves source-dependent differences without independently choosing among them.

Strangulated inguinal hernia — skin warning

• High-yield fact: New overlying erythema at an inguinal hernia is identified in CLEAN V3 as a warning for possible strangulation and compromised bowel perfusion.

احمرار الجلد الجديد فوق Inguinal Hernia قد يشير إلى Strangulation وليس مجرد تغير جلدي.

Gastrointestinal & Abdominopelvic Emergencies

Supplementary Clinical Points

Topic: Small-bowel obstruction: source population estimate

Source-reported proportion: CLEAN V3 quotes approximately 60% postoperative-adhesion contribution to small-bowel obstruction in a developed-world population; the figure is context-specific and should not be generalized to every setting.

نسبة 60% في المصدر تتعلق بسياق سكاني محدد، ولا تُطبّق تلقائيًا على كل مريض بانسداد الأمعاء.

Source evidence: C01361: DOCX paragraph 1361

Gastrointestinal & Abdominopelvic Emergencies. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 90

Peptic ulcer disease - Helicobacter pylori [#029]

High-yield fact: H. pylori is a major cause of peptic ulcer disease and chronic gastric inflammation.

تُعد H. pylori سببًا مهمًا للقرحات الهضمية والتهاب المعدة المزمن.

EDIC / Emergency & Critical Care

05 — Neurology & Ophthalmology

Cerebrovascular Emergencies: Recognition, Imaging and Reperfusion

Topic 1: Acute stroke: first assessment and mimics

- **Core concept:** Treat sudden focal neurologic dysfunction as potential acute stroke and activate the stroke pathway while stabilizing vital functions.
- **Clinical trigger:** Establish last-known-well, point-of-care glucose, focused neurologic severity, and anticoagulant or bleeding history immediately.
- **Management priority:** Arrange urgent brain imaging to distinguish hemorrhage from ischemia and determine whether vascular or advanced imaging is needed.
- **Key discriminator:** Hypoglycemia can mimic a focal neurologic deficit and requires bedside testing.
- **Exam trap:** A normal early noncontrast CT does not exclude an evolving ischemic stroke.
- **Clinical Pearl:** A low NIHSS score can coexist with a disabling deficit; the score is not a stroke rule-out test.

Stroke حالة time-critical: حدّد last-known-well وافحص glucose مباشرة، لأنه قد يقلد عجزًا بؤريًا. CT السريع المبكر stroke لكنه لا يستبعد ischemia وhemorrhage يفرق بين.

Topic 2: Stroke localization: vascular territories and brainstem

- **High-yield fact:** Anterior cerebral artery stroke preferentially causes contralateral leg weakness and sensory loss, with relative face and arm sparing.
- **High-yield fact:** The middle cerebral artery is identified in CLEAN V3 as the most commonly affected vessel in ischemic stroke.
- **Key discriminator:** The NIH Stroke Scale favors anterior-circulation deficits and may underestimate posterior-circulation stroke.
- **Clinical trigger:** Lateral medullary symptoms raise concern for posterior-circulation ischemia; combined upper- and lower-extremity deficits can indicate corticospinal tract involvement.
- **Clinical Pearl:** Multiple simultaneous infarcts raise suspicion for an embolic source and warrant investigation of a shared origin.
- **Exam trap:** In suspected locked-in syndrome, test eye-command responses; failure to speak or move alone does not prove coma.

Localize قبل وصف العلاج: ACA و posterior circulation يميل للتأثير على الرجل المقابلة، قد لا يظهر واضحًا في NIHSS. لاحظ علامات brainstem أو locked-in.

Topic 3: Imaging pathway, TIA and early ischemia

- **Management priority:** Use noncontrast brain imaging urgently and pursue vessel imaging or specialized imaging when suggested by the stroke pathway.
- **High-yield fact:** Diffusion-weighted MRI is sensitive for early ischemic lesions in suspected TIA.
- **Exam trap:** A negative MRI does not independently exclude TIA or replace vascular evaluation.
- **Number/threshold:** The AQRAB-cited NICE pathway requests TIA assessment within 24 hours and considers aspirin 300 mg daily when appropriate and without contraindication.
- **Clinical Pearl:** CLEAN V3 notes that TIA carries particularly high recurrence risk in the first two days.
- **Differential point:** Symptomatic internal carotid stenosis of 70% or more is described as a setting in which carotid endarterectomy can reduce subsequent stroke risk.
- **Differential point:** For TIA or ischemic stroke associated with nonvalvular atrial fibrillation, CLEAN V3 describes oral anticoagulation with a suitable NOAC in patients without contraindications; this differs from a carotid-stenosis pathway.

TIA سلبيًا أو تحسن العجز سببًا لتأخير MRI حتى لو اختفت الأعراض ليست حالة مطمئنة. لا تجعل vascular prevention أو assessment.

Topic 4: Intravenous reperfusion: selection and drug regimens

- **Core concept:** Eligibility for IV thrombolysis depends on patient features, last-known-well, brain imaging and the applicable stroke protocol.
- **Number/threshold:** AQRAB cites IV thrombolysis within 4.5 hours for conventionally eligible patients and a door-to-needle goal of 60 minutes.
- **Number/threshold:** The cited stroke tenecteplase regimen is 0.25 mg/kg IV once, maximum 25 mg; it is distinct from myocardial-infarction dosing.
- **Number/threshold:** The cited alteplase regimen is 0.9 mg/kg IV, maximum 90 mg; 10% is given as bolus and the remainder over 60 minutes.
- **Number/threshold:** The source pathway requires blood pressure below 185/110 mmHg before IV thrombolysis.
- **Clinical Pearl:** Higher baseline stroke severity and CT signs of infarction are associated with hemorrhagic risk in the trial cited by CLEAN V3.
- **Exam trap:** Do not substitute ischemic-stroke pretreatment pressure goals for those used in ICH or after reperfusion.
- **Clinical Pearl:** CLEAN V3 describes emergency exchange transfusion targeting HbS below 30% in an ischemic-stroke patient with sickle cell disease; this is a disease-specific pathway, not routine reperfusion for all stroke.

كأنها كل القرار. المريض يحتاج وقت البداية والتصوير والاستطباب والموانع، ولا تخلط hours لا تحفظ 4.5 MI. مع جرعة stroke جرعة

Topic 5: Mechanical thrombectomy and large-vessel occlusion

- **Clinical trigger:** Early symptomatic internal carotid or proximal middle cerebral artery occlusion warrants prompt endovascular-team assessment.
- **Number/threshold:** Selected large-vessel-occlusion patients may undergo thrombectomy assessment up to 24 hours with advanced-imaging selection in the cited pathway.
- **Management priority:** Activate appropriate stroke-center resources and avoid preventable imaging or transfer delays.
- **Exam trap:** Onset time alone is not the entire late-window selection criterion; the specific imaging and clinical protocol matter.
- **Clinical Pearl:** Thrombolysis and thrombectomy are different reperfusion decisions; eligibility for one does not eliminate assessment for the other.

بسرعة. بعض المرضى Endovascular therapy يغيّر المسار إلى تقييم proximal MCA occlusion أو ICA وجود advanced imaging. قد يُختارون حتى 24 ساعة عبر

Topic 6: Blood-pressure targets: distinguish stroke pathways

- **Key discriminator:** Pretreatment thrombolysis pressure, selected intracerebral hemorrhage pressure, and post-reperfusion pressure are separate clinical decisions.
- **Number/threshold:** Before IV thrombolysis, AQRAB and CLEAN V3 cite a pressure threshold below 185/110 mmHg; CLEAN V3 separately mentions nicardipine infusion at 5 mg/hour for an indicated high-pressure example.
- **Number/threshold:** For selected ICH patients presenting with systolic pressure 150–220 mmHg, AQRAB describes smooth reduction toward 140 mmHg while avoiding pressure below 130 mmHg.
- **Exam trap:** After IVT or EVT, do not routinely pursue systolic pressure below 140 mmHg simply by importing the ICH target.
- **Common mistake:** Treating an example of 195/105 mmHg in a non-thrombolysis patient as proof that the IV-thrombolysis threshold applies universally.
- **Management priority:** Nicardipine is described as a preferred controllable agent in selected neurocritical hypertensive emergencies.

لا يوجد BP target واحد لكل Stroke. يجب التفريق بين المريض المرشح لـ IVT، ومن لديه ICH، وغير المرشح، وبعد reperfusion.

Topic 7: Intracerebral hemorrhage, anticoagulants and deterioration

- **Clinical trigger:** Acute focal deficit with intracranial blood on urgent imaging redirects treatment from ischemic reperfusion to hemorrhage management.
- **Management priority:** Assess blood-pressure control, anticoagulant exposure and reversal availability using the intracerebral hemorrhage pathway.
- **High-yield fact:** CLEAN V3 identifies chronic hypertensive vasculopathy as a leading mechanism for spontaneous ICH.
- **High-yield fact:** CLEAN V3 describes immediate factor VIII replacement in a patient with hemophilia A and high concern for intracranial bleeding after head injury.
- **Red flag:** Falling consciousness or a new pupil abnormality raises concern for secondary intracranial deterioration.
- **Exam trap:** The selected ICH pressure target should not be generalized to every ischemic stroke patient.

إلى مسار thrombolysis يحتاج تقييم السبب وارتفاع الضغط والأدوية المضادة للتخثر. لا تنقل أهداف ICH النزف، ولا تفسّر تدهور الوعي بأنه benign.

Topic 8: Subarachnoid hemorrhage: recognition and diagnostic boundaries

- **Clinical trigger:** Thunderclap headache, abrupt severe pain or associated altered consciousness requires evaluation for subarachnoid hemorrhage.
- **Key discriminator:** Depressed consciousness is not a typical feature of uncomplicated migraine and increases concern for a secondary cause.
- **Number/threshold:** A normal noncontrast CT within 6 hours can support exclusion of nontraumatic SAH only in selected adults with normal neurologic examination and suitable image quality and interpretation.
- **Common mistake:** Extending the six-hour CT rule to neurologically abnormal patients or all causes of headache.
- **Management priority:** If clinical concern remains, complete the indicated further diagnostic assessment rather than relying on one negative early test.
- **Clinical Pearl:** The source distinguishes the timing of CSF xanthochromia assessment for suspected SAH from the timing of lumbar puncture in suspected meningitis.

خلال 6 CT حتى لو تحسن الألم. قاعدة SAH workup مع تغير وعي أو عصبي يحتاج Thunderclap headache ساعات لا تشمل كل المراجعين.

Topic 9: Malignant MCA edema and raised intracranial pressure

- **Clinical trigger:** CLEAN V3 describes malignant middle-cerebral-artery edema developing commonly 2–5 days after stroke onset.

- **Number/threshold:** An approximately 80% mortality rate is quoted in the CLEAN V3 malignant-MCA source example; it is context-specific and not a universal prognostic estimate.
- **Red flag:** Falling level of consciousness, abnormal pupils or extensor posturing can indicate dangerous intracranial pressure or herniation.
- **High-yield fact:** The intracranial pressure-volume relationship is nonlinear and becomes steep as compensatory reserve is exhausted.
- **Management priority:** Maintain oxygenation and cerebral perfusion, monitor the examination, and seek urgent neurocritical support.
- **Exam trap:** The AQRAB source cautions against treating all raised-pressure syndromes with steroids or routine hyperventilation.

مراقبة الوعي والحدقات والتروية أهم من edema. كبير قد يتدهور المريض خلال أيام بسبب infarction بعد الاعتماد على صورة واحدة مبكرة.

Topic 10: Posterior circulation stroke versus peripheral vertigo

- **Clinical trigger:** Vertigo with abnormal gaze or neurologic signs may reflect posterior circulation stroke rather than benign inner-ear disease.
- **Key discriminator:** Use HINTS only in continuous acute vestibular syndrome with nystagmus and an examiner trained in its interpretation.
- **Key discriminator:** Unidirectional horizontal nystagmus and an abnormal head-impulse response favor a peripheral vestibular process in an appropriate clinical setting.
- **Exam trap:** A negative cranial CT is insufficient to exclude all posterior circulation infarction.
- **Differential point:** Vestibular neuritis typically lacks hearing loss; labyrinthitis can produce vertigo together with hearing impairment.
- **Management priority:** Assess timing, triggering pattern, hearing, gait and focal deficits before labeling a patient with peripheral vertigo.

HINTS يُستخدم فقط في dizzy المستمر مع Acute Vestibular Syndrome ؛ ليس فحصًا لكل مريض nystagmus ومن فاحص مدرب. CT سلبي لا ينفي posterior stroke.

Seizures, Altered Consciousness and Acute Encephalopathy

Topic 11: Status epilepticus recognition and immediate actions

- **Number/threshold:** A convulsive seizure lasting at least 5 minutes, or recurrent seizures without recovery between them, is treated as status epilepticus in the source pathway.
- **Management priority:** Support airway, breathing and circulation, check bedside glucose and administer an effective benzodiazepine promptly.
- **Clinical trigger:** Hypoglycemia and interruption of antiseizure medication are important correctable precipitants.
- **Exam trap:** Do not wait for prolonged seizure duration or an IV line when an effective alternative administration route is available.
- **Red flag:** Untreated status epilepticus may itself compromise ventilation and circulation; do not withhold appropriate seizure treatment out of fear alone.
- **Clinical Pearl:** Correct glucose and investigate metabolic, toxic, infectious, traumatic and pregnancy-related contributors in parallel.

سريعًا وأعط glucose وABC يبدأ القرار عند 5 دقائق، وليس بعد 30 دقيقة. عالج Status epilepticus مناسب route إذا هناك IV دون انتظار benzodiazepine.

Topic 12: First-line benzodiazepines and dose safety

- **Management priority:** Benzodiazepines are first-line emergency treatment in convulsive status epilepticus.

- **Number/threshold:** AQRAB cites an adult lorazepam example of 4 mg IV over 2 minutes, with appropriate respiratory monitoring.
- **High-yield fact:** CLEAN V3 describes repeating a benzodiazepine dose under the chosen protocol if the first adequate dose fails.
- **Common mistake:** Underdosing or repeatedly delaying treatment while awaiting tests.
- **Exam trap:** The displayed 4-mg regimen is a selected adult example, not a blanket pediatric or route-independent prescription.

First-line هو benzodiazepine بجرعة فعالة وسريعة. راقب التنفس وسجل الجرعات السابقة، لكن لا تجعل الخوف من sedation على السيطرة لتأخير seizure سببًا.

Topic 13: Second-line therapy and refractory status

- **Clinical trigger:** Persistent seizure after an adequate benzodiazepine warrants prompt second-line antiseizure loading.
- **Number/threshold:** A selected adult levetiracetam loading regimen in the sources is 60 mg/kg IV, maximum 4,500 mg; AQRAB gives an infusion time of 10 minutes.
- **Differential point:** CLEAN V3 separately lists pediatric levetiracetam examples with different maximum doses and timing; these must not be interchanged with the adult regimen.
- **High-yield fact:** Refractory status epilepticus persists despite appropriate first- and second-line agents and requires critical-care escalation.
- **Management priority:** CLEAN V3 discusses continuous anesthetic infusions such as midazolam or propofol under advanced-care management.
- **Exam trap:** An antiseizure loading dose is not the same as a maintenance prescription.

ولا تكرر المهدئات بلا second-line loading المناسب، انتقل إلى benzodiazepine إذا لم يتوقف الاختلاج بعد خطة. في refractory status يلزم ICU وEEG.

Topic 14: Nonconvulsive seizures, EEG and delayed awakening

- **Clinical trigger:** Failure to recover consciousness after convulsions stop raises concern for nonconvulsive status epilepticus.
- **Management priority:** Arrange EEG and specialist assessment if persistent altered mentation cannot be explained by a normal postictal course.
- **Key discriminator:** Absence of movement after paralysis or deep sedation does not prove electrical seizure termination.
- **High-yield fact:** EEG supports seizure diagnosis, status-epilepticus management and evaluation of unexplained impaired consciousness.
- **Exam trap:** Do not automatically label prolonged unresponsiveness as a postictal state.

فكر في seizure، بعد altered consciousness توقف الحركات لا يساوي توقف النشاط الكهربائي. إذا استمر حسب السياق، واطلب EEG وNCSE.

Topic 15: First seizure, reversible causes and clinical assessment

- **Core concept:** CLEAN V3 describes CT and serum glucose/electrolyte assessment when evaluating an adult first seizure for important structural or metabolic causes.
- **Clinical trigger:** Bedside hypoglycemia testing is an immediate priority because the cause is rapidly reversible.
- **Differential point:** Consider infection, trauma, stroke, toxic exposure, medication withdrawal and pregnancy-related seizures as context-specific causes.
- **Clinical Pearl:** A posterior shoulder dislocation can follow violent contractions during convulsions.
- **Number/threshold:** The CLEAN V3 cited NICE head-injury pathway calls for CT head within 1 hour after a post-traumatic seizure.
- **Exam trap:** A first seizure cannot be assumed idiopathic before evaluating treatable triggers and red flags.

First seizure يحتاج تحديد السبب القابل للعلاج مثل glucose أو electrolyte abnormalities. افحص trauma والعدوى والأدوية، ودوّن الحالة بعد النوبة.

Topic 16: Posterior reversible encephalopathy syndrome (PRES)

- **Clinical trigger:** Acute headache, seizures, encephalopathy or visual symptoms with significant hypertension suggest posterior reversible encephalopathy syndrome.
- **Core concept:** PRES is associated with disturbed cerebral autoregulation and endothelial dysfunction, producing vasogenic edema, often parieto-occipital.
- **High-yield fact:** Calcineurin inhibitors such as tacrolimus and ciclosporin are identified as potential triggers through endothelial toxicity.
- **Differential point:** PRES and ischemic stroke are not interchangeable despite overlapping neurologic complaints.
- **Management priority:** Evaluate the precipitant and hypertension in the patient-specific neurologic context.

Hypertension مع seizure أو visual disturbance قد يوجّه إلى PRES. لا تخلطه تلقائيًا مع السكتة أو الصداع الأولي دون تقييم.

Topic 17: Altered consciousness: glucose, delirium and mimics

- **Core concept:** Delirium is a rapid or fluctuating impairment of attention and awareness and usually reflects underlying illness.
- **Management priority:** Assess ABCs, glucose, vital signs, neurologic findings and reversible metabolic, infectious or toxic causes.
- **Key discriminator:** Establish the patient’s prior cognitive baseline and time course with collateral history.
- **Exam trap:** Do not treat a normal earlier CT as proof against evolving stroke, encephalitis or systemic deterioration.
- **Differential point:** Normal-pressure hydrocephalus is described with gait disturbance, urinary dysfunction and cognitive change.
- **Clinical Pearl:** Gait disturbance is often an early prominent symptom of normal-pressure hydrocephalus.

Delirium يختلف عن dementia المزمنة، ولا يستدعي البحث عن سبب طبي التغير الحاد أو المتذبذب بالانتباه. glucose وstroke والعدوى تنس.

Topic 18: Wernicke encephalopathy and Korsakoff pattern

- **Core concept:** Wernicke encephalopathy is linked to thiamine deficiency and classically produces confusion, ataxia and ocular motor disturbance.
- **Exam trap:** Absence of the full classic triad does not exclude Wernicke encephalopathy.
- **High-yield fact:** Korsakoff syndrome can produce severe memory disturbance and confabulation following thiamine-deficiency illness.
- **Number/threshold:** CLEAN V3 quotes a Pabrinex regimen of 2–3 pairs three times daily for 3–5 days, followed by 1 pair daily for a further 3–5 days or while improving; this is a formulation- and local-pathway-specific example.
- **Differential point:** The broader evaluation of altered consciousness must not be reduced to a single nutritional diagnosis.

Wernicke triad نادر أن يكون كاملًا: confusion، ataxia، ocular findings. علاج thiamine الوارد بالمصدر محدد بالبروتوكول ولا يُخلط مع غيره.

Headache, Meningitis, Encephalitis and Neurocritical Red Flags

Topic 19: Secondary headache: red flags and targeted workup

- **Clinical trigger:** Thunderclap onset, focal deficit, altered consciousness, fever or meningismus, pregnancy/postpartum, cancer, immunocompromise, anticoagulation, trauma or papilledema are secondary-headache red flags.
- **Management priority:** Use neurologic and ocular examination, appropriate imaging and cause-specific workup according to the presentation.
- **Key discriminator:** Loss of consciousness is unusual for uncomplicated migraine and should trigger reassessment for secondary causes.
- **Exam trap:** A normal head CT cannot independently exclude meningitis, cerebral venous thrombosis or dissection.
- **Clinical Pearl:** An established migraine diagnosis does not erase a new or changing red flag.

أو thunderclap onset، fever، focal deficit، pregnancy، cancer استبعد Migraine قبل تسمية الصداع anticoagulation. صورة طبيعية لا تستبعد كل الأسباب الخطيرة.

Topic 20: Migraine recognition and acute ED treatment

- **High-yield fact:** CLEAN V3 uses POUND to recall pulsatile pain, 4–72-hour duration, unilateral pain, nausea/vomiting and disabling severity.
- **Management priority:** For a presumed primary headache after red-flag assessment, use source-described nonopioid acute therapy and reassess symptoms and diagnosis.
- **Number/threshold:** AQRAB provides a selected regimen of ibuprofen 400 mg orally and metoclopramide 10 mg after contraindication review.
- **Number/threshold:** CLEAN V3 separately cites oral sumatriptan 50 mg for selected migraine attacks.
- **High-yield fact:** Metoclopramide is presented as an ED option for migraine, particularly with nausea or vomiting.
- **Exam trap:** Elevated blood pressure in pain alone does not justify ignoring headache red flags or the clinical examination.

عالج النوبة وتأكد من الاستجابة .red flags لكن ليس وسيلة لإلغاء ،Migraine يساعد في تذكر نمط POUND قبل الخروج.

Topic 21: Cluster headache and tension-type differentiation

- **Key discriminator:** Differentiate primary headache syndromes by time course, distribution and associated symptoms rather than applying the label "tension" to any tight-feeling pain.
- **High-yield fact:** CLEAN V3 describes tension-type headache as a common recurrent headache type.
- **Management priority:** Cited acute cluster-headache options include high-concentration oxygen and selected triptan routes after appropriate risk assessment.
- **Number/threshold:** CLEAN V3 describes 100% oxygen at 6–12 L/min for a selected cluster-headache regimen; other source text uses the term high-flow without a universal setting.
- **Clinical Pearl:** The AQRAB OSCE approach emphasizes identifying the pattern, excluding dangerous alternatives, treating the attack and planning prevention.

عالي التركيز مذكور لعلاج النوبة Oxygen. حدّد النمط وفكّر بمخاطر الصداع الثانوي أولًا في Cluster headache ضمن سياق المصدر.

Topic 22: Giant cell arteritis and vision-threatening headache

- **Clinical trigger:** A presentation suggestive of giant cell arteritis with developing visual loss calls for urgent treatment and specialist assessment.
- **Management priority:** CLEAN V3 specifically describes intravenous methylprednisolone when vision loss is developing in strongly suspected giant cell arteritis.

CLINICAL NUMBERS · TOPIC CONTEXT

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0.25 mg/kg, 25 mg	needle goal of 60 minutes. Number/threshold: The cited stroke tenecteplase regimen is 0.25 mg/kg IV once, maximum 25 mg; it is distinct from myocardial-infarction dosing.
0.9 mg/kg, 90 mg	from myocardial-infarction dosing. Number/threshold: The cited alteplase regimen is 0.9 mg/kg IV, maximum 90 mg; 10% is given as bolus and the remainder over 60 minutes.
4 mg	Number/threshold: AQRAB cites an adult lorazepam example of 4 mg IV over 2 minutes, with appropriate respiratory monitoring.
60 mg/kg	loading. Number/threshold: A selected adult levetiracetam loading regimen in the sources is 60 mg/kg IV, maximum 4,500 mg; AQRAB gives an infusion time of 10 minutes.
50 mg	after contraindication review. Number/threshold: CLEAN V3 separately cites oral sumatriptan 50 mg for selected migraine attacks. High-yield fact: Metoclopramide is presented as an ED option for migraine, particularly with nausea or vomiting.
40–60 mg	Number/threshold: CLEAN V3 also cites prednisolone 40–60 mg orally daily for another GCA presentation; the oral and IV source scenarios should not be collapsed.
10 mg	must also consider Listeria risk, age and host factors. Number/threshold: A source adjunctive dexamethasone example is 10 mg IV every 6 hours for up to 4 days when indicated by the relevant meningitis pathway.
600 mg	High-yield fact: Close contacts with relevant exposure to Neisseria meningitidis may need chemoprophylaxis. Number/threshold: CLEAN V3 cites rifampin 600 mg orally twice daily for 4 doses after direct mucosal secretion exposure in a healthcare-worker example.
500 mg	exposure in a healthcare-worker example. Number/threshold: Another CLEAN V3 UK public-health example cites ciprofloxacin 500 mg as a single dose to eligible contacts; agent selection is source/protocol dependent.
10 mg/kg	pathway. Number/threshold: AQRAB gives aciclovir 10 mg/kg IV every 8 hours as a selected HSV example, with weight, renal function and fluid adjustment; each dose is infused over at least 1 hour.
16 mg	compression. Number/threshold: The AQRAB-cited NICE example gives dexamethasone 16 mg orally (or equivalent parenteral dose) promptly, then 16 mg daily in suspected metastatic compression with neurologic signs.
16 mg	Number/threshold: The AQRAB-cited NICE example gives dexamethasone 16 mg orally (or equivalent parenteral dose) promptly, then 16 mg daily in suspected metastatic compression with neurologic signs. Number/threshold: AQRAB quotes MRI as soon as possible, within 24 hours under the cited malignant-compression
0.5 g, 1 g	relapse treatment. Number/threshold: AQRAB cites oral methylprednisolone 0.5 g daily for 5 days or selected IV 1 g daily for 3–5 days in severe or intolerant cases under specialist-pathway supervision.

CLINICAL NUMBERS · TOPIC CONTEXT

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50 mg	protection needs. Number/threshold: CLEAN V3 cites prednisolone 50 mg orally daily for 10 days and highlights treatment within 72 hours in its Bell-palsy example.
30 mg/kg	Differential point: CLEAN V3 describes levetiracetam 30 mg/kg IV over 5 minutes in a status- epilepticus case after first-line benzodiazepine failure. AQRAB p.140 separately describes a
0.9 mg/kg, 90 mg	08. Stroke tenecteplase example: 0.25 mg/kg IV, maximum 25 mg. 09. Stroke alteplase example: 0.9 mg/kg IV, maximum 90 mg. 10. Before IV thrombolysis the cited pressure goal is <185/110 mmHg.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 99

Stroke localization - hemispatial neglect [#039]

High-yield fact: Hemispatial neglect commonly follows a lesion in the nondominant parietal cortex.

الإهمال النصفي غالبًا ما يرتبط بإصابة الفص الجداري غير المسيطر؛ وهو غير الضعف الحركي البسيط.

- **Number/threshold:** CLEAN V3 also cites prednisolone 40–60 mg orally daily for another GCA presentation; the oral and IV source scenarios should not be collapsed.
- **Exam trap:** The corticosteroid route and treatment pathway depend on the presence and severity of threatened visual function.
- **Differential point:** Older age with a new or changed headache is a red flag requiring targeted secondary-headache evaluation.

وعدم التأخير بالعلاج كما في GCAجديد مع تهديد البصر خصوصًا عند كبار السن يحتاج التفكير بـ Headache المصدر، مع مراجعة مسار specialist.

Topic 23: Bacterial meningitis: early treatment and lumbar puncture

- **Clinical trigger:** Fever, meningismus, headache, abnormal consciousness or a concerning infection syndrome warrants prompt meningitis assessment.
- **Management priority:** Stabilize airway, circulation and seizures, obtain relevant cultures promptly and start appropriate empiric IV antimicrobials without clinically significant delay.
- **Key discriminator:** Perform LP first only if safe and if it will not delay treatment; routine CT before every LP is not recommended in the cited AQRAB pathway.
- **Red flag:** Focal neurologic signs, abnormal pupils, profoundly reduced or declining consciousness and concern for mass effect require LP safety assessment and imaging first.
- **Exam trap:** A delayed LP or indicated CT does not justify postponing time-critical antimicrobials.
- **Clinical Pearl:** Organism and age/host factors determine the empirical antimicrobial pathway.

في suspected bacterial meningitis العلاج time-critical: خذ blood cultures إن أمكن دون تأخير، ولا تنتظر CT أو إذا سيؤخران LP antibiotics.

Topic 24: CSF interpretation: bacterial, viral and fungal patterns

- **High-yield fact:** CLEAN V3 describes bacterial meningitis with high opening pressure, marked neutrophilic pleocytosis, elevated protein and a low CSF-to-blood glucose ratio.
- **Number/threshold:** A bacterial-source example cites WBC above 1,000/mm³, protein above 1 g/L and CSF glucose below 40% of serum glucose.
- **High-yield fact:** A source viral-meningitis example has lymphocytic WBC below 500/mm³, mild protein increase and a CSF/serum glucose ratio above 0.6.
- **Clinical Pearl:** CLEAN V3 identifies cryptococcal meningitis in patients with advanced HIV as a cause requiring different targeted assessment.
- **Exam trap:** CSF patterns must be interpreted with the clinical setting; do not delay initial therapy in a strongly suspected serious infection.

CSF pattern يدعم التفريق ولا يحسمه وحده: bacterial غالبًا neutrophils وprotein مرتفع و glucose منخفض؛ viral غالبًا lymphocytes وglucose محافظ عليه.

Topic 25: Meningitis drug examples and contact prophylaxis

- **Number/threshold:** An adult bacterial-meningitis example cites ceftriaxone 2 g IV every 12 hours; agent selection must also consider Listeria risk, age and host factors.
- **Number/threshold:** A source adjunctive dexamethasone example is 10 mg IV every 6 hours for up to 4 days when indicated by the relevant meningitis pathway.
- **Management priority:** Antimicrobials should not be delayed for imaging, LP or corticosteroid administration.
- **High-yield fact:** Close contacts with relevant exposure to Neisseria meningitidis may need chemoprophylaxis.
- **Number/threshold:** CLEAN V3 cites rifampin 600 mg orally twice daily for 4 doses after direct mucosal secretion exposure in a healthcare-worker example.
- **Number/threshold:** Another CLEAN V3 UK public-health example cites ciprofloxacin 500 mg as a single dose to eligible contacts; agent selection is source/protocol dependent.

- **Exam trap:** Do not confuse meningitis treatment dosing with prophylaxis dosing for exposed contacts.

الجرعات المذكورة هنا أمثلة بالغين ضمن بروتوكولات المصدر وليست اختيارات عامة لكل مريض. افصل العلاج للمريض عن chemoprophylaxis للمخالطين.

Topic 26: Encephalitis and HSV: recognition and initial treatment

- **Core concept:** Encephalitis involves brain inflammation and often presents with altered cognition or behavior, seizures, fever or focal neurologic deficits.
- **Clinical trigger:** New psychiatric or behavioral change with neurologic abnormalities should prompt evaluation for an encephalitic process.
- **Management priority:** Stabilize, investigate with MRI/CSF/EEG when indicated, and begin empiric IV aciclovir promptly when infectious encephalitis is suspected.
- **Number/threshold:** AQRAB cites an ideal aciclovir initiation window within 6 hours under its named 2026 guideline pathway.
- **Number/threshold:** AQRAB gives aciclovir 10 mg/kg IV every 8 hours as a selected HSV example, with weight, renal function and fluid adjustment; each dose is infused over at least 1 hour.
- **Exam trap:** Do not wait for an initial PCR result if suspicion remains high; specialist reassessment may be needed after early negative testing.
- **Differential point:** If bacterial meningitis has not been excluded, empiric antibacterial treatment may also be indicated.

سريعًا حسب المصدر aciclovir ابدأ encephalitis. قد يشير إلى seizure أو fever تغير الشخصية أو السلوك مع إذا الاشتباه مرتفع PCR ولا تنتظر.

Topic 27: Acute raised ICP and impending herniation

- **Red flag:** A new abnormal pupil, falling consciousness or extensor posturing can signal rising intracranial pressure or impending herniation.
- **Core concept:** The intracranial pressure-volume relation becomes steep when compensatory CSF and vascular shifts are exhausted.
- **High-yield fact:** CLEAN V3 notes a strong relationship between PaCO2 and cerebral blood flow within the cited physiological range.
- **Management priority:** Maintain adequate oxygenation and perfusion, repeat neurologic observations and obtain urgent expert support.
- **Exam trap:** Routine universal steroids or prophylactic hyperventilation are not supported as interchangeable treatments in the provided source review.

سريعًا. ليست كل وسائل neurosurgical escalation يستدعي brain injury تدهور الوعي أو تغير الحدقات مع خفض الضغط مناسبة لكل سبب.

Spinal Cord, Peripheral Nerve and Neuromuscular Emergencies

Topic 28: Acute spinal cord compression: recognize the level and cause

- **Clinical trigger:** Progressive weakness, sensory level, gait change, sphincter dysfunction, severe spinal pain, cancer, fever, trauma or anticoagulation raises concern for cord compression.
- **Management priority:** Perform a focused motor, sensory, reflex and sphincter assessment, obtain urgent MRI and contact spine/neurosurgical services.
- **Differential point:** Compression may arise from malignancy, epidural infection, hemorrhage, trauma, degenerative disease or inflammatory pathology.
- **Clinical Pearl:** Lhermitte-type shock-like sensations on neck flexion support possible cervical cord involvement.
- **Exam trap:** Do not automatically apply a malignancy-specific steroid regimen to other causes of spinal neurologic deficit.

- **High-yield fact:** CLEAN V3 describes central cord syndrome as disproportionate upper-extremity neurologic deficit, and Brown-Séquard syndrome as a hemicord injury pattern.

ابدأ spinal emergency evaluation. يحتاج bladder/bowel symptoms مع sensory level أي ضعف متزايد أو MRI المناسبة. ولا تؤخر neurological examination

Topic 29: Cauda equina, conus medullaris and early urinary symptoms

- **Clinical trigger:** Back pain with saddle sensory change, altered bladder/bowel or sexual function, or progressive neurologic deficit requires urgent cauda equina assessment.
- **High-yield fact:** Urinary retention, with or without overflow incontinence, is a classic presenting feature of cauda equina syndrome.
- **Key discriminator:** A reassuring postvoid residual measurement does not independently exclude early or incomplete cauda equina syndrome.
- **Differential point:** Conus medullaris and cauda equina syndromes require careful motor/reflex localization; a single upper-motor-neuron sign is not a complete rule.
- **Management priority:** Obtain urgent MRI and specialist review when the clinical pattern is concerning.
- **Common mistake:** Waiting for complete incontinence or a single confirmatory symptom before escalation.

Cauda equina قد تكون incomplete: اسأل عن urinary sensation وليس incontinence وصعوبة بدء التبول. PVR منخفض وحده لا ينفيها فقط.

Topic 30: Metastatic spinal cord compression: distinct steroid pathway

- **Clinical trigger:** Cancer with new severe back pain or a neurologic deficit raises suspicion for metastatic spinal cord compression.
- **Number/threshold:** The AQRAB-cited NICE example gives dexamethasone 16 mg orally (or equivalent parenteral dose) promptly, then 16 mg daily in suspected metastatic compression with neurologic signs.
- **Number/threshold:** AQRAB quotes MRI as soon as possible, within 24 hours under the cited malignant-compression pathway and sooner if symptoms worsen.
- **Management priority:** Obtain urgent spine oncology/neurosurgical assessment for definitive treatment.
- **Exam trap:** Do not import the metastatic-compression dexamethasone dose into all nonmalignant cord syndromes.

الجرعة dexamethasone 16 mg المذكورة تخص suspected metastatic compression مع neurological signs، وليست وصفة لكل back pain أو cauda equina.

Topic 31: Guillain–Barré syndrome: progression and respiratory risk

- **Core concept:** Guillain–Barré syndrome is an acute immune-mediated polyradiculoneuropathy, often after infection, with progressive weakness and reduced reflexes.
- **Key discriminator:** Symmetric progressive limb weakness with depressed deep tendon reflexes supports the typical pattern, though variants can differ.
- **Clinical trigger:** Weak cough, poor secretion clearance, bulbar dysfunction, autonomic instability or worsening respiratory mechanics requires early airway and ICU input.
- **High-yield fact:** CLEAN V3 identifies acute inflammatory demyelinating polyneuropathy as the commonest GBS variant and Campylobacter as a notable antecedent infection.
- **Exam trap:** Normal oxygen saturation does not exclude dangerous respiratory muscle weakness.
- **Clinical Pearl:** CSF protein and electrodiagnostic tests may be nondiagnostic early, and serial clinical assessment is important.
- **High-yield fact:** CLEAN V3 provides elevated CSF protein as a supporting GBS finding; an early normal CSF result does not reliably rule out the clinical syndrome.

bulbar function طبيعي لا يطمئن؛ قيّم SpO2. مع ضعف متناظر ونقص المنعكسات infection قد يبدأ بعد GBS والتنفس بشكل متكرر. وcough

Topic 32: GBS: immunotherapy and contraindication traps

- **Management priority:** IV immunoglobulin or plasma exchange is described for appropriately selected, clinically significant GBS.
- **High-yield fact:** AQRAB explicitly states that routine corticosteroids do not improve GBS recovery.
- **Red flag:** CLEAN V3 flags succinylcholine-related hyperkalemia risk in GBS.
- **Number/threshold:** CLEAN V3 describes IVIg eligibility in severe GBS or within a cited four-week symptom period; do not detach that example from specialist assessment.
- **Exam trap:** Do not delay airway assessment while awaiting CSF protein elevation or nerve-conduction confirmation.

ليست علاجًا مناعيًا فعالًا حسب المصدر، plasma exchange. Steroids أو IVIg الشديد قد يفيد GBS في

hyperkalemia. قد يسبب succinylcholineو

Topic 33: Myasthenic crisis: fatigability, bulbar function and rescue

- **Core concept:** Myasthenia gravis impairs neuromuscular transmission through antibodies against postsynaptic acetylcholine receptors or related junctional proteins.
- **Clinical trigger:** Fluctuating diplopia, ptosis, dysphagia, weak cough or fatigable limb and respiratory weakness raises concern for myasthenia and possible crisis.
- **Management priority:** Assess airway protection, secretion handling and serial respiratory mechanics; escalate promptly for bulbar or respiratory deterioration.
- **High-yield fact:** IVIg or plasma exchange is described as disease-directed rescue treatment in selected myasthenic crises.
- **Clinical Pearl:** CLEAN V3 notes a thymoma association in some myasthenia patients and mentions contrast CT chest within appropriate evaluation.
- **Exam trap:** A reassuring saturation or isolated inspiratory-force reading is insufficient when bulbar function is deteriorating.

لا يضمن سلامة SpO2 قد يشمل العين والبلع والتنفس. ارتفاع fatigable weakness تعطي Myasthenia gravis

ventilatory muscles.

Topic 34: Myasthenia versus Lambert–Eaton syndrome

- **Key discriminator:** Myasthenia gravis classically causes fatigability that worsens with repeated muscle use.
- **Key discriminator:** Lambert–Eaton myasthenic syndrome may show facilitation or improvement of strength with repeated activity.
- **High-yield fact:** CLEAN V3 identifies anti-acetylcholine-receptor transmission impairment as a key MG mechanism.
- **Clinical Pearl:** Activity-related diplopia with fluctuating generalized weakness is a source-described MG clue.
- **Exam trap:** Neuromuscular cause should not be excluded by absence of a focal brain lesion when the pattern is fatigable.

قد تتحسن القوة بعد استعمال متكرر. الفارق مهم عند LEMS عادة تتعب العضلات مع التكرار، بينما MG

اختيار التشخيص.

Topic 35: Botulism: descending weakness and antitoxin urgency

- **Clinical trigger:** Symmetric cranial neuropathies, bulbar findings and descending weakness raise concern for botulism.

- **Key discriminator:** A descending cranial/bulbar pattern helps distinguish suspected botulism from typical ascending GBS or fatigable MG.
- **Management priority:** Contact public-health and specialist services promptly, obtain specimens when feasible without delaying treatment, and initiate antitoxin according to the local pathway.
- **Red flag:** Bulbar dysfunction and secretion problems may threaten breathing before obvious limb weakness.
- **Source limitation:** AQRAB explicitly provides no independent botulism-antitoxin dose; none is invented here.

Botulism قد يظهر بضعف قحفي متناظر ونزول الضعف. لا تنتظر الفحوص لتبدأ antitoxin pathway عند الاشتباه العالي.

Topic 36: Multiple sclerosis relapse versus pseudo-relapse

- **Core concept:** Multiple sclerosis is a central demyelinating disease affecting brain, spinal cord and optic nerves.
- **Key discriminator:** A true relapse is a new or worsening neurologic syndrome persisting beyond 24 hours without an alternative trigger such as fever or infection.
- **Differential point:** Pseudo-relapse reproduces prior symptoms in association with physiological triggers without necessarily representing new inflammation.
- **Management priority:** Assess neurologic disability and exclude infection or alternative diagnoses before initiating relapse treatment.
- **Number/threshold:** AQRAB cites oral methylprednisolone 0.5 g daily for 5 days or selected IV 1 g daily for 3–5 days in severe or intolerant cases under specialist-pathway supervision.
- **Exam trap:** High-dose corticosteroids should not be started solely because a prior MS diagnosis exists.

MS relapse بسبب أعراض قديمة تكرر مجرد وليس ساعة 24 من أكثر غالبًا يستمر جديد fever أو infection. استبعد العدوى قبل steroids.

Topic 37: Peripheral facial weakness: Bell palsy and central mimics

- **Key discriminator:** Forehead sparing favors a central facial-weakness pattern; involvement of the forehead supports a peripheral seventh-nerve lesion in a compatible presentation.
- **Exam trap:** A facial-nerve distribution alone is not sufficient to exclude stroke when other neurologic red flags are present.
- **Management priority:** Assess onset, accompanying cranial-nerve deficits, stroke features, ear disease and corneal protection needs.
- **Number/threshold:** CLEAN V3 cites prednisolone 50 mg orally daily for 10 days and highlights treatment within 72 hours in its Bell-palsy example.
- **Red flag:** Failure of eyelid closure requires attention to eye protection while the diagnosis is clarified.

Bell palsy يحتاج تقييم علامات مطلق للسكتة rule-out. لكنه ليس central palsy، Forehead sparing قد يرجح stroke وحماية العين إذا الإغلاق ضعيف.

Topic 38: Vertigo: neuritis, labyrinthitis and Ménière pattern

• High-yield fact: The Epley manoeuvre is used in benign paroxysmal positional vertigo (BPPV) to reposition displaced otoliths from a semicircular canal toward the utricle.

مناورة Epley مخصصة لـBPPV، وهدفها إعادة otoliths إلى موقعها.

- **Core concept:** Vertigo evaluation begins with timing, triggers, associated hearing change, gait and neurologic findings.
- **Key discriminator:** Vestibular neuritis typically has sustained severe vertigo without hearing loss or tinnitus.
- **Key discriminator:** Labyrinthitis may combine vertigo with unilateral hearing loss or tinnitus.
- **High-yield fact:** CLEAN V3 describes Ménière disease with intermittent vertigo and unilateral reduced hearing or tinnitus.

- **Exam trap:** Do not use HINTS as a universal screening examination for every transient or positional dizziness complaint.
- **Management priority:** Assess central causes, clinical course and hearing before committing to a benign vestibular diagnosis.

التفريق يبدأ من timing وtriggers وhearing. Vestibular neuritis غالبًا دون hearing loss، بينما قد labyrinthitis يرافقها hearing loss.

Topic 39: Trigeminal neuralgia and peripheral nerve localization

- **Clinical trigger:** Brief severe electric-shock facial pain triggered by touch or chewing in the trigeminal distribution is source-described trigeminal neuralgia.
- **Management priority:** CLEAN V3 identifies carbamazepine with neurologic referral as a treatment approach in a compatible case.
- **Key discriminator:** Infraorbital nerve sensory loss affects cheek, upper lip and side of nose through CN V2.
- **High-yield fact:** The supraorbital nerve supplies forehead and upper eyelid sensation through V1.
- **Clinical Pearl:** A positive Froment sign supports ulnar nerve dysfunction; compression against the humerus can cause radial nerve palsy.
- **Exam trap:** Localize peripheral-nerve findings anatomically rather than assuming a central brain lesion.

قد يحرضه اللمس أو المضغ. انتبه لتوزع العصب بدل V2/V3 تعطي ألمًا كهربائيًا في Trigeminal neuralgia وصف كل ألم وجه بأنه migraine.

Topic 40: Autonomic dysreflexia after spinal injury

- **Clinical trigger:** Sudden marked hypertension, bradycardia or flushing above a preexisting spinal-injury level suggests autonomic dysreflexia.
- **Number/threshold:** CLEAN V3 associates this emergency especially with spinal cord injury at or above T6.
- **Core concept:** Noxious stimuli below the lesion can trigger uncontrolled sympathetic responses.
- **Management priority:** Identify and remove the provoking stimulus, such as bladder distension, stool impaction or constrictive clothing, while managing the emergency.
- **Exam trap:** Do not assume that hypertension following spinal injury is always primary essential hypertension.

قد يكون autonomic flushing أو bradycardia أو أعلى، ارتفاع ضغط مفاجئ مع T6 بعد إصابة نخاعية عند dysreflexia. ابحث عن bladder/bowel triggers.

Ophthalmic and Neuro-ophthalmic Emergencies

Topic 41: Sudden painless monocular vision loss: CRAO

- **Clinical trigger:** Central retinal artery occlusion typically presents with sudden, painless, severe monocular visual loss.
- **Key discriminator:** A pale retina with a cherry-red macula is the classic funduscopic finding described in CLEAN V3.
- **High-yield fact:** The central retinal artery is supplied through the ophthalmic artery, a branch of the internal carotid artery.
- **Management priority:** CLEAN V3 calls for transfer to a stroke center with neurology and ophthalmology capability in a relevant CRAO case.
- **Number/threshold:** An AQRAB critical-window infographic describes a very short retinal ischemia window of generally less than 90–100 minutes; it is presented as a physiologic teaching point, not an independently verified intervention cutoff.
- **Exam trap:** Do not confuse CRAO with retinal detachment solely because both may produce painless monocular vision loss.

للعين: فقد بصر مفاجئ غير مؤلم في عين واحدة يستوجب stroke هو CRAO وليس stroke-center pathway، مجرد موعد عيادة روتيني.

Topic 42: Central retinal vein occlusion: fundus and clinical distinction

- **Clinical trigger:** Sudden painless monocular blurring or loss of vision can be caused by central retinal vein occlusion.
- **Key discriminator:** Dilated tortuous retinal veins, widespread retinal hemorrhages and blurred optic-disc margins support CRVO.
- **High-yield fact:** CLEAN V3 distinguishes ischemic and nonischemic forms of central retinal vein occlusion.
- **Clinical Pearl:** A relative afferent pupillary defect may be present in a severe source-described CRVO case.
- **Exam trap:** Do not interpret the vein-occlusion fundus pattern as the pale retina with cherry-red macula of arterial occlusion.

CRVO مع يظهر غالبًا بفقد بصر غير مؤلم retinal hemorrhages وtortuous veins. هذا يختلف عن cherry-red spot في CRAO.

Topic 43: Retinal detachment and flashes/floaters

- **Clinical trigger:** New flashes or floaters followed by progressive peripheral field loss described as a curtain suggest retinal detachment.
- **Key discriminator:** Retinal detachment typically does not cause pain, unlike acute angle-closure glaucoma or anterior uveitis.
- **High-yield fact:** Blunt ocular trauma can cause retinal hemorrhage, tear or detachment; posterior vitreous detachment is a common source-described antecedent of retinal tears.
- **Management priority:** CLEAN V3 recommends ophthalmologic assessment within 24 hours when symptoms raise concern for retinal detachment.
- **Clinical Pearl:** Ultrasound is described in CLEAN V3 as a supportive diagnostic test in an appropriate retinal detachment case.
- **Exam trap:** Do not dismiss new floaters with field loss because the eye is not red or painful.

Flashes وfloaters ثم curtain يرجح على المجال البصري retinal detachment. الألم ليس سمة معتادة، ولا يؤخر ذلك فحص ophthalmology.

Topic 44: Acute angle-closure glaucoma: painful pressure emergency

- **Clinical trigger:** Sudden painful visual loss in a red eye raises concern for acute angle-closure glaucoma.
- **Core concept:** Blocked aqueous-humor outflow through the trabecular pathway causes a rapid rise in intraocular pressure.
- **Number/threshold:** CLEAN V3 cites a usual intraocular-pressure interval of 10–21 mmHg and describes markedly increased pressure in angle closure.
- **Management priority:** CLEAN V3 lists topical beta-blockers and IV acetazolamide as first-line ED measures in the relevant source case.
- **Number/threshold:** A cited single-dose acetazolamide example is 500 mg IV or orally for selected acute angle closure.
- **Exam trap:** Do not treat all red eyes as conjunctivitis before evaluating pain, vision and pressure.

Painful red eye مع فقد الرؤية وارتفاع IOP يشير إلى angle-closure glaucoma ويحتاج خفض الضغط وتقييم ophthalmology عاجل.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 106

2 g	Number/threshold: An adult bacterial-meningitis example cites ceftriaxone 2 g IV every 12 hours; agent selection must also consider Listeria risk, age and host factors.
500 mg	relevant source case. Number/threshold: A cited single-dose acetazolamide example is 500 mg IV or orally for selected acute angle closure.
16 mg	cord pressure bladder dysfunction each 16 mg dexamethasone cited No automatic metastatic Do not copy steroid dose Source steroid
10 mg/kg	28. Suspected HSV encephalitis: give IV aciclovir without waiting for PCR. 29. Cited encephalitis aciclovir: 10 mg/kg IV every eight hours, adjusted. 30. Cauda equina can be incomplete; a low residual urine does not exclude it.

Topic 45: Anterior uveitis and traumatic iritis

- **Clinical trigger:** Red painful eye, photophobia, blurred vision, constricted pupil and cloudy anterior chamber suggest anterior uveitis.
- **Key discriminator:** Pain with consensual light stimulation and anterior-chamber cell-and-flare favor iritis.
- **High-yield fact:** Anterior uveitis is inflammation of the iris and adjacent anterior uveal structures.
- **Differential point:** Traumatic iritis may follow blunt or penetrating ocular injury.
- **Clinical Pearl:** Concomitant genital ulceration and an inflamed eye is a source clue to a systemic uveitis association.
- **Exam trap:** Photophobia and reduced acuity make a simple benign surface diagnosis insufficient.

Photophobia مع painful consensual response وcell-and-flare يدعم anterior uveitis. هذا ليس نمط chalazion أو retinal detachment.

Topic 46: Optic neuritis: color loss, pain and demyelination

- **Clinical trigger:** Subacute monocular visual loss, pain and impaired color perception suggest optic neuritis.
- **Key discriminator:** An afferent pupillary defect and reduced red-color discrimination are source-described findings.
- **High-yield fact:** Inflammatory demyelination of the optic nerve may occur in multiple sclerosis.
- **Management priority:** CLEAN V3 discusses pulse methylprednisolone in a selected acute optic-neuritis presentation; appropriateness depends on specialist assessment and the actual cause.
- **Differential point:** CLEAN V3 also notes possible autoimmune, infectious, metabolic and nutritional causes of optic-neuropathy presentations.
- **Exam trap:** Loss of acuity with eye pain is not automatically an ocular-surface disorder.

Optic neuritis يسبب نقص بصر غالبًا أحادي مع ألم وحساسية لتغير red color. قد يرتبط بـMS ويحتاج تفريق الأسباب ومتابعة عصبية/عينية.

Topic 47: Retrobulbar hematoma and orbital compartment syndrome

- **Clinical trigger:** Proptosis, severe ocular pain and threatened vision after trauma raise concern for orbital compartment syndrome.
- **Core concept:** A retrobulbar hematoma can increase pressure behind the globe and compromise optic-nerve perfusion.
- **Management priority:** Lateral canthotomy and cantholysis are identified in CLEAN V3 as an emergency decompression procedure in an appropriate case.
- **Red flag:** A painful proptotic eye is an immediate threat to vision, not an elective ophthalmology complaint.
- **Exam trap:** Do not rely solely on an external lid finding when the orbit may be under pressure.

Painful proptosis بعد إصابة مع تهديد للرؤية قد يكون orbital compartment syndrome. المصدر يذكر lateral canthotomy/cantholysis كإجراء إنقاذ عند الاستطباب.

Topic 48: Chalazion and low-acuity eyelid disorders

- **Core concept:** A chalazion is a sterile inflammatory nodule arising from a meibomian gland in the eyelid.
- **Key discriminator:** It typically appears as a painless, firm or nontender eyelid nodule.
- **Number/threshold:** CLEAN V3 describes warm compresses and eyelid hygiene over 2–4 weeks in a selected chalazion example.
- **Differential point:** Loss of vision, marked photophobia or an acutely painful eye does not follow the uncomplicated chalazion pattern.

Chalazion عادة nontender firm lid nodule من Meibomian gland، ويختلف كليًا عن painful vision-threatening red eye.

Comparison Tables

Comparisons are limited to source-grounded distinctions already presented in the numbered topics. They introduce no new numerical recommendations.

Acute stroke imaging and decisions

Feature	Ischemic stroke	Intracerebral hemorrhage	Exam point
Imaging	Noncontrast CT may be initially normal	Blood is identified on imaging	Same presentation; different management
Time-sensitive pathway	Reperfusion selection by onset and imaging	Hemorrhage care and anticoagulant-reversal assessment	Do not import treatments across
Blood pressure	Pretreatment IVT source threshold <185/110	Selected ICH source approach toward SBP 140, avoiding <130	Different pathway and clinical setting

Vertigo localization

Feature	Peripheral vestibular process	Posterior circulation stroke	Exam point
Typical clue	Neuritis lacks hearing loss; labyrinthitis may cause it	Focal signs, gait or central eye movement findings	Consider timing and trigger first
Bedside HINTS	Trained examiner, continuous acute vestibular syndrome only	Central pattern may be detected in appropriate AVS	Not a generic dizziness test
Imaging	No reflex imaging rule established by label	Early negative CT does not reliably exclude	Do not over-reassure from CT

Convulsive and nonconvulsive status

Feature	Convulsive status	Nonconvulsive status	Exam point
Presentation	Ongoing convulsions or recurrence without recovery	Persistent altered awareness without overt convulsion	Motor stillness may conceal seizures
Initial action	ABC, glucose, prompt benzodiazepine	EEG and specialist assessment when suspected	Do not wait for consciousness recovery without evaluation
Common trap	Waiting for 30 minutes	Assuming prolonged postictal state	Both are time-critical

CSF interpretation

Feature	Bacterial meningitis pattern	Viral meningitis pattern	Exam point
White cells	Often neutrophil predominant and marked	Often lymphocyte predominant	Patterns are supportive, not absolute
Protein	Typically substantially increased	Often mildly increased	Interpret with time course and host
CSF glucose	Often reduced relative to blood	Often preserved	Antibiotic timing is not decided by LP delay

Neuromuscular weakness

Feature	GBS	Myasthenic crisis	Exam point
Motor pattern	Progressive, often symmetric; reduced reflexes	Fluctuating fatigability, often ocular/bulbar	The pattern changes the differential
Immunotherapy	IVIg or plasma exchange	IVIg or plasma exchange	Both still require ventilatory assessment
Critical pitfall	Normal SpO2 can hide ventilatory weakness	Bulbar failure can precede hypoxemia	Trend cough, swallowing and mechanics

PASS 9 — Source-backed clinical recovery | Domain 05

Neurology & Ophthalmology

dosing or medical-correctness certification.

Pituitary apoplexy

Clinical trigger: Sudden severe headache with visual impairment (including possible bitemporal field loss) and cranial-nerve palsy, especially CN III, is the source-described pituitary apoplexy presentation.

صداع شديد مفاجئ مع ضعف رؤية أو شلل CN III يوجّه في المصدر إلى pituitary apoplexy.

Hyphema in sickle cell disease

Clinical Pearl: CLEAN V3 cautions that acetazolamide, while used to lower intraocular pressure in some hyphema patients, may worsen the situation in patients with sickle cell disease.

لا تنقل علاج hyphema بين المرضى تلقائيًا؛ يحذّر المصدر من acetazolamide عند وجود sickle cell disease.

CT evaluation of early subarachnoid hemorrhage

Key discriminator: The CLEAN V3 discussion of CT sensitivity for suspected SAH depends on presentation within 6 hours and timely expert neuroradiology interpretation; neither qualifier should be omitted.

قيمة CT المبكر في SAH مرتبطة بزمن الفحص وجودة تفسيره، وليس رقم 6 ساعات وحده.

Critical illness myopathy

Risk discriminator: CLEAN V3 describes high-dose corticosteroid exposure together with neuromuscular blocking agents in critical illness as an important risk context for acute quadriplegic/critical illness myopathy.

يذكر المصدر وجود خطر Critical Illness Myopathy عند اجتماع high-dose steroids مع neuromuscular blockers في مرضى العناية.

Neurology & Ophthalmology

Precisely source-linked additions after the priority-safety review. Not independent guideline verification.

Topic: First unprovoked seizure: recurrence estimate

High-yield fact: CLEAN V3 cites a recurrence estimate of up to 45% within 2 years after a first unprovoked seizure in adults, with most recurrences occurring during the first year. This is a source-specific study estimate, not an individual risk prediction.

الرقم 45% وارد كتقدير في المصدر بعد first unprovoked seizure؛ لا تستخدمه كتنبؤ قطعي لمريض بعينه.

> Do not merge case-specific, adult and pediatric regimens into a single dose. Source-dependent conflicts remain unresolved.

EDIC AUDIT PASS 11 | DOMAIN 05 | SOURCE-ONLY CONTEXT RESOLUTION

Stroke thrombolysis: unverified source MCQ restriction

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

Exam-source caution: CLEAN V3 and AQRAB p.405 include “gastrointestinal hemorrhage in the preceding 21 days” as the answer to a thrombolysis contraindication question. The excerpt does not specify a complete current eligibility pathway, severity, treatment type or exceptions. Retain it only as source-question content; do not use it alone to make a treatment decision.

رقم 21 يومًا ورد في إجابة سؤال عن Thrombolysis، لكن لا يكفي وحده للحكم على أهلية مريض فعلي دون بقية المعايير.

EDIC AUDIT PASS 11 | DOMAIN 05 | SOURCE-ONLY CONTEXT RESOLUTION

Status epilepticus: source-specific loading examples

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

Differential point: CLEAN V3 describes levetiracetam 30 mg/kg IV over 5 minutes in a status-epilepticus case after first-line benzodiazepine failure. AQRAB p.140 separately describes a selected adult example 60 mg/kg IV up to 4,500 mg over 10 minutes; AQRAB p.416 has a four-year-old child receiving 40 mg/kg over 5 minutes with a pediatric explanatory range of 30–60 mg/kg and maximum 3 g. These are source-defined contexts—not one averaged protocol.

لا تجمع جرعات Levetiracetam المختلفة في رقم واحد: العمر، المرحلة العلاجية، ومدة الإعطاء تختلف بين المقاطع.

Source-based case: CLEAN V3 has a prolonged-seizure example mentioning fosphenytoin 20 phenytoin equivalents/kg; another case gives phenytoin 1,000 mg over 20 minutes. Because infusion safety details, patient weight and contraindications are not supplied by these lines, neither is reproduced as a complete stand-alone order.

المصدر يذكر أمثلة لـFosphenytoin وPhenytoin لكنها ليست أوامر علاج كاملة؛ تُحفظ كأمثلة اختبارية فقط.

Painful versus painless acute vision loss

Feature	Acute angle-closure glaucoma	Retinal detachment	Exam point
Symptoms	Painful red eye with visual loss	Usually painless flashes/floaters then visual curtain	Pain alone changes differential
Mechanism	Acute elevation of IOP	Neurosensory retina detaches	Different urgent management
Immediate focus	IOP-lowering assessment and ophthalmology	Urgent retinal assessment	Do not treat both as conjunctivitis

Retinal vascular events

Feature	Central retinal artery occlusion	Central retinal vein occlusion	Exam point
Visual symptoms	Sudden painless profound monocular loss	Sudden painless variable monocular loss	Both require recognition
Fundus clue	Pale retina, cherry-red macula	Tortuous veins and diffuse hemorrhages	Distinguish artery and vein patterns
Pathway	CLEAN V3 identifies stroke-center assessment	Ophthalmic vascular assessment	Do not confuse with detachment

Spinal compression cause

Feature	Metastatic cord compression	Cauda equina with other cause	Exam point
Primary concern	Cancer, neurologic decline and cord pressure	Back pain with saddle or bladder dysfunction	Urgent spine assessment for each
Source steroid	16 mg dexamethasone cited for MSCC with neurologic signs	No automatic metastatic steroid regimen	Do not copy steroid dose across
Diagnostic pitfall	Do not delay MRI/spine team	Normal residual urine alone does not exclude	Time-critical neurologic evaluation

Rapid Recall: Must-Know Facts

40 concise restatements of clinical facts already included in this Domain; no independent new clinical content.

01. Suspected stroke: record last-known-well and activate stroke assessment early.
02. Bedside glucose is essential because hypoglycemia mimics stroke.
03. An early normal CT does not exclude ischemic stroke.
04. NIHSS may underestimate posterior-circulation deficits.
05. ACA stroke classically emphasizes contralateral leg deficit.
06. A low NIHSS score does not exclude a disabling stroke.
07. The AQRAB IV thrombolysis pathway cites a conventional 4.5-hour window.
08. Stroke tenecteplase example: 0.25 mg/kg IV, maximum 25 mg.
09. Stroke alteplase example: 0.9 mg/kg IV, maximum 90 mg.
10. Before IV thrombolysis the cited pressure goal is <185/110 mmHg.
11. Selected LVO cases may need thrombectomy evaluation up to 24 hours.
12. TIA still requires urgent vascular assessment despite symptom resolution.
13. Do not apply the ICH blood-pressure target to every ischemic stroke.
14. Sudden thunderclap headache demands secondary-headache assessment.
15. The six-hour CT rule for SAH applies only to selected appropriate adults.
16. Seizure activity lasting five minutes triggers status management.
17. Status: check glucose and administer a benzodiazepine promptly.
18. Selected adult lorazepam example: 4 mg IV over two minutes.
19. Selected levetiracetam loading: 60 mg/kg, maximum 4,500 mg.
20. Persistent unresponsiveness after convulsions may be nonconvulsive status.
21. PRES can combine seizures, visual symptoms and hypertension.
22. Wernicke triad may be incomplete despite true thiamine deficiency.

23. Red flags override a prior diagnosis of migraine.
24. Bacterial meningitis: do not delay antibiotics for postponed CT/LP.
25. LP safety depends on focal signs, pupils and consciousness trajectory.
26. Bacterial CSF often has neutrophils, high protein and low glucose.
27. Viral CSF more often has lymphocytes and preserved glucose.
28. Suspected HSV encephalitis: give IV aciclovir without waiting for PCR.
29. Cited encephalitis aciclovir: 10 mg/kg IV every eight hours, adjusted.
30. Cauda equina can be incomplete; a low residual urine does not exclude it.
31. Metastatic cord compression neurologic signs: cited dexamethasone 16 mg.
32. GBS: serial bulbar and respiratory assessment matters more than SpO2.
33. GBS immune options include IVIg or plasma exchange, not routine steroids.
34. Myasthenia classically causes fatigable weakness and bulbar risk.
35. LEMS may improve strength after repeated muscle activity.
36. Descending cranial-bulbar weakness suggests possible botulism.
37. MS pseudo-relapse may reproduce deficits during infection or fever.
38. CRAO: sudden painless monocular vision loss, often a cherry-red macula.
39. Retinal detachment: flashes, floaters and a progressive visual curtain.
40. Acute angle closure: painful visual loss with high intraocular pressure.

Conflict Resolution Register — Domain 05

The following source-dependent discrepancy was retained without choosing between incompatible wording. Internal source inconsistencies are distinguished from unequivocal treatment contradictions.

1. Convulsive status epilepticus duration terminology

- **CLEAN V3:** CLEAN V3 para 1795 contains a historical wording defining generalized convulsive status as 30 minutes or recurrent seizures over 30 minutes; para 1792 separately uses >=5 minutes.
- **AQRAB:** AQRAB PDF p. 140 explicitly instructs treating >=5 minutes or recurrent seizures without recovery as status epilepticus and not waiting 30 minutes.
- **Nature:** The word "definition" is applied to incompatible duration cutoffs in the retained sources. The newer operational treatment threshold and older definition can reflect different constructs, but no external reconciliation is made. Classification: Conflict — source-dependent.

Context and protocol differences (not confirmed direct conflicts)

- **Issue:** 1. Steroid use across neurologic syndromes

CLEAN V3: CLEAN V3 includes steroids for GCA, MS and an ambiguous severe encephalitis case.

AQRAB: AQRAB defines distinct steroid indications for MS relapse and malignant cord compression, and cautions against routine use in GBS or generic raised ICP.

Classification: source-context/protocol difference. Context-specific indications, not a blanket between-file contradiction.

- **Issue:** 2. SAH diagnostic CT sensitivity

CLEAN V3: CLEAN V3 para 1700 gives an approximate overall CT sensitivity of 90% for SAH and para 1702 notes improvement with early CT and expert interpretation.

AQRAB: AQRAB p. 141 allows a normal CT <=6 hours to support exclusion only for selected neurologically normal adults under quality criteria.

Classification: source-context/protocol difference. Different populations, timing and testing conditions; not combined into one sensitivity.

- **Issue:** 3. Stroke blood-pressure thresholds

CLEAN V3: CLEAN V3 includes a pre-IVT BP threshold plus a non-IVT case with BP 195/105.

AQRAB: AQRAB p. 139 separates pretreatment IVT, ICH and post-IVT/EVT pressure recommendations.

Classification: source-context/protocol difference. Different decision stages, not a direct disagreement.

- **Issue:** 4. GBS diagnostic testing and respiratory numbers

CLEAN V3: CLEAN V3 includes a single negative inspiratory force example (para 2014) without enough trajectory or full context.

AQRAB: AQRAB pp. 148–149 emphasizes serial respiratory mechanics and says SpO2 can remain normal.

Classification: source-context/protocol difference. One isolated value does not set a universal intubation threshold.

- **Issue:** 5. Ceftriaxone wording for meningitis

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 113

40 mg/kg	selected adult example 60 mg/kg IV up to 4,500 mg over 10 minutes; AQRAB p.416 has a four- year-old child receiving 40 mg/kg over 5 minutes with a pediatric explanatory range of 30–60 mg/ kg and maximum 3 g. These are source-defined contexts—not one averaged protocol.
1,000 mg	Source-based case: CLEAN V3 has a prolonged-seizure example mentioning fosphenytoin 20 phenytoin equivalents/kg; another case gives phenytoin 1,000 mg over 20 minutes. Because infusion safety details, patient weight and contraindications are not supplied by these lines,
0.25 mg/kg, 25 mg	07. The AQRAB IV thrombolysis pathway cites a conventional 4.5-hour window. 08. Stroke tenecteplase example: 0.25 mg/kg IV, maximum 25 mg. 09. Stroke alteplase example: 0.9 mg/kg IV, maximum 90 mg.
4 mg	17. Status: check glucose and administer a benzodiazepine promptly. 18. Selected adult lorazepam example: 4 mg IV over two minutes. 19. Selected levetiracetam loading: 60 mg/kg, maximum 4,500 mg.
60 mg/kg, 4,500 mg	18. Selected adult lorazepam example: 4 mg IV over two minutes. 19. Selected levetiracetam loading: 60 mg/kg, maximum 4,500 mg. 20. Persistent unresponsiveness after convulsions may be nonconvulsive status.
2 g	CLEAN V3: CLEAN V3 para 1839 gives 2 g IV with no repeated schedule in that isolated statement; other source context is truncated.

CLEAN V3: CLEAN V3 para 1839 gives 2 g IV with no repeated schedule in that isolated statement; other source context is truncated.

AQRAB: AQRAB p. 143 explicitly provides an adult example of 2 g IV every 12 hours with host-dependent additions.

Classification: source-context/protocol difference. Difference is completeness of dosing context; no unsupported reconciliation.

Internal source note: ocular symptom wording

CLEAN V3 para 1933 applies the "curtain coming down" description to CRAO, while paras 1875, 1907–1908 describe it as a typical retinal-detachment clue. This phrase alone is not used to diagnose CRAO in the integrated material; distinct fundus and stroke-pathway findings are retained. Classification: Conflict — source-dependent (internal ambiguity; not a verified between-file conflict).

Merge Quality Note — Domain 05

Counting unit: each authored English point beneath the 48 numbered Topic headings, once. Arabic explanatory paragraphs, comparison-table cells, footer text, source notes and Rapid Recall restatements do not increase the point count. Origin codes are assigned conservatively from identifiable supporting passages, not inferred original authorship.

- **Final integrated clinical topics:** 48
- **Integrated English clinical points:** 274
- **Authored points supported by CLEAN V3 only:** 134
- **Authored points supported by AQRAB only:** 104
- **Authored points directly supported by both source files:** 36
- **Unresolved between-file definition/wording discrepancy:** 1
- **Confirmed direct incompatible inter-file treatment directives:** 0
- **Protocol or context differences recorded:** 5
- **Additional internal source ambiguity flagged:** 1
- **Excluded/reassigned source fragment groups logged:** 8
- **Comparison tables:** 8
- **Rapid Recall facts (not new points):** 40

Incomplete in source — not included

- **CLEAN V3 para 1649:** Afterload pressure-volume-loop material is misclassified inside stroke recognition — Reassigned to cardiovascular scope; not inserted into this neurologic diagnosis.
- **CLEAN V3 paras 1668–1669, 1685:** Case answers or isolated statements depend on absent images and clinical options — Incomplete in source — not included.
- **CLEAN V3 para 1795:** Historical 30-minute status definition presented without separating operational versus older terminology — Not converted into a treatment-delay instruction; tracked in Conflict Resolution Register.
- **CLEAN V3 para 1819:** Intravenous corticosteroid treatment line attached to a poorly specified viral encephalitis example — Incomplete in source — not included as an infectious-encephalitis treatment recommendation.
- **CLEAN V3 para 1906:** Pulsed methylprednisolone stated without full indication/dose/risk context — No specific steroid dose inserted for optic neuritis.
- **CLEAN V3 para 1922:** Penicillin G dose appears in an ocular-uveitis case without a recoverable infection diagnosis — Incomplete in source — not included.
- **CLEAN V3 paras 2091–2093, 2099–2100, 2106:** Orphan numeric fragments or corrupted threshold text without clinical context — Incomplete in source — not included.
- **AQRAB PDF p. 140:** Hyperlink/image placeholder named ping lacks usable educational detail — Incomplete in source — not included.

Cross-domain content allocation

- **CLEAN V3 para 1649:** Arterial afterload and cardiac pressure-volume loops. Reserved for: Cardiovascular Emergencies.
- **CLEAN V3 paras 1695, 1708, 1710–1711:** Fluid response, heatstroke and blunt cerebrovascular trauma cases. Reserved for: Resuscitation / Trauma domains.
- **CLEAN V3 paras 1748, 1801–1802:** Neonatal seizure causes, pregnancy seizures, pediatric second-line seizure regimens. Reserved for: Pediatrics / Obstetrics domains.
- **CLEAN V3 para 1916; AQRAB PDF p. 147:** Ocular trauma is cross-referenced, with detailed head-and-face trauma in Domain 6. Reserved for: Trauma & Environmental Emergencies.
- **CLEAN V3 paras 2036–2038, 2061, 2074–2079:** Back pain, geriatric falls, SDOH, psychiatry and unrelated peripheral injuries. Reserved for: Musculoskeletal / Mixed Presentations / Ethics domains.

Quality limitation: source-linked authored counts are not a machine-verified census of every statement on every original PDF image or Word paragraph; they reflect the reviewed and integrated domain output. The sources were not checked against current external guidelines and no external medical information was intentionally added.

Source integrity: the original two files are unmodified. Any incomplete, image-dependent, misleading or internally inconsistent statement was either withheld or described in the register instead of being silently repaired.

Neurology & Ophthalmology | Pass 8 | Explicit CLEAN V3 source locations

Neurogenic thoracic outlet syndrome

• High-yield fact: Neurogenic thoracic outlet syndrome results from brachial-plexus compression in the thoracic-outlet region in the supplied source.

الفكرة المميزة هي ضغط brachial plexus عند Thoracic Outlet، وليس مشكلة وعائية بالضرورة.

Neurology & Ophthalmology

Supplementary Clinical Points

Topic: Herpes zoster ophthalmicus: source mechanism

Clinical Pearl: AQRAB describes herpes zoster ophthalmicus as potentially vision-threatening disease associated with varicella-zoster reactivation in the trigeminal ganglion.

HZO يرتبط بإعادة نشاط VZV في trigeminal ganglion وقد يهدد الرؤية.

Source evidence: A425-33: AQRAB PDF p.425, block 33

Neurology & Ophthalmology. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

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Pterygium - ocular surface disease [#033, 034]

High-yield fact: Pterygium is triangular conjunctival tissue that can extend toward the cornea and is distinct from an acute painful corneal ulcer.

الظفرة Pterygium نسيج ملتحمي مثلث قد يمتد نحو القرنية؛ وهي غير قرحة القرنية المؤلمة الحادة.

Herpes simplex keratitis - fluorescein findings [#035, 036]

High-yield fact: Branching (dendritic) fluorescein-staining corneal lesions support herpes simplex keratitis.

الآفات المتشجّرة التي تتصبغ بالـFluorescein تدعم تشخيص التهاب القرنية الهربسي؛ لا تعاملها كخدش بسيط.

Ocular hemorrhage - anatomic localization [#040]

High-yield fact: A visible ocular hemorrhage should be localized anatomically before deciding urgency.

حدّد مكان النزف العيني تشريحيًا قبل تحديد خطورته؛ نزف تحت الملتحمة يختلف عن نزف الحجرة الأمامية.

EDIC / Emergency & Critical Care

06 — Trauma, Burns & Environmental Emergencies

I. Initial Trauma Assessment & Hemorrhage Control

Topic 1: Trauma primary survey and time-critical threats

- **Core concept:** Use a structured primary survey: airway and cervical-spine precautions when indicated, breathing, circulation and hemorrhage control, disability and exposure with temperature protection.
- **Management priority:** Treat immediate threats during the primary survey rather than waiting for imaging or completing a secondary survey.
- **Clinical trigger:** Tension pneumothorax, hemothorax, tamponade and major external bleeding require rapid recognition and action.
- **Common mistake:** A reassuring first examination does not exclude evolving thoracic, abdominal or pediatric traumatic injury.
- **Clinical Pearl:** Reassess vital functions and response after every intervention; identify both the injury and the mechanism.

في Trauma ابدأ بـABCDE وعالج التهديد فورًا. لا تجعل التصوير أو الفحص الثانوي يؤخر السيطرة على النزف أو علاج خطر صدري واضح.

Topic 2: Hemorrhage recognition and the physiology of compensated shock

- **Core concept:** Major hemorrhage can remain clinically important while blood pressure is initially preserved; follow perfusion and serial physiology.
- **High-yield fact:** The source describes spleen and liver as the most commonly injured organs in adult blunt abdominal trauma.
- **Clinical trigger:** Tachycardia, perfusion changes, changing mental status or a concerning mechanism warrant reassessment for occult blood loss.
- **Management priority:** Hemorrhage control and a blood-product pathway take priority in major ongoing traumatic bleeding; avoid automatic crystalloid boluses.
- **Exam trap:** Do not explain post-trauma hypotension as spinal/neurogenic shock without compatible neurologic findings and exclusion of bleeding.

قد يحافظ المريض على BP رغم نزف كبير، خاصة الطفل أو المسن. راقب perfusion باستمرار ولا تعتبر إصابة النخاع تفسيرًا جاهزًا لأي هبوط ضغط بعد Trauma.

Topic 3: External hemorrhage and tourniquet priorities

- **Management priority:** Control severe extremity hemorrhage and use a tourniquet when indicated; scene safety and evacuation remain part of the initial pathway.

- **Key discriminator:** External bleeding can be addressed directly; internal hemorrhage requires recognition, resuscitation and definitive source control.
- **Common mistake:** A normal early pressure reading is not proof that bleeding is controlled.
- **Clinical Pearl:** Rapid transport does not replace bleeding control or reassessment of circulation.

إذا كان النزف شديدًا من الطرف، فكّر بـTourniquet عند الحاجة. إسعاف النزف ليس مجرد رفع الضغط، بل الوصول إلى source control.

Topic 4: Traumatic cardiac arrest and chest decompression

- **Core concept:** Traumatic cardiac arrest is managed through the reversible traumatic causes rather than assuming the same sequence as a primary medical arrest.
- **Management priority:** CLEAN V3 describes bilateral finger thoracostomies and blood-product resuscitation in the supplied traumatic-arrest scenario.
- **Clinical trigger:** Thoracic obstructive causes and exsanguination deserve immediate attention during traumatic arrest.
- **Exam trap:** Do not let resuscitation maneuvers substitute for correcting treatable thoracic obstruction or hemorrhage.

في Traumatic Arrest لا تنسَ السبب الرضي القابل للعلاج، خصوصًا النزف أو الضغط داخل الصدر. وصف bilateral thoracostomy مرتبط بسيناريو المصدر ولا يُحوَّل إلى قاعدة بلا سياق.

Topic 5: FAST, CT and the unstable trauma patient

- **Core concept:** FAST is one assessment tool, not a stand-alone exclusion test for all intra-abdominal and retroperitoneal injuries.
- **Key discriminator:** A negative FAST does not exclude bowel injury or retroperitoneal hemorrhage.
- **Management priority:** An unstable patient may need operative or procedural control without waiting for CT.
- **Exam trap:** Do not use negative FAST to discharge a patient whose mechanism or course remains concerning.
- **Clinical Pearl:** Monitoring is an active component of selected nonoperative solid-organ management, not an automatic discharge order.

FAST السلبي لا ينفي إصابة bowel أو retroperitoneum. إذا كان المريض غير مستقر، القرار يعتمد على التروية والنزف لا على انتظار CT.

Topic 6: Damage-control surgery and bleeding-focused escalation

- **Clinical trigger:** Penetrating abdominal trauma with profound instability is a source scenario requiring expedited exploratory laparotomy.
- **Management priority:** Urgent damage-control laparotomy is explicitly identified for a source trauma scenario in CLEAN V3.
- **Core concept:** Anatomic injury, physiologic instability and ongoing bleeding determine urgency.
- **Common mistake:** Do not delay definitive source control while pursuing tests unlikely to change the urgent decision.

إذا كانت هناك Penetrating Injury مع انهيار دوري، فالقرار ليس إكمال كل الصور بل التحرك إلى surgical control حسب السيناريو.

Topic 7: Trauma-associated coagulopathy and the lethal diamond

- **High-yield fact:** CLEAN V3 describes the lethal diamond: hypothermia, acidosis, coagulopathy and hypocalcemia.
- **Management priority:** Control bleeding, avoid unnecessary hypothermia and use the applicable blood-product protocol.

- **Clinical Pearl:** In a source case with active bleeding and an isolated prolonged viscoelastic R time, fresh-frozen plasma factor replacement was selected.
- **Exam trap:** A viscoelastic-tracing treatment choice depends on the precise pattern and clinical setting; do not generalize a question-specific answer.

يذكّرك Lethal Diamond أن النزف والبرد والحماض واضطراب التخثر ونقص الكالسيوم قد تتفاقم معًا. أما قرار FFP فيعتمد على نمط الاختبار والسياق.

II. Head, Brain & Spinal Trauma

Topic 8: Initial head injury assessment and neurological warning signs

- **Core concept:** Head-injury assessment uses mechanism, consciousness, serial GCS, examination and associated injury signs rather than an isolated test.
- **Clinical trigger:** Altered consciousness, posturing, repeated vomiting and deterioration require timely reassessment and appropriate imaging.
- **High-yield fact:** Decorticate posturing is described as arm flexion/adduction with lower-extremity extension.
- **Exam trap:** Intoxication may impair capacity and obscure head injury; an isolated normal-looking initial examination is insufficient.
- **Common mistake:** A negative initial CT does not by itself establish that every head-injury risk has resolved.

عند Head Injury تابع GCS وتغير الوعي والفحص المتكرر. السكر أو الثمالة أو CT أولي مطمئن لا يعوّض المتابعة السريرية عند وجود Red Flags.

Topic 9: NICE-derived head CT triggers: keep their original conditions

- **Number/threshold:** CLEAN V3 cites CT within 1 hour when initial ED GCS is 12 or lower.
- **Number/threshold:** The source also cites GCS below 15 at 2 hours after injury and at least two vomiting episodes in adults as 1-hour CT triggers.
- **Number/threshold:** For certain patients with loss of consciousness or amnesia, the source describes 8-hour pathways tied to age at least 65 years or retrograde amnesia greater than 30 minutes.
- **Number/threshold:** Other source items discuss anticoagulation and injury mechanism; those abbreviated stems must not be merged into a universal imaging rule.
- **Exam trap:** Different timing statements reflect different eligibility conditions; the underlying question details must remain attached.

لا تحفظ CT خلال ساعة أو 8 ساعات دون سبب القرار. اربط كل threshold بـGCS والقيء والعمر وLOC/Amnesia، ولا تدمج سيناريوهين في قاعدة واحدة.

Topic 10: Diffuse axonal injury and post-traumatic coma

- **Clinical trigger:** Immediate prolonged coma after high-velocity injury with relatively limited early CT findings suggests diffuse axonal injury in the supplied material.
- **Number/threshold:** A source description uses severe depressed consciousness, GCS below 8, as part of the high-velocity DAI scenario.
- **Key discriminator:** The neurologic condition may appear disproportionately severe relative to early visible CT findings.
- **Common mistake:** Do not assume a relatively unrevealing early scan excludes significant traumatic brain injury.

DAI قد يسبب غيبوبة شديدة مع CT أولي لا يفسر شدة الحالة. تذكّر الفرق بين الصورة وبين العجز العصبي الحقيقي.

Topic 11: ICP waveforms, compliance and cerebral perfusion pressure

- **Number/formula:** Cerebral perfusion pressure (CPP) = mean arterial pressure (MAP) - intracranial pressure (ICP).
- **High-yield fact:** The source describes an ICP waveform with P1 (percussion), P2 (tidal) and P3 (dicrotic) peaks.
- **Key discriminator:** The supplied waveform description links ICP pulse morphology with intracranial compliance; it does not provide a stand-alone waveform diagnostic cutoff.
- **Number/threshold:** CLEAN V3 cites CPP around 55 mmHg in a source question as carrying ischemic risk; this is not presented here as a universal management target.
- **Number/threshold:** Lundberg A plateau waves are described as ICP rises approximately 50-100 mmHg lasting about 5-20 minutes.

CPP = MAP - ICP. ارتفاع ICP يقلل ضغط التروية الدماغية حتى لو كان MAP مناسبًا، والـPlateau waves تشير إلى poor intracranial compliance.

Topic 12: TBI treatments and steroid trap

- **Management priority:** The source identifies prevention of secondary brain injury and attention to cerebral perfusion in significant TBI.
- **Exam trap:** CLEAN V3 specifically warns against high-dose corticosteroids to improve severe-TBI outcomes or lower ICP.
- **Clinical Pearl:** Do not turn a single source-case hypertonic-saline regimen into a universal bolus for every head injury.
- **Common mistake:** Focus on the actual lesion, physiology and source criteria rather than treating the scan description alone.

من الفخاخ الامتحانية استعمال Steroids لتحسين Severe TBI؛ المصدر يحذّر منها. أرقام hypertonic saline في سيناريو محدد ليست بروتوكولًا عامًا.

Topic 13: Mild TBI follow-up and delayed sequelae

- **Clinical Pearl:** Post-concussive symptoms may persist for months after mild TBI.
- **Number/threshold:** One CLEAN V3 study summary reports symptoms in up to 30% at 3 months and 15% at 1 year after mild TBI.
- **High-yield fact:** Post-traumatic hypopituitarism is described even after apparently mild injury and a normal initial CT, including delayed presentation around 6 months.
- **Common mistake:** Do not equate a normal initial scan with guaranteed absence of later functional or endocrine consequences.

بعد Mild TBI قد تستمر الأعراض، وقد تظهر اضطرابات هرمونية متأخرة. الأرقام نسب دراسات مذكورة بالمصدر وليست قاعدة تنبؤ لكل مريض.

Topic 14: Head trauma in hemophilia and anticoagulation context

- **Red flag:** Hemophilia with head trauma requires particular concern for intracranial bleeding, even when initial findings are limited.
- **Number/threshold:** The source hemophilia A head-injury case specifies Factor VIII replacement to 100% activity.
- **Clinical Pearl:** CLEAN V3 lists anticoagulation among context-dependent head CT considerations; the whole pathway must be reviewed.
- **Exam trap:** A shortened stem saying that hemophilia alone is not an imaging trigger must not be read as permission to ignore suspected CNS bleeding.

Hemophilia مع Head Trauma يستدعي حذرًا خاصًا. قيمة Factor VIII المذكورة مرتبطة بحالة محددة، وقرار CT يحتاج السياق السريري الكامل.

Topic 15: Cervical-spine protection and airway access in trauma

- **Core concept:** Apply cervical-spine precautions when the injury mechanism and findings justify them while protecting oxygenation and ventilation.
- **Management priority:** For a source scenario requiring spinal immobilization, CLEAN V3 describes jaw thrust as the first airway maneuver.
- **Key discriminator:** CLEAN V3 also states that routine immobilization may be unnecessary in selected presentations without blunt injury or cord findings.
- **Exam trap:** The two statements apply to different trauma contexts and must not be simplified as either always or never immobilize.

حماية C-spine تعتمد على mechanism والفحص. لا تحوّل عبارة "قد لا نحتاج immobilization" إلى قاعدة تنطبق على كل رضّ، ولا تؤخر تأمين Airway.

Topic 16: Spinal cord syndromes and spinal versus neurogenic shock

- **Clinical trigger:** Neurologic deficits after injury raise concern for spinal cord involvement and require careful assessment.
- **Key discriminator:** CLEAN V3 flags a case consistent with central cord syndrome but supplies too little detail to expand the syndrome-specific examination.
- **Differential point:** The source warns against labeling hemorrhagic hypotension as spinal shock without compatible flaccidity and areflexia.
- **Exam trap:** Neurologic loss of reflex activity and hemodynamic shock are different concepts; avoid conflating terms in trauma assessment.

وجود عجز عصبي يغير تقييم Spinal Trauma، لكن انخفاض BP وحده لا يثبت Neurogenic Shock. قيِّم النزف أولًا وتأكد من علامات النخاع.

III. Thoracic Injury

Topic 17: Tension pneumothorax: clinical priority

- **Clinical trigger:** Thoracic trauma with escalating respiratory and hemodynamic compromise requires immediate consideration of tension pneumothorax.
- **Management priority:** When tension physiology is suspected in an unstable patient, decompression is prioritized over waiting for imaging.
- **Common mistake:** Fluid alone does not relieve tension pneumothorax obstruction.
- **Differential point:** Distinguish tension pneumothorax from bleeding into the chest, tamponade and other causes of traumatic shock.

Tension Pneumothorax مشكلة ضغط وانسداد داخل الصدر، وليس نقص سوائل فقط. إذا كان المريض غير مستقر، لا تجعل الصورة تؤخر Decompression.

Topic 18: Hemothorax and need for surgical escalation

- **Core concept:** Traumatic hemothorax is a potential cause of respiratory compromise and hemorrhagic shock.
- **Clinical trigger:** Persistent chest drainage and hemodynamic instability require assessment for ongoing bleeding and definitive thoracic control.
- **Number/threshold:** AQRAB supplemental review gives continuous output of 250 mL/hour for 3 hours as an exploratory-thoracotomy indicator in an adult hemothorax question.
- **Exam trap:** This quoted output criterion belongs to that source question; do not treat it as the only indication for surgery.

وجود Hemothorax قد يعني نزفًا مستمرًا. رقم 250 mL/h لثلاث ساعات ورد في سؤال محدد، ويُقرأ مع حالة المريض وسياق التدخل.

Topic 19: Flail chest and pulmonary contusion

- **Number/threshold:** Flail chest is described as at least three adjacent ribs each fractured in at least two places, producing a free segment.
- **Key discriminator:** Paradoxical movement of the chest-wall segment is the characteristic sign described by CLEAN V3.
- **High-yield fact:** Underlying pulmonary contusion contributes importantly to respiratory compromise.
- **Management priority:** The source highlights pain control and pulmonary hygiene when considering admission for significant chest injury.
- **Common mistake:** Do not focus on rib appearance alone while overlooking impaired ventilation or pulmonary injury.

في Flail Chest تذكّر fracture pattern والحركة paradoxical، لكن تدهور التنفس قد يكون مرتبطًا بالـPulmonary Contusion أيضًا.

Topic 20: Blunt cardiac injury and myocardial contusion

- **High-yield fact:** CLEAN V3 describes the right ventricle as commonly affected in cardiac contusion due to its anterior position behind the sternum.
- **Clinical trigger:** Significant blunt chest impact with concerning rhythm or perfusion findings raises cardiac-injury concern.
- **Red flag:** Ventricular fibrillation appears in a source question on blunt cardiac injury and arrhythmic complications.
- **Clinical Pearl:** Cardiac injury can coexist with lung, rib or vascular trauma and should be assessed in the full thoracic context.

الـRight Ventricle قريب من القص لذلك تذكره عند Blunt Cardiac Injury. لا تهمل اضطراب النظم إذا كان الصدر تعرض لرض شديد.

Topic 21: Diaphragmatic injury and occult thoracoabdominal trauma

- **Red flag:** CLEAN V3 flags left-sided diaphragmatic rupture as a serious injury with risk of abdominal-organ herniation and strangulation.
- **Clinical Pearl:** A negative FAST does not exclude bowel and other injuries that may evolve or require later reassessment.
- **Management priority:** Use physiologic instability and injury suspicion to guide timely surgical consultation.
- **Exam trap:** Do not treat an initially unrevealing examination as definitive exclusion of delayed thoracoabdominal injury.

Diaphragmatic Rupture قد يؤدي إلى herniation واختناق الأحشاء. تذكّر أن FAST لا يستبعد كل الإصابات خلف الصفاق أو بالأمعاء.

Topic 22: Chest-drain safe triangle and thoracic procedure anatomy

- **High-yield fact:** The source safe-triangle boundaries are the lateral border of pectoralis major, anterior border of latissimus dorsi and an inferior line around the fifth intercostal space.
- **Clinical Pearl:** CLEAN V3 repeats each anatomic boundary as a separate item; these have been merged into one anatomical description.
- **Red flag:** The superficial long thoracic nerve is highlighted as vulnerable in emergency thoracic procedures.
- **Common mistake:** Do not count repeated mentions of a single safe-triangle border as three separate clinical facts.

احفظ حدود Safe Triangle كمعلم تشريحي واحد: pectoralis major أمامًا وlatissimus dorsi خلفًا والخط السفلي قرب 5th ICS.

IV. Abdominal, Pelvic & Limb Trauma

Topic 23: Blunt abdominal trauma: solid-organ versus hollow-viscus injury

- **High-yield fact:** CLEAN V3 names the spleen and liver as commonly injured organs in adult blunt abdominal trauma.
- **Key discriminator:** A negative FAST does not exclude bowel injury or retroperitoneal bleeding.
- **Management priority:** Selected nonoperative solid-organ injuries require observation and reassessment rather than automatic discharge.
- **Red flag:** Persistent shock or worsening findings after abdominal trauma require reconsideration of concealed bleeding or bowel injury.

Blunt Abdominal Trauma في تذكّر spleen/liver، لكن سلبية FAST لا تنفي إصابة الأمعاء أو retroperitoneal hemorrhage.

Topic 24: Pelvic injury and hemorrhage control

- **Core concept:** Pelvic trauma may produce major bleeding and may coexist with injuries elsewhere.
- **Management priority:** Use indicated pelvic stabilization, blood resuscitation and definitive source-control pathways.
- **Common mistake:** Do not repeatedly manipulate an unstable pelvis or assume the pelvis is necessarily the only bleeding site.
- **Clinical Pearl:** Assess urinary tract and other associated injuries after initial circulation priorities are addressed.

Pelvic Trauma يحتاج تثبيتًا مناسبًا مع تقييم النزف؛ لا تكرر ضغط أو تحريك الحوض، ولا تنسَ مصادر نزف أخرى.

Topic 25: Suspected urethral injury

- **Clinical trigger:** Blood at the urethral meatus after blunt abdominal or pelvic trauma raises concern for urethral injury.
- **Management priority:** The source specifies retrograde urethrogram before a blind urinary catheterization decision.
- **Clinical Pearl:** Further bladder assessment may require dedicated imaging, including cystography in the supplied sequence.
- **Exam trap:** Do not place a urinary catheter blindly when physical findings suggest urethral disruption.

Blood at meatus بعد Trauma علامة تستدعي تقييم urethra؛ لا تضع Foley بشكل أعمى قبل التصوير الملائم.

Topic 26: Compartment syndrome following limb injury

- **Clinical trigger:** Escalating pain, pain on passive stretch and a tense compartment after fracture or limb injury suggest compartment syndrome.
- **Key discriminator:** Distal pulses can remain palpable; waiting for pulselessness is a dangerous delay.
- **Number/threshold:** AQRAB supplemental material cites absolute compartment pressure above 30 mmHg or diastolic pressure minus compartment pressure below 30 mmHg as measurement-based triggers.
- **Management priority:** Obtain immediate surgical assessment when compartment syndrome is suspected.
- **Exam trap:** A single pressure measurement does not replace the clinical trajectory or specialist decision.

Pain on passive stretch مع tense compartment مهم حتى لو pulses موجودة. ΔP أقل من 30 mmHg وارد في الملحق، لكن لا تؤخر surgical review حتى يفقد المريض النبض.

Topic 27: Traumatic peripheral nerve and soft-tissue patterns

- **High-yield fact:** Anterior interosseous nerve is a motor branch of the median nerve and may be injured by trauma or compression.
- **Clinical trigger:** Injury around the fibular head may involve the common peroneal nerve.

CLINICAL NUMBERS · TOPIC CONTEXT

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• High-yield fact: CLEAN V3 describes exposed dentin as having an ivory-yellow appear fracture and cites calcium-hydroxide covering in that specific case.

1000 mL	exceed approximately 15-20% total body surface area. Management priority: The same source case selects an initial 1000 mL warmed Hartmann solution or 0.9% saline bolus, but does not establish a universal regimen.

Domain 06 | Trauma, Burns & Environmental Emergencies

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

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Blunt chest trauma - pulmonary contusion [#048]

Key discriminator: Pulmonary contusion can evolve after blunt trauma, requiring monitoring of oxygenation despite initially limited findings.

قد تتطور كدمة الرئة بعد رضّ الصدر؛ راقب الأكسجة حتى إن بدت العلامات الأولية محدودة.

Cervical spine injury - central cord syndrome [#049]

High-yield fact: Central cord syndrome causes arm weakness that is typically more pronounced than leg weakness.

في Central cord syndrome يكون ضعف الطرفين العلويين عادة أوضح من ضعف الطرفين السفليين.

- **Key discriminator:** A Morel-Lavallee lesion is flagged as a recognizable closed soft-tissue injury in the supplied question bank.
- **Clinical Pearl:** Recheck neurovascular findings after limb trauma and immobilization rather than relying only on bony imaging.

بعد إصابة الطرف لا تفحص العظم فقط. التروية والإحساس والحركة قد تكشف Nerve Injury أو إصابة Soft Tissue غير واضحة بالأشعة.

V. Burns, Wounds & Exposure-Related Tissue Injury

Topic 28: Major thermal burns: early assessment and resuscitation

- **Core concept:** Major burn injury requires reassessment of circulation, exposure and the extent of tissue involvement.
- **Number/threshold:** An AQRAB source question describes formal adult fluid-resuscitation concern when major burns exceed approximately 15-20% total body surface area.
- **Management priority:** The same source case selects an initial 1000 mL warmed Hartmann solution or 0.9% saline bolus, but does not establish a universal regimen.
- **Common mistake:** Do not apply one question-specific volume to every burn patient without accounting for extent and clinical response.

في Major Burns قيّم TBSA% والدورة الدموية. قيمة 1000 mL واردة ضمن حالة بالمصدر وليست وصفة ثابتة لكل مريض.

Topic 29: Circumferential burn and escharotomy

- **Clinical trigger:** A circumferential charred extremity with threatened distal circulation may become constricted by burn eschar.
- **Management priority:** AQRAB explicitly selects escharotomy for a source case involving a severely burned circumferential upper extremity before prolonged transfer.
- **Key discriminator:** Escharotomy relieves constrictive eschar pressure; established irreversible gangrene is a different stage of limb damage.
- **Exam trap:** Do not delay assessment of perfusion because the skin injury appears isolated.

إذا كان الحرق circumferential مع تدهور perfusion، فكر بالحاجة إلى Escharotomy حسب خبرة الفريق. لا تخلط تحرير الضغط مع علاج نسيج أصبح غير قابل للحياة.

Topic 30: Electrical burns and deep-tissue injury

- **High-yield fact:** Small external lesions can conceal significant deep muscle and neurovascular injury after electrical exposure.
- **Clinical trigger:** Assess for rhabdomyolysis, compartment syndrome, trauma from falls and cardiac effects.
- **Common mistake:** Do not estimate injury severity from the visible skin-burn area alone.
- **Management priority:** Significant electrical burns or deep-tissue complications require specialty escalation.

Electrical Injury قد يحرق العضلات عميقًا رغم أن مساحة الجلد صغيرة. افحص CK والتروية و compartment signs بحسب الحالة.

Topic 31: Puncture wounds and infection-risk context

- **Clinical trigger:** A deep puncture wound in a patient with diabetes is flagged in AQRAB as a higher-concern presentation for empiric antibiotic decisions.
- **Key discriminator:** A puncture mechanism can carry deeper contamination and injury even when the skin opening is small.

- **Exam trap:** Do not assume every puncture has the same antibiotic indication or the same risk as an uncomplicated superficial wound.

الجرح الصغير من Puncture قد يكون أعمق مما يبدو، خصوصًا عند Diabetes. قرار Antibiotics يعتمد على الخطر والسياق وليس الشكل الخارجي فقط.

Topic 32: Wound review: pressure ulcer, ischemic lesion and infection

- **High-yield fact:** A source stage-2 sacral pressure injury without invasive infection was managed with wound care and moist dressings.
- **Key discriminator:** Ischemic arterial ulcers are described as more distal, including digits, with vascular context.
- **Red flag:** Radiographic cortical erosion in a diabetic foot wound raises concern for osteomyelitis in the supplied case.
- **Management priority:** Avoid extrapolating case-specific antibiotic choices to all wounds; assess for deep infection and the need for admission.

ميّز Pressure Ulcer من Ischemic Ulcer ومن Deep Infection. وجود cortical erosion مع جرح قدم سكري يغيّر التقييم نحو osteomyelitis.

Topic 33: Animal exposures, rabies prophylaxis and stinging injury

- **Number/threshold:** CLEAN V3 cites rabies vaccine on days 0 and 3 after exposure for a previously vaccinated immunocompetent patient, plus local wound care.
- **Clinical Pearl:** The regimen is explicitly for the previously vaccinated subgroup, not a general post-exposure schedule.
- **High-yield fact:** Fire-ant stings can range from local symptoms to systemic reactions in the cited material.
- **Exam trap:** A routine sting-care example does not replace recognition of systemic allergic manifestations.

جدول Rabies المذكور يخص من تلقّى التطعيم سابقًا، فلا تعممه. وSting قد يكون موضعيًا أو يرافقه رد فعل جهازي.

VI. Environmental Emergencies

Topic 34: Accidental hypothermia: assessment and severity

- **Core concept:** Assess ABCs, core temperature, glucose, trauma/intoxication and cardiac rhythm in cold-exposed patients.
- **Number/threshold:** AQRAB chapter review defines hypothermia as core temperature below 35 degrees Celsius.
- **Number/threshold:** CLEAN V3 describes severe hypothermia below 28 degrees Celsius and notes distal esophageal core measurement as useful in severe cases.
- **Management priority:** Remove wet clothing, insulate, minimize unnecessary movement and select rewarming by severity and circulatory stability.
- **Red flag:** Severe hypothermia carries afterdrop and dysrhythmia risk; gentle handling and positioning matter.
- **Exam trap:** Core temperature contributes to the pathway but should not be the only resuscitation decision variable.

في Hypothermia لا تعتمد على حرارة الجلد؛ قِس Core Temperature وتعامل بلطف مع المريض البارد جدًا، واختر التدفئة حسب stability.

Topic 35: Hypothermia rewarming, arrest and delayed complications

- **Management priority:** In unstable hypothermia or hypothermic arrest, AQRAB directs consultation with an ECMO/ECLS-capable pathway.
- **Clinical Pearl:** Pulmonary edema may develop after severe cold exposure and resuscitation even after apparent early stabilization.
- **High-yield fact:** The material describes risk of dysrhythmias and afterdrop during handling and rewarming.

- **Common mistake:** Do not stop observation solely because temperature and symptoms initially improve.

بعد Rewarming راقب المريض: التحسن الأولي لا ينفي pulmonary edema أو اضطراب النظم لاحقًا. عند shock/arrest استدعِ مسار ECMO/ECLS المناسب.

Topic 36: Frostbite: prevent refreezing and protect tissue

- **Management priority:** Remove constricting items, prevent further cold exposure and avoid rubbing, snow or direct dry heat.
- **Key discriminator:** Rapid water-bath thawing is appropriate only when refreezing can be prevented.
- **Management priority:** After thawing, use loose dry dressings, analgesia and perfusion assessment; consider specialist-directed management for deep injury.
- **Common mistake:** Do not confuse reversible frostnip with deeper frostbite before reassessment after warming.
- **Number/threshold:** CLEAN V3 mentions frostbite grading systems up to grade 4; no single complete grade-to-treatment rule is supplied.

في Frostbite أهم فخ هو thaw ثم refreeze؛ لا تبدأ التذويب إذا كان سيتجمد مجددًا. احمِ النسيج بعد التدفئة وافحص عمق الإصابة والتروية.

Topic 37: Heatstroke versus heat exhaustion

- **Core concept:** Heatstroke is an emergency hyperthermic syndrome with central nervous system dysfunction.
- **Key discriminator:** Confusion or other CNS dysfunction distinguishes heatstroke from uncomplicated heat exhaustion in the supplied material.
- **Clinical trigger:** An exertional collapse with altered mentation and high core temperature requires immediate cooling.
- **Exam trap:** A recorded temperature below 40 degrees Celsius after cooling does not exclude preceding heatstroke.
- **High-yield fact:** Classic heatstroke may occur during heat waves with impaired cooling rather than exertion alone.

الفكرة الأساسية هي خلل Heat + CNS Dysfunction يعني Heatstroke حتى لو الرقم بعد بدء التبريد صار أقل. الوعي وليس مجرد ارتفاع الحرارة.

Topic 38: Heatstroke cooling and systemic complications

- **Management priority:** Begin active whole-body cooling immediately after rapid ABC assessment; do not wait for complete laboratory testing.
- **Management priority:** Cold-water immersion is preferred for exertional heatstroke when practical in the supplied source.
- **High-yield fact:** Assess glucose, electrolytes, CK, renal and hepatic function, coagulation and urine output for evolving complications.
- **Exam trap:** Antipyretics do not treat environmental hyperthermia in the source pathway.
- **Clinical Pearl:** The AQRAB chapter review cites 38 degrees Celsius as one cooling-stop example, not a substitute for monitoring and protocol.

Cooling هو العلاج الزمني الحاسم في Heatstroke. لا تنتظر CK أو وظائف الكلى؛ ابدأ التبريد مع مراقبة مضاعفات rhabdomyolysis والتخثر.

Topic 39: Electrical injury: scene safety, ECG and concealed injury

- **Management priority:** De-energize the electrical source and make the scene safe before touching the patient.
- **Clinical trigger:** Assess burns, entry/exit marks, falls or blast trauma, neurologic findings, distal perfusion and compartment syndrome.
- **Management priority:** AQRAB describes ECG evaluation after significant exposure and monitoring with arrest, syncope, abnormal ECG, high-voltage exposure or cardiac risk.

- **Exam trap:** A small skin lesion does not exclude severe internal injury.
- **Clinical Pearl:** Lichtenberg figures after lightning are described as transient markings, not conventional thermal burns.

أمّن الكهرباء أولًا؛ ثم افحص Heart + Trauma + Deep Muscle. لا تتوقع حجم الإصابة من حجم العلامات الجلدية.

Topic 40: Lightning strike and reverse triage

- **Clinical trigger:** Lightning may cause cardiac or respiratory arrest despite limited external burns.
- **Management priority:** Prompt CPR and advanced resuscitation are highlighted; with multiple victims the source mentions reverse-triage principles.
- **Differential point:** Look for concurrent neurologic, auditory, ocular and blunt-trauma complications.
- **Common mistake:** Do not treat lightning as only a superficial thermal-burn event.

في Lightning قد ترى جلدًا شبه طبيعي مع Arrest. التقييم يتجاوز الجلد إلى القلب والأعصاب والسمع والبصر والرضوض.

Topic 41: Decompression sickness and arterial gas embolism

- **Clinical trigger:** New neurologic, vestibular, respiratory, cutaneous or joint symptoms after diving raise concern for dysbaric illness.
- **Key discriminator:** Arterial gas embolism may present very rapidly after ascent with neurologic or cardiopulmonary findings.
- **Management priority:** Stabilize ABCs, give high-concentration oxygen, document dive profile and obtain early diving-medicine/hyperbaric advice.
- **Management priority:** Recompression with hyperbaric oxygen is definitive for serious decompression illness in the supplied material.
- **Exam trap:** Normal initial chest imaging or symptomatic improvement with oxygen does not rule out the need for recompression assessment.

بعد Dive، أي ضعف أو دوار أو ضيق نفس جديد يحتاج تقييم Decompression Illness. التحسن بالأكسجين لا يلغي Hyperbaric consult.

Topic 42: Diving transport and documentation pitfalls

- **Clinical Pearl:** Record dive depth, duration, ascent pattern, breathing gas, repetitive dives and exact symptom timing.
- **Management priority:** Do not delay evacuation for nonessential testing when serious decompression illness is suspected.
- **Common mistake:** Avoid unplanned air transport to altitude without specialist advice because lower ambient pressure may worsen injury.
- **Exam trap:** The source explicitly advises against routine in-water recompression as first aid.

تاريخ Dive Profile جزء من التشخيص، وقرار النقل يحتاج تنسيقًا مع Hyperbaric Service. لا تعتبر تحسن الأعراض كافيًا لإنهاء التقييم.

Topic 43: Altitude illness: AMS versus HACE versus HAPE

- **Key discriminator:** Headache with nausea or fatigue after ascent suggests AMS; ataxia or confusion suggests HACE; disproportionate dyspnea and hypoxemia suggest HAPE.
- **Clinical trigger:** New mental-status change at altitude is an escalation warning for cerebral edema rather than simple mountain sickness.
- **Management priority:** Stop ascent and prioritize descent with assistance and oxygen for severe or worsening illness.

- **Clinical Pearl:** Portable hyperbaric support may bridge a delay when descent is impossible but is not a reason to delay descent.
- **Exam trap:** Pulse oximetry at altitude requires context and cannot be the only diagnostic criterion.

AMS = headache وأعراض عامة، HAPE = respiratory compromise، HACE = confusion/ataxia. عند الشدة أو التدهور، Descent أولوية.

Topic 44: Drowning: hypoxic physiology and early resuscitation

- **Core concept:** Drowning causes respiratory impairment from submersion, with hypoxemia generally preceding cardiac arrest.
- **Management priority:** Ensure rescuer safety, restore ventilation/oxygenation and provide CPR with rescue breaths and compressions when arrest occurs.
- **High-yield fact:** CLEAN V3 describes alveolar surfactant disruption and atelectasis as mechanisms of drowning-related lung injury.
- **Common mistake:** Do not postpone CPR to drain water from lungs or attempt abdominal thrusts to expel it.
- **Clinical Pearl:** Associated hypothermia, aspiration injury and traumatic mechanisms must be evaluated.

Drowning هو Hypoxic Event؛ Rescue Breaths مهمة بجانب CPR. محاولة إخراج الماء يدويًا لا تسبق الأكسجة والتهوية.

Topic 45: Drowning reassessment and disposition

- **Red flag:** Persistent hypoxemia, respiratory distress, altered mentation or resuscitation requirement favors hospital admission and escalation.
- **Clinical Pearl:** Respiratory complications may evolve after an initially reassuring assessment.
- **Management priority:** Base observation and disposition on oxygen requirement, examination and clinical trajectory.
- **Common mistake:** Do not routinely administer prophylactic antibiotics or corticosteroids without a separate indication.
- **Exam trap:** Avoid relying on imprecise labels such as near drowning or dry drowning to decide disposition.

بعد إنقاذ الغرق افحص SpO2 وwork of breathing والوعي مرات متكررة. التحسن اللحظي لا ينفي إصابة رئوية متأخرة.

Topic 46: Radiation exposure versus radioactive contamination

- **Key discriminator:** Radiation exposure is not synonymous with radioactive material remaining on or in the patient.
- **Management priority:** Treat life-threatening injuries and ABC problems first; decontamination must not delay lifesaving resuscitation or surgery.
- **Management priority:** Use PPE, controlled zones, removal/bagging of clothing and gentle decontamination when indicated.
- **Clinical Pearl:** For significant exposure, involve radiation experts and consider serial CBC/lymphocyte measurements as the source advises.
- **Exam trap:** Potassium iodide is specific to radioiodine risk, not a general antidote for radiation exposure.

Irradiated لا تعني Contaminated. ابدأ بعلاج الخطر المهدد للحياة، ثم نظّم إزالة التلوث بالتعاون مع Radiation Safety.

VII. Special Populations & Trauma Decision Traps

Topic 47: Pediatric trauma and occult shock

- **Core concept:** Children can preserve blood pressure until a late stage of shock; assess mental status, pulses, perfusion and work of breathing.

- **Number/threshold:** AQRAB chapter review cites systolic pressure below 70 + (2 x age in years) as a warning threshold for ages 1-10 years.
- **Management priority:** Control bleeding, prevent hypothermia and use a pediatric primary survey without delay for imaging.
- **Exam trap:** Do not apply adult imaging or resuscitation thresholds to children automatically.
- **Red flag:** Injury inconsistent with developmental stage or history warrants safeguarding assessment after stabilization.

الطفل قد يكون في Shock رغم BP طبيعي؛ راقب CRT والنبض والوعي. العمر يدخل في تفسير الضغط وقرارات التصوير.

Topic 48: Pediatric trauma resuscitation amounts: source-context caution

- **Number/threshold:** AQRAB chapter-matched review mentions older examples of crystalloid 20 mL/kg and packed red cells 10 mL/kg.
- **Exam trap:** The same passage explicitly cautions that those older examples require protocol review and should not delay blood-based resuscitation.
- **Management priority:** Clinical perfusion and ongoing hemorrhage assessment determine escalation; neither value is presented here as an independent current order.

20 mL/kg و10 mL/kg وردتا كمثالين قديمين مع تنبيه واضح بالمصدر. احفظ السياق ولا تحوّلهما إلى وصفة ثابتة.

Topic 49: Trauma during pregnancy: maternal-first priorities

- **Core concept:** Maternal stabilization is central to fetal resuscitation; standard trauma primary-survey priorities still apply.
- **Clinical trigger:** Consider concealed hemorrhage, placental abruption, uterine injury and fetomaternal hemorrhage even with initially stable vital signs.
- **Management priority:** Do not withhold indicated maternal diagnostic imaging solely because of pregnancy when it changes care.
- **Clinical Pearl:** After initial maternal evaluation, involve obstetrics and assess the fetus according to gestation and available resources.
- **Number/threshold:** The AQRAB-cited EAST 2025 guidance conditionally recommends 4-6 hours of formal observation after trauma in viable pregnancy, with longer evaluation for concerning findings.

في Pregnancy Trauma إنعاش الأم هو أول حماية للجنين. لا تمنع imaging ضروريًا بسبب الحمل، وراقب الأم والجنين وفق viability والسياق.

Topic 50: Pregnancy trauma, Rh status and obstetric injury

- **Clinical trigger:** An Rh-negative pregnant patient after blunt abdominal trauma raises concern about fetomaternal hemorrhage.
- **Management priority:** AQRAB advises consideration of Rh(D) immune globulin with gestation-specific assessment and local dosing protocol.
- **Red flag:** Placental abruption and other obstetric complications may require assessment despite reassuring initial maternal observations.
- **Exam trap:** Do not confuse a normal maternal pressure reading with exclusion of placental or fetal injury.

بعد abdominal trauma في الحمل افحص Rh status وfetomaternal bleeding إذا كان ذلك مستطبًا. استقرار الأم لا يغني عن تقييم المشيمة والجنين.

Topic 51: Geriatric trauma, frailty and anticoagulants

- **Core concept:** Low-energy falls can cause significant head, cervical-spine, rib, pelvic or long-bone injuries in older adults.
- **Key discriminator:** Absence of marked tachycardia or hypotension does not exclude important injury or shock.
- **Clinical trigger:** Review anticoagulants and antiplatelets, baseline function, cognition, fall mechanism and collateral history.
- **Number/threshold:** AQRAB chapter review identifies systolic pressure below 110 mmHg as a concerning geriatric-trauma signal, not as a universal isolated diagnostic rule.
- **Management priority:** Reassess neurologic/perfusion findings and evaluate safe mobility, function and support before discharge.

في Geriatric Trauma قد تكون آلية سقوط بسيطة مع نزف أو كسر مهم. لا تعتمد على Tachycardia أو BP فقط، وراجع Anticoagulants وBaseline.

High-Yield Comparison Tables

Comparisons below only consolidate differences already present in the two supplied files.

Tension Pneumothorax vs Hemothorax

Feature	Tension pneumothorax	Traumatic hemothorax	Exam point
Primary threat	Obstructive tension physiology	Bleeding into pleural cavity	Identify mechanism of shock
Immediate action	Decompression if unstable	Resuscitation and bleeding-source control	Do not wait for CT in unstable patient
Incorrect shortcut	Fluids alone	Imaging alone during ongoing shock	Action follows physiology

Heatstroke vs Heat Exhaustion

Feature	Heatstroke	Heat exhaustion	Exam point
CNS status	CNS dysfunction	No heatstroke-type CNS dysfunction	Neurologic deficit changes urgency
Temperature	High core temperature may fall after cooling	Heat symptoms without heatstroke syndrome	Post-cooling <40 C does not exclude heatstroke
First priority	Rapid active whole-body cooling	Reassess symptoms and alternative causes	Do not delay heatstroke cooling for labs

Systemic Hypothermia vs Frostbite

Feature	Hypothermia	Frostbite	Exam point
Core problem	Low core-body temperature	Localized freezing tissue injury	Different rewarming pathways
Key action	Gentle handling and severity-specific rewarming	Prevent refreezing, protect thawed tissue	Avoid rubbing or unsafe thawing
Warning	Arrhythmia, afterdrop	Tissue ischemia and deep injury	Do not treat them as one diagnosis

AMS vs HACE / HAPE

Feature	Acute mountain sickness	HACE / HAPE	Exam point
Presentation	Headache, nausea, fatigue after ascent	Ataxia/confusion (HACE); dyspnea/hypoxemia (HAPE)	Neurologic and respiratory red flags
Escalation	Stop ascent and reassess	Assisted descent and oxygen for severe disease	Do not continue ascent with deterioration
Pitfall	Confusing symptoms with ordinary fatigue	Over-relying on SpO2 alone	Clinical context matters at altitude

PASS 9 — Source-backed clinical recovery | Domain 06

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 132

1000 mL	Source statements: AQRAB supplemental page provides an initial 1000 mL warmed fluid bolus in an adult source scenario, and separately notes formal resuscitation for >15-20% TBSA.

Trauma, Burns & Environmental Emergencies

dosing or medical-correctness certification.

Posterior sternoclavicular dislocation

Red flag: Posterior sternoclavicular dislocation threatens nearby mediastinal structures; CLEAN V3 describes urgent chest CT angiography in the relevant presentation.

خلع sternoclavicular للخلف قد يهدد تراكيب المنصف؛ يذكر المصدر urgent CTA عند الاشتباه المناسب.

Possible basilar skull fracture

Procedural caution: CLEAN V3 highlights possible communication between the nasal cavity and intracranial space after a skull-base fracture, creating a risk of intracranial passage of a nasally introduced tube.

عند الاشتباه بكسر قاعدة الجمجمة، انتبه لخطر مرور الأنبوب الأنفي باتجاه داخل القحف.

Radiation Exposure vs Contamination

Feature	Exposure	Contamination	Exam point
Meaning	Irradiated without necessarily retaining radioactive material	Radioactive material on or in the patient	These are not synonyms
Priorities	Stabilize ABCs and injury	ABCs, PPE, containment, controlled decontamination	Life-saving care comes first
Common error	Unnecessary routine isolation	Assuming a survey reading equals absorbed dose	Consult radiation experts

Pediatric vs Geriatric Trauma

Feature	Pediatric trauma	Geriatric trauma	Exam point
Compensation	Hypotension often late	Tachycardia may be blunted	Normal-range vitals do not exclude injury
Special checks	Perfusion, development, safeguarding	Frailty, anticoagulants, cognition	Age modifies risk assessment
Pitfall	Applying adult thresholds	Assuming low-energy fall is safe	Mechanism plus trajectory

Decompression Sickness vs Arterial Gas Embolism

Feature	Decompression sickness	Arterial gas embolism	Exam point
Typical clue	Symptoms after ascent: joints, skin, neurologic or vestibular	Rapid neurologic or cardiopulmonary presentation near ascent	Timing helps but overlap exists
Immediate route	Oxygen, serial neurologic exam, hyperbaric consultation	Oxygen, stabilization, urgent hyperbaric consultation	Improvement on oxygen does not clear either
Pitfall	Unplanned altitude transport	Delayed specialist referral	Avoid routine in-water recompression

Rapid Recall: Must-Know Facts

Forty short memory prompts derived from the integrated body above; no new information is introduced here.

01. Treat traumatic life threats during the ABCDE primary survey.
02. Normal blood pressure does not exclude early traumatic hemorrhage.
03. Control external bleeding; tourniquet use depends on severity and setting.
04. Traumatic arrest requires evaluation of reversible traumatic causes.
05. FAST-negative does not exclude bowel or retroperitoneal injury.
06. An unstable abdominal trauma patient may need intervention before CT.
07. Control the bleeding source; do not rely on crystalloids alone.
08. The lethal diamond adds hypocalcemia to hypothermia, acidosis and coagulopathy.
09. GCS and clinical evolution determine head-injury urgency.
10. Source NICE pathway: GCS 12 or less -> CT within 1 hour.
11. Source NICE pathway: two or more adult vomiting episodes -> 1-hour CT consideration.
12. Do not combine distinct 1-hour and 8-hour head CT eligibility pathways.
13. DAI can cause severe coma with relatively limited early CT changes.
14. CPP equals MAP minus ICP.
15. Lundberg A waves reflect dangerously reduced intracranial compliance.
16. CLEAN V3 warns against high-dose corticosteroids for severe TBI outcomes.
17. Hemophilia-associated head injury deserves special bleeding concern.
18. Use cervical-spine precautions according to mechanism and findings.
19. Traumatic hypotension is not automatically neurogenic shock.
20. Suspected unstable tension pneumothorax prioritizes decompression.
21. Source adult hemothorax question: 250 mL/hour for 3 hours is a surgery clue.
22. Flail chest: at least three adjacent ribs fractured in two places.
23. The pulmonary contusion may drive flail-chest respiratory compromise.
24. Blunt cardiac injury may affect the anterior right ventricle.
25. Diaphragmatic rupture can lead to organ herniation and strangulation.
26. Blood at the meatus -> investigate urethral injury before blind catheterization.

27. Compartment syndrome can exist with palpable pulses.
28. Pain on passive stretch is an important compartment syndrome clue.
29. Burn eschar may impair limb perfusion; assess need for escharotomy.
30. Deep electrical injury may greatly exceed external skin findings.
31. Hypothermia is a core temperature below 35 C in AQRAB.
32. Severe hypothermia requires gentle handling and rhythm assessment.
33. Do not thaw frostbite when refreezing is likely.
34. Heatstroke: hyperthermia plus CNS dysfunction -> active cooling now.
35. Antipyretics do not replace cooling for environmental hyperthermia.
36. After diving, new neurologic signs warrant early hyperbaric advice.
37. HACE suggests confusion or ataxia; HAPE suggests dyspnea/hypoxemia.
38. Drowning resuscitation prioritizes ventilation and oxygenation.
39. Radiation exposure is not automatically contamination.
40. Pediatric, pregnant and geriatric trauma require context-specific assessment.

Conflict Resolution Register

No confirmed direct clinical contradiction between CLEAN V3 and AQRAB was established for this Domain from the compared source passages. Context-dependent differences and incomplete cases are documented separately, not silently resolved.

Contextual and source-dependent cautions (not counted as confirmed inter-file conflicts)

1. Cervical-spine precautions

- **Source statements:** CLEAN V3 includes a limited situation where routine immobilization is unnecessary; elsewhere jaw thrust with immobilization is chosen. AQRAB uses cervical protection for indicated traumatic mechanisms.
- **Nature:** Different clinical circumstances; not a direct contradiction.

Source evidence: CLEAN V3 ¶2169,2332; AQRAB PDF pp. 292-293.

2. Head-injury CT timing

- **Source statements:** CLEAN V3 lists 1-hour and 8-hour pathways with different eligibility criteria; some shortened stems omit important context.
- **Nature:** Protocol-context/fragment caution; do not collapse into a single rule.

Source evidence: CLEAN V3 ¶2189,2193-2204,2352,2362.

3. Major-burn fluid example

- **Source statements:** AQRAB supplemental page provides an initial 1000 mL warmed fluid bolus in an adult source scenario, and separately notes formal resuscitation for >15-20% TBSA.
- **Nature:** Question-specific adult example; not a universal regimen.

Source evidence: AQRAB PDF p. 409.

4. Observation after trauma in pregnancy

- **Source statements:** AQRAB describes a conditionally recommended 4-6-hour observation for viable pregnancy and longer observation with concerning findings.
- **Nature:** Gestation and clinical status matter; not a fixed discharge rule.

Source evidence: AQRAB PDF pp. 295, 332-333.

5. Heatstroke temperature interpretation

- **Source statements:** A shortened CLEAN V3 case labels heat exhaustion using temperature below 40 C; AQRAB warns that a temperature below 40 C measured after cooling does not exclude heatstroke.
- **Nature:** Different time points and CNS-status context: both descriptions remain source-dependent.

Source evidence: CLEAN V3 ¶2273; AQRAB PDF p. 217.

6. Drowning observation duration

- **Source statements:** CLEAN V3 gives abbreviated 4-6-hour and 6-hour observation examples, whereas AQRAB emphasizes oxygenation, symptoms and trajectory rather than a universal time.

- **Nature:** Risk-selection and protocol-context difference; neither passage establishes a universal discharge rule.

Source evidence: CLEAN V3 ¶2271,2274; AQRAB PDF p. 221.

Internal source ambiguity

A portion of CLEAN V3 contains shortened question-answer phrases, including head-CT disposition and isolated numerical answers, whose original clinical conditions are not preserved. These were not used to invent universal thresholds or artificial cross-file conflicts.

Metric	Result
Final integrated clinical topics	51
Integrated English clinical points	224
CLEAN V3-only points (edited source attributions)	68
AQRAB-only points (edited source attributions)	132
Jointly supported merged points	24
Short Arabic topic explanations	51
High-yield comparison tables	7
Rapid Recall facts	40
Confirmed direct inter-file conflicts	0
Protocol/context cautions logged	6
Unusable/incomplete source groups excluded	8
External medical information added	0

Counting rule: each authored clinical point is assigned one source disposition (CLEAN only / AQRAB only / both). These are editorial trace counts for this Domain, not an exhaustive count of every statement in the original documents. Comparison and Rapid Recall restatements are excluded from point totals.

Incomplete in source — not included

1. Uninterpretable INR/head CT item. [CLEAN V3 ¶2176]
2. Ceftriaxone/azithromycin phrase detached from chest-trauma question. [CLEAN V3 ¶2177]
3. Picture-dependent herpetic-whitlow classification without original exam context. [CLEAN V3 ¶2321]
4. Isolated urinalysis 20 RBC/hpf decision without complete case. [CLEAN V3 ¶2349]
5. Unattributed platelet threshold >50,000/uL. [CLEAN V3 ¶2357]
6. Case-specific 3% saline bolus without complete TBI physiology or regimen context. [CLEAN V3 ¶2350]
7. Isolated genital-amputation packaging statement without full protocol. [CLEAN V3 ¶2331]
8. Miscellaneous orthopedic overuse items outside acute trauma scope. [CLEAN V3 ¶2314,2318-2320,2329-2340]

Image-only source panels were visually sampled alongside text. This Domain does not reproduce the original infographic artwork; only material sufficiently supported by the supplied documents is restated.

No external references, books, web content, drug regimens or medical guideline updates were used. The named guideline citations inside AQRAB/CLEAN V3 are reported only as claims in those files.

Trauma, Burns & Environmental Emergencies | Pass 8 | Explicit CLEAN V3 source locations

Fat embolism syndrome — recognition pattern

• High-yield fact: The cited CLEAN V3 presentation associates respiratory distress, confusion and petechial rash with fat embolism syndrome.

الجمع بين ضيق النفس وتغير الوعي وPetechiae هو نمط تمييز يذكره المصدر لـFat Embolism.

Sea-urchin injury — spine-associated envenomation

• High-yield fact: CLEAN V3 describes sea-urchin injuries as caused by toxin-coated spines; treatment details require the individual injury context.

إصابة Sea Urchin قد تنجم من أشواك مغطاة بالسم؛ مصدرنا لا يكفي لتعميم خطوات علاج كاملة.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 136

Wound closure - tissue adhesive selection [#041, 042]

High-yield fact: Tissue adhesive is suitable for selected low-tension wounds; joints, mucosa, and high-tension edges limit its use.

لاصق الجلد يناسب جروحًا مختارة قليلة الشد، وله قيود قرب المفاصل والمخاطيات وحواف الجروح المشدودة.

Laceration irrigation - practical technique [#043]

High-yield fact: Wound irrigation requires adequate pressure to dislodge debris while avoiding additional tissue damage.

ريّ الجرح يتطلب ضغطًا كافيًا لإزالة الملوثات دون إحداث أذية إضافية للأنسجة.

Supplement IDs 043 | PreTest Emergency Medicine 5e (2021) | PDF p. 657 | Question 501

Stingray injury - envenomation and retained spine [#045]

High-yield fact: Stingray injuries cause puncture wounds and envenomation; assess for retained spine and secondary infection.

إصابات Stingray قد تجمع بين جرح وخزي وسم؛ ابحث عن شوكة متبقية واحتمال عدوى ثانوية.

Brown recluse spider injury - diagnostic caution [#047]

High-yield fact: A brown recluse diagnosis cannot reliably be made from an ulcer alone without verified exposure.

لا يمكن تشخيص لدغة Brown recluse من وجود قرحة وحدها دون قرينة تعرض موثوقة.

EDIC / Emergency & Critical Care

07 — Infectious Diseases & Sepsis

I. Sepsis Recognition, Definitions & Source Control

Topic 1: Recognizing sepsis as infection-associated organ dysfunction

- **Core concept:** The AQRAB source defines sepsis as acute organ dysfunction associated with suspected or confirmed infection.
- **Number/threshold:** Its Sepsis-3 table specifies acute SOFA increase of at least 2 points in the presence of infection.
- **Clinical trigger:** New confusion, impaired perfusion, oliguria, hypoxemia or organ injury with suspected infection should raise concern even if blood pressure is normal.
- **Exam trap:** Neither high fever nor isolated leukocytosis establishes sepsis, and absent fever does not exclude it.
- **High-yield fact:** The supplied CLEAN V3 review describes pneumonia as a common sepsis source, followed by intra-abdominal infection in cited epidemiology.

Sepsis ليس حرارة مرتفعة فقط؛ اسأل هل العدوى سببت Organ Dysfunction؟ افحص الوعي والتنفس والتروية والبول واتجاه Lactate.

Topic 2: Septic shock: retain distinct definition frameworks

- **Number/threshold:** Sepsis-3 in AQRAB: vasopressors needed to maintain MAP at least 65 mmHg AND lactate greater than 2 mmol/L after adequate fluid resuscitation.
- **Number/threshold:** CLEAN V3 quotes a CMS-style severe-sepsis/shock framework using persistent hypotension after 30 mL/kg crystalloid OR lactate at least 4 mmol/L.
- **Key discriminator:** The two quoted definitions use different criteria and should not be merged into one cutoff.
- **Common mistake:** Do not infer that an isolated lactate elevation is specific for sepsis, or that a normal blood pressure proves adequate perfusion.

Sepsis-3 وCMS ليسا نفس التعريف. احفظ اسم المعيار مع الأرقام؛ لا تخلط Lactate >2 مع ≤4 دون معرفة الإطار.

Topic 3: Clinical screening and the perfusion-focused approach

- **High-yield fact:** AQRAB presents NEWS/NEWS2, MEWS or SIRS as screening approaches and warns against relying on qSOFA alone to exclude sepsis.
- **Clinical trigger:** Mottling, delayed capillary refill, altered mentation, reduced urine output or a worsening lactate trajectory warrants reassessment.
- **Key discriminator:** Shock may be compensated, with seemingly normal blood pressure.

- **Exam trap:** A respiratory rate of 22/min appears as an isolated score-related value in CLEAN V3 and is not a standalone sepsis diagnosis.

SIRS وNEWS وqSOFA أدوات فحص وليست تشخيصًا نهائيًا. أهم شيء تفسير Perfusion وOrgan Dysfunction بشكل سريري.

Topic 4: Cultures, lactate and time-critical diagnostics

- **Management priority:** Obtain blood cultures promptly and before antimicrobials when feasible without clinically meaningful delay.
- **High-yield fact:** Measure lactate and repeat it when elevated or when abnormal perfusion persists, as described in AQRAB.
- **Clinical Pearl:** Select investigations according to syndrome and suspected source rather than ordering every test indiscriminately.
- **Common mistake:** Do not delay treatment of shock to preserve culture yield or to await complete source identification.

خذ Blood Cultures قبل المضاد إذا أمكن دون تأخير مؤثر. Lactate يفيد في تقييم الخطورة والمتابعة وليس اختبارًا نوعيًا للعدوى.

Topic 5: Antimicrobial timing by clinical severity

- **Management priority:** AQRAB calls for immediate antimicrobials in septic shock or probable/definite sepsis.
- **Key discriminator:** Possible sepsis without shock may allow a rapid, time-limited diagnostic assessment in the source pathway.
- **Common mistake:** Do not translate the source advice into an identical antibiotic-timing rule for every febrile patient.
- **Clinical Pearl:** Choose an empiric regimen according to source, previous organisms, allergy, organ function and local resistance, then narrow when evidence supports this.

في Septic Shock أو العدوى المرجّحة الخطيرة لا تؤخر antibiotics. أما Possible Sepsis دون Shock فالمصدر يسمح بتقييم سريع ومحدد الوقت.

Topic 6: Crystalloid resuscitation and reassessment

- **Core concept:** The AQRAB sepsis pathway recommends individualized crystalloid resuscitation with frequent dynamic reassessment.
- **Clinical trigger:** Persistent hypoperfusion after a fluid challenge requires reconsideration of response, overload, vasodilation and the infected source.
- **High-yield fact:** CLEAN V3 explicitly states that hydroxyethyl starches are discouraged in sepsis despite a misleading answer-stem phrase saying they are recommended.
- **Exam trap:** Do not continue fixed-volume fluids without reviewing the cardiac/renal context, lung congestion and response.

السوائل في Septic Shock تُعطى مع إعادة تقييم متكررة لـPerfusion والاحتقان؛ لا تستمر تلقائيًا بجرعات ثابتة لمجرد بقاء الضغط منخفضًا.

Topic 7: Vasopressors and MAP targets in septic shock

- **Management priority:** The sources identify norepinephrine (noradrenaline) as first-line vasopressor for persistent septic-shock hypotension.
- **Number/threshold:** A mean arterial pressure target of at least 65 mmHg appears in the supplied adult shock pathways.
- **Key discriminator:** Fluid responsiveness and vasodilatory shock physiology are not established by blood pressure alone.
- **Common mistake:** A CLEAN V3 metaraminol 0.5 mg bolus belongs to one particular examination scenario and is not treated here as a universal shock order.

Norepinephrine/Noradrenaline هو الخيار الأول المذكور عند استمرار Septic Hypotension بعد التقييم المناسب. MAP ≥65 هدف مذكور في المصدر ويحتاج سياق المريض.

Topic 8: Source control: drainage, obstruction and necrotizing foci

- **Core concept:** Draining an abscess, relieving infected obstruction, debriding dead tissue or removing an infected focus may be necessary alongside resuscitation and antibiotics.
- **Management priority:** Identify a source requiring an intervention early; AQRAB summarizes a suggested early source-control target, ideally within 6 hours after diagnosis in its cited pathway.
- **Exam trap:** Do not repeatedly change antibiotics while an undrained or obstructed septic focus remains untreated.
- **Clinical Pearl:** Perform source control and circulation/antimicrobial treatment in parallel where feasible.

Antibiotics لا تفتح Abscess ولا تزيل Obstruction. إذا بقي المصدر موجودًا، تغيير المضاد وحده لا يعالج المشكلة الأساسية.

Topic 9: Sepsis-related organ support and treatment pitfalls

- **High-yield fact:** CLEAN V3 describes a restrictive red-cell transfusion reference threshold around hemoglobin 7 g/dL for an otherwise healthy adult in the cited sepsis discussion.
- **Management priority:** The supplied source includes hydrocortisone among adjuncts considered in persistent shock despite appropriate resuscitation, without a complete universal dosing algorithm.
- **Exam trap:** A CLEAN V3 question-stem claim that routine glutamine is recommended is contradicted by the accompanying explanation advising against routine supplementation in sepsis/multiorgan failure.
- **Clinical Pearl:** Reassess oxygenation, renal function, circulation and evolving organ injury during treatment.

تثبيت مريض Sepsis يشمل دعم الأعضاء وليس Antibiotic فقط. واحذر تحويل سؤال عن Glutamine أو Transfusion إلى توصية عامة دون سياقه.

Topic 10: Pediatric sepsis: recognition and fluid context

- **Core concept:** Pediatric sepsis assessment emphasizes perfusion, mental status, work of breathing and organ function rather than waiting for hypotension.
- **Number/threshold:** CLEAN V3 quotes 10-20 mL/kg isotonic balanced crystalloid over 5-10 minutes in a cited pediatric septic-shock source pathway.
- **Clinical Pearl:** Pediatric fluid selection and repeated boluses require reassessment for congestion and response rather than blind repetition.
- **Number/threshold:** CLEAN V3 mentions Phoenix pediatric sepsis cardiovascular scoring with at least one point in a specific definitional discussion.
- **Exam trap:** Do not generalize isolated hydrocortisone 3 mg/kg in a shortened pediatric exam case as a universal dose.

الطفل قد يحافظ على BP رغم نقص التروية؛ لا تجعل جرعة Adult Fluid أو Adult Septic Shock Definition تُطبق عليه تلقائيًا.

Topic 11: Neonatal infection: early-onset definition and empiric therapy

- **Clinical trigger:** A neonate with systemic illness or shock warrants prompt assessment for infection among other neonatal causes.
- **Number/threshold:** One CLEAN V3 passage describes early-onset neonatal sepsis within 72 hours, while another uses the first 7 days and notes a 72-hour convention.
- **Management priority:** CLEAN V3 describes blood cultures and empiric ampicillin plus gentamicin in a specified early-onset neonatal case.
- **Exam trap:** The discrepant time-window definitions must not be collapsed into a falsely precise universal cutoff.

في Neonatal Sepsis توجد صياغتان للـEarly-onset window: أول 72 ساعة أو أول 7 أيام. احفظ الاختلاف مع المصدر، ولا تحسمه دون بروتوكول.

Topic 12: Toxic shock syndrome recognition

- **Clinical trigger:** Rapid erythema, severe pain beyond examination findings, systemic toxicity, hypotension and organ injury in an infected wound can suggest streptococcal toxic shock.
- **High-yield fact:** CLEAN V3 associates streptococcal toxic shock with group A streptococcal infection, especially necrotizing fasciitis or myositis.
- **Differential point:** Staphylococcal toxic shock is described in association with retained foreign material, wounds, surgery or skin/soft-tissue sources.
- **Management priority:** The CLEAN V3 source includes IV vancomycin and clindamycin for a source staphylococcal toxic-shock scenario; source control remains critical.

Rapidly spreading soft-tissue infection مع Pain out of proportion وShock قد يشير إلى Streptococcal Toxic Shock؛ ركّز على Source Control سريعًا.

II. Skin, Soft-Tissue, Bone & Joint Infections

Topic 13: Cellulitis, abscess and infection depth

- **Core concept:** Differentiate diffuse nonpurulent cellulitis, a purulent collection and deep necrotizing disease by depth, progression and systemic illness.
- **Clinical trigger:** Localized warmth, erythema, swelling and pain support cellulitis, whereas a localized collection prompts drainage assessment.
- **Management priority:** An abscess needs a drainage decision; antibiotic choice depends on illness severity and clinical source.
- **Exam trap:** CLEAN V3 notes that routine dual antibiotics do not improve uncomplicated cellulitis outcomes in its cited material.

Cellulitis منتشر في الجلد، أما Abscess فقد يحتاج Drainage. وجود سمية عامة أو ألم سريع التفاقم يعني أن التقييم يجب أن يتجاوز عدوى سطحية.

Topic 14: Necrotizing fasciitis: do not wait for late signs

- **Red flag:** Rapidly progressive severe pain, edema extending beyond erythema, bullae, ecchymosis, anesthesia, crepitus or systemic toxicity raises concern for necrotizing infection.
- **Management priority:** Urgent surgical assessment, broad empiric IV antimicrobials, resuscitation and operative debridement are emphasized in both sources.
- **Exam trap:** A low LRINEC score, absent tissue gas or reassuring early imaging cannot safely overrule strong clinical suspicion.
- **High-yield fact:** Antibiotics alone do not replace removal of necrotic tissue.

Pain out of proportion أو الانتشار السريع مع Toxicity أهم من انتظار Crepitus أو صورة تؤكد Gas. الاستشارة الجراحية لا تتأخر.

Topic 15: Fournier gangrene and perineal necrotizing infection

- **Clinical trigger:** Severe perineal or scrotal infection with rapidly worsening pain, skin changes or systemic toxicity raises concern for Fournier gangrene.
- **Management priority:** CLEAN V3 supplies a case selecting IV piperacillin-tazobactam plus vancomycin with urgent urologic debridement; this is a case regimen, not an automatic universal order.
- **Clinical Pearl:** The initial appearance may resemble a minor abscess before rapid progression.
- **Common mistake:** Do not delay surgical source control solely for imaging or a change in antibiotic regimen.

Perineal Pain أو Scrotal Infection يتدهور بسرعة؟ فكّر بـFournier ولا تنتظر CT إذا كانت علامات Necrotizing Infection قوية.

Topic 16: Septic arthritis and urgent arthrocentesis

- **Core concept:** Septic arthritis is an urgent infection capable of rapid irreversible joint destruction if inadequately treated.
- **Management priority:** Urgent aspiration for cell count, Gram stain and culture with appropriate specialty involvement is emphasized in the supplied texts.
- **Number/threshold:** CLEAN V3 repeatedly cites synovial WBC above 50,000 cells/mm3 and neutrophil predominance above 90% as concerning patterns, but lower counts can occur.
- **Exam trap:** Crystals may coexist with bacterial infection; do not use a crystal result to exclude septic arthritis.
- **Clinical Pearl:** In a clinically unstable patient, treatment and consultation must not be delayed for ideal sampling conditions.

Hot swollen joint = Septic Arthritis حتى يثبت غير ذلك. وجود Crystals لا يلغي Infection، ولا يجوز انتظار تدهور المفصل قبل أخذ العينة.

Topic 17: Overlying cellulitis and the aspiration pathway

- **Source-dependent caution:** CLEAN V3 lists cellulitis directly over the intended arthrocentesis entry site as a contraindication to that path due to inoculation risk.
- **Management priority:** AQRAB continues to prioritize urgent diagnostic joint aspiration and specialist input for suspected septic arthritis.
- **Key discriminator:** The source positions are not necessarily contradictory: entry-site choice and whether aspiration is needed are distinct decisions.
- **Exam trap:** Do not interpret a hazardous entry route as an instruction to abandon evaluation of a potentially infected joint.

المصدر الأول يعتبر Cellulitis فوق موقع الإبرة مانعًا للطريق المباشر؛ والمصدر الثاني يؤكد ضرورة Aspirate للـSeptic Joint. الفرق في مكان الدخول لا في أهمية التشخيص.

Topic 18: Osteomyelitis in children and adults

- **Core concept:** Focal bone pain, fever, bacteremia, injection drug use or recent intervention can signal bone infection.
- **High-yield fact:** CLEAN V3 describes hematogenous spread as the dominant mechanism in childhood osteomyelitis.
- **High-yield fact:** Staphylococcus aureus is identified as a common organism in the AQRAB bone-infection companion.
- **Clinical Pearl:** CLEAN V3 associates Salmonella osteomyelitis with sickle-cell disease and highlights Pseudomonas in puncture-wound or injection-related contexts.
- **Management priority:** Use clinical assessment and blood cultures; AQRAB favors MRI when advanced imaging is needed and warns that early plain films may be unrevealing.

Focal Bone Pain مع حمى أو Bacteremia قد يعني Osteomyelitis، خصوصًا مع IVDU أو Sickle Cell Disease أو عوامل خطورة أخرى.

Topic 19: Spinal epidural abscess, discitis and vertebral infection

- **Clinical trigger:** Focal back pain with fever, neurologic change or risk factors such as diabetes, IVDU, kidney disease, cancer or recent procedures raises concern for spinal infection.
- **Key discriminator:** Spinal epidural abscess, vertebral osteomyelitis and discitis are serious infectious alternatives to uncomplicated back pain.
- **Management priority:** Urgent imaging and specialist involvement are warranted for neurologic compromise or high clinical concern.

- **Common mistake:** A modest inflammatory marker or lack of dramatic fever should not close the differential.

Back Pain مع Fever أو عوامل خطورة ليس دائمًا ألم عضلات؛ Epidural Abscess قد يهدد النخاع ويحتاج تصويرًا وتقييمًا عاجلًا.

Topic 20: Preseptal versus orbital cellulitis

- **Key discriminator:** Preseptal cellulitis involves tissues anterior to the orbital septum and typically spares ocular movements and vision.
- **Red flag:** Orbital involvement affects post-septal orbital tissue and raises concern when eye movement or vision is impaired.
- **Clinical Pearl:** CLEAN V3 identifies cavernous sinus thrombosis as a serious potential complication of orbital infection.
- **Management priority:** CLEAN V3 provides oral amoxicillin-clavulanate as one mild, cooperative-patient preseptal scenario, not a regimen for orbital disease.

Preseptal أمام Orbital Septum؛ Orbital Cellulitis خلفه وقد يهدد Vision وEye Movements. ليس كل تورم جفن بنفس الخطورة.

Topic 21: Hand, bite and puncture-related deep infection

- **Clinical trigger:** A bite over a hand joint, deep puncture or small opening with disproportionate pain needs assessment for tendon, joint, foreign-body or bone involvement.
- **Clinical Pearl:** CLEAN V3 notes that cat teeth can inoculate structures deeply and that paronychia may progress to a drainable abscess.
- **High-yield fact:** Pseudomonas is cited in a puncture-related foot osteomyelitis examination item, but its stated percentage is not reused as a universal estimate.
- **Exam trap:** Antibiotics do not remove a foreign body or replace needed drainage.

Animal Bite أو Clenched-fist Wound فوق Joint قد يصل إلى Tendon أو Bone رغم جرح صغير. تقييم العمق أهم من مظهر الجلد وحده.

III. Urinary, Genital & Other Focal Infections

Topic 22: Uncomplicated UTI versus asymptomatic bacteriuria

- **Key discriminator:** CLEAN V3 describes asymptomatic bacteriuria as positive urine culture without urinary symptoms and includes a case in which no further antibiotic was indicated.
- **Common mistake:** CLEAN V3 warns that dipstick testing alone is unreliable as a sole UTI diagnostic decision tool in adults 65 years and older.
- **Clinical Pearl:** Interpret urine tests in the setting of fever, urinary symptoms, competing diagnoses and clinical trajectory.
- **Management priority:** The source discourages fluoroquinolones as first-line agents for uncomplicated UTI in its cited resistance discussion.

Urine Culture إيجابي وحده لا يعني علاجًا دائمًا. اربط التحليل بالأعراض، خصوصًا عند المسنين، وتجنب علاج Asymptomatic Bacteriuria بلا سبب في المصدر.

Topic 23: Pyelonephritis, abscess and infected obstruction

- **Clinical trigger:** Fever/flank symptoms with shock, infected stone or poor improvement after therapy raises concern for complicated infection or an undrained focus.
- **Management priority:** CLEAN V3 describes interventional-radiology drainage for a renal abscess in a particular nonresponding case.
- **Clinical Pearl:** AQRAB emphasizes relief of infected obstruction and source control as well as antimicrobials.

- **High-yield fact:** CLEAN V3 names pyelonephritis as an important source of septic shock in pregnancy in a cited clinical context.

Pyelonephritis مع Obstruction أو Renal Abscess قد يحتاج Drainage وليس Antibiotics فقط. استمرار المرض رغم العلاج يستدعي البحث عن Source.

Topic 24: Genital infections, PID and disseminated gonorrhea

- **High-yield fact:** CLEAN V3 describes Neisseria gonorrhoeae as the cause of gonorrhea and gonococcemia as bacteremic dissemination from a genital or extragenital site.
- **Clinical trigger:** Dysuria accompanied by vaginal discharge warrants consideration of gonorrhea/chlamydia rather than uncomplicated cystitis alone.
- **Management priority:** AQRAB and CLEAN V3 describe site- and syndrome-specific antimicrobial pathways for PID and other genital infections.
- **Exam trap:** Older single-case STI treatment doses in CLEAN V3 should not be silently promoted to a universal current regimen.

Dysuria مع Vaginal Discharge لا تُفسّر كـUTI فقط؛ فكر بـSTI. أما Gonococcemia فتسبب Systemic Infection خارج الموقع الأصلي.

Topic 25: Spontaneous bacterial peritonitis as a sepsis source

- **Number/threshold:** CLEAN V3 cites ascitic fluid polymorphonuclear cells greater than 250/mm3 in a specific spontaneous bacterial peritonitis case.
- **Clinical trigger:** Suspected infected ascites in a systemically unwell patient prompts diagnostic paracentesis assessment.
- **Management priority:** CLEAN V3 lists cefotaxime 2 g IV as an examination-case treatment example; it is not generalized to every patient.
- **Exam trap:** Do not transfer this ascitic fluid threshold to unrelated body-fluid or septic-arthritis tests.

SBP قد يظهر كـSepsis مع بطن مريض لديه Ascites؛ التشخيص يعتمد على Ascitic PMN وليس انتظار مزرعة إيجابية فقط.

IV. Transmissible, Viral, Opportunistic & Travel Infections

Topic 26: Respiratory diphtheria and public-health urgency

- **Clinical trigger:** Suspected respiratory diphtheria requires attention to airway, respiratory isolation and public-health notification.
- **Management priority:** The source describes antitoxin plus antibiotics, specimen collection when feasible without delaying treatment, and monitoring for cardiac and neurologic complications.
- **Clinical Pearl:** Antitoxin neutralizes circulating toxin but not toxin already bound to tissues in the supplied description.
- **Exam trap:** A negative early test does not necessarily rule out a strongly suspected clinical syndrome.

في Suspected Diphtheria لا تنتظر نتيجة الفحص كي تعزل المريض وتحمي Airway. Antitoxin يتعامل مع Circulating Toxin لذلك التأخير مهم.

Topic 27: Viral encephalitis and influenza: infectious priorities

- **Clinical trigger:** Fever plus new confusion, behavior change, seizure or focal neurologic findings may indicate HSV encephalitis.
- **Management priority:** AQRAB advocates prompt IV acyclovir for suspected encephalitis, without waiting for MRI, LP or PCR confirmation.

CLINICAL NUMBERS · TOPIC CONTEXT

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0.5 mg	alone. Common mistake: A CLEAN V3 metaraminol 0.5 mg bolus belongs to one particular examination scenario and is not treated here as a universal shock order.
3 mg/kg	specific definitional discussion. Exam trap: Do not generalize isolated hydrocortisone 3 mg/kg in a shortened pediatric exam case as a universal dose.
2 g	assessment. Management priority: CLEAN V3 lists cefotaxime 2 g IV as an examination-case treatment example; it is not generalized to every patient.
4.5 g	Number/threshold: AQRAB lists adult source-edition examples: cefepime 2 g IV every 8 h, meropenem 1 g IV every 8 h, and piperacillin-tazobactam 4.5 g IV every 6 h. Key discriminator: The same source notes that cefepime does not reliably provide anaerobic coverage.

- **Clinical Pearl:** An early negative HSV CSF PCR may not exclude disease; the source recommends reassessment and repeat testing with specialist advice when suspicion persists.
- **Management priority:** For severe/progressive, hospitalized or high-risk suspected influenza, AQRAB prioritizes early antiviral assessment without waiting for a rapid test result.

Fever مع New Seizure أو Behavioral Change قد يشير إلى HSV Encephalitis؛ لا تؤخر IV Acyclovir بانتظار MRI أو LP إذا كان الاشتباه قويًا.

Topic 28: COVID-19: infection severity and bacterial coinfection

- **Core concept:** The source recommends assessing COVID-19 by oxygenation, respiratory work, perfusion and organ dysfunction rather than by a positive test alone.
- **Management priority:** Eligible high-risk outpatients may need time-sensitive antiviral consideration with interaction and organ-function checks under the applicable source pathway.
- **Exam trap:** Do not prescribe routine antibacterial treatment for presumed viral infection without suspicion of bacterial coinfection.
- **Red flag:** Hypoxemia, respiratory distress, confusion, shock or organ injury warrants escalation and an alternative-diagnosis assessment.

COVID Test إيجابي لا يفسّر تلقائيًا Shock أو PE أو ACS. عالج حسب شدة المرض، ولا تعطِ Antibiotics دون اشتباه بكتيري مستقل.

Topic 29: Tuberculosis: suspicion-driven airborne isolation

- **Clinical trigger:** Chronic cough, hemoptysis, weight loss, sweats, TB exposure or immunosuppression prompts consideration of active pulmonary TB.
- **Management priority:** Suspected infectious pulmonary or laryngeal TB requires early airborne precautions and coordination with infection control.
- **Key discriminator:** A positive IGRA/TST does not distinguish latent TB infection from active contagious disease.
- **Exam trap:** Normal chest radiography may not exclude TB in advanced immunosuppression.

Cough مزمن مع Weight Loss أو Night Sweats أو عوامل خطورة = ضع Airborne Precautions مبكرًا. لا تنتظر Chest X-ray ليبدأ العزل.

Topic 30: HIV: assess immune status and competing infections

- **Core concept:** The source instructs assessment of HIV therapy, adherence, CD4, viral load, prophylaxis, previous infections and interactions during an acute presentation.
- **Number/threshold:** AQRAB gives CD4 below 200 cells/mm3 as a marker increasing concern for opportunistic infections, not a diagnosis by itself.
- **Clinical Pearl:** CLEAN V3 associates cerebral toxoplasmosis with ring-enhancing lesions and surrounding edema in an opportunistic-infection case.
- **Management priority:** Stabilize, investigate syndrome-specific causes and use isolation when transmissible infection is suspected; involve specialists with advanced immunosuppression.

اسأل عن CD4 وViral Load وART والالتزام والوقاية، لكن لا تنسَ أن مريض HIV قد يصاب بأسباب شائعة أيضًا.

Topic 31: Pneumocystis pneumonia in advanced HIV: regimen context

- **Clinical trigger:** In a patient with advanced immune suppression and respiratory disease, the source raises Pneumocystis jirovecii pneumonia as an opportunistic differential.
- **Number/threshold:** The cited PJP example uses TMP-SMX trimethoprim component 15-20 mg/kg/day divided into three doses for 21 days.

- **Number/threshold:** The supplied adjunctive steroid example uses PaO2 below 70 mmHg or A-a gradient above 35 mmHg in selected hypoxemic HIV-PJP patients.
- **Exam trap:** The source steroid taper and antibiotic schedule are condition-specific examples requiring review of kidney function, potassium, allergy and applicable protocol.

PJP Doses المذكورة تخص HIV-PJP وفق شدة ونقص أكسجة محددين. لا تستخدم Steroid Taper لكل مريض لديه Pneumonia.

Topic 32: Immunocompromised patient and febrile neutropenia

- **Clinical trigger:** Cancer treatment, neutropenia, transplant medicines, steroids, biologics or advanced HIV may blunt classic infection signs.
- **Management priority:** AQRAB cites initial empiric antibacterial treatment within 1 hour of triage for the defined cancer-with-febrile-neutropenia group.
- **High-yield fact:** CLEAN V3 links mucositis in a neutropenic case to viridans streptococcal bacteremia with severe systemic complications.
- **Exam trap:** Do not apply outpatient low-risk criteria to unstable patients or assume a normal temperature excludes severe infection.

Neutropenic Patient قد يكون Severely Infected دون Fever أو WBC مرتفع. أولوية مضاد مبكر تعتمد على الحالة والبروتوكول.

Topic 33: Solid-organ transplant: infection versus rejection and drug effects

- **Core concept:** Transplant recipients may present with infection, rejection, drug toxicity, surgical problems or unrelated emergencies.
- **Clinical trigger:** Fever, graft-region pain, oliguria, diarrhea, dyspnea or new organ dysfunction requires prompt assessment even when inflammatory features are muted.
- **Management priority:** Obtain transplant timeline, baseline function, immunosuppression, prophylaxis and relevant cultures and contact the transplant team early.
- **Exam trap:** AQRAB advises against unilaterally stopping immunosuppressants or starting anti-rejection therapy without specialist coordination.

Transplant بعد Fever أو Graft Dysfunction مباشرة كـRejection؛ فكر أيضًا بعدوى أو Drug Toxicity ولا توقف Immunosuppression من نفسك.

Topic 34: Imported fever and severe malaria

- **Clinical trigger:** Fever after travel or residence in malaria-endemic regions warrants malaria consideration regardless of prophylaxis or an initially nonspecific presentation.
- **Management priority:** Obtain urgent thick/thin smears or locally accepted testing and repeat if strong suspicion persists after one negative result.
- **Red flag:** Altered consciousness, shock, respiratory distress, acidosis, renal failure or hypoglycemia suggests severe malaria in the supplied review.
- **Management priority:** AQRAB identifies prompt admission and IV artesunate for severe malaria, with species-specific and geographic treatment decisions requiring specialist input.

Travel History جزء من Fever Assessment؛ Malaria ممكن حتى لو استخدم Prophylaxis أو ظهرت الأعراض متأخرة. Severe Malaria تحتاج علاجًا إسعافيًا.

Topic 35: Tickborne disease and Lyme carditis

- **Clinical trigger:** Outdoor exposure, compatible geography and erythema migrans, facial palsy, meningitis or arthritis raise concern for Lyme disease.

- **Red flag:** Syncope, palpitations or conduction abnormality can indicate Lyme carditis.
- **Management priority:** AQRAB recommends ECG, admission and monitoring with IV treatment consideration for severe suspected carditis/high-grade block.
- **Exam trap:** Early Lyme serology can be negative and absent remembered tick exposure does not exclude infection.

Lyme Carditis قد يظهر بـSyncope أو AV Block. لا يشترط تذكّر Tick Bite ولا ينتظر ECG أو التقييم حتى تظهر Rash.

Topic 36: Travel-related and foodborne differential patterns

- **High-yield fact:** CLEAN V3 describes Bacillus cereus as a foodborne spore-forming organism associated with preformed toxins.
- **High-yield fact:** CLEAN V3 identifies Zika as an Aedes-associated mosquito-borne flavivirus.
- **High-yield fact:** CLEAN V3 associates infant botulism with ingestion of spores and descending neurologic/cranial-nerve signs in botulinum illness.
- **Clinical Pearl:** An exposure pattern may redirect investigations when common tests or clinical presentation are nonspecific.

يوجه التشخيص: السفر، الطعام، الحيوانات والقراد ليست تفاصيل ثانوية في Exposure History أو Fever أو Diarrhea.

V. Antimicrobials, Isolation & Safe Prescribing

Topic 37: Empiric antibiotics: match site, host and resistance

- **Core concept:** Empiric antimicrobial choices require infection-site, host, allergy, renal-function, resistance and severity context.
- **Management priority:** CLEAN V3 identifies piperacillin-tazobactam as an antipseudomonal penicillin and describes broad regimens in selected severe infection cases.
- **Clinical Pearl:** AQRAB emphasizes reviewing the antibiotic plan as cultures and the clinical response become available.
- **Exam trap:** Two named broad-spectrum regimens need not provide interchangeable coverage for every source.

لا يوجد Antibiotic واحد مناسب لكل عدوى. تحدد Site وSeverity وAllergy وRenal Function وResistance الاختيار، ثم ضيّق التغطية عند توفر النتائج.

Topic 38: Antibiotic source examples: retain precise limits

- **Number/threshold:** AQRAB lists adult source-edition examples: cefepime 2 g IV every 8 h, meropenem 1 g IV every 8 h, and piperacillin-tazobactam 4.5 g IV every 6 h.
- **Key discriminator:** The same source notes that cefepime does not reliably provide anaerobic coverage.
- **Clinical Pearl:** Vancomycin has no single universal dosing schedule in the source; review host factors and monitoring.
- **Exam trap:** These doses are historical/source-edition examples and are not independently verified current prescribing advice.

جرعات Cefepime وMeropenem وPiperacillin-Tazobactam في المصدر أمثلة Adult تحتاج مراجعة Kidneys وAllergy وIndication؛ ليست وصفة شاملة.

Topic 39: Necrotizing infection antibiotics and source control

- **Management priority:** CLEAN V3 advocates adding clindamycin in a necrotizing-infection regimen and identifies urgent debridement with broad coverage as definitive care.

- **High-yield fact:** An uncomplicated cellulitis regimen should not be extrapolated to suspected necrotizing disease.
- **Common mistake:** Repeated antibiotics without needed operation, drainage or removal of infected material leaves a treatable source uncontrolled.
- **Clinical Pearl:** Early multidisciplinary surgical involvement matters more than a superficially reassuring scan.

Clindamycin يُذكر في Necrotizing Infection ضمن نظام أوسع، لكن أهم فخ أن Antibiotics لا تعوّض Surgery.

Topic 40: Antibiotic adverse effects and medication interactions

- **High-yield fact:** CLEAN V3 links fluoroquinolones, including ciprofloxacin/levofloxacin, with tendinitis and tendon rupture.
- **High-yield fact:** AQRAB warns that carbapenems may reduce valproate effectiveness and seizure control.
- **Key discriminator:** CLEAN V3 reports QT shortening with isavuconazole rather than QT prolongation as a distinguishing antifungal feature.
- **Exam trap:** Avoid selecting an antimicrobial solely from its organism label without checking adverse effects, comedications and organ function.

قبل Antibiotics راجع Drug Interactions، وخصوصًا Fluoroquinolone Tendon Risk وCarbapenem مع Valproate، وتأثير بعض Antifungals على QT.

Topic 41: Antifungal mechanisms and QT considerations

- **High-yield fact:** Polyenes such as amphotericin B and nystatin bind membrane ergosterol in the source pharmacology review.
- **Key discriminator:** CLEAN V3 identifies isavuconazole as a triazole associated with dose-dependent shortening of QTc.
- **High-yield fact:** CLEAN V3 describes echinocandins as not associated with QT prolongation in its comparative QT discussion.
- **Exam trap:** Do not treat all antifungal classes as having the same electrophysiologic or membrane target effects.

Polyenes ترتبط بـErgosterol، وIsavuconazole مذكور مع QT Shortening. احفظ الفارق بين classes بدل اعتبار كل Antifungal له تأثير QT نفسه.

Topic 42: Beta-lactamase mechanisms and resistant Gram negatives

- **High-yield fact:** CLEAN V3 states that avibactam lacks activity against Ambler class B metallo-beta-lactamases.
- **Clinical Pearl:** The CLEAN V3 explanation describes susceptible fluoroquinolones or aminoglycosides as possible carbapenem-sparing agents for selected ESBL infections.
- **Key discriminator:** Susceptibility testing and infection context are required before extrapolating a carbapenem-sparing statement.
- **Exam trap:** Two CLEAN V3 answer-stem fragments say that these non-beta-lactam drugs have no role; their associated explanatory text says the opposite under susceptibility conditions.

في Resistant Organisms لا تعتمد على كلمة ESBL فقط؛ افهم نوع Resistance وما إذا كانت Sensitivity تسمح بخيار أقل اتساعًا.

Topic 43: Infection isolation and exposure prevention

- **Management priority:** Apply suspected-TB airborne precautions early and appropriate diphtheria respiratory precautions with infection-control guidance.
- **High-yield fact:** CLEAN V3 cites prompt meningococcal close-contact chemoprophylaxis and mentions ciprofloxacin in a selected source example.

- **Key discriminator:** CLEAN V3 differentiates sterilization, which removes all viable microorganisms including spores, from ordinary disinfection in its definitions.
- **Clinical Pearl:** Exposure protection, specimen handling and public-health notification are part of disease management when transmissibility matters.

Isolation يبدأ عند الاشتباه الملائم لا بعد تأكيد المختبر. TB يحتاج Airborne، وDiphtheria يتطلب Precautions وإبلاغ الجهات المختصة.

Topic 44: Antibiotics versus observation: avoid treating every positive result

- **Key discriminator:** COVID-19 or influenza alone does not establish bacterial coinfection requiring routine antibiotics.
- **Management priority:** Selected asymptomatic bacteriuria requires no further antibiotic in the CLEAN V3 source example.
- **Clinical Pearl:** Prescribing and de-escalation should be guided by infection probability, source, microbiology and clinical course.
- **Exam trap:** Do not confuse a laboratory detection with a clinical syndrome requiring treatment.

Positive Viral Test أو Urine Culture لا يعني تلقائيًا Antibiotics. اربط العلاج بوجود Bacterial Infection وسياق الأعراض.

High-Yield Comparison Tables

Each table consolidates information already stated above. Numerical criteria retain their source and setting.

Sepsis-3 vs CMS-style septic shock definitions

Feature	Sepsis-3 in AQRAB	CMS-style in CLEAN V3	Exam point
Clinical basis	Sepsis + vasopressors for MAP >=65	Severe sepsis + persistent hypotension after 30 mL/kg OR lactate >=4	Definitions are not interchangeable
Lactate	>2 mmol/L after adequate fluids	>=4 mmol/L is one cited criterion	Always label the source framework
Risk	Combining thresholds	Combining thresholds	Do not invent one hybrid definition

Sepsis vs uncomplicated infection

Feature	Sepsis	Localized uncomplicated infection	Exam point
Organ dysfunction	Present by source definition	Not automatically present	Look for acute dysfunction
Hemodynamics	May have shock or hypoperfusion	Often stable	Normal BP does not exclude sepsis
Decision	Urgent organ support and source evaluation	Syndrome- and risk-based care	No fever-only diagnosis

Cellulitis vs abscess vs necrotizing infection

Feature	Cellulitis	Abscess / necrotizing infection	Exam point
Pattern	Diffuse erythema/warmth	Purulence or rapid deep tissue progression	Establish depth and trajectory
Key action	Risk-based antimicrobial choice	Drain abscess / immediate surgery for necrotizing	Antibiotics do not replace source control
Warning	Stable superficial infection	Pain out of proportion, bullae, shock	Do not wait for crepitus

Preseptal vs orbital cellulitis

Feature	Preseptal	Orbital (postseptal)	Exam point
Anatomy	Anterior to orbital septum	Within orbit behind septum	Depth matters
Eye movement/vision	Typically unaffected	May be impaired	Concern for serious complication

Infectious Diseases & Sepsis

Precisely source-linked additions after the priority-safety review. Not independent guideline verification.

Topic: Sepsis MAP target in selected older adults

Number/threshold: AQRAB cites an SSC general initial MAP target of 65 mmHg for adult sepsis, with a suggested 60–65 mmHg range in some adults aged 65 years or older. This is conditional and not a universal goal for every shock mechanism.

الـMAP 60–65 هنا لبعض كبار السن في سياق Sepsis، وليس هدفاً ثابتاً لكل أنواع Shock.

Topic: Adult pertussis: source-cited treatment example

Source-dependent regimen: CLEAN V3 cites azithromycin 500 mg once daily for 3 days as an adult pertussis treatment example. The extracted line does not supply all allergy, interaction or local-protocol qualifications; do not treat it as a universal prescription.

الجرعة مذكورة في المصدر كمثال خاص ببالغ مصاب بـPertussis؛ لا تُعمّم دون سياق الدواء الكامل.

> Do not merge case-specific, adult and pediatric regimens into a single dose. Source-dependent conflicts remain unresolved.

EDIC AUDIT PASS 11 | DOMAIN 07 | SOURCE-ONLY CONTEXT RESOLUTION

GAS pharyngitis: incomplete maximum-dose context

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

Source limitation: CLEAN V3 gives penicillin VK 50 mg/kg/day in four divided doses for 10 days for GAS pharyngitis without drug allergy. Neither source provides a complete maximum-dose and patient-eligibility pathway for that extracted pediatric-looking regimen; it is therefore not included as an executable dosing recommendation.

وردت جرعة Penicillin VK بنص المصدر، لكن الحد الأقصى والسياق الكامل غير موجودين؛ لا أُكمل الجرعة بالتخمين ولا أعتبرها أمرًا علاجيًا جاهزًا.

EDIC AUDIT PASS 11 | DOMAIN 07 | SOURCE-ONLY CONTEXT RESOLUTION

Pertussis: treatment versus postexposure prophylaxis

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

Source-dependent difference: CLEAN V3 cites adult pertussis treatment with azithromycin 500 mg daily for 3 days. AQRAB p.401 and CLEAN V3 para 1040 separately cite postexposure prophylaxis of adult contacts: azithromycin 500 mg orally on day 1, then 250 mg daily on days 2–5. The source contexts differ, so this is not automatically a same-indication contradiction. Do not replace one with the other.

نظام الثلاثة أيام ورد مع علاج Pertussis، ونظام الخمسة أيام ورد مع الوقاية بعد المخالطة؛ لا يصح تبديل الاستطبابين أو اعتبارهما جرعة واحدة.

EDIC AUDIT PASS 11 | DOMAIN 07 | SOURCE-ONLY CONTEXT RESOLUTION

PID therapy: parenteral versus outpatient pathway

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

Key discriminator: CLEAN V3 gives a parenteral PID example with ceftriaxone 1 g IV, doxycycline 100 mg PO/IV and metronidazole 500 mg PO/IV; the isolated source line does not establish a full repeat schedule for every drug.

الجرعات هنا واردة في سياق Parenteral PID، ولذلك لا نُحوّلها وحدها إلى وصفة كاملة.

Source-dependent regimen: AQRAB p.133 separately describes an eligible nonpregnant adult outpatient pathway: ceftriaxone 500 mg IM once, followed by doxycycline 100 mg and metronidazole 500 mg orally twice daily for 14 days. The same source emphasizes pregnancy testing, exclusion of ectopic pregnancy/torsion, admission indications, contraindications and response reassessment.

الـ500 mg IM هنا للمريضة البالغة غير الحامل ضمن Outpatient PID، وليس بديلًا تلقائيًا عن علاج الحالات المنومة.

Feature	Preseptal	Orbital (postseptal)	Exam point
Treatment context	Mild oral-case example	Requires evaluation for serious orbital disease	No shared routine regimen

Septic arthritis vs crystal arthritis

Feature	Septic arthritis	Crystal arthritis	Exam point
Fluid result	Inflammatory/neutrophilic; culture assessment	Crystals may be detected	Crystals do not exclude infection
Priority	Urgent aspiration and treatment evaluation	Assess mimics and coexistence	Do not miss joint destruction
Synovial WBC	Often >50,000/mm3; lower values possible	No isolated exclusion threshold	Interpret clinically

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 152

2 g, 1 g	Number/threshold: AQRAB lists adult source-edition examples: cefepime 2 g IV every 8 h, meropenem 1 g IV every 8 h, and piperacillin-tazobactam 4.5 g IV every 6 h.
100 mg, 500 mg	Key discriminator: CLEAN V3 gives a parenteral PID example with ceftriaxone 1 g IV, doxycycline 100 mg PO/IV and metronidazole 500 mg PO/IV; the isolated source line does not establish a full repeat schedule for every drug.
500 mg, 100 mg	Source-dependent regimen: AQRAB p.133 separately describes an eligible nonpregnant adult outpatient pathway: ceftriaxone 500 mg IM once, followed by doxycycline 100 mg and metronidazole 500 mg orally twice daily for 14 days. The same source emphasizes pregnancy
500 mg	outpatient pathway: ceftriaxone 500 mg IM once, followed by doxycycline 100 mg and metronidazole 500 mg orally twice daily for 14 days. The same source emphasizes pregnancy testing, exclusion of ectopic pregnancy/torsion, admission indications, contraindications and

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 152

Cryptosporidiosis - immunocompromised patients [#055]

High-yield fact: Cryptosporidium can cause prolonged watery diarrhea in immunocompromised patients. .Cryptosporidium.

الإسهال المائي المطوّل عند مثبط المناعة قد يكون بسبب Cryptosporidium.

TB active infection vs latent TB

Feature	Active infectious TB	Latent TB	Exam point
Clinical features	May have cough, sweats, weight loss	No active transmissible syndrome established	Use clinical context
Isolation	Airborne precautions when suspected	Positive IGRA/TST alone not proof of infectiousness	Do not wait for confirmation
Tests	Chest imaging, sputum NAAT/culture	IGRA/TST does not distinguish active disease	Coordinate microbiology

Routine infection vs immune-suppressed infection

Feature	Typical host	Immunocompromised host	Exam point
Fever/WBC	May be prominent	May be muted	Do not rule out severe infection
Organisms	Common syndromes	Opportunistic plus common infections	Ask immune defect and prophylaxis
Care	Syndrome-directed	Early specialist/empiric pathway	Do not delay unstable cases

Blood cultures and antibiotics: severe versus possible sepsis

Feature	Shock / probable sepsis	Possible sepsis without shock	Exam point
Antibiotic timing	Immediate according to AQRAB	Rapid time-limited evaluation may be appropriate	Different source severity pathways
Cultures	Before drugs only if no meaningful delay	Directed investigation	Do not prioritize cultures over resuscitation
Source control	Find early	Evaluate according to source	Do not delay needed drainage

Rapid Recall: Must-Know Facts

Forty concise restatements of the Domain body. No external or new medical facts appear in this section.

01. Sepsis = infection-associated acute organ dysfunction.
02. Sepsis-3 uses acute SOFA increase >=2.
03. Sepsis-3 shock: vasopressor for MAP >=65 AND lactate >2 after fluids.
04. CMS-style shock criteria in CLEAN V3 differ from Sepsis-3.
05. Normal blood pressure does not exclude hypoperfusion.
06. Fever absence does not exclude serious infection.
07. Do not use qSOFA alone to exclude sepsis.
08. Track mental status, perfusion, urine output and lactate trend.
09. Cultures before antibiotics only when this does not meaningfully delay therapy.
10. Immediate antibiotics: septic shock or probable/definite sepsis in the source pathway.
11. Possible sepsis without shock may permit rapid focused assessment.
12. Reassess crystalloid response; avoid automatic repeat boluses.
13. Hydroxyethyl starch is discouraged in the supplied sepsis explanation.
14. Norepinephrine is the cited first-line septic-shock vasopressor.

15. Source control may require drainage, surgery or obstruction relief.
16. Pain out of proportion and rapid progression suggest necrotizing infection.
17. A reassuring scan or LRINEC does not rule out necrotizing infection.
18. Debridement is not replaced by antibiotics.
19. Fournier gangrene requires immediate surgical escalation.
20. A hot swollen joint requires septic arthritis evaluation.
21. Synovial WBC >50,000/mm3 is concerning, not an exclusive cutoff.
22. Crystals do not exclude simultaneous septic arthritis.
23. Overlying cellulitis affects needle-entry choice, not infection urgency.
24. Childhood osteomyelitis is commonly hematogenous.
25. S. aureus is a common osteomyelitis organism.
26. Back pain with systemic findings warrants consideration of spinal infection.
27. Preseptal infection is anterior to the orbital septum.
28. Orbital cellulitis may threaten eye function.
29. Positive urine culture without symptoms does not automatically require antibiotics.
30. Infected obstruction needs source control.
31. Diphtheria suspicion triggers airway precautions, isolation and antitoxin pathway.
32. HSV encephalitis: do not delay empiric acyclovir for MRI or LP.
33. Positive viral test does not exclude bacterial infection or a separate emergency.
34. Suspected active pulmonary TB requires early airborne isolation.
35. A positive IGRA/TST cannot distinguish active from latent TB.
36. CD4 <200 increases opportunistic infection concern but does not diagnose it.
37. Cancer febrile-neutropenia pathway cites antibiotics within 1 hour of triage.
38. Single negative malaria smear may require repeat testing when suspicion persists.
39. Syncope or high-grade block is a Lyme carditis warning.
40. Antibiotic choice follows source, severity, allergy, kidneys and susceptibility.

Conflict Resolution Register

No independently confirmed direct clinical disagreement across the two supplied files was established after distinguishing separate definitional frameworks and clinical contexts. Source-dependent differences remain visible below, rather than being resolved externally.

Contextual / protocol differences — retained, not treated as proven inter-file conflicts

1. Different septic shock definitions

- **Source statements:** AQRAB Sepsis-3: vasopressors for MAP >=65 plus lactate >2 after fluids. CLEAN V3 quotes CMS-style hypotension after 30 mL/kg or lactate >=4.
- **Nature:** Distinct definition systems, not evidence of a direct contradiction.

2. Neonatal early-onset sepsis cutoff

- **Source statements:** CLEAN V3 contains both a within-72-hour description and a first-7-days description, noting the 72-hour convention.
- **Nature:** Internal source variation; do not synthesize a single cutoff.

3. Arthrocentesis through cellulitis

- **Source statements:** CLEAN V3 describes overlying cellulitis as unsafe for the intended needle pathway; AQRAB prioritizes joint aspiration when septic arthritis is suspected.
- **Nature:** Choice of entry site differs from whether urgent aspiration is indicated.

4. Sepsis antimicrobial timing

- **Source statements:** AQRAB differentiates shock/probable sepsis from possible sepsis without shock.

- **Nature:** Different risk populations, not one unconditional time rule.

5. Unreconciled source-era antibiotic examples

- **Source statements:** AQRAB lists adult source-edition doses and warns to review latest protocol, while CLEAN V3 contains additional case-specific antibiotic regimens.
- **Nature:** Dose indications and publication-era context are incomplete; not merged into a universal table.

6. Neutropenic patient definitions

- **Source statements:** AQRAB 1-hour antibiotic target is attributed to the defined cancer/febrile-neutropenia population.
- **Nature:** Cannot automatically generalize to every immunocompromised patient.

7. Synovial WBC as diagnostic marker

- **Source statements:** CLEAN V3 cites >50,000/mm3 with neutrophil predominance but acknowledges cases with a lower WBC.
- **Nature:** A supportive threshold, not a rule-out boundary.

Internal source ambiguities requiring caution

- **1 — Hydroxyethyl starch answer-stem reversal:** CLEAN V3 ¶2441: the stem says recommended; the explanatory clause says strongly recommended AGAINST.
- **2 — Routine glutamine answer-stem reversal:** CLEAN V3 ¶2476: the stem says recommended for critically ill patients; the explanatory clause advises against routine use.
- **3 — ESBL carbapenem-sparing answer-stem reversal:** CLEAN V3 ¶2712: the stem says no role; explanatory text allows susceptible fluoroquinolones/aminoglycosides in selected cases.
- **4 — Neonatal early-onset window:** CLEAN V3 ¶2454,2465: 72 hours versus first 7 days.

Metric	Result
Final integrated clinical topics	44
Integrated English clinical points	180
CLEAN V3-only points	66
AQRAB-only points	92
Jointly supported points	22
Short direct Arabic explanations	44
Comparison tables	8
Rapid Recall facts	40
Confirmed direct inter-file conflicts	0
Contextual/protocol differences logged	7
Internal source ambiguity examples	4
Incomplete / unusable source groups	10
External medical facts introduced	0

Unit of count: one authored clinical point with one dominant source disposition (CLEAN-only, AQRAB-only or joint). Figures measure retained edited points, not all raw sentences. Rapid Recall, table restatements and Arabic explanations are excluded from the English-point denominator. Counts are editorial and auditable by topic source labels.

Incomplete in source — not included

1. Unanchored pediatric IV-antibiotics-answer phrase whose diagnosis is omitted. [CLEAN V3 ¶2439]
2. Pseudomonas osteomyelitis phrase with mixed organism labels. [CLEAN V3 ¶2481]
3. Isolated antibiotic choice without syndrome or susceptibility context. [CLEAN V3 ¶2578-2581,2595,2599-2601]
4. STI/sexual-assault dose examples with differing source-era regimens and incomplete applicability. [CLEAN V3 ¶2549-2551,2618,2645-2646]
5. Single numeric fragments without meaningful denominator, e.g. 10% or 30%-50%. [CLEAN V3 ¶2555,2640]
6. Positive-Prehn-sign differential stated without adequate diagnostic context. [CLEAN V3 ¶2535]
7. Puncture-wound osteomyelitis percentage detached from population and sampling design. [CLEAN V3 ¶2517]
8. Methemoglobinemia dose fragment unrelated to infection-domain scope. [CLEAN V3 ¶2693]

9. Broad clinical pearls with contradictory answer stems not restated as direct therapeutic truth. [CLEAN V3 ¶2441,2476,2712]

10. Repeated infographic navigation, attribution paragraphs and literal unexpanded question fragments. [AQRAB PDF pp.190-215,394-456]

Original source infographic pages in the relevant infection and sepsis range were inspected. Neither original artwork nor references to outside web pages were introduced into the integrated medical body.

This PDF is a source-limited study synthesis, not an independently guideline-validated clinical protocol. Unresolved differences are flagged rather than decided with outside knowledge.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 156

Chlamydial urethritis - culture-negative dysuria [#051]

Key discriminator: Dysuria with sterile routine urine culture can be caused by chlamydial urethritis.

عسر البول مع زرع بول روتيني سلبي لا يستبعد التهاب إحليل ناجمًا عن Chlamydia.

Necrotizing otitis externa - high-risk presentation [#053]

High-yield fact: Severe persistent external-ear pain in diabetes or immunosuppression raises concern for necrotizing otitis externa.

ألم الأذن الخارجية الشديد والمستمر لدى السكري أو مثبط المناعة يرفع احتمال Necrotizing otitis externa.

Ebola - transmission and exposure history [#054]

High-yield fact: Ebola spreads primarily through direct contact with infected blood or body fluids; exposure history is essential.

في Ebola، تاريخ التعرض والتماس المباشر للدم أو سوائل الجسم المصابة عنصران أساسيان.

EDIC / Emergency & Critical Care

08 — Toxicology & Pharmacology

I. Initial Poisoning Assessment, Recognition & Decontamination

Topic 1: The poisoned patient: stabilize before naming the agent

- **Core concept:** A poisoning assessment begins with scene safety, airway and breathing, circulation, bedside glucose, temperature, ECG, and focused neurologic assessment.
- **Clinical trigger:** Ventilatory failure, shock, dysrhythmia, seizure, severe hyperthermia or hypoglycemia requires immediate physiologic stabilization.
- **Management priority:** Establish the substance, amount, route, time, formulation, possible co-ingestants, intent and tolerance; obtain collateral information or packaging when safe.
- **High-yield fact:** Use serial vital signs, mental status and ECG monitoring; prolonged-release or mixed ingestion can evolve after a reassuring first examination.
- **Common mistake:** Waiting for a urine toxicology screen or confirmatory drug concentration before supporting ventilation or circulation.
- **Clinical Pearl:** Contact a poison center or medical toxicologist early for severe, mixed, unknown or unusual exposure.

في poisoned patient ابدأ بـ ABCs وglucose وECG وtemperature. تحديد اسم السم مهم، لكنه لا يسبق علاج فشل التنفس أو shock أو seizures.

Topic 2: Recognize toxidromes without overcommitting to one substance

- **Core concept:** A toxidrome narrows the differential by clustered findings but does not identify one drug with certainty.
- **Key discriminator:** Opioid: CNS depression and hypoventilation, often miosis; sedative-hypnotic: somnolence, ataxia and reduced ventilation.
- **Key discriminator:** Sympathomimetic: agitation, mydriasis, diaphoresis, tachycardia, hypertension or hyperthermia.
- **Key discriminator:** Antimuscarinic: delirium, dry flushed skin or mucosa, mydriasis, urinary retention and reduced bowel sounds.
- **Key discriminator:** Cholinergic: secretions, bronchorrhea, bronchospasm, miosis, fasciculations and weakness.
- **Red flag:** Mixed ingestion, late presentation, older age and coexisting illness can blur characteristic findings.

Toxidrome هو نمط متجمع من العلامات وليس تحليلًا يثبت المادة. mixed ingestion قد يخفي الصورة الكلاسيكية؛ ركّز على التنفس والحدقات والتعرق والحرارة والحركة العصبية.

Topic 3: Opioid versus sedative-hypnotic clinical patterns

- **Key discriminator:** Both syndromes depress consciousness; respiratory depression is the priority and pupils are supportive rather than conclusive.

- **Clinical trigger:** Opioid exposure may present without the full classic triad of coma, miosis and bradypnea.
- **Differential point:** Sedative-hypnotics may cause slurred speech, ataxia and nystagmus; co-ingested opioids can deepen respiratory depression.
- **Management priority:** Provide assisted ventilation when indicated while evaluating for reversible causes and mixed exposure.
- **High-yield fact:** Buprenorphine is described in CLEAN V3 as a high-affinity partial mu-receptor agonist with a ceiling effect.
- **Exam trap:** A negative routine urine screen may miss fentanyl or synthetic opioids and does not exclude clinical poisoning.

حافظ على التهوية أولًا وحدد إن hypoventilation وحدها. أهم خطر مشترك هو miosis ؛ القرار ليس مبنيًا على كان Naloxone مناسبًا.

Topic 4: Sympathomimetic versus antimuscarinic toxidromes

- **Key discriminator:** Diaphoresis favors a sympathomimetic syndrome; dry mucosa, reduced bowel sounds and urinary retention favor antimuscarinic toxicity.
- **Clinical trigger:** Agitation, mydriasis and hyperthermia after an unknown exposure require evaluation for both syndromes and other mimics.
- **Management priority:** Check glucose, ECG, temperature, hydration and evolving complications while providing appropriate sedation or cooling.
- **Exam trap:** Mydriasis is not diagnostic by itself and may also accompany serotonin toxicity.
- **Red flag:** Seizures, dangerous hyperthermia, dysrhythmia or progressive agitation should escalate monitoring and management.

كلاهما قد يسبب agitation وmydriasis وtachycardia. الفرق الامتحاني الأساسي: sympathomimetic غالبًا sweaty؛ antimuscarinic غالبًا dry مع urinary retention.

Topic 5: Serotonin toxicity: recognize neuromuscular clues

- **Core concept:** Serotonin toxicity combines mental-status change, autonomic activation and neuromuscular hyperactivity.
- **Key discriminator:** Clonus, hyperreflexia, tremor, myoclonus and diaphoresis are important discriminators; diarrhea may accompany the syndrome.
- **Clinical trigger:** CLEAN V3 identifies risk from linezolid with SSRI therapy, Hypericum perforatum with serotonergic antidepressants, and citalopram plus dextromethorphan.
- **Management priority:** Stop serotonergic agents; provide supportive care, benzodiazepine sedation and cooling, with cyproheptadine as a specialist-guided adjunct in selected cases.
- **Red flag:** Severe hyperthermia, seizures, rigidity, rhabdomyolysis or organ dysfunction require urgent escalation.
- **Exam trap:** Serotonin toxicity can overlap with antimuscarinic or sympathomimetic findings; prioritize the neuromuscular examination.

وجود clonus وhyperreflexia مع serotonergic exposure يوجّه بقوة إلى serotonin syndrome، ولا يكفي وجود fever أو agitation دون فحص المنعكسات.

Topic 6: Neuroleptic malignant syndrome and serotonin syndrome

- **Core concept:** Neuroleptic malignant syndrome is a life-threatening reaction linked to dopamine-blocking medications or changes in dopaminergic treatment.
- **Key discriminator:** NMS typically evolves over days with pronounced generalized rigidity and reduced reflexes; serotonin toxicity often begins within hours with clonus and hyperreflexia.
- **Clinical trigger:** Fever, marked rigidity, altered mental status, autonomic instability and elevated CK should prompt assessment for NMS.
- **Management priority:** Stop the causative medication, provide cooling, fluids and supportive care; selected severe cases may require specialist-directed dantrolene or bromocriptine.
- **Exam trap:** Do not label fever and rigidity after antipsychotic exposure as uncomplicated agitation.

في NMS فكّر بـ lead-pipe rigidity بعد dopamine blockade، بينما clonus وhyperreflexia مع بداية سريعة يرجحان serotonin toxicity؛ كلاهما قد يكون خطيرًا.

Topic 7: Poisoning ECG: sodium-channel blockade and QT prolongation

- **High-yield fact:** QRS widening with terminal R in aVR suggests toxic sodium-channel blockade such as TCA poisoning.
- **Number/threshold:** The AQRAB teaching risk markers for TCA poisoning are QRS >100 ms for seizure risk and >160 ms for ventricular dysrhythmia risk; these are not absolute cutoffs.
- **Key discriminator:** Prolonged QTc raises concern for torsades with methadone, antipsychotics, antidepressants and other exposures.
- **Management priority:** Life-threatening sodium-channel cardiotoxicity is treated with sodium bicarbonate, serial ECG and electrolyte/pH monitoring according to the cited source protocols.
- **Clinical Pearl:** CLEAN V3 identifies intravenous magnesium sulfate 2 g in a sotalol-associated torsades example.
- **Common mistake:** Assuming a normal initial ECG excludes delayed toxicity from extended-release agents.

الـECG أحيانًا يحدد الخطر قبل اسم الدواء: wide QRS يوجّه إلى sodium-channel blockade، وQT الطويل يزيد خطر torsades أعد ؛ ECG عند تطور الحالة.

Topic 8: Acid-base and osmol gaps in toxicology

- **Number/formula:** Anion gap = Na - (Cl + HCO3); osmol gap = measured serum osmolality minus calculated osmolality, accounting for laboratory method and ethanol.
- **Key discriminator:** Salicylates can cause respiratory alkalosis combined with high-anion-gap metabolic acidosis.
- **Clinical trigger:** High-anion-gap acidosis after suspected methanol or ethylene glycol exposure should prompt urgent evaluation and treatment.
- **Exam trap:** Normal osmol gap does not exclude late toxic-alcohol poisoning, and an elevated osmol gap is nonspecific.
- **Differential point:** Isopropanol produces ketosis often without the toxic-metabolite high-anion-gap acidosis of methanol or ethylene glycol.
- **Clinical Pearl:** Severe lactic acidosis in a smoke inhalation victim raises concern for cyanide but is not by itself specific.

احسب AG وosmol gap قد ينخفض؛ لكن لا تجعل الرقم وحده يحسم التشخيص مع تقدم toxic alcohol metabolism بينما يرتفع AG.

Topic 9: Targeted laboratory testing and limitations of screening

- **Management priority:** Select blood gas/lactate, electrolytes, kidney/liver tests, ECG, acetaminophen and salicylate levels, osmolality and agent-specific levels according to presentation.
- **High-yield fact:** Timing, formulation, renal function and clinical course determine how a measured drug concentration should be interpreted.
- **Exam trap:** Routine urine drug screens have false positives/negatives, detect selected drug classes and do not assess severity.
- **Red flag:** A negative drug screen must not delay clinical treatment when toxidrome and physiological deterioration indicate poisoning.
- **Clinical Pearl:** Consider targeted CK, troponin, urinalysis or pregnancy testing when the exposure and complications make results actionable.

اختبر ما سيغير القرار: glucose وECG وblood gas وAPAP/salicylate الـ. عند اللزومurine drug screen لا يحدد الشدة ولا ينفي جميع السموم.

Topic 10: Gastrointestinal decontamination: activated charcoal and whole-bowel irrigation

- **Core concept:** Activated charcoal is selective for potentially serious adsorbable ingestion, commonly within approximately one hour, with later use only in selected circumstances.
- **Contraindication:** Avoid charcoal in an unprotected airway, bowel obstruction/ileus or significant aspiration risk and do not use it routinely for alcohols, metals, caustics or hydrocarbons.
- **Clinical trigger:** Whole-bowel irrigation may be considered for selected sustained-release/enteric-coated products, iron, lithium or body packers.
- **Contraindication:** Avoid whole-bowel irrigation with ileus, obstruction, perforation, major hemodynamic instability or an unprotected airway.
- **Common mistake:** Inducing vomiting, giving ipecac or attempting chemical neutralization after caustic ingestion.
- **Clinical Pearl:** Discuss high-risk decontamination decisions with the poison center instead of applying a universal procedure.

Activated charcoal ليس علاجًا روتينيًا لكل التسممات. احسب وقت الابتلاع وخطر aspirationأما ؛ lithium وiron والكحول فلا يستفيدون منه مثل المواد القابلة للامتصاص.

Topic 11: Skin, eye and responder decontamination

- **Management priority:** Wear suitable PPE and remove contaminated clothing and jewelry before ongoing clinical contact when secondary contamination is possible.
- **High-yield fact:** For dry powder on skin, brush off the material before washing; avoid spreading contamination.
- **Management priority:** Chemical eye exposure requires immediate copious irrigation, with ocular pH reassessment when appropriate.
- **Common mistake:** Allowing contaminated clothes or pesticide residue to expose healthcare staff.
- **Clinical Pearl:** Decontamination and medical stabilization may need to occur in parallel.

احمِ الفريق أولًا، انزع الملابس الملوثة ونظف الجلد. إصابة العين contaminated clothing أو pesticide في عاجل ولا تنتظر تحديد المادة irrigation الكيميائية تحتاج.

II. Analgesics, Alcohols, Sedatives and Major Metabolic Poisons

Topic 12: Acetaminophen exposure: timing, risk assessment and nomogram

- **Number/threshold:** For a single acute ingestion with reliable timing, use a serum acetaminophen concentration obtained at four hours or later with the locally adopted Rumack-Matthew nomogram.
- **Exam trap:** The nomogram is inappropriate for unknown timing or repeated supratherapeutic ingestion.
- **Management priority:** Evaluate acetaminophen concentration, AST/ALT, INR, glucose, kidney function and acid-base status according to exposure and clinical severity.
- **Red flag:** Early minimal symptoms or a late low concentration do not exclude evolving hepatic injury.
- **Clinical Pearl:** CLEAN V3 describes early severe lactic acidosis in massive ingestion before fulminant hepatic failure from NAPQI-associated mitochondrial toxicity.
- **Common mistake:** Delaying N-acetylcysteine in a high-risk late or uncertain presentation while waiting for inappropriate nomogram interpretation.

في single acute ingestion مستوى، مع وقت معروف APAP لا يوضع على 4 ساعات قبل Rumack-Matthew nomogram. لا تستخدمه في staggered ingestion أو وقت غير معروف.

Topic 13: N-acetylcysteine: treatment course and stopping criteria

- **Number/threshold:** AQRAB lists a common IV NAC 300 mg/kg total over 21 hours: 150 mg/kg over 1 hour, 50 mg/kg over 4 hours, then 100 mg/kg over 16 hours; alternative validated protocols exist.
- **Management priority:** Start NAC promptly when indicated; early treatment is emphasized in both sources.
- **High-yield fact:** The AQRAB-cited ACMT stopping framework requires undetectable APAP (<10 micrograms/mL), improving AST/ALT, INR <2 and improving prognostic markers.
- **Exam trap:** Do not stop NAC based only on elapsed infusion duration.
- **Red flag:** Evolving hepatic failure, encephalopathy, acidosis, hypoglycemia or renal injury requires critical care and hepatology/transplant consultation.

NAC بقاء على يعتمد القرار. ساعة 21 انتهاء بعد تلقائيًا يتوقف لا APAP واتجاه ALT/ASTو INR وبقية
المذكورة بالمصدر التدهور مؤشرات.

Topic 14: Salicylate poisoning: respiratory drive and mixed acid-base disturbance

- **Clinical trigger:** Tinnitus, vomiting, tachypnea, sweating, hyperthermia or confusion after salicylate ingestion suggests significant toxicity.
- **Key discriminator:** Direct respiratory-center stimulation can cause respiratory alkalosis alongside high-anion-gap metabolic acidosis.
- **High-yield fact:** Serum salicylate concentration must be interpreted serially with acid-base status, acuity, renal function and clinical trajectory.
- **Red flag:** Worsening mental status, pulmonary edema, acidemia or renal failure indicates severe toxicity.
- **Exam trap:** A falling salicylate concentration alone does not prove clinical improvement if pH or mentation deteriorates.

Salicylate يرفع minute ventilation مبكرًا فيحدث respiratory alkalosis، ومع السمية تتطور metabolic
acidosis قد يظهر ؛ pH قريبًا من الطبيعي رغم اضطرابين خطيرين.

Topic 15: Salicylate poisoning: alkalinization, ventilation and dialysis

- **Management priority:** Correct volume and potassium deficits and use IV bicarbonate for significant toxicity with frequent blood gas, electrolyte and urine pH monitoring.
- **Number/threshold:** AQRAB describes urine alkalinization to approximately pH 7.5-8.0; a bicarbonate example is 1-2 mEq/kg under monitoring.
- **Exam trap:** Intubation may precipitate deterioration if it removes compensatory high minute ventilation; if unavoidable, preserve ventilation and involve experienced clinicians.
- **Management priority:** Seek early hemodialysis input for severe neurologic or pulmonary toxicity, refractory acid-base disturbance, renal failure or deterioration.
- **Number/threshold:** CLEAN V3 cites severe salicylate examples including >100 mg/dL or >90 mg/dL with renal impairment as extracorporeal consultation triggers, not an isolated decision rule.

الخطأ الأخطر هو فقد compensatory hyperventilation بعد sedation أو intubation يزيد دخول ؛ هذا قد
السماح إلى CNS. على الحفاظ خطة يحتاج القرار minute ventilation.

Topic 16: NSAID overdose and high-dose ibuprofen

- **Core concept:** Most NSAID poisoning is managed supportively; very large ingestions can cause serious systemic toxicity.
- **Red flag:** CLEAN V3 lists seizures, coma, acidosis, acute kidney injury and hyperkalemia in massive ibuprofen overdose.
- **Management priority:** Assess vital signs, mental status, renal function, gastrointestinal injury and acid-base/electrolyte complications.
- **Exam trap:** Routine NSAID serum concentrations are generally not useful for deciding management.
- **Clinical Pearl:** NSAIDs may provoke bronchospasm in susceptible patients with asthma, a distinct medication adverse effect from overdose.

غالبًا NSAID overdose يحتاج supportive care، لكن الجرعات الهائلة قد تسبب seizures وacidosis وAKI
وshock وصفة دون متاح الدواء أن لمجرد تطمئن لا ؛.

Topic 17: Methanol: vision-threatening toxic alcohol

- **Clinical trigger:** Visual disturbance with severe high-anion-gap acidosis after an unknown alcohol or garage-fluid exposure strongly suggests methanol poisoning.
- **High-yield fact:** CLEAN V3 lists windshield washer fluid, model fuel, carburetor cleaner and adulterated ethanol among potential methanol sources.

- **Management priority:** Start alcohol-dehydrogenase blockade with fomepizole when strongly suspected rather than waiting for confirmatory concentrations in a deteriorating patient.
- **Management priority:** Correct severe acidosis, consider folinic acid as an adjunct and discuss hemodialysis under source-cited EXTRIP criteria.
- **Exam trap:** A normal osmol gap late after exposure does not exclude methanol toxicity.
- **Clinical Pearl:** Folinic acid is an adjunct and does not replace fomepizole, resuscitation or indicated extracorporeal treatment.

Methanol. لا تنتظر وتعرض محتمل للكحول السام يثير high-anion-gap acidosis مع visual symptoms وجود المستوى المؤكد إذا كانت الصورة شديدة.

Topic 18: Ethylene glycol: renal, calcium and metabolite clues

- **Clinical trigger:** Suspect ethylene glycol with compatible exposure, high-anion-gap acidosis, renal injury, hypocalcemia or calcium oxalate crystals.
- **Exam trap:** The absence of urinary calcium oxalate crystals does not exclude poisoning.
- **Management priority:** Use timely alcohol-dehydrogenase blockade and assess severity for hemodialysis with toxicology/nephrology advice.
- **Clinical Pearl:** CLEAN V3 notes a possible lactate gap because glycolate may interfere with some lactate-oxidase analyzers.
- **High-yield fact:** CLEAN V3 mentions thiamine and pyridoxine as metabolic adjuncts to ethylene glycol management.
- **Red flag:** Clinical deterioration, severe acidosis or progressive kidney injury requires urgent escalation.

Ethylene glycol قد يسبب AKI وhypocalcemia مع high AG acidosis وجود ؛ calcium oxalate crystals مفيد لكنه ليس شرطًا للتشخيص.

Topic 19: Isopropanol versus methanol and ethylene glycol

- **Key discriminator:** Isopropanol exposure can cause greater CNS depression than ethanol, gastritis and an osmolar gap with ketosis but usually without high-anion-gap metabolic acidosis.
- **Differential point:** Methanol favors visual/CNS injury; ethylene glycol favors kidney injury, hypocalcemia and oxalate-related findings.
- **Management priority:** Isopropanol management is generally supportive, with dialysis considered in selected severe cases; fomepizole is not routinely indicated.
- **Exam trap:** Do not equate a positive osmolar gap with the same toxic metabolite burden or treatment for every alcohol.
- **Clinical Pearl:** Calculated gaps vary over time after alcohol metabolism and cannot alone substitute for clinical reassessment.

Isopropanol غالبًا يسبب ketosis مع osmolar gap دون high-AG acidosis المعتادة في methanol/ethylene glycol. هذه مقارنة كلاسيكية تمنع اختيار antidote خطأً.

Topic 20: Fomepizole, cofactor adjuncts and extracorporeal removal

- **Core concept:** Fomepizole inhibits conversion of methanol and ethylene glycol to toxic metabolites.
- **Number/threshold:** AQRAB gives fomepizole 15 mg/kg IV as a loading example, not a complete regimen for all alcohol exposures.
- **Management priority:** Hemodialysis decisions depend on clinical severity, acidosis, organ injury and source-cited EXTRIP guidance with nephrology/toxicology input.
- **High-yield fact:** Fomepizole dosing may need adjustment during hemodialysis according to the specific protocol.
- **Clinical Pearl:** Methanol folinic-acid and ethylene-glycol cofactor discussions are adjuncts and do not replace definitive blockade or supportive care.

Fomepizole يعطّل alcohol dehydrogenase في toxic alcohols أثناء ؛ المناسبة dialysis قد يلزم تعديل جرعته وفق البروتوكول المذكور بالمصدر.

Topic 21: Alcohol withdrawal seizures and delirium

- **Core concept:** Withdrawal after reduction or cessation of heavy alcohol use may progress from tremor and autonomic activation to seizures or delirium tremens.
- **Number/threshold:** The AQRAB source describes withdrawal seizures within approximately 7-48 hours after stopping heavy intake.
- **Management priority:** Benzodiazepines are first line for moderate to severe withdrawal; monitor mental status, ventilation and vital signs.
- **High-yield fact:** Provide thiamine when deficiency risk exists, correct hypoglycemia without delay and assess electrolytes, infection, trauma or other seizure causes.
- **Exam trap:** CIWA-Ar is unreliable in delirium or patients unable to communicate; antipsychotics do not substitute for benzodiazepines for withdrawal seizure prevention.

Alcohol withdrawal قد يحدث حتى مع blood alcohol إيجابي. Seizure أو delirium بعد التوقف يحتاج مراقبة وتقييم أسباب أخرى، وليس مجرد sedation ثم discharge.

Topic 22: Opioid respiratory depression and naloxone titration

- **Management priority:** Support airway, oxygenation and assisted ventilation while giving naloxone when opioid toxicity is likely.
- **Number/threshold:** AQRAB adult dosing examples: naloxone 0.2-2 mg IV/IM/IO or 2-4 mg intranasal, repeating every 2-3 minutes when indicated.
- **Number/threshold:** CLEAN V3 separately presents a clinical sequence beginning with 0.4 mg IV followed by 0.8 mg IV if no response after about 60 seconds; this falls within the broader dose range.
- **High-yield fact:** The AQRAB infusion example is approximately two-thirds of the effective reversal dose per hour for recurrent toxicity.
- **Exam trap:** A patient who initially responds may re-sedate; extended-release opioids or recurrent hypoventilation require repeat dosing, infusion or longer observation.
- **Clinical Pearl:** A negative routine urine drug screen does not exclude fentanyl or synthetic opioid poisoning.

هدف Naloxone هو عودة ventilation الكافية وليس إيقاظ المريض تمامًا. بعد تحسن أولي قد يعود respiratory depression إذا كان opioid أطول مفعولًا.

Topic 23: Sedative-hypnotics and the limits of flumazenil

- **Core concept:** Sedative-hypnotic poisoning often causes somnolence, impaired coordination and respiratory depression; most management is supportive.
- **Management priority:** Protect the airway and monitor ventilation, blood pressure, temperature, glucose and possible co-ingestants.
- **Number/threshold:** AQRAB gives flumazenil 0.2 mg IV titration to a cumulative 1 mg as a selected-case example, not routine empiric therapy.
- **Contraindication:** Avoid flumazenil in unknown/mixed overdose, chronic benzodiazepine exposure, seizure disorder or suspected TCA/proconvulsant co-ingestion.
- **Clinical Pearl:** CLEAN V3 highlights selected iatrogenic procedural-sedation overdose as a setting where reversal may be considered.
- **Exam trap:** Supplemental oxygen may preserve SpO2 despite inadequate ventilation.

Flumazenil ليس antidote لكل مريض غائب عن الوعي. chronic benzodiazepine use أو mixed overdose أو seizure risk قد يحول الإعطاء إلى خطر.

III. Antidepressants, Cardiotoxic Drugs and Specialist Antidotes

Topic 24: Antimuscarinic poisoning and cautious physostigmine use

- **Clinical trigger:** Agitated delirium with dry flushed mucosa, mydriasis, tachycardia, urinary retention and decreased bowel sounds suggests antimuscarinic toxicity.
- **Management priority:** Evaluate glucose, ECG, body temperature and co-ingestants; use cooling and supportive benzodiazepine sedation when appropriate.

CLINICAL NUMBERS · TOPIC CONTEXT

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300 mg/kg, 150 mg/kg, 50 mg/kg	Number/threshold: AQRAB lists a common IV NAC 300 mg/kg total over 21 hours: 150 mg/kg over 1 hour, 50 mg/kg over 4 hours, then 100 mg/kg over 16 hours; alternative validated protocols exist.
100 mg/kg	Number/threshold: AQRAB lists a common IV NAC 300 mg/kg total over 21 hours: 150 mg/kg over 1 hour, 50 mg/kg over 4 hours, then 100 mg/kg over 16 hours; alternative validated protocols exist. Management priority: Start NAC promptly when indicated; early treatment is emphasized in both sources.
15 mg/kg	Core concept: Fomepizole inhibits conversion of methanol and ethylene glycol to toxic metabolites. Number/threshold: AQRAB gives fomepizole 15 mg/kg IV as a loading example, not a complete regimen for all alcohol exposures.
0.2-2 mg, 2-4 mg	toxicity is likely. Number/threshold: AQRAB adult dosing examples: naloxone 0.2-2 mg IV/IM/IO or 2-4 mg intranasal, repeating every 2-3 minutes when indicated.
0.4 mg, 0.8 mg	2-3 minutes when indicated. Number/threshold: CLEAN V3 separately presents a clinical sequence beginning with 0.4 mg IV followed by 0.8 mg IV if no response after about 60 seconds; this falls within the broader dose range.
0.2 mg, 1 mg	possible co-ingestants. Number/threshold: AQRAB gives flumazenil 0.2 mg IV titration to a cumulative 1 mg as a selected-case example, not routine empiric therapy.
1-2 mEq/kg, 50-150 mEq	pressure, pH, sodium and potassium. Number/threshold: AQRAB cites a bicarbonate bolus of 1-2 mEq/kg IV and a source adult range of 50-150 mEq, titrated to response rather than a fixed universal dose.
2-10 mg, 1-15 mg/h	activation. Number/threshold: AQRAB lists glucagon 2-10 mg IV as a specialist-guided initial example and 1-15 mg/h as a possible infusion adjunct.
2 g	specialist monitoring. Number/threshold: AQRAB adult toxicology examples cite calcium chloride 2 g IV (central access preferred) or calcium gluconate 6 g IV, guided by hemodynamics and ionized calcium.
6 g	Number/threshold: AQRAB adult toxicology examples cite calcium chloride 2 g IV (central access preferred) or calcium gluconate 6 g IV, guided by hemodynamics and ionized calcium. Red flag: Severe cardiogenic shock may prompt discussion of VA-ECMO; routine lipid emulsion for CCB poisoning is
5 g	Management priority: Give pyridoxine along with airway, seizure and cardiopulmonary support. Number/threshold: AQRAB describes gram-for-gram pyridoxine when ingestion is known, or 5 g IV as an adult severe-toxicity initial example when amount is unknown.

- **High-yield fact:** Physostigmine is a centrally active acetylcholinesterase inhibitor discussed for selected severe, likely pure antimuscarinic delirium.
- **Contraindication:** Avoid physostigmine with QRS widening, suspected TCA/sodium-channel blockade, seizure risk or uncertain mixed ingestion.
- **Exam trap:** CLEAN V3 notes that physostigmine can worsen a cholinergic organophosphate crisis and should not be used prophylactically.
- **Clinical Pearl:** Urinary retention itself may contribute to agitation and requires assessment.

في antimuscarinic delirium، قد يفيد physostigmine بحالات مختارة فقط؛ وجود wide QRS أو suspected TCA يمنع التعامل معه كحل روتيني.

Topic 25: Tricyclic antidepressant overdose and sodium-channel cardiotoxicity

- **Core concept:** TCA poisoning can cause fast sodium-channel blockade, wide QRS, hypotension, seizures and ventricular dysrhythmia.
- **Key discriminator:** A terminal R wave in aVR and marked QRS widening raise concern for sodium-channel blockade.
- **Management priority:** Treat life-threatening cardiotoxicity with IV sodium bicarbonate while monitoring QRS, pressure, pH, sodium and potassium.
- **Number/threshold:** AQRAB cites a bicarbonate bolus of 1-2 mEq/kg IV and a source adult range of 50-150 mEq, titrated to response rather than a fixed universal dose.
- **Exam trap:** CLEAN V3 identifies phenytoin as potentially worsening conduction in suspected TCA overdose; avoid reflexive use of sodium-channel-blocking antiarrhythmics.
- **Common mistake:** Using physostigmine or flumazenil reflexively when mixed TCA overdose is possible.

ECG؛ الـwide QRS وseizures وhypotension مع anticholinergic signs قد يجمع TCA overdose يحدد الأولوية ولا تنتظر تأكيد المستوى.

Topic 26: SSRI, SNRI and bupropion overdose

- **Core concept:** Antidepressant toxicity depends on the exact agent, formulation, co-ingestants and serotonergic or cardiotoxic mechanism.
- **Clinical trigger:** Clonus, hyperreflexia, diaphoresis, fever and agitation after a serotonergic medication suggests serotonin syndrome.
- **Red flag:** Bupropion and some modified-release antidepressant ingestions may cause delayed seizures or cardiotoxicity despite reassuring early observations.
- **Management priority:** Use serial ECG, glucose and electrolyte assessment, cooling or sedation when indicated, and agent-specific toxicology consultation.
- **Clinical Pearl:** CLEAN V3 reports serotonin interactions involving linezolid, St Johns wort and dextromethorphan.
- **Exam trap:** Do not use an early normal ECG as the sole basis for discharge after a high-risk modified-release ingestion.

كسم واحد: SSRI/SNRI قد يسبب serotonin syndrome؛ bupropion قد يتأخر معه antidepressants لا تعامل كل seizure وcardiotoxicity.

Topic 27: Beta-blocker overdose: cardiac and metabolic patterns

- **Clinical trigger:** Bradycardia, hypotension, conduction block, hypoglycemia or altered mentation after beta-blocker ingestion raises concern for toxicity.
- **High-yield fact:** Propranolol can cause seizures and sodium-channel cardiotoxicity; not all beta-blocker overdoses are physiologically identical.
- **Management priority:** Severe poisoning may require vasopressors, high-dose insulin with dextrose monitoring and other targeted therapies under toxicology/critical-care supervision.
- **High-yield fact:** CLEAN V3 describes glucagon acting through adenylate cyclase without relying on beta-receptor activation.
- **Number/threshold:** AQRAB lists glucagon 2-10 mg IV as a specialist-guided initial example and 1-15 mg/h as a possible infusion adjunct.
- **Exam trap:** Atropine or pacing alone may not resolve severe drug-induced shock.

فقط: قد يصاحبها hypotension وAV block، وبعض الأنواع مثل propranolol BB overdose لا يعني bradycardia قد تسبب seizures وwide QRS.

Topic 28: Calcium-channel blocker overdose: bradycardia versus vasodilatory shock

- **Key discriminator:** CCB poisoning can combine hypotension, bradycardia, AV block and hyperglycemia.
- **Differential point:** Dihydropyridine agents may predominantly produce vasodilatory hypotension, with possible reflex tachycardia.
- **Management priority:** Life-threatening CCB toxicity may require IV calcium, vasopressors and high-dose insulin with specialist monitoring.
- **Number/threshold:** AQRAB adult toxicology examples cite calcium chloride 2 g IV (central access preferred) or calcium gluconate 6 g IV, guided by hemodynamics and ionized calcium.
- **Red flag:** Severe cardiogenic shock may prompt discussion of VA-ECMO; routine lipid emulsion for CCB poisoning is not supported in the cited AQRAB pathway.
- **Exam trap:** CLEAN V3 notes that pronounced QRS/QT abnormalities in an otherwise typical CCB presentation should raise suspicion for co-ingestants.

CCB overdose قد يظهر bradycardia مع hyperglycemia، بينما dihydropyridines قد تسبب vasodilatory shock شديدًا حتى دون بطء قلب بارز.

Topic 29: High-dose insulin euglycemia therapy in cardiotoxic overdose

- **Core concept:** High-dose insulin is described as an important treatment for severe beta-blocker or calcium-channel blocker cardiotoxicity.
- **Number/threshold:** AQRAB gives a specialist-directed example of regular insulin 1 unit/kg IV bolus followed by 1-10 units/kg/hour infusion, with dextrose as required.
- **Management priority:** Continuously reassess blood pressure, glucose, potassium, ionized calcium and fluid balance.
- **Exam trap:** Do not apply the toxicology insulin regimen without protocolized glucose/potassium replacement and critical-care monitoring.
- **Clinical Pearl:** Vasopressors and calcium may be co-treatments; high-dose insulin does not replace stabilization or agent-specific expert advice.

HDI في BB/CCB toxicity علاج إنعاش تخصصي، وليس جرعة insulin أخطر أخطائه العادية. hypoglycemia وhypokalemia إذا لم تتم المراقبة المتكررة.

Topic 30: Digoxin poisoning and digoxin-specific antibody fragments

- **Clinical trigger:** Nausea, visual changes, confusion, AV block, bradyarrhythmia or ventricular dysrhythmia following digoxin exposure suggests toxicity.
- **Red flag:** Severe hyperkalemia, life-threatening arrhythmia, shock or cardiac arrest warrant urgent digoxin-specific Fab consultation/administration.
- **Management priority:** Monitor ECG, hemodynamics, electrolytes and renal function; interpret drug levels with elapsed time and exposure history.
- **Number/threshold:** AQRAB provides source examples of Fab vial calculation: acute known ingestion 1 vial per 0.5 mg and chronic concentration-based calculation as [digoxin ng/mL x weight kg] / 100.
- **Exam trap:** Routine post-Fab total digoxin concentrations are not useful for standard reassessment.
- **Clinical Pearl:** Potassium may fall after Fab therapy and needs monitoring.

في digoxin toxicity اربط ECG والـK والـrenal function والأعراض؛ المستوى وحده قد يكون مضللًا، ولا تفسّر post-Fab level كأنه المستوى الحر المعتاد.

Topic 31: Lithium toxicity: acute versus chronic presentation

- **Core concept:** Lithium has a narrow therapeutic range and renal clearance; acute ingestion may start with GI symptoms while chronic accumulation commonly has neurologic findings.

- **Clinical trigger:** Coarse tremor, ataxia, dysarthria, confusion, hyperreflexia or seizures with lithium exposure and impaired renal clearance.
- **High-yield fact:** Volume depletion, diuretics, ACE inhibitors and NSAIDs can contribute to higher lithium concentrations.
- **Key discriminator:** Chronic lithium therapy may cause nephrogenic diabetes insipidus (arginine-vasopressin resistance) with concentrating defects and hypernatremia.
- **Management priority:** Obtain serial, timed lithium levels, renal function, sodium, ECG and neurologic assessment; correct depletion with isotonic fluid when indicated.
- **Red flag:** Serious neurologic or cardiac effects, selected high levels or poor clearance require early hemodialysis consultation using the cited EXTRIP framework.
- **Exam trap:** Do not use diuretics to enhance lithium elimination.

Lithium في الكلى؛ يُطرح عبر chronic toxicity قد تكون tremor وataxia والارتباك شديدة حتى لو لم يكن serum level مرتفعًا جدًا، لذلك لا تفصل الرقم عن الحالة.

Topic 32: Valproate overdose and hyperammonemia

- **Key discriminator:** Valproate overdose may produce altered consciousness and hyperammonemia even when liver enzymes are not markedly abnormal.
- **Red flag:** Severe poisoning can include hypotension, cerebral edema, metabolic acidosis, hepatotoxicity and pancreatitis.
- **Management priority:** Monitor valproate level, ammonia, liver tests, glucose, acid-base status and platelets according to severity.
- **Clinical Pearl:** L-carnitine is described as an adjunct in severe valproate toxicity or associated hepatic injury under specialist guidance.
- **Management priority:** Discuss extracorporeal treatment in severe deterioration with toxicology/nephrology, not by level alone.

Valproate قد يسبب encephalopathy مع hyperammonemia حتى عندما لا تكون LFTs تابع؛ شديدة الارتفاع ammonia والوعي ولا تعتمد على drug level وحده.

Topic 33: Isoniazid overdose: refractory seizures and pyridoxine

- **Clinical trigger:** Recurrent or refractory generalized seizures with severe lactic metabolic acidosis after suspected isoniazid ingestion.
- **Management priority:** Give pyridoxine along with airway, seizure and cardiopulmonary support.
- **Number/threshold:** AQRAB describes gram-for-gram pyridoxine when ingestion is known, or 5 g IV as an adult severe-toxicity initial example when amount is unknown.
- **Red flag:** Coma, respiratory failure and severe acidosis require monitored resuscitation and toxicology consultation.
- **Clinical Pearl:** CLEAN V3 relates INH-associated refractory seizures to depletion of vitamin B6-related GABA metabolism.

Seizures لـ INH toxicity؛ تثير TB medication بعد lactic acidosis مع benzodiazepines لا تستجيب بشكل كافٍ Pyridoxine هو antidote النوعي.

Topic 34: Sulfonylurea poisoning and recurrent hypoglycemia

- **Core concept:** Sulfonylurea exposure can cause prolonged or recurrent hypoglycemia.
- **Management priority:** Correct hypoglycemia with appropriate glucose and use octreotide to suppress continued insulin release when indicated.
- **High-yield fact:** Glucose alone may stimulate additional endogenous insulin release and be followed by rebound hypoglycemia.
- **Exam trap:** Do not discharge on the basis of a single corrected glucose measurement; monitor for recurrence.
- **Clinical Pearl:** CLEAN V3 identifies glyburide as a sulfonylurea medication.

Dextrose يعالج hypoglycemia لكن، فورًا sulfonylurea هنا؛ قد يسبب عودة انخفاض السكر Octreotide يحد من إفراز insulin المتكرر.

Topic 35: Theophylline toxicity: seizures, arrhythmias and metabolic effects

- **Clinical trigger:** Vomiting, tremor, agitation, refractory seizures or tachyarrhythmias after theophylline exposure.
- **High-yield fact:** Hypokalemia, hyperglycemia, lactic acidosis and severe neurologic/cardiac toxicity may accompany overdose.
- **Management priority:** Monitor ECG, electrolytes, glucose, acid-base status and serial clinical course.
- **Red flag:** Severe seizures, unstable arrhythmias or systemic toxicity may require extracorporeal-treatment consultation under EXTRIP guidance.
- **Common mistake:** Treating tachycardia as isolated anxiety without evaluating an exposure and metabolic consequences.

Theophylline overdose قد يسبب refractory seizures وtachyarrhythmias مع hypokalemia اطلب تقييمًا ؛ تخصصيًا سريعًا عندما تتفاقم الأعراض.

Topic 36: Antipsychotic overdose: ECG, sedation and extrapyramidal effects

- **Core concept:** Antipsychotic overdose may cause CNS depression, hypotension, antimuscarinic effects, seizures, conduction abnormalities or extrapyramidal findings.
- **Management priority:** Obtain ECG and monitor QTc/QRS, rhythm, mental status, BP, glucose and electrolytes with agent-specific observation.
- **Exam trap:** Not all antipsychotics have the same torsades risk; a normal early ECG does not exclude later toxicity.
- **High-yield fact:** CLEAN V3 describes akathisia as severe restlessness after antipsychotic initiation or a dose increase.
- **Clinical Pearl:** Acute medication-induced dystonia may be treated with anticholinergic agents such as diphenhydramine, benztropine or procyclidine in source-specific examples.
- **Red flag:** Fever, generalized rigidity and autonomic instability after dopamine blockade require urgent consideration of NMS.

Antipsychotic toxicity تسبب بعض الأدوية: تختلف حسب الدواء QT prolongation، وقد تظهر hypotension أو delirium أو seizures. NMS طارئ منفصل يجب تمييزه.

Topic 37: Sotalol, QT-prolonging drugs and torsades

- **Clinical trigger:** Prolonged QTc or polymorphic ventricular tachycardia after sotalol, antipsychotic or other QT-prolonging exposure.
- **Number/threshold:** CLEAN V3 describes IV magnesium sulfate 2 g in a sotalol-associated torsades case.
- **Management priority:** Remove the contributing agent where possible, correct potassium and magnesium disturbances and manage unstable arrhythmia through appropriate resuscitation.
- **Exam trap:** A drug-related QT interval finding is not equivalent to a TCA-type wide-QRS sodium-channel blockade.
- **Clinical Pearl:** Serial ECG is required when delayed absorption or ongoing symptoms remain possible.

QT prolongation بعد دواء قلبي أو نفسي قد يتطور إلى torsades؛ راقب ECG وأصلح اضطرابات الشوارد بدل الاكتفاء باسم الدواء.

IV. Stimulants, Caustics, Inhaled Agents, Metals & Biological Exposures

Topic 38: Cocaine and sympathomimetic poisoning

- **Clinical trigger:** Agitation, hypertension, tachycardia, sweating, hyperthermia, chest pain or seizure after stimulant exposure.
- **Management priority:** Provide a safe monitored setting and benzodiazepine sedation for dangerous agitation; rapidly cool life-threatening hyperthermia.
- **Red flag:** Evaluate for coronary ischemia/vasospasm, dysrhythmia, rhabdomyolysis, metabolic acidosis and sodium-channel QRS widening.
- **Management priority:** AQRAB supports appropriate vasodilators for coronary vasospasm and bicarbonate for life-threatening sodium-channel cardiotoxicity.

- **Clinical Pearl:** CLEAN V3 contains case-specific nitroglycerin plus benzodiazepine treatment for cocaine-associated chest pain.
- **Exam trap:** Avoid assuming that chest pain is anxiety or that physical restraint without sedation is adequate care.

Cocaine toxicity قد يجمع chest pain وagitation وhyperthermia وseizures. Benzodiazepine sedation وcooling مهمان، لكن وجود ألم صدري يحتاج تقييم ischemia.

Topic 39: Hallucinogens and dissociative agents

- **Core concept:** Hallucinogen or dissociative exposure may involve an unknown substance, adulterants or mixed ingestion.
- **Management priority:** Use a safe low-stimulation environment and carefully titrated sedation for dangerous agitation.
- **Clinical trigger:** Seizure, severe hyperthermia, altered consciousness, chest pain or dysrhythmia requires emergency reassessment.
- **Differential point:** Consider serotonergic, sympathomimetic, antimuscarinic and nontoxicologic medical mimics.
- **Exam trap:** A psychiatric-appearing presentation can conceal hypoxia, head injury, hypoglycemia or hyperthermia.

عند intoxication مجهولة مع agitation، لا تفترض أن اسم street drug افحص دقيق. glucose وtemperature وtrauma وECG عند اللزوم.

Topic 40: Caustic ingestion: airway injury and prohibited interventions

- **Core concept:** Caustic ingestion can cause airway edema, esophageal injury, perforation and mediastinal or abdominal complications.
- **Clinical trigger:** Stridor, drooling, dyspnea, chest/abdominal pain or suspected concentrated corrosive exposure requires urgent reassessment.
- **Management priority:** Stabilize airway and circulation and arrange early ENT/GI/surgical input when injury is significant.
- **Contraindication:** Do not induce emesis, administer activated charcoal routinely, attempt neutralization or undertake blind gastric decontamination.
- **Exam trap:** Absence of visible oropharyngeal burns does not exclude esophageal injury.

Caustic ingestion قد يصيب esophagus رغم غياب oral burns. لا induce vomiting ولا neutralization ولا charcoalفي فكّر مبكرًا ؛ airway وGI/surgical consultation.

Topic 41: Button-battery ingestion: an esophageal emergency

- **Core concept:** An esophageal button battery can cause rapid local electrical tissue injury and delayed perforation or fistula.
- **Management priority:** Obtain urgent AP/lateral localization imaging and arrange emergent endoscopic removal of an esophageal battery.
- **Key discriminator:** The double-rim/halo or lateral step-off imaging signs help distinguish a button battery from a coin.
- **Exam trap:** Do not delay esophageal battery removal because of fasting status or a proposed oral bridge.
- **Clinical Pearl:** Honey is discussed only for guideline-eligible children as a conditional bridge and is not suitable in infants younger than 12 months in the supplied material.

Button battery في esophagus تحتاج removal إسعافيًا ولا يجوز انتظار fasting؛ honey إن ورد بالمصدر هو bridge مشروط جدًا ولا يعرقل الإزالة.

Topic 42: Hydrocarbon ingestion and aspiration

- **Core concept:** Hydrocarbon toxicity varies with viscosity, volatility and chemical composition; low-viscosity products can be aspirated.
- **Clinical trigger:** Coughing, choking, vomiting or hypoxemia after hydrocarbon exposure suggests aspiration injury.

- **Management priority:** Remove exposure, evaluate oxygenation, respiratory findings and ECG, and monitor symptomatic or significant exposures for evolving pneumonitis.
- **Contraindication:** Do not induce vomiting or routinely administer activated charcoal because of aspiration risk.
- **Exam trap:** Antibiotics and corticosteroids are not routine for uncomplicated chemical pneumonitis without another supported indication.

أهم خطر من hydrocarbon low-viscosity هو aspiration pneumonitis. لا induce emesis ولا charcoal لمجرد أن المريض ابتلع مادة كيميائية.

Topic 43: Carbon monoxide poisoning and pulse oximetry limitations

- **Clinical trigger:** Headache, nausea, dizziness, syncope or confusion after heater, generator, exhaust or smoke exposure; multiple affected household members are a clue.
- **Key discriminator:** Standard pulse oximetry can be falsely reassuring; measure carboxyhemoglobin using blood co-oximetry.
- **Management priority:** Remove from exposure, give high-concentration oxygen and evaluate neurologic and cardiac injury.
- **Clinical Pearl:** Hyperbaric oxygen consultation is described for selected high-risk circumstances, but indications vary with clinical and local source protocols.
- **Red flag:** Loss of consciousness, pregnancy, severe acidosis, neurologic deficit or cardiac involvement raises concern for severe exposure.

Standard SpO2 قد يبدو طبيعيًا رغم CO poisoning. التاريخ البيئي (عدة أفراد لديهم headache) وCOHb بالـco-oximetry أهم من قراءة pulse oximeter وحدها.

Topic 44: Cyanide poisoning after smoke inhalation

- **Clinical trigger:** Severe smoke exposure followed by collapse, seizures, shock or marked lactic acidosis is compatible with cyanide toxicity.
- **Management priority:** Provide resuscitation and hydroxocobalamin promptly in a compelling life-threatening presentation without waiting for cyanide level.
- **Number/threshold:** AQRAB gives adult hydroxocobalamin 5 g IV, potentially repeated once to 10 g total according to the source protocol.
- **Exam trap:** High lactate supports concern but does not independently prove cyanide toxicity.
- **Clinical Pearl:** Hydroxocobalamin can interfere with laboratory assays and dialysis blood-leak detectors.

في enclosed-space fire مع shock أو collapse أو severe lactate، فكّر في cyanide حتى لو مستوى cyanide غير متاح. العلاج النوعي لا ينتظر فحصًا متأخرًا.

Topic 45: Methemoglobinemia: saturation gap and methylene blue

- **Clinical trigger:** Cyanosis with blood described as chocolate-brown, pulse-oximetry plateau near 85%, and preserved arterial PaO2 suggests methemoglobinemia.
- **Key discriminator:** Blood co-oximetry, rather than standard pulse oximetry, identifies methemoglobin.
- **Number/threshold:** Both sources describe methylene blue 1-2 mg/kg IV for clinically indicated cases; AQRAB also discusses repeat dosing and a commonly cited total ceiling of 5-7 mg/kg.
- **Number/threshold:** CLEAN V3 states a threshold of >30% methemoglobin or symptomatic poisoning in a source-specific treatment summary.
- **Red flag:** Review G6PD deficiency and serotonergic medications because methylene blue may fail or cause harm in these settings.
- **Exam trap:** Normal PaO2 does not prove adequate oxygen carriage in dyshemoglobinemia.

التشخيص ؛ methemoglobinemia يوجّه إلى chocolate-brown blood طبيعي و PaO2 قريب من 85% مع SpO2 وحده. SpO2 وليس بالـco-oximetry

Topic 46: Iron overdose: deceptive latency and deferoxamine

- **Core concept:** Serious iron overdose may cause early GI symptoms, a deceptively quiet interval, then acidosis, shock, hepatic injury or coagulopathy.
- **Management priority:** Assess elemental iron amount, timed serum iron, serial clinical state, acid-base status and kidney/liver function.
- **High-yield fact:** Activated charcoal does not bind iron; whole-bowel irrigation may be considered in significant tablet ingestion with expert advice.
- **Management priority:** Consider deferoxamine for significant systemic toxicity or severe clinical findings under toxicology guidance.
- **Exam trap:** CLEAN V3 notes a latent phase roughly 6-24 hours in a described iron-ingestion scenario; a quiet interval is not a discharge criterion.
- **Clinical Pearl:** An early drug level or a normal abdominal radiograph alone cannot exclude severe exposure.

Iron poisoning قد يبدأ GI irritation ثم latent period يبدو فيه المريض أفضل قبل shock أو hepatic failure.

تحسن الأعراض لا ينفي تطور السمّية.

Topic 47: Heavy metals: lead, arsenic, mercury and thallium

- **Core concept:** Metal toxicity requires plausible exposure history and appropriately validated metal-specific testing.
- **Management priority:** Consider lead, arsenic and mercury with matching exposure and clinical findings; specialist consultation guides chelation decisions.
- **Clinical Pearl:** CLEAN V3 mentions lead lines seen on radiography in a lead exposure context.
- **High-yield fact:** AQRAB associates thallium poisoning with early GI symptoms, subsequent neuropathic pain and delayed alopecia; Prussian blue is discussed with toxicology advice.
- **Exam trap:** Provoked urine metal testing is not valid as a general diagnostic shortcut and empirical chelation may be harmful.

Heavy-metal symptoms لا تشخّص؛ غير نوعية exposure من اختبار provoked urine أو من الأعراض وحدها.

Chelation تختلف حسب المعدن وتحتاج اختصاصيًا.

Topic 48: Organophosphate and carbamate poisoning: airway and atropinization

- **Core concept:** Organophosphates inhibit acetylcholinesterase and cause muscarinic secretions, bronchospasm, bradycardia, nicotinic fasciculations and weakness.
- **Management priority:** Protect responders, remove contaminated clothing and support airway, suction and ventilation.
- **Number/threshold:** AQRAB adult atropine example is 1-2 mg IV, with rapidly repeated/doubled doses about every 5 minutes until bronchial secretions and ventilation improve.
- **Management priority:** Pralidoxime may help selected organophosphate exposures; source regimens and evidence vary by compound and toxicology advice.
- **Exam trap:** Use bronchial drying and respiratory improvement as atropine endpoints; pupil size alone is not the goal.
- **Red flag:** Respiratory failure may arise from secretions, bronchospasm or delayed neuromuscular weakness.

هذا wet patient: bronchorrhea هي هدف علاج Atropine، وليس اتساع الحدقة. اهتم بسلامة الطاقم والتهوية،

وحده atropine لا يعالجه weakness لأن النيكوتينيك.

Topic 49: Toxic plants and mushroom ingestion

- **Clinical trigger:** Vomiting or diarrhea starting more than approximately six hours after wild-mushroom ingestion raises concern for amatoxin toxicity.
- **High-yield fact:** An apparent period of symptomatic improvement may precede progressive hepatic or renal failure.
- **Management priority:** Seek poison-center identification assistance and serial liver tests, INR, glucose, renal and acid-base monitoring when significant exposure is suspected.
- **Red flag:** Coagulopathy, hypoglycemia or evolving hepatic injury warrants critical-care and transplant-capable-center discussion.
- **Exam trap:** Color or an unverified photograph alone cannot reliably identify a mushroom or establish a safe discharge interval.

Wild mushrooms مع GI symptoms قد تنذر بـ amatoxin liver injury متأخرة أكثر من 6 ساعات حتى لو خف الإسهال لاحقًا.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 172

2 g	prolonging exposure. Number/threshold: CLEAN V3 describes IV magnesium sulfate 2 g in a sotalol-associated torsades case. Management priority: Remove the contributing agent where possible, correct potassium and magnesium
5 g, 10 g	presentation without waiting for cyanide level. Number/threshold: AQRAB gives adult hydroxocobalamin 5 g IV, potentially repeated once to 10 g total according to the source protocol.
1-2 mg/kg	Key discriminator: Blood co-oximetry, rather than standard pulse oximetry, identifies methemoglobin. Number/threshold: Both sources describe methylene blue 1-2 mg/kg IV for clinically indicated cases; AQRAB also discusses repeat dosing and a commonly cited total ceiling of 5-7 mg/kg.
5-7 mg/kg	Number/threshold: Both sources describe methylene blue 1-2 mg/kg IV for clinically indicated cases; AQRAB also discusses repeat dosing and a commonly cited total ceiling of 5-7 mg/kg. Number/threshold: CLEAN V3 states a threshold of >30% methemoglobin or symptomatic poisoning in a source-
1-2 mg	ventilation. Number/threshold: AQRAB adult atropine example is 1-2 mg IV, with rapidly repeated/doubled doses about every 5 minutes until bronchial secretions and ventilation improve.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 172

Organophosphate poisoning - pralidoxime [#058]

Key discriminator: Pralidoxime can reactivate acetylcholinesterase before aging in organophosphate poisoning.

يساعد Pralidoxime على إعادة تنشيط Acetylcholinesterase قبل حدوث Aging في تسمم الفوسفات العضوية.

Iron poisoning - multisystem toxicity [#062]

Key discriminator: Major iron ingestion can injure the gastrointestinal tract, liver and cardiovascular system.

الابتلاع السام للحديد قد يؤذي الجهاز الهضمي والكبد والدورة الدموية، لا المعدة فقط.

Digitalis / foxglove poisoning [#063]

High-yield fact: Foxglove (Digitalis) contains cardiac glycosides that can cause characteristic dysrhythmias.

يحتوي نبات Foxglove على جليكوزيدات قلبية قد تسبب اضطرابات نظم مميزة.

Topic 50: Medication-induced movement disorders and acute dystonia

- **Core concept:** Dopamine-blocking medicines such as metoclopramide or antipsychotics can cause extrapyramidal adverse effects.
- **Management priority:** CLEAN V3 gives anticholinergic examples such as procyclidine, diphenhydramine or benztropine for acute dystonic reactions.
- **Number/threshold:** A cited CLEAN V3 acute-dystonia example uses benztropine 2 mg IM; context-specific rather than universal.
- **Key discriminator:** Akathisia is described as marked restlessness; NMS has fever, diffuse rigidity and systemic instability.
- **Exam trap:** Differentiate drug-induced movement disorders from toxin-related seizures or severe systemic hyperthermia.

Acute dystonia بعد dopamine-blocking antiemetics أو antipsychotics قد تحتاج anticholinergic treatment ؛ لا تخلطها مع NMS ذي hyperthermia وrigidity العامة.

Topic 51: Inhalant exposures, nitrous oxide and delayed neurologic effects

- **High-yield fact:** CLEAN V3 describes chronic nitrous oxide exposure causing functional vitamin B12 inactivation with neuropathy and elevated homocysteine/methylmalonic acid.
- **Management priority:** CLEAN V3 gives intramuscular vitamin B12 with folate as treatment in a chronic exposure example.
- **Red flag:** Hydrogen sulfide exposure can endanger rescuers, and absence of a characteristic odor does not establish safety.
- **Management priority:** Remove from exposure, protect the team and support ventilation and oxygenation for severe inhalational illness.
- **Clinical Pearl:** CLEAN V3 identifies phosgene as an industrial chemical exposure with potential relevance to inhalational toxicity.

Chronic N2O exposure قد يؤدي إلى functional vitamin B12 deficiency مع neuropathy تسمم بعض ؛ الغازات الأخرى قد يتطلب حماية المنقذين أولًا.

Topic 52: Local anesthetic systemic toxicity and lipid rescue

- **Core concept:** Severe local anesthetic systemic toxicity is an agent-specific setting for intravenous lipid emulsion rescue.
- **Number/threshold:** AQRAB lists an example 20% lipid-emulsion bolus of 1.5 mL/kg for patients below 70 kg, followed by protocol-dependent infusion or additional treatment.
- **Management priority:** Use the source-described LAST checklist and specialist resuscitation approach with careful dosing and monitoring.
- **Exam trap:** Lipid emulsion is not a universal antidote for every cardiotoxic or sedative ingestion.
- **Clinical Pearl:** CLEAN V3 mentions ester local anesthetics as more associated with PABA-related allergic reactions than the general class.

LAST severe قد يحتاج lipid emulsion ضمن protocol متخصص؛ لا تستخدمه روتينيًا مع كل drug overdose. حافظ على airway وcirculation أولًا.

Topic 53: Disposition, recurrence and antidote safety after stabilization

- **Core concept:** Disposition depends on the agent, formulation, time since exposure, serial examination, ECG, laboratory trajectory and available antidote/monitoring resources.
- **Exam trap:** Do not apply one observation window to every poison or discharge after a single apparent response.

- **Management priority:** Recheck clinical status and targeted tests for delayed absorption, rebound hypoglycemia, recurrent respiratory depression or evolving organ injury.
- **High-yield fact:** CLEAN V3 cites observation of at least 24 hours for MAOI overdose because symptom onset can be delayed; retain this as its source-specific statement.
- **Clinical Pearl:** Agent-specific antidote preparation, concentration, monitoring, repeat doses and stopping criteria require the actual toxicology protocol; a single dose is not a complete plan.

تحسن العلامات الأولية لا يعني انتهاء الخطر: delayed absorption، recurrent hypoglycemia، recurrent opioid sedation، وlate liver injury كلها تحتاج خطة مراقبة مناسبة.

Comparison Tables

High-yield differentiators derived only from the preceding integrated topics.

Sympathomimetic versus Antimuscarinic

Feature	Sympathomimetic	Antimuscarinic	Exam point
Skin	Usually sweaty / diaphoretic	Usually dry and flushed	Sweaty versus dry is a major clue
Pupils	Mydriasis	Mydriasis	Shared feature is not diagnostic
Bowel/bladder	Not the defining clue	Reduced bowel sounds and retention	Look for antimuscarinic retention
Immediate danger	Agitation, seizures, hyperthermia	Delirium, seizures, hyperthermia	Check ECG and temperature

Serotonin Toxicity versus NMS

Feature	Serotonin Toxicity	NMS	Exam point
Onset	Often within hours of serotonergic change	Often develops over days after dopamine blockade	Exposure and timing
Neuromuscular	Clonus, hyperreflexia, tremor	Generalized lead-pipe rigidity, reduced reflexes	Best discriminator
GI and skin	Diarrhea, diaphoresis common	Less prominent GI hyperactivity	Supportive not exclusive
Management	Stop serotonergic drug; sedation/cooling	Stop dopamine blocker; supportive care	Treat and monitor both severe syndromes

Opioid versus Sedative-Hypnotic Poisoning

Feature	Opioid	Sedative-Hypnotic Poisoning	Exam point
Breathing	Bradypnea/apnea prominent	Hypoventilation may occur	Support ventilation first
Pupils	Miosis may occur; not universal	No single defining pupil finding	Never rely on pupils alone
Neurologic	Depressed consciousness	Somnolence, ataxia, slurred speech	Assess mixed ingestion
Reversal	Naloxone when indicated	Flumazenil only selected cases	Different antidote risks

Methanol versus Ethylene Glycol

Feature	Methanol	Ethylene Glycol	Exam point
Organ clue	Visual / optic injury	Renal injury, hypocalcemia	Vision versus kidney
Acid-base	High-AG metabolic acidosis	High-AG metabolic acidosis	Both can cause life-threatening acidosis
Other findings	Possible elevated osmol gap	Possible oxalate crystals	Neither clue required
Treatment basis	Fomepizole, dialysis assessment; folinic adjunct	Fomepizole, dialysis assessment; cofactor notes	Do not wait for send-out levels in severe suspicion

Methanol/Ethylene Glycol versus Isopropanol

Feature	Methanol/Ethylene Glycol	Isopropanol	Exam point
Acid-base	Toxic-metabolite high-AG acidosis common	Usually ketosis without same high-AG acidosis	Do not equate alcohols
Organ injury	Methanol optic; ethylene glycol renal	CNS depression and gastritis	Different dominant patterns

PASS 9 — Source-backed clinical recovery | Domain 08

dosing or medical-correctness certification.

Fentanyl-associated rigidity

Exam trigger: CLEAN V3 links a rapid large fentanyl bolus (5 micrograms/kg in the cited case) to a wooden/rigid chest syndrome; the number is a case exposure, not a recommended dose.

يعرض المصدر rigid chest بعد bolus سريع من fentanyl؛ جرعة الحالة ليست توصية للإعطاء.

Pupils do not establish a toxidrome

Exam trap: AQRAB warns that opioid-associated miosis may be absent and that anticholinergic and sympathomimetic mydriasis can overlap.

لا تشخّص toxidrome من الحدقة وحدها؛ قد تغيب miosis في opioid poisoning وتتداخل الأنماط.

Toxicology & Pharmacology

Precisely source-linked additions after the priority-safety review. Not independent guideline verification.

Topic: Sodium-channel blocker / TCA poisoning: bicarbonate targets

Number/threshold: In AQRAB's ADULT sodium-channel-blocker/TCA table, specialist-directed sodium bicarbonate treatment follows QRS and hemodynamic response. The table cites a blood pH target about 7.50–7.55 and warns against pH >7.55 or Na >155 mEq/L, with potassium and serial ECG monitoring. These numbers belong to the TCA/Na-channel-blocker row, not the salicylate row.

هذه الحدود من جدول TCA/Na-channel blocker مع Sodium Bicarbonate، وليست من جدول Salicylates. راقب ECG وK وسياق العلاج.

Topic: Lithium toxicity: source-cited extracorporeal-treatment criteria

Number/threshold: AQRAB's EXTRIP summary recommends extracorporeal treatment in impaired kidney function with lithium >4 mEq/L, or decreased consciousness, seizures or life-threatening dysrhythmia at any level; it suggests treatment when lithium >5 mEq/L, significant confusion, or predicted time >36 h to fall below 1 mEq/L despite optimal care. It favors intermittent hemodialysis when indicated and calls for repeat lithium levels for rebound. These are cited criteria, not a self-standing prescription.

مصدر Lithium يربط Dialysis بمستوى الدواء ووظيفة الكلى والأعراض العصبية والقلبية معاً؛ الأرقام وحدها لا تكفي.

> Do not merge case-specific, adult and pediatric regimens into a single dose. Source-dependent conflicts remain unresolved.

EDIC AUDIT PASS 11 | DOMAIN 08 | SOURCE-ONLY CONTEXT RESOLUTION

Toxic mushrooms: delayed onset discriminator

> The statements below reproduce source-specific clinical contexts. They are not a current independently validated prescription protocol. All facts come exclusively from CLEAN V3 and AQRAB.

High-yield fact: CLEAN V3 states that severe mushroom poisoning often begins with GI symptoms more than 6 hours after ingestion. AQRAB p.269 separately describes delayed GI illness followed by possible hepatic, renal or coagulation complications, even after apparent improvement.

بعض أخطر تسمم الفطر تبدأ أعراضه الهضمية متأخرة؛ التحسن الظاهري لا يثبت انتهاء السمية.

Feature	Methanol/Ethylene Glycol	Isopropanol	Exam point
Fomepizole	Used for strongly suspected toxic metabolites	Not generally used	Antidote is toxin-specific
Osmol gap	May be high early, normal late	May be high	Gap alone is nonspecific

Beta-Blocker versus Calcium-Channel Blocker

Feature	Beta-Blocker	Calcium-Channel Blocker	Exam point
Heart rate	Often bradycardia / AV block	Bradycardia or vasodilatory shock	Agent phenotype matters
Glucose	Hypoglycemia can occur	Hyperglycemia is a clue	Metabolic discriminator
Special clue	Propranolol: seizure / QRS	Dihydropyridines: vasodilatory shock	Do not assume uniform pattern
Severe treatment	Vasopressor, insulin; glucagon adjunct	Calcium, vasopressor, high-dose insulin	Specialist glucose/K monitoring

TCA versus CCB Cardiotoxicity

Feature	TCA	CCB Cardiotoxicity	Exam point
ECG	Wide QRS, terminal R in aVR	Bradycardia, conduction block; wide QRS less typical	Consider co-ingestion if atypical
Metabolic	Seizures, hypotension	Hyperglycemia often present	Use bedside glucose and serial ECG
Treatment emphasis	Sodium bicarbonate for Na-channel block	Calcium, vasopressor, high-dose insulin	Do not interchange protocols

Acetaminophen versus Salicylate Overdose

Feature	Acetaminophen	Salicylate Overdose	Exam point
Main clue	Often initially mild; delayed hepatic injury	Tinnitus, tachypnea, acid-base disorder	Different time courses
Key measurement	Timed APAP at >=4 h for eligible nomogram	Serial salicylate, pH and clinical status	Drug level timing matters
Antidote strategy	N-acetylcysteine as indicated	Bicarbonate alkalinization; dialysis in severe cases	Specific protocols
Exam trap	Do not stop NAC by clock alone	Do not remove compensatory ventilation	Different fatal errors

Carbon Monoxide versus Methemoglobinemia

Feature	Carbon Monoxide	Methemoglobinemia	Exam point
Typical story	Heater/generator, smoke; multiple victims	Oxidizing drugs; chocolate-brown blood	Exposure pattern matters
Pulse oximetry	May read falsely normal	Often near 85%, poorly responsive	Standard SpO2 can mislead
Confirmation	Co-oximetry for COHb	Co-oximetry for MetHb	Order the appropriate assay
Therapy	High-concentration oxygen	Methylene blue in selected cases	Different treatments

Cholinergic versus Antimuscarinic Toxicity

Feature	Cholinergic	Antimuscarinic Toxicity	Exam point
Secretions	Wet: bronchorrhea, salivation	Dry skin/mucosa, retention	Wet versus dry
Muscles	Fasciculation and weakness	No defining fasciculation pattern	Nicotinic effects matter
Management	Atropine titrated to secretions; consider oxime	Supportive; selected physostigmine	Do not confuse antidotes
Staff safety	Pesticide contamination may be hazardous	Depends on exposure	Protect responders first

Rapid Recall: Must-Know Facts

1. Prioritize ABCs, bedside glucose, ECG and temperature before naming the poison.
2. A toxidrome narrows the differential; mixed ingestion can blur it.
3. Opioid toxicity is a ventilation emergency, even without pinpoint pupils.

4. Sympathomimetic poisoning is typically sweaty; antimuscarinic toxicity is dry.
5. Cholinergic poisoning produces bronchorrhea and often fasciculations.
6. Serotonin toxicity: clonus and hyperreflexia are key discriminators.
7. NMS favors pronounced generalized rigidity and reduced reflexes.
8. TCA QRS widening and terminal R in aVR suggest sodium-channel block.
9. AQRAB cites QRS >100 ms and >160 ms as TCA risk markers.
10. Routine urine drug screen does not reliably exclude serious poisoning.
11. Activated charcoal is selective; protect the airway and check agent.
12. Do not induce emesis or neutralize caustic substances.
13. For timed single acute APAP ingestion, use a level at >=4 hours.
14. The Rumack-Matthew nomogram does not apply to unknown-time/repeated APAP ingestion.
15. Do not stop NAC solely because 21 hours have elapsed.
16. Salicylates often cause respiratory alkalosis plus AG metabolic acidosis.
17. Preserve compensatory hyperventilation if salicylate intubation is unavoidable.
18. Tinnitus, confusion or pulmonary edema are salicylate danger clues.
19. Methanol threatens vision; ethylene glycol threatens kidneys.
20. A normal late osmol gap does not exclude toxic-alcohol poisoning.
21. Isopropanol commonly causes ketosis without toxic-metabolite AG acidosis.
22. Fomepizole is an example for suspected methanol/ethylene glycol, not all alcohols.
23. Alcohol-withdrawal seizure is described in the source within about 7-48 h.
24. Naloxone is titrated to adequate breathing, not full awakening.
25. Recurrent opioid depression may need repeat naloxone or infusion.
26. Empiric flumazenil can be dangerous in mixed or dependent patients.
27. Selected physostigmine requires caution with QRS widening or TCA suspicion.
28. Bupropion toxicity can show delayed seizures and cardiotoxicity.
29. Beta-blocker toxicity can include hypoglycemia; CCB toxicity may cause hyperglycemia.
30. High-dose insulin for BB/CCB shock requires glucose and potassium monitoring.
31. Digoxin-specific Fab is considered for life-threatening dysrhythmia or shock.
32. Lithium toxicity severity depends on renal function and clinical pattern.
33. Lithium may cause nephrogenic diabetes insipidus and hypernatremia.
34. Valproate toxicity can produce hyperammonemia with modest LFT abnormalities.
35. Refractory seizures after INH exposure call for pyridoxine.
36. Sulfonylurea hypoglycemia may recur; octreotide reduces insulin secretion.
37. Cocaine toxicity: assess ischemia, hyperthermia, seizures and QRS widening.
38. Esophageal button battery requires emergent removal; do not await fasting.
39. Standard SpO2 is unreliable for carbon monoxide and methemoglobinemia.
40. Iron or amatoxin poisoning may deteriorate after an apparent improvement.

Conflict Resolution Register

Confirmed direct contradictions between the two supplied files after topic-based review: 0. Differences below were retained as contextual, source-specific or incompletely resolved variations; they are not automatically treated as mutually exclusive clinical recommendations.

Source-dependent variations and context

Topic	CLEAN V3	AQRAB v18	Nature / handling
Naloxone repeat-dose examples	CLEAN V3: 0.4 mg IV then 0.8 mg at about 60 seconds if no response.	AQRAB: 0.2-2 mg IV/IM/IO or 2-4 mg IN; repeat every 2-3 minutes.	Different example doses and reassessment intervals; compatible overall range, not proven contradiction.
Salicylate dialysis thresholds	CLEAN V3 lists level examples including >100 mg/dL or >90 with renal impairment.	AQRAB places emphasis on clinical deterioration, neurologic/pulmonary signs, and acid-base/renal status.	Threshold examples versus clinical indications; neither replaces the other.

Topic	CLEAN V3	AQRAB v18	Nature / handling
Ethylene glycol cofactors	CLEAN V3 states that all patients should receive thiamine and also mentions pyridoxine.	AQRAB emphasizes fomepizole, acidosis correction, renal function, and dialysis; not an equivalent universal cofactor directive.	Difference in strength of advice / coverage; source-dependent, no external resolution.
Fomepizole regimen	CLEAN V3 contains source-specific clinical examples.	AQRAB gives 15 mg/kg loading example and says dialysis adjustment is required.	Incomplete regimen if copied as one number; retained only as a context-specific loading example.
Iron toxicity and symptom-free intervals	CLEAN V3 emphasizes vomiting and a latent phase in separate question-derived fragments.	AQRAB emphasizes serial observation and that a quiet interval does not exclude deterioration.	Apparent tension in emphasis, not a verified incompatible clinical claim.

No cross-file clinical claim was selected as correct using knowledge external to the supplied documents.

Metric	Recorded value
Final integrated topics	53
Integrated core English points	290
CLEAN V3 only points	34
AQRAB only points	190
Shared points merged	66
Comparison tables	10
Rapid Recall facts	40
Confirmed direct cross-file conflicts	0
Contextual / source-dependent variations logged	5
Unusable, internally contradictory or incomplete source groups logged	9

Counting method: each final English body bullet is one integrated point; manually assigned AQRAB-only, CLEAN-only or supported by both at drafting time. Arabic explanations, table cells and Rapid Recall are not additional points. Classification describes the finished section, not a complete automated audit of every original paragraph.

Toxicology & Pharmacology

Supplementary Clinical Points

Topic: Isopropyl alcohol: distinguishing clinical pattern

Key discriminator: CLEAN V3 describes greater CNS depression after isopropyl alcohol than ethanol, while AQRAB identifies ketosis without acidosis as a distinguishing laboratory clue. The comparisons are source teaching points, not measured severity predictions for a particular patient.

الميزة المذكورة هي CNS depression مع ketosis دون acidosis؛ لا تجعل شدة التسمم متوقعة من اسم المادة وحده.

Source evidence: C02786: DOCX paragraph 2786 | A416-02: AQRAB PDF p.416, block 2

Toxicology & Pharmacology. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 179

0.4 mg, 0.2-2 mg	Different example doses and CLEAN V3: 0.4 mg IV then 0.8 AQRAB: 0.2-2 mg IV/IM/IO or 2- Naloxone repeat-dose reassessment intervals;
4 mg	Naloxone repeat-dose reassessment intervals; mg at about 60 seconds if no 4 mg IN; repeat every 2-3 examples compatible overall range, not
15 mg/kg	Incomplete regimen if copied AQRAB gives 15 mg/kg loading CLEAN V3 contains source- as one number; retained only

EDIC / Emergency & Critical Care

09 — Pediatrics & Neonatology

I. Initial Assessment, Shock, Airway & Medication Safety

Topic 1: The deteriorating child: appearance, breathing and circulation

- **Core concept:** Assess pediatric instability with appearance, work of breathing, circulation, mental status, age-appropriate vital signs and trajectory, followed by ABCDE.
- **Clinical trigger:** An unusually quiet or exhausted child, poor feeding, altered interaction, delayed capillary refill or respiratory fatigue may be deteriorating.
- **Management priority:** Measure weight in kilograms, check glucose when appropriate, resuscitate immediate threats and reassess after every intervention.
- **Common mistake:** Reassuring oneself with one normal blood pressure, oxygen saturation or temporary improvement despite worsening clinical appearance.
- **Clinical Pearl:** Caregiver concern and developmentally appropriate communication contribute to pediatric assessment.
- Clinical discriminator: The pictured Pediatric Assessment Triangle (PAT) is a rapid visual first-look assessment (the panel specifies under 30 seconds) covering appearance, work of breathing, and circulation to skin; an abnormal component prompts immediate assessment. [AQRAB original PDF p. 275, visual panel]
- Number/threshold: The AQRAB illustration provides teaching reference ranges for an awake, resting neonate: heart rate 100-180/min, respiratory rate 30-60/min, and systolic blood pressure 60-90 mmHg. These are source-table age bands, not an independent triage rule. [AQRAB original PDF p. 275, visual panel]

في أول نظرة استخدم PAT: appearance والتنفس والتروية الجلدية. قيم العلامات الحيوية تختلف مع العمر؛ الجدول هنا للتذكّر وليس لتجاهل العلامات السريرية.

والوعي perfusion وwork of breathing وappearance حتى قبل الانهيار؛ راقب BP الطفل قد يحافظ على باستمرار، ولا تعتمد على قراءة واحدة.

Topic 2: Compensated pediatric shock: do not wait for hypotension

- **High-yield fact:** Children can maintain blood pressure until late in shock; perfusion and mental status may reveal decompensation earlier.
- **Key discriminator:** Cool extremities, weak pulses, delayed capillary refill and poor interaction suggest inadequate perfusion even with a normal pressure.
- **Management priority:** Treat the mechanism of shock, monitor response to fluids and other therapy, and reassess respiratory status and possible fluid overload.
- **Number/threshold:** AQRAB cites systolic blood pressure below 70 + 2 x age in years as a warning threshold for children aged 1-10 years; interpret within the cited trauma context.
- **Exam trap:** Do not transplant an adult threshold or fluid algorithm into every pediatric case.

في compensated shock افحص. قد يبقى الضغط طبيعيًا رغم ضعف التروية pulsesو capillary refill والوعي وurine output مع سبب الصدمة.

Topic 3: Pediatric fluid treatment and repeated reassessment

- **Management priority:** For a child with suspected shock, fluid decisions depend on cause, physiology and response; give monitored aliquots rather than repeating boluses automatically.
- **High-yield fact:** AQRAB cites older pediatric trauma examples of 20 mL/kg crystalloid and 10 mL/kg packed cells, explicitly requiring protocol review rather than uncritical universal use.
- **Red flag:** Increased tachypnea or respiratory distress after fluid administration should prompt reassessment for evolving disease or fluid overload.
- **Common mistake:** Treating the initial blood pressure as the sole resuscitation endpoint or delaying hemorrhage control in trauma.

Fluid bolus بعد كل؛ ليس وصفة متكررة دون تقييم aliquot افحص perfusion وbreathing وعلامات overload، وفكّر في طبيعة shock.

Topic 4: Pediatric airway anatomy and positioning

- **Core concept:** Pediatric airways differ from adults in anatomy and oxygen reserve, making obstruction and rapid desaturation important concerns.
- **Key discriminator:** Young children have a relatively larger tongue, more anterior glottis and short trachea; infants may need a shoulder roll because of the prominent occiput.
- **Management priority:** Choose age-appropriate equipment and optimize position, suction and preoxygenation before attempting an advanced airway.
- **Exam trap:** A high, anterior glottis and short trachea increase technical difficulty and the risk of endobronchial intubation.

عند الرضيع الـoccipitأكبر والـairway عدّل؛ أعلى وأكثر أمامية positioning وقد تحتاج shoulder roll، ولا تعتمد على عمق أنبوب الكبار.

Topic 5: Failed pediatric intubation and rescue oxygenation

- **Management priority:** Prepare a primary and rescue airway plan with skilled support, and prioritize bag-mask oxygenation over repeated intubation attempts.
- **Clinical trigger:** Desaturation or bradycardia during laryngoscopy requires immediate oxygenation-focused reassessment and airway troubleshooting.
- **High-yield fact:** Tube size and depth estimates are only starting points; reassess chest movement, bilateral ventilation and tube position after placement or movement.
- **Key discriminator:** Continuous quantitative waveform capnography supports tracheal placement confirmation and subsequent monitoring.
- **Exam trap:** A functioning bag-mask technique should not be abandoned solely to pursue an advanced airway.

بعد فشل المحاولة تصبح الأولوية oxygenation وليس تكرار laryngoscopyاستخدم ؛ BMV الفعال واطلب المساعدة وخطة إنقاذ مبكرة.

Topic 6: Pediatric procedural sedation and airway rescue

- **Core concept:** Pediatric procedural sedation requires an indication, consent, assessment of airway risk, assigned monitoring and rescue capability.
- **Key discriminator:** Capnography may identify hypoventilation before oxygen saturation drops, particularly with supplemental oxygen.
- **Management priority:** Document prior sedation problems, comorbidities, medication history and recovery toward an appropriate baseline before discharge.
- **Common mistake:** Treating fasting status alone as an absolute reason to delay an urgent procedure without assessing aspiration and airway risks.

Domain 09 — Pediatric Airway

Pass 5 | Source-page-verified additions, not universal dosing guidance

Number / equipment sizing: The AQRAB pediatric-airway infographic gives the age-based uncuffed endotracheal-tube internal-diameter example as 4 + (age in years ÷ 4) mm; for a cuffed tube, it subtracts 0.5 mm. Select and confirm actual equipment for the specific child rather than treating the formula as a complete airway plan.

المعادلة في صورة AQRAB تقديرية لاختيار حجم الأنبوب لدى الأطفال، وليست بديلًا عن تقييم الطفل والتأكد من وضعية الأنبوب وتجهيز مقاسات بديلة.

Number / depth estimation: The same image presents an oral tube-depth estimate at the teeth of about 3 × endotracheal-tube internal diameter (cm when diameter is in mm); its role is an illustrative sizing guide, not a stand-alone confirmation of tracheal placement.

عمق الأنبوب المحسوب تقديري؛ لا يكفي وحده لإثبات أن الأنبوب داخل الرغامى في المكان الصحيح.

Context / oxygenation: The illustrated preparation pathway describes preoxygenation with 100% oxygen for at least 3 minutes or eight vital-capacity breaths, and mentions nasal oxygen during apnea as a source-specific technique. It does not override the individual airway and respiratory physiology assessment.

النقطة الامتحانية هي التحضير للأكسجة قبل RSI، مع عدم تحويل المثال المصور إلى بروتوكول ثابت لكل طفل.

Added after the pediatric-airway topic. These image-specific statements were not represented at comparable numerical detail in the Pass 4 text. No external guideline check was performed.

- **Red flag:** Progressive airway obstruction, laryngospasm or ventilatory failure during sedation requires escalation from airway maneuvers to assisted ventilation as indicated.
- Red flag: The original pediatric-sedation panel flags age below 3 months, anticipated difficult airway, OSA/obesity, active respiratory illness, hemodynamic instability, raised intracranial pressure and neuromuscular disease as pre-sedation risk factors. [AQRAB original PDF p. 283, visual panel]
- Clinical Pearl: The pictured recovery checklist calls for return toward baseline mental status, acceptable vital signs and protective airway reflexes; a validated discharge score (e.g., PADSS) and caregiver instructions are described. [AQRAB original PDF p. 283, visual panel]

قبل sedation انتبه إلى عمر الرضيع والأمراض التنفسية وصعوبة airway. وبعد الإجراء القرار ليس الوقت وحده؛ المهم العودة للوضع الأساسي وسلامة التنفس والتروية.

يجب مراقبة التنفس أثناء sedation. لا ينفي hypoventilation supplementary oxygen طبيعي مع SpO2 وجود والوعي والتروية مع فريق قادر على rescue.

Topic 7: Analgesia and safe weight-based medication calculation

- **Core concept:** Prescribe pediatric drugs using measured kilograms, the indication, mg/kg amount, maximum single/daily dose, route, interval and formulation.
- **Management priority:** Independently check concentration and resulting mL volume, especially for high-alert drugs, infusions, neonatal patients and unfamiliar products.
- **High-yield fact:** Use a leading zero for doses below one (0.5 mg), not a trailing zero (5.0 mg).
- **Common mistake:** Confusing pounds with kilograms, or milligrams with milliliters; a unit error can materially multiply the administered dose.
- **Clinical Pearl:** Check allergies, age restrictions, kidney/liver function, interactions and product-specific limits; consult pharmacy or seniors when uncertain.
- Common mistake: The AQRAB pediatric-safety figure instructs clinicians to record weight in kilograms, use a length-based resuscitation tape if weight is unavailable, and double-check medication dose, concentration and route. [AQRAB original PDF p. 275, visual panel]

احسب الجرعة على kg وتأكد من التركيز والطريق، وميّز الوزن المقاس عن تقدير الطول في الحالات الإسعافية.

احسب الجرعة من kg ثم mg/kg ثم max dose ثم concentration وصولًا إلى mL، ولا تنقل حجم حقنة مبنيًا على تركيز مختلف.

Topic 8: Pediatric anaphylaxis: route and concentration are inseparable

- **Management priority:** Pediatric anaphylaxis requires immediate intramuscular epinephrine into the mid-anterolateral thigh; antihistamines and steroids do not replace it.
- **Number/threshold:** AQRAB cites 0.01 mg/kg IM of 1 mg/mL epinephrine, capped at 0.5 mg per dose, in its ASCIA weight-based pathway.
- **Number/threshold:** CLEAN V3 quotes a pediatric repeat interval of 5-10 minutes as needed; AQRAB includes 5-minute and 5-15-minute pathways depending on its cited protocol.
- **Exam trap:** Do not administer the undiluted IM preparation as a routine intravenous bolus to a patient with a pulse.
- **Red flag:** Progressive upper-airway edema or persistent circulatory compromise calls for expert airway and monitored refractory-anaphylaxis support.

أعطِ IM Epinephrine 1 mg/mL الخاص للعضل مع 0.1 mg/mL بالفخذ فورًا، لكن لا تخلط في anaphylaxis بـIV/IO arrest.

II. Pediatric & Neonatal Resuscitation

Topic 9: Pediatric arrest: respiratory failure before circulatory collapse

- **Core concept:** Pediatric cardiac arrest often follows progressive respiratory failure or shock rather than an isolated primary rhythm event.
- **High-yield fact:** CLEAN V3 describes asystole and pulseless electrical activity as common initial pediatric arrest rhythms.
- **Management priority:** Start high-quality CPR, attach defibrillation monitoring, identify shockable versus non-shockable rhythm and address reversible causes.
- **Clinical Pearl:** Effective bag-mask ventilation can remain an appropriate airway strategy when it provides adequate chest rise and oxygenation.
- **Common mistake:** Interrupting compressions while obtaining IV access or repeatedly pursuing an invasive airway.

انتظار CPR المبكر أهم من الفعالة و ventilation ؛ لذلك shock الأطفال يسبقه فشل تنفسي أو كثير من arrest تدهور متأخر.

Topic 10: Pediatric arrest medications and route safety

- **Number/threshold:** The cited pediatric arrest epinephrine dose is 0.01 mg/kg IV/IO using 0.1 mg/mL, maximum 1 mg; route and concentration must be checked.
- **Management priority:** For initial non-shockable arrest, AQRAB presents early epinephrine alongside CPR and reversible-cause treatment.
- **High-yield fact:** The supplied 2025 pediatric algorithm favors IV or IO over endotracheal drug delivery when feasible.
- **Exam trap:** Do not routinely administer calcium or sodium bicarbonate during pediatric arrest without a specific indication.
- **Clinical Pearl:** A single ETCO2 threshold must not be used as the only basis for terminating resuscitation.

جرعة arrest هي Epinephrine 0.01 mg/kg IV/IO بتركيز 0.1 mg/mL، وليست IM anaphylaxis preparation ؛ تحقق من كل الحسابات.

Topic 11: Post-resuscitation pediatric stabilization

- **Management priority:** After ROSC, stabilize oxygenation, ventilation, circulation, temperature and neurologic monitoring.
- **Common mistake:** Using a single early post-arrest neurologic finding to make a definitive prognosis.
- **Clinical Pearl:** Continue a structured search for the precipitating respiratory, circulatory, metabolic or other reversible condition.

بعد ROSC تابع: لا يتوقف العلاج عند عودة النبض oxygenation وventilation وperfusion وtemperature والدماغ.

Topic 12: Neonatal resuscitation: first steps and the first minute

- **Core concept:** Neonatal resuscitation prioritizes effective ventilation over compressions when the infant is apneic, gasping or persistently bradycardic.
- **Number/threshold:** The supplied AHA/AAP 2025 pathway begins ventilation within 60 seconds for apnea/gasping or persistent HR <100/min after initial steps.
- **Management priority:** Prepare a trained team, warm and position the newborn, assess heart rate, and correct leak or obstruction when ventilation does not raise the pulse.
- **Common mistake:** Starting compressions before establishing visible lung inflation or routinely suctioning without suspected obstruction.

قبل mask leak وحركة الصدر وصحح HR في الوليد غير المتنفس الأولوية لتهوية فعالة خلال 60 ثانية؛ افحص أي تصعيد.

Topic 13: Neonatal compressions and response to ventilation

- **Number/threshold:** Start coordinated 3:1 compressions-to-ventilation if HR remains <60/min after 30 seconds of ventilation that visibly inflates the lungs.

- **Key discriminator:** No heart-rate rise after ventilatory effort commonly requires correction of mask seal, head position or airway obstruction.
- **Management priority:** Use ECG for rapid heart-rate assessment when available and titrate oxygen according to the supplied transition pathway.
- **Exam trap:** The newborn 3:1 compression ratio is not the adult 30:2 ratio.

لا تبدأ compressions لمجرد HR منخفض إذا كان mask ventilation من 30 ثانية بعد؛ غير فعال effective ventilation وHR <60 استخدم 3:1.

Topic 14: Newborn temperature and immediate safety

- **Number/threshold:** AQRAB gives a neonatal temperature target of 36.5-37.5°C during delivery-room care.
- **Management priority:** Anticipate high-risk delivery, prepare equipment and personnel, prevent heat loss and assess heart rate alongside breathing.
- **High-yield fact:** The supplied resuscitation pathway discourages routine suction of clear or meconium-stained fluid unless an obstruction is suspected.
- **Clinical Pearl:** The first intervention should correct the physiologic failure rather than delay oxygenation for nonessential procedures.

Hypothermia جهّز؛ بعد الولادة قد يزيد التعقيد warmth وdrying وtemperature monitoring ضمن خطوات الانتقال.

III. Pediatric Cardiac and Respiratory Emergencies

Topic 15: Congenital cardiac disease and infant heart-failure clues

- **Clinical trigger:** Feeding diaphoresis, poor growth, cyanosis, tachypnea or shock in an infant can indicate congenital heart disease or heart failure.
- **Management priority:** Stabilize airway, breathing and circulation while considering duct-dependent physiology and early specialist involvement.
- **High-yield fact:** CLEAN V3 identifies congenital structural disease including ventricular septal defect as a pediatric heart-failure mechanism.
- **Exam trap:** Not every infant with feeding difficulty and tachypnea has isolated lower-respiratory disease.

Cardiac قد يكون shock أو cyanosis أو poor growth مع feeding أثناء التعرق؛ وليس مجرد مشكلة تنفسية اطلب Pediatric Cardiology مبكرًا.

Topic 16: Sinus tachycardia versus pediatric SVT

- **Key discriminator:** Distinguish sinus tachycardia from SVT using abrupt onset, variability, P-wave appearance and age-specific rate context.
- **Management priority:** Identify hypotension, poor perfusion, altered consciousness and cardiopulmonary compromise before choosing rhythm treatment.
- **Clinical Pearl:** A very rapid regular rhythm in a stable child requires monitoring and rhythm confirmation, not automatic interpretation by adult cutoffs.
- **Common mistake:** Treating physiologic sinus tachycardia as SVT without assessing the underlying illness.

sinus tachycardia انظر إلى؛ وبداية مفاجئة؛ regular غالبًا SVT يتبدل مع الحمى والبكاء والحالة؛ وP onset waves لا إلى الرقم فقط.

Topic 17: Stable pediatric SVT: vagal maneuver and adenosine

- **Management priority:** For stable probable narrow-complex SVT, consider an age-appropriate vagal maneuver before adenosine.
- **Number/threshold:** The AQRAB AHA 2025 pathway cites adenosine 0.1 mg/kg rapid IV/IO (maximum 6 mg), followed if needed by 0.2 mg/kg (maximum 12 mg).
- **Exam trap:** Adenosine is not a routine treatment for an irregular wide-complex rhythm.
- **Clinical Pearl:** Use a rapidly delivered dose and flush with ECG and clinical monitoring.

إذا كان SVT منتظمًا ومستقرًا، قد تبدأ بـ vagal maneuver ثم adenosine rapid push المناسب مع flush ووزن وتركيز مؤكدين.

Topic 18: Unstable pediatric SVT and cardioversion

- **Management priority:** Cardiopulmonary compromise during probable SVT calls for synchronized cardioversion with ongoing resuscitation.
- **Number/threshold:** The cited pathway begins synchronized cardioversion at 0.5-1 J/kg and increases to 2 J/kg if ineffective.
- **High-yield fact:** CLEAN V3 mentions a specialist-discussed slow IV amiodarone 5 mg/kg over more than 20 minutes after unsuccessful cardioversions in one pathway.
- **Exam trap:** Refractory and wide-complex rhythms require expert support; do not apply the stable narrow-complex algorithm indiscriminately.

إذا ترافق SVT مع shock أو altered perfusion فالقرار هو synchronized cardioversion، وليس انتظار الأدوية دون دعم ABC.

Topic 19: The febrile infant: age matters more than reassurance

- **Number/threshold:** AQRAB describes the AAP febrile-infant pathway for well-appearing term infants aged 8-60 days with temperature >=38.0°C and its defined exclusions.
- **Key discriminator:** Ill appearance, prematurity, immunocompromise and instability require a different urgent assessment pathway.
- **Management priority:** Use age-band evaluation for urine, blood, inflammatory markers, CSF and safe disposition when indicated.
- **Exam trap:** Antipyretic response or an initially well-appearing infant does not exclude serious bacterial infection.
- Clinical trigger: The AQRAB pediatric-fever infographic flags neonates younger than 28 days, infants younger than 3 months, prematurity below 37 weeks, under-immunization, recent antibiotics within 48-72 hours and hyperpyrexia above 40°C as context-dependent risk features; these are not interchangeable with formal pathway inclusion criteria. [AQRAB original PDF p. 297, visual panel]

هذه عوامل قد ترفع الانتباه في fever عند الأطفال، لكن لا تعني تطبيق نفس pathway على كل الأعمار أو الحالات.

Febrile infant بعمر 8-60 يوم له شروط دخول محددة بالـAAP pathway ؛ المظهر الجيد وحده لا يلغي خطر serious infection.

Topic 20: Fever, feeding failure and neonatal infection warning signs

- **Red flag:** Lethargy, poor feeding, temperature instability or abnormal perfusion in a young infant may signal serious infection even without high fever.
- **Management priority:** Assess age, immunization, comorbidity, appearance, source of infection and physiologic instability promptly.
- **Common mistake:** Using a nonsterile bag urine specimen as the sole evidence for UTI diagnosis when sterile culture collection is required.
- **Clinical Pearl:** Communicate specific return precautions and reassessment plans when a child is discharged after a fever evaluation.

تقييم fever مرتفعة. sepsis من أهم قد تكون الرضيع الصغير لدى الحرارة وتغيّر الرضاعة وضعف الخمول يعتمد على المظهر والعمر والعلامات.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 188

0.5 mg, 5.0 mg	infusions, neonatal patients and unfamiliar products. High-yield fact: Use a leading zero for doses below one (0.5 mg), not a trailing zero (5.0 mg). Common mistake: Confusing pounds with kilograms, or milligrams with milliliters; a unit error can materially
0.01 mg/kg, 1 mg/mL, 0.5 mg	anterolateral thigh; antihistamines and steroids do not replace it. Number/threshold: AQRAB cites 0.01 mg/kg IM of 1 mg/mL epinephrine, capped at 0.5 mg per dose, in its ASCIA weight-based pathway.
0.01 mg/kg, 0.1 mg/mL	Number/threshold: The cited pediatric arrest epinephrine dose is 0.01 mg/kg IV/IO using 0.1 mg/mL, maximum 1 mg; route and concentration must be checked.
0.01 mg/kg	Clinical Pearl: A single ETCO2 threshold must not be used as the only basis for terminating resuscitation. جرعة هي 0.1 بتركيز ، ولي‌ست ؛ arrest Epinephrine 0.01 mg/kg IV/IO mg/mL IM anaphylaxis preparation الحسابات لكل من تحقق .
0.1 mg/kg	before adenosine. Number/threshold: The AQRAB AHA 2025 pathway cites adenosine 0.1 mg/kg rapid IV/IO (maximum 6 mg), followed if needed by 0.2 mg/kg (maximum 12 mg).
0.2 mg/kg	Number/threshold: The AQRAB AHA 2025 pathway cites adenosine 0.1 mg/kg rapid IV/IO (maximum 6 mg), followed if needed by 0.2 mg/kg (maximum 12 mg). Exam trap: Adenosine is not a routine treatment for an irregular wide-complex rhythm.
5 mg/kg	ineffective. High-yield fact: CLEAN V3 mentions a specialist-discussed slow IV amiodarone 5 mg/kg over more than 20 minutes after unsuccessful cardioversions in one pathway.
0.15 mg/kg	severity-based pediatric pathways. Number/threshold: CLEAN V3 quotes single-dose oral dexamethasone 0.15 mg/kg in one croup pathway; AQRAB notes protocol-specific ceilings and dosing.
0.1 mg/mL, 1 mg/mL	pathway anaphylaxis treatment or reverse 0.1 mg/mL in cited arrest 1 mg/mL in cited anaphylaxis Check mg/kg, mg/mL, total mg Concentration

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 188

Kawasaki disease - pediatric vasculitis [#068]

High-yield fact: Kawasaki disease is a pediatric vasculitis with prolonged fever and mucocutaneous findings.

التهاب أوعية عند الأطفال يترافق مع حمى مطولة ومظاهر مخاطية جلدية؛ انتبه لمضاعفاته. Kawasaki

Topic 21: Croup: barking cough, stridor and graded severity

- **Core concept:** Viral croup typically presents with barking cough, hoarseness and inspiratory stridor; parainfluenza is repeatedly identified in the sources.
- **Key discriminator:** Stridor at rest, work of breathing, fatigue, oxygenation and ability to feed/handle secretions guide severity.
- **Management priority:** Corticosteroid treatment and nebulized epinephrine for significant distress are described in severity-based pediatric pathways.
- **Number/threshold:** CLEAN V3 quotes single-dose oral dexamethasone 0.15 mg/kg in one croup pathway; AQRAB notes protocol-specific ceilings and dosing.
- **Exam trap:** Persistent or recurrent stridor at rest after therapy warrants further observation and escalation, not routine discharge.

Barking cough وhoarseness وinspiratory stridor توجه إلى Croup، لـ الاستجابة المؤقتة لكنnebulized epinephrine لا تعني زوال الخطر.

Topic 22: Dangerous stridor: epiglottitis versus uncomplicated croup

- **Key discriminator:** A barking cough and viral prodrome favor croup; drooling, toxic appearance, muffled voice or tripod posture raise concern for epiglottitis or another critical upper-airway process.
- **Management priority:** Keep the child calm, minimize distressing examination and involve experienced airway support before risky procedures or imaging.
- **Differential point:** Bacterial tracheitis and deep-neck infection remain important causes of severe stridor and airway compromise.
- **Exam trap:** Normal oxygen saturation or an unremarkable posterior pharynx does not exclude a dangerous supraglottic obstruction.
- Imaging sign: AQRAB illustrates the lateral-neck “thumbprint” sign with epiglottitis and an AP-neck “steeple” sign with croup; the drawings are teaching illustrations, not reasons to postpone airway management in an unstable child. [AQRAB original PDF p. 300, visual panel]

airway يحتاج المستقر غير الطفل لكن الرسومات، في croupـب Steepleو epiglottitisـب يرتبط Thumbprint قبل التصوير. assessment

Drooling وtripod وmuffled voice مع toxic appearance ليست Croup المهيّج فحص البلعوم بسيط؛ تجنّب ونسّق expert airway.

Topic 23: Bronchiolitis: feeding and work of breathing drive decisions

- **Core concept:** Bronchiolitis is a common infant lower-respiratory illness, often associated with RSV and predominantly managed supportively.
- **Management priority:** Assess work of breathing, hydration, feeding safety, oxygenation and apnea risk; support oxygen and fluids/feeding as indicated.
- **High-yield fact:** CLEAN V3 identifies age under three months as an important risk factor for severe bronchiolitis.
- **Common mistake:** Automatically giving antibiotics or asthma medications without a specific indication or missing poor feeding and exhaustion.
- **Red flag:** Apnea, profound fatigue, impaired consciousness or worsening hypoxemia requires escalation.
- High-yield fact: The AQRAB source figure describes bronchiolitis as predominantly affecting children younger than 2 years, commonly following a 1-3 day upper-respiratory prodrome, with tachypnea, retractions, wheeze or crackles; dehydration is highlighted as a complication. [AQRAB original PDF p. 273, visual panel]

Bronchiolitis يبدأ غالبًا بأعراض URI ثم تعب تنفس وصعوبة رضاعة؛ dehydration جزء مهم من تقييم الخطورة.

Bronchiolitis في لا يكفي سماع wheezeراقب ؛ feeding، apnea، oxygenation، work of breathing وعلامات dehydration.

Topic 24: Asthma versus bronchiolitis versus foreign-body wheeze

- **Differential point:** A wheezing child may have asthma, bronchiolitis, pneumonia, aspirated foreign body or anaphylaxis.
- **Key discriminator:** Age, symptom onset, response to bronchodilator, fever, asymmetric air entry and systemic findings help separate diagnoses.
- **Management priority:** Use the child-specific asthma pathway when asthma is likely and supportive care when bronchiolitis is the better fit.
- **Exam trap:** Applying asthma drugs automatically to bronchiolitis or labeling sudden focal wheeze as a recurrent asthma attack.

كل wheeze ليس Asthmaافصل بين ؛ Bronchiolitis وforeign body وpneumonia وanaphylaxis بحسب العمر والبداية والـexam.

Topic 25: Impending respiratory failure: the quiet chest

- **Clinical trigger:** Silent chest, altered consciousness, cyanosis, severe fatigue or apnea signals possible impending respiratory failure.
- **Key discriminator:** Reduced wheeze with worsening work of breathing or drowsiness can mean falling airflow rather than successful treatment.
- **Management priority:** Escalate airway, oxygenation and ventilation support promptly and obtain skilled critical-care assistance.
- **Common mistake:** Assuming a child is recovering because audible wheeze has diminished.

الطفل الذي كان wheezy و ثم أصبح صامتًا drowsy انخفاض؛ قد يكون أسوأ لا أفضل air movement مع exhaustion إنذار خطير.

Topic 26: Pediatric pneumonia and parapneumonic effusion

- **Core concept:** Pediatric pneumonia severity is assessed with respiratory distress, oxygenation, feeding, perfusion and evidence of pleural complications.
- **Key discriminator:** AQRAB cites chest ultrasound for characterizing moderate-to-large parapneumonic effusions when available, often before CT/MRI.
- **Management priority:** Small uncomplicated effusions may be observed; drainage decisions for more serious effusions depend on size, respiratory compromise and purulence.
- **Clinical Pearl:** Where drainage is needed, the supplied source discusses small-bore drainage with intrapleural tPA in selected pediatric pathways, with specialist individualization.

في pneumonia مع moderate/large effusion استخدم ultrasound التصريف؛ لتقدير الحجم والتعقيد إذا توفر يعتمد على الضائقة والصديد والسياق.

IV. Pediatric GI, GU, Metabolic and Neurologic Emergencies

Topic 27: Neonatal bilious vomiting and bowel obstruction

- **Clinical trigger:** Bilious vomiting in a neonate requires urgent evaluation for malrotation, volvulus or intestinal obstruction.
- **Management priority:** Assess perfusion, hydration and abdominal findings, begin stabilization and urgently involve pediatric surgery.
- **Exam trap:** A soft early abdomen or initially reassuring assessment does not exclude time-critical obstruction.

القيء الأخضر Bilious في newborn هو Surgical red flag الأولى الصورة أو البطن لين أن لمجرد تطمئن لا ؛ غير واضحة.

Topic 28: Pyloric stenosis versus bilious obstruction

- **Key discriminator:** Pyloric stenosis classically produces progressive projectile non-bilious vomiting, unlike bilious vomiting from intestinal obstruction.

- **High-yield fact:** The supplied pyloric-stenosis pattern includes hypochloremic, hypokalemic metabolic alkalosis.
- **Management priority:** Correct dehydration and electrolyte abnormalities before definitive pyloric intervention.
- **Common mistake:** Proceeding directly to definitive treatment without first assessing and stabilizing the fluid/electrolyte state.

Projectile non-bilious vomiting عند الرضيع يوحي بـ pyloric stenosis، أما bilious vomiting فيوجّه إلى obstruction أخطر ويغيّر الأولوية.

Topic 29: Intussusception: intermittent pain or lethargy

- **Clinical trigger:** Episodic abdominal pain, pallor, lethargy or bloody stool in a child should raise intussusception concern.
- **Key discriminator:** Ultrasound is a preferred investigation; a normal abdominal radiograph does not reliably exclude intussusception.
- **Management priority:** Assess for perforation, peritonitis, instability and appropriate surgical/radiologic reduction support.
- **Exam trap:** Absence of the full classical symptom triad must not end evaluation.

Intussusception قد يظهر على شكل episodic pain أو pallor أو lethargy حتى دون triad كاملة؛ ultrasound أهم من الاعتماد على plain film.

Topic 30: Pediatric appendicitis: ultrasound and ongoing probability

- **Core concept:** Pediatric appendicitis evaluation combines clinical probability, serial examination and imaging instead of using one test alone.
- **Management priority:** Ultrasound is a radiation-sparing initial study for many children with suspected appendicitis.
- **Exam trap:** A nonvisualized appendix or equivocal ultrasound does not safely terminate evaluation when concern persists.

في suspected appendicitis، ultrasound غير الحاسم لا ينفي المرض إذا بقي clinical suspicion ؛ أعد الفحص واختر imaging أو consult وفق الحالة.

Topic 31: Gastroenteritis: oral versus intravenous rehydration

- **Management priority:** Use oral rehydration solution for uncomplicated dehydration when the child is able to drink and not in shock.
- **High-yield fact:** Continue breastfeeding and age-appropriate feeding during rehydration when clinically appropriate.
- **Clinical trigger:** Severe dehydration, shock, inability to drink or persistent vomiting calls for escalation to monitored IV/IO fluids.
- **Common mistake:** Repeating isotonic boluses without reassessment for fluid response and overload.

ORS هو الأساس عندما يستطيع الطفل الشرب ولا توجد shock. إذا فشلت enteral rehydration أو ظهرت علامات severe dehydration فأعد تقييم IV/IO.

Topic 32: Diarrhea: bloody stool, infection testing and antibiotics

- **Clinical trigger:** Bloody stool, severe abdominal pain, systemic illness, very young age or public-health concern requires targeted evaluation.
- **Management priority:** Most uncomplicated watery diarrhea is treated with rehydration and safety-netting rather than routine stool tests and antibiotics.
- **Clinical Pearl:** Antimicrobial choice depends on suspected pathogen, severity and risk rather than being applied to all diarrheal presentations.
- **Exam trap:** Do not substitute antibiotic treatment for fluid and perfusion assessment.

Watery diarrhea بدون red flags لا يحتاج stool culture أو antibiotics دائمًا؛ bloody stool أو systemic illness يغيّر المسار.

Topic 33: Pediatric UTI: specimen quality and population limits

- **Core concept:** Pediatric UTI diagnosis depends on age, symptoms, urine collection quality, urinalysis and appropriate culture.
- **Number/threshold:** AQRAB cites an AAP 2026 pediatric UTI guidance population of 8 days to 5 years, with specified exclusions such as complex urinary anomalies.
- **Exam trap:** A bag-collected urine culture must not be used alone to establish UTI in a child unable to provide a clean-catch specimen.
- **Management priority:** Tailor antibiotic treatment to site, severity, culture, allergies and renal function.

Bag urine قد يضلل بسبب contamination ولا يثبت UTI على القادر غير الطفل عند clean catch. راجع طريقة الجمع وحدود الـguideline.

Topic 34: Testicular torsion and infected urinary obstruction

- **Clinical trigger:** Abrupt severe unilateral testicular pain or a child with infected obstruction requires urgent urologic evaluation.
- **Red flag:** Atypical duration, preserved cremasteric reflex, normal urinalysis or equivocal ultrasound does not reliably exclude torsion.
- **Management priority:** Obtain urgent surgical input when suspicion is high instead of waiting for falsely reassuring imaging.
- **Differential point:** CLEAN V3 separately notes hydrocele as a common source of painless pediatric scrotal swelling.

Sudden unilateral scrotal pain هو time-critical torsion حتى لو Doppler لا تؤخر؛ غير حاسم urology consult عندما الشك عالٍ.

Topic 35: Pediatric DKA: insulin timing and cerebral-injury vigilance

- **Management priority:** Pediatric DKA treatment uses measured fluids, insulin, potassium and serial neurologic/metabolic monitoring.
- **Number/threshold:** The cited CPS example starts insulin 0.05-0.1 units/kg/h after an hour of fluids when potassium is above 3.0 mmol/L, without an IV insulin bolus.
- **Red flag:** Suspected cerebral injury warrants immediate clinical treatment without waiting for head CT.
- **Exam trap:** Do not apply adult DKA dosing or bolus practices automatically to a child.

في Pediatric DKA راقب potassium وfluids قبل insulin؛ المصدر يشدد على عدم إعطاء IV bolus للإنسولين وعلى أعراض cerebral injury.

Topic 36: Pediatric status epilepticus: five-minute action threshold

- **Number/threshold:** The cited NICE pathway treats continuous convulsion for five minutes or longer as an emergency requiring a status epilepticus pathway.
- **Management priority:** Check ABC, bedside glucose, and reversible causes; use the approved fastest safe benzodiazepine route and count prehospital doses.
- **High-yield fact:** If seizures persist after two appropriate benzodiazepine doses, follow the local second-line pathway and prepare PICU support.
- **Exam trap:** Do not wait for the historical 30-minute definition or continue giving benzodiazepines without escalating.

إذا استمر convulsive seizure خمس دقائق ابدأ العلاج، واحسب جرعات benzodiazepine غير التكرار السابقة؛ المحدود يعرّض التنفس للخطر.

Topic 37: Febrile seizure: simple versus concerning features

- **Number/threshold:** AQRAB defines the cited simple febrile seizure pattern as generalized, less than 15 minutes, once in 24 hours and typically age 6-60 months, after suitable evaluation.
- **Key discriminator:** Focal seizures, prolonged activity, recurrent events or delayed return to baseline change risk assessment.

• **Clinical Pearl:** Families may be advised about recurrence; younger age, family history and short fever duration before seizure increase recurrence likelihood.
• **Exam trap:** Fever plus convulsions is not automatically a benign febrile seizure; consider CNS infection, ingestion or metabolic disorder.

Simple febrile seizure تُعرّف في المصدر حسب العمر والمدة والتعميم وعدم التكرر، لكن لا يلغي fever meningitis أو metabolic causes.

Topic 38: Neurologic deterioration: glucose, trauma and CNS infection

• **Management priority:** With seizure or altered mental status, stabilize ABC, check glucose early and consider trauma, infection, metabolic disorder, poisoning or raised intracranial pressure.
• **Red flag:** Persistent altered mental status, meningism, focal deficit or severe headache warrants targeted urgent assessment.
• **Common mistake:** Explaining persistent neurologic abnormality by fever or a preceding seizure without checking for additional causes.

قبل تسمية altered consciousness بأنه postictal افحص glucose وكرر والحرارة والرض والعدوى والتسمم، neurological assessment.

V. Trauma, Musculoskeletal, Hematologic & Safeguarding

Topic 39: Pediatric trauma primary survey and temperature protection

• **Core concept:** Pediatric trauma assessment starts with a structured primary survey and immediate treatment of airway, breathing and hemorrhage threats.
• **Management priority:** Protect the cervical spine when indicated, prevent hypothermia, control bleeding and repeat the assessment after each intervention.
• **Key discriminator:** Tachycardia, weak pulses, delayed refill and altered mental state may indicate compensated shock before hypotension.
• **Red flag:** CLEAN V3 describes SCIWORA as a pediatric spinal injury pattern that can occur despite initially unrevealing radiographs.
• **Exam trap:** Do not apply adult trauma decision thresholds indiscriminately to children.

اعمل ABCDE مع C-spine وhemorrhage control من البداية، وراقب temperature ؛ لا تنتظر imaging لتصحيح life threats.

Topic 40: Pediatric head injury and possible abusive trauma

• **Clinical trigger:** Injury severity, inconsistent history or developmental mismatch warrants evaluation for non-accidental trauma without presuming intent.
• **Management priority:** Suspected acute abusive head trauma may require urgent head CT and specialist child-protection and pediatric assessment.
• **Number/threshold:** CLEAN V3 cites CT head within one hour for a specific high-risk suspected non-accidental injury scenario in its NICE-derived pathway.
• **Common mistake:** Deferring immediate medical stabilization while seeking forensic details, or accepting a normal examination as proof of safety.

في head injury مع قصة غير متوافقة مع التطور أو إصابات غير مفسرة، عالج التهديد أولًا ثم وثّق بموضوعية وفعّل safeguarding.

Topic 41: Occult physeal injury and Salter-Harris classification

• **Core concept:** Pediatric growth plates are vulnerable to injury; Salter-Harris classification describes patterns involving the physis.
• **Key discriminator:** Persistent focal physeal tenderness may remain clinically important despite unremarkable initial radiographs.

- **Management priority:** Immobilize when indicated, document neurovascular function and arrange reassessment or further investigation.
- **Exam trap:** A normal first x-ray does not rule out an occult pediatric physeal or nondisplaced fracture.

الـgrowth plate قد تُصاب رغم x-ray أولية طبيعية؛ إذا بقيت focal tenderness احمِ الطرف ورتّب follow-up.

Topic 42: Supracondylar fracture and compartment danger signs

- **Clinical trigger:** Displaced supracondylar elbow fracture can threaten the brachial circulation or adjacent nerves.
- **Management priority:** Document pulses, perfusion, motor and sensory function before and after splinting and seek timely orthopedic input.
- **Red flag:** Pain out of proportion, increasing analgesia requirement, tense swelling or evolving neurologic/vascular signs raises compartment syndrome concern.
- **Common mistake:** Concluding the child is safe based on appearance alone without a structured neurovascular examination.

في supracondylar fracture افحص median/AIN وradial وulnar nerves والنبضات قبل وبعد splint؛ worsening pain أو swelling إنذار.

Topic 43: Safeguarding: developmental mismatch is a clinical clue

- **Clinical trigger:** A nonambulatory infant with a fracture, changing history or injuries of different ages warrants careful child-protection assessment.
- **Management priority:** Treat urgent injuries and current safety first; use neutral open questions and document objective findings and exact spontaneous statements.
- **Differential point:** Consider accidental injury and medical mimics alongside possible abuse; no single sign by itself proves maltreatment.
- **Common mistake:** Performing repeated leading forensic interviews or assuming a normal single radiograph excludes risk.

وجود fracture في nonambulatory infant خذ history أو تغير القصة يرفع الشك ولا يثبت الإساءة وحده؛ محايدة ووثّق ألفاظ المرافق.

Topic 44: Skeletal survey and the safe disposition decision

- **Number/threshold:** AQRAB cites skeletal survey for children under two years with increased concern for physical abuse, and selected older children after specialist review.
- **Number/threshold:** When significant concern persists, follow-up skeletal imaging after about two to three weeks may reveal healing injuries.
- **Management priority:** Involve pediatric radiology or child-protection specialists and follow local reporting and safe-disposition processes.
- **Exam trap:** A reporting concern does not require definitive proof before appropriate safeguarding referral.

عند suspicion of physical abuse في الطفل الصغير قد تحتاج skeletal survey مع ومتابعة بعد 2-3 أسابيع، فريق حماية وقرار خروج آمن.

Topic 45: Pediatric epistaxis and dental trauma

- **Management priority:** For routine anterior pediatric epistaxis associated with local irritation or URI, CLEAN V3 describes firm nasal compression for about 5-10 minutes.
- **Clinical Pearl:** CLEAN V3 discusses soft diet and dental review after a tender but immobile incisor in a particular source case; the decision is case-dependent.
- **Exam trap:** Isolated dental-trauma timing figures in CLEAN V3 depend on the specific avulsion scenario and should not be generalized across all teeth.

في anterior epistaxis استخدم ضغطًا مباشرًا على الأنف، ولا تخلط تعليمات dental trauma بين أسنان دائمة ولبنية دون سياق.

Topic 46: Pediatric sickle cell: recognize four high-risk systems

- **High-yield fact:** AQRAB identifies fever, chest complications, neurologic events and splenic emergencies as four important risk areas in pediatric sickle cell disease.
- **Clinical trigger:** Fever, respiratory symptoms, focal neurologic findings or sudden abdominal/splenic concerns should prompt focused urgent assessment.
- **Management priority:** Evaluate the serious complication pattern and obtain appropriate pediatric and hematology support.

في Sickle Cell child، احفظ fever وchest وbrain وspleen مع؛ لا تتعامل كمناطق خطر fever أو weakness كأعراض بسيطة دون تقييم.

Topic 47: Child and adolescent behavioral safety and self-harm

- **Management priority:** CLEAN V3 specifies safeguarding and mental-health assessment before disposition after deliberate self-harm in a child.
- **High-yield fact:** Care planning should use developmentally appropriate communication, family context when safe and a suitable follow-up/safety plan.
- **Common mistake:** Discharging before addressing safety concerns merely because the acute physical examination has normalized.

أي deliberate self-harm يحتاج safeguarding وmental-health evaluation قبل dischargeعلى؛ القرار ليس مبنيًا تراجع الأعراض فقط.

Topic 48: Neonatal jaundice and clinical context

- **High-yield fact:** CLEAN V3 identifies physiological jaundice as a common explanation for neonatal hyperbilirubinemia.
- **Exam trap:** That isolated source statement does not provide a safe bilirubin treatment threshold or an age-specific jaundice pathway; no such threshold is inferred here.
- **Management priority:** Evaluate the newborn clinically and avoid replacing a complete neonatal assessment with an isolated label.

Physiologic jaundice مذكور كسبب شائع، لكن لا تحوّل ذلك إلى قاعدة تطمئن لكل طفل دون عمر ظهور الحالة أو العلامات السريرية.

Topic 49: Atraumatic pediatric limp and adolescent growth pain

- **Differential point:** CLEAN V3 lists transient synovitis, Legg-Calve-Perthes disease and slipped capital femoral epiphysis as important atraumatic pediatric hip-pain or limp causes.
- **Management priority:** Use age, mechanism, focal tenderness, gait and neurovascular examination to direct investigation or follow-up.
- **High-yield fact:** CLEAN V3 associates Osgood-Schlatter disease with knee pain during adolescent growth spurts, particularly in active youth.
- **Management priority:** For the selected Osgood-Schlatter pattern, the source describes activity modification and quadriceps stretching.
- **Common mistake:** Ignoring persistent focal bony tenderness simply because an early radiograph is normal.

في pediatric limp لا تخلط؛ راجع العمر وموضع الألم وقدرة المشي transient synovitis أو Perthes أو SCFE مع إصابة رياضية تلقائيًا.

Topic 50: Pediatric esophageal foreign body and dysphagia clues

- **High-yield fact:** CLEAN V3 identifies the proximal esophagus at the cricopharyngeus as a common pediatric foreign-body impaction site.
- **Differential point:** Stomatitis, pharyngitis and peritonsillar abscess are listed among infectious causes of pediatric dysphagia.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 195

40 mg/kg	Source-dependent regimen: AQRAB's four-year-old status-epilepticus question gives levetiracetam 40 mg/kg over 5 minutes after failure of two benzodiazepine doses; its accompanying pediatric explanation lists 30–60 mg/kg over 5 minutes, maximum 3 g. This
30–60 mg/kg, 3 g	levetiracetam 40 mg/kg over 5 minutes after failure of two benzodiazepine doses; its accompanying pediatric explanation lists 30–60 mg/kg over 5 minutes, maximum 3 g. This pediatric example must not be conflated with the separate selected adult 60 mg/kg
60 mg/kg	accompanying pediatric explanation lists 30–60 mg/kg over 5 minutes, maximum 3 g. This pediatric example must not be conflated with the separate selected adult 60 mg/kg (maximum 4.5 g) regimen documented in the master.
0.01 mg/kg, 1 mg, 0.5 mg	dosing dosing and mL 0.01 mg/kg, max 1 mg in cited 0.01 mg/kg, capped at 0.5 mg Same mg/kg does not imply Dose

- **Management priority:** First determine whether airway obstruction, respiratory compromise or inability to handle secretions threatens immediate safety.
- **Exam trap:** The isolated study-guide text does not supply a universal imaging or endoscopic-removal rule for every pediatric esophageal foreign body.

قد يكون dysphagia عند الطفل. لكن esophageal foreign body موضع شائع لاحتباس الـcricopharyngeus ؛ افحص السياق قبل الافتراض stomatitis أو pharyngitis بسبب.

Comparison Tables

Pediatric arrest vs neonatal resuscitation

Feature	Pediatric arrest	Neonatal resuscitation	Exam point
Main initial problem	Often respiratory failure or shock before arrest	Failure to establish effective ventilation at birth	Treat the physiology and age-specific pathway
Airway strategy	Effective bag-mask ventilation may be appropriate	Establish visible lung inflation before compressions	Do not pursue an advanced airway at expense of oxygenation
Compression ratio	Use current pediatric arrest algorithm	3:1 after HR <60 following 30 s effective ventilation	Never transfer adult 30:2 to the neonate

Croup vs threatening supraglottic obstruction

Feature	Croup	Supraglottic obstruction	Exam point
Typical clues	Barking cough, hoarseness, inspiratory stridor	Drooling, toxic appearance, muffled voice or tripod	Secretions and appearance are key
Initial action	Severity assessment, steroid +/- nebulized epinephrine per pathway	Minimize agitation and prepare expert-controlled airway	Do not force oral examination
Disposition clue	Observe if persistent/recurrent resting stridor	Airway-focused expert disposition	Improvement of one sign is not enough

Asthma vs bronchiolitis in children

Feature	Asthma	Bronchiolitis	Exam point
Differential	Wheeze may reflect bronchospasm	Often viral infant lower-airway illness	Age, history and presentation matter
Management	Age-specific asthma pathway	Mostly oxygenation, hydration and feeding support	Avoid routine asthma drugs for bronchiolitis
Red flags	Silent chest, fatigue, altered consciousness	Apnea, exhaustion, poor feeding or hypoxemia	Watch for respiratory failure

Sinus tachycardia vs SVT

Feature	Sinus tachycardia	SVT	Exam point
Onset and course	Often variable with underlying illness	Often abrupt, sustained and regular	Use history and serial rhythm assessment
ECG clues	P waves and rate variation may help	P-wave visibility and regularity help	Age-specific rates and P waves matter
Management axis	Treat cause and reassess	Stable: vagal/adenosine; unstable: synchronized cardioversion	Assess perfusion before rhythm treatment

Bilious emesis vs pyloric stenosis

Feature	Bilious emesis	Pyloric stenosis	Exam point
Vomiting	Green/bilious	Progressive projectile non-bilious	Neonatal bilious vomiting is a surgical red flag
Major concern	Malrotation/volvulus or obstruction	Pyloric obstruction and electrolyte depletion	Do not delay urgent surgical assessment
Acid-base	Requires evaluation	Hypochloremic, hypokalemic alkalosis in cited pattern	Correct fluids/electrolytes before definitive treatment

PASS 9 — Source-backed clinical recovery | Domain 09

Pediatrics & Neonatology

dosing or medical-correctness certification.

Palivizumab and RSV

Key discriminator: AQRAB identifies palivizumab as RSV immune prophylaxis, not a vaccine or treatment of the current RSV episode.

Palivizumab للوقاية المناعية بحسب المصدر، وليس علاجًا للنوبة الحالية.

Early repeat concussion risk

Source-specific clinical pearl: CLEAN V3 reports greater recurrence of concussion during the next 10 days, especially with rapid return to play; its quoted 80% same-season figure is source-specific.

يربط المصدر تكرر concussion بالعودة المبكرة للرياضة؛ النسبة المذكورة تخص سياق السؤال.

Pediatrics & Neonatology

Precisely source-linked additions after the priority-safety review. Not independent guideline verification.

Topic: Pediatric status epilepticus: source-specific levetiracetam example

Source-dependent regimen: AQRAB's four-year-old status-epilepticus question gives levetiracetam 40 mg/kg over 5 minutes after failure of two benzodiazepine doses; its accompanying pediatric explanation lists 30–60 mg/kg over 5 minutes, maximum 3 g. This pediatric example must not be conflated with the separate selected adult 60 mg/kg (maximum 4.5 g) regimen documented in the master.

جرعة السؤال تخص طفلاً بعد فشل جرعتين من Benzodiazepine. لا تخلط نطاق الأطفال مع جرعة البالغين أو الحد الأقصى المختلف.

Topic: SCIWORA: pediatric age and safe disposition

High-yield fact: CLEAN V3 describes spinal cord injury without radiographic abnormality (SCIWORA) especially in children younger than 8 years, with spinal precautions and transfer to pediatric trauma expertise in the cited scenario.

الطفل قد يُظهر Spinal Cord Injury رغم صور أولية غير كاشفة؛ العمر هنا وارد في وصف المصدر ولا يغني عن الفحص السريري.

Topic: Retropharyngeal abscess: source age pattern

Clinical trigger: CLEAN V3 describes retropharyngeal abscess as a potentially life-threatening pediatric airway infection, commonly presented in children aged 3–5 years in its teaching passage. Transfer to an operating-room airway plan is described in a source case rather than as a universal instruction.

العمر 3–5 سنوات نقطة تمييز امتحانية بالمصدر؛ قرار Airway يتم حسب شدة الحالة وليس العمر وحده.

> Do not merge case-specific, adult and pediatric regimens into a single dose. Source-dependent conflicts remain unresolved.

Simple febrile seizure vs convulsive status

Feature	Simple febrile seizure	Convulsive status	Exam point
Duration	<15 minutes in a simple pattern	Ongoing convulsion >=5 minutes is emergency threshold	Do not wait 15 minutes to treat ongoing convulsions
Pattern	Generalized, one in 24 hours, age 6-60 months	May be prolonged or recurrent despite benzodiazepine	Review prehospital doses
Priority	Evaluate for mimics and safe disposition	ABC/glucose, benzodiazepine and second-line escalation	Fever does not exclude CNS infection

Compensated shock vs decompensated physiology

Feature	Compensated shock	Decompensated shock	Exam point
Blood pressure	May remain normal	Hypotension can be a late sign	Blood pressure alone is insufficient
Perfusion	Weak pulse, altered refill or mentation	Progressive poor perfusion/organ signs	Trend response after intervention
Treatment	Target cause with reassessment	Escalate circulatory/respiratory support	Avoid reflex repeated fluids

Pediatric arrest epinephrine vs anaphylaxis epinephrine

Feature	Arrest epinephrine	Anaphylaxis epinephrine	Exam point
Route	IV/IO for cited cardiac-arrest pathway	IM anterolateral thigh for anaphylaxis	Never copy an IV bolus into IM treatment or reverse
Concentration	0.1 mg/mL in cited arrest dosing	1 mg/mL in cited anaphylaxis dosing	Check mg/kg, mg/mL, total mg and mL
Dose	0.01 mg/kg, max 1 mg in cited arrest pathway	0.01 mg/kg, capped at 0.5 mg in cited ASCIA pathway	Same mg/kg does not imply same preparation or route

Occult physeal injury vs benign imaging

Feature	Occult physeal injury	Benign imaging	Exam point
X-ray	May be normal with occult physeal fracture	Normal film without concerning examination	Persistent focal tenderness overrides false reassurance
Clinical priority	Protect and arrange follow-up	Review developmental and injury context	Document neurovascular status

Rapid Recall: Must-Know Facts

1. Pediatric physiology may deteriorate before blood pressure falls.
2. Appearance, work of breathing and circulation are the first pediatric visual clues.
3. Reassess perfusion and breathing after every fluid aliquot.
4. A quiet, exhausted child may be sicker than a crying child.
5. Pediatric airway positioning differs because of the large infant occiput.
6. Prioritize bag-mask oxygenation over repeated failed laryngoscopy.
7. Continuous waveform capnography helps confirm tracheal placement.
8. Normal SpO2 on oxygen does not exclude sedation-induced hypoventilation.
9. Pediatric drugs require kg, mg/kg, max dose, mg/mL and mL verification.
10. IM epinephrine for anaphylaxis is not the IV/IO arrest preparation.
11. Pediatric arrest frequently follows respiratory failure or shock.
12. Asystole and PEA are common initial pediatric arrest rhythms in CLEAN V3.
13. Early epinephrine is emphasized for non-shockable pediatric arrest.
14. Avoid routine arrest calcium or bicarbonate without a specific indication.
15. Do not stop resuscitation based on one ETCO2 value alone.
16. Neonatal apnea/gasping or HR <100 after initial steps requires ventilation.
17. A newborn should receive effective ventilation within the first 60 seconds.
18. Start neonatal 3:1 compressions only if HR <60 after 30 seconds of effective ventilation.
19. Neonatal temperature target in AQRAB: 36.5-37.5°C.
20. Feeding diaphoresis and poor growth can indicate congenital heart disease.

21. Stable narrow-complex pediatric SVT may respond to vagal maneuvers or adenosine.
22. Unstable pediatric SVT requires synchronized cardioversion.
23. AQRAB cites cardioversion 0.5-1 J/kg, then 2 J/kg if ineffective.
24. Febrile-infant algorithms apply only to their defined age and eligibility criteria.
25. Antipyretic response does not exclude serious bacterial infection.
26. Croup causes barking cough, hoarseness and inspiratory stridor.
27. Drooling, toxic appearance or tripod posture suggests a dangerous airway alternative.
28. Bronchiolitis management is primarily respiratory and hydration support.
29. A silent chest with fatigue can signal impending respiratory failure.
30. Pediatric pneumonia with substantial effusion may benefit from chest ultrasound.
31. Bilious vomiting in a neonate is an urgent surgical red flag.
32. Progressive projectile non-bilious vomiting suggests pyloric stenosis.
33. Intussusception can present with episodic pain or lethargy without a full triad.
34. A nondiagnostic ultrasound does not exclude appendicitis when suspicion persists.
35. Oral rehydration is preferred when a child can drink and is not in shock.
36. Bag urine culture alone cannot establish pediatric UTI.
37. Suspected testicular torsion warrants urgent surgical input despite equivocal imaging.
38. Pediatric DKA insulin is not administered as a routine IV bolus in the cited pathway.
39. Treat convulsive status after approximately five minutes; count earlier benzodiazepine doses.
40. Persistent physeal tenderness can justify immobilization despite a normal first x-ray.

Conflict Resolution Register

Direct, mutually incompatible clinical claims confirmed between the two documents in this section: 0. The source-specific differences below are preserved without external adjudication; they are not automatically true direct conflicts.

Source-dependent context and protocol variations

Topic	CLEAN V3	AQRAB v18	Nature / handling
Repeat IM epinephrine interval	CLEAN V3: pediatric IM repeat dose every 5-10 min when needed.	AQRAB: 5 min in its ASCIA pathway, and a general 5-15-min protocol example.	Different source-dependent protocols; not made into a single universal interval.
Pediatric fluid volume	CLEAN V3: a 20 mL/kg IV saline option appears in a bronchiolitis case with limited context.	AQRAB: fluids by mechanism and response; older trauma fluid examples explicitly need protocol review.	Case-specific number versus response-guided resuscitation, not a general bolus rule.
Croup dexamethasone	CLEAN V3: oral 0.15 mg/kg single-dose example.	AQRAB: severity-based steroid pathway and a protocol-specific maximum of 12 mg, without a single universal dose in the core summary.	Different components of cited pathways; no dose chosen to supersede another.
Newborn ventilation details	CLEAN V3: five initial inflation breaths and example inflation pressures appear in one neonatal pathway.	AQRAB 2025 pathway: begin ventilation within 60 seconds; compressions only after 30 seconds of effective ventilation.	Different source pathway granularity; device/pressure examples not assumed universally interchangeable.
Pediatric asthma guideline year	CLEAN V3 carries historical question-derived principles.	AQRAB states both GINA 2025 reference text and a 2026 strategy update in the pediatric respiratory pages.	Source-era difference; no external update used to select one.
Pediatric UTI guideline scope	CLEAN V3 includes undifferentiated pediatric urologic facts.	AQRAB cites an AAP 2026 population of 8 days to 5 years with exclusions.	A newer and narrower source-specific framework, not automatically generalizable.

Classification for any genuine unresolved cross-file conflict: Conflict — source-dependent. No unverified protocol was selected as more correct than the other.

Metric	Recorded value
Final integrated clinical topics	50
Integrated core English points	204
CLEAN V3 only points	26
AQRAB only points	150

Metric	Recorded value
Shared concepts merged in point form	28
Comparison tables	9
Rapid Recall facts	40
Confirmed direct cross-file conflicts	0
Source-dependent / context differences documented	6
Confirmed incomplete/unusable source groups	6

Counting method: every final English body bullet is one point. Each point was manually attributed to CLEAN V3 only, AQRAB only or both during drafting. Arabic explanations, tables and Rapid Recall are not counted twice. Counts refer to edited output, not a claim of full machine proof of every original sentence.

Incomplete source material and documented scope limits

Incomplete in source — not included. [CLEAN V3 para 3157] Bronchiolitis admission line ends with a bare "1." and lacks the numbered criteria; it is not turned into an admission rule.

Incomplete in source — not included. [CLEAN V3 para 3160] Nebulized adrenaline "0.5 mg/kg" lacks a fully specified preparation and condition; the dose is excluded.

Incomplete in source — not included. [CLEAN V3 para 3174] "Racemic epinephrine q15 mins, HFNC, heliox" is tied to a cropped case and not used as a general croup sequence.

Incomplete in source — not included. [CLEAN V3 para 3177] An isolated cuffed tube-size answer without full age or calculation context is not converted into a formula.

Incomplete in source — not included. [CLEAN V3 para 3199] Avulsed-tooth prognosis and replantation-timing percentages lack a confirmed permanent/primary tooth context and are excluded.

Incomplete in source — not included. [CLEAN V3 para 3202] Refractory pediatric SVT amiodarone is stated only as a specialist-discussed example; it is not a routine unsupervised pathway.

Incomplete in source — not included. [CLEAN V3 para 3173] A 2017 malaria mortality share is historical and not retained as a current pediatric risk statistic.

Incomplete in source — not included. [AQRAB PDF pp. 273-274, 277-279, 286-287, 290-291, 296-297, 300, 303-304, 306-307] Original image/infographic panels were visually checked; relevant recognizable facts integrated from source while unrelated image fragments were not imported.

Incomplete in source — not included. [AQRAB PDF pp. 394-440] Some selected recall lines are question-stem-dependent or truncated; only independently supported pediatric claims were retained.

Source-limitation count: six incomplete/unusable CLEAN V3 examples were excluded; one refractory SVT example was retained only in specialist-qualified context and two entries document visual/appendix inspection, not exclusions.

Visual source review: Pediatrics and neonatal panels within AQRAB PDF pp. 272-323 were examined for usable clinical information. Unrelated clipped panels, repeated illustrations, URLs, production labels and reference-only material were not appended to the content.

External-information confirmation: This document introduces no medical facts from the internet, independent clinical reference books, memory of guidelines or outside sources. All medical content is source-limited to the two supplied documents.

Clinical-use limitation: The documents are examination review material, not independent validation of current bedside protocols. Critical treatment decisions require locally approved protocols.

addendum count is not part of the original Merge Quality Note editorial tally. No external clinical source was consulted.

Pediatrics & Neonatology | Pass 8 | Explicit CLEAN V3 source locations

Biliary atresia — obstructive neonatal jaundice

• High-yield fact: CLEAN V3 describes biliary atresia as a disorder of intrahepatic and extrahepatic bile ducts leading to obstructive jaundice and, untreated, progressive cirrhosis.

عند اليرقان الانسدادي عند الرضيع، يذكر المصدر Biliary Atresia وعواقب تأخر علاجها.

Spasmodic croup — recurrent attacks

• High-yield fact: Recurrent episodes of croup-like cough, with or without respiratory distress, suggest spasmodic croup in CLEAN V3.

تكرر نوبات السعال المشابهة لـCroup هو المفتاح المذكور لـSpasmodic Croup.

Pediatrics & Neonatology

Supplementary Clinical Points

Topic: Croup: parainfluenza subtype

High-yield fact: CLEAN V3 identifies parainfluenza virus type 1 as a commonly cited principal cause of croup; this is an etiologic review fact, not a test result establishing the cause in an individual child.

يربط المصدر croup مع Parainfluenza type 1؛ التشخيص السريري لا يثبت الفيروس بعينه.

Source evidence: C03167: DOCX paragraph 3167

Pediatrics & Neonatology. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 202

0.15 mg/kg	pathway and a protocol- Different components of cited CLEAN V3: oral 0.15 mg/kg Croup dexamethasone specific maximum of 12 mg, pathways; no dose chosen to
12 mg	CLEAN V3: oral 0.15 mg/kg Croup dexamethasone specific maximum of 12 mg, pathways; no dose chosen to single-dose example.
0.5 mg/kg	criteria; it is not turned into an admission rule. Incomplete in source — not included. [CLEAN V3 para 3160] Nebulized adrenaline "0.5 mg/kg" lacks a fully specified preparation and condition; the dose is excluded.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 202

Hemolytic uremic syndrome - Shiga toxin [#070]

Key discriminator: Shiga-toxin-producing E. coli can precede hemolytic uremic syndrome in children with bloody diarrhea. .EHEC.

قد تسبق عدوى E. coli المنتجة لسم Shiga متلازمة HUS بعد إسهال دموي عند الأطفال.

Nephrotic syndrome - pediatric presentation [#072]

High-yield fact: Childhood nephrotic syndrome may first present with periorbital swelling and proteinuria.

الوذمة حول العينين مع بروتين البول قد تكون التظاهر الأول للمتلازمة النفروزية عند الطفل.

Pediatric sulfonylurea ingestion - hypoglycemia [#073]

Key discriminator: Small accidental sulfonylurea ingestions in children may cause severe or recurrent hypoglycemia.

حتى الابتلاع العرضي الصغير لأدوية Sulfonylurea عند الطفل قد يسبب نقص سكر شديدًا أو ناكسًا.

EDIC / Emergency & Critical Care

10 — Obstetrics & Gynecology

I. Early Pregnancy & Gynecologic Emergencies

Topic 1: Sexually Transmitted Infections & Pelvic Inflammatory Disease

- Core concept: PID is a clinical diagnosis with a deliberately low treatment threshold in at-risk patients with pelvic/lower abdominal pain and cervical-motion, uterine, or adnexal tenderness when no better cause is identified.
- Exam trap: Do not delay empiric treatment for STI test results when criteria are met.
- Management priority: Evaluate pregnancy and urgent mimics including ectopic pregnancy, appendicitis, and ovarian torsion.
- Management priority: Obtain gonorrhea/chlamydia testing and assess for complications such as tubo-ovarian abscess.
- Number/threshold: Hospitalize for pregnancy, severe illness, abscess, inability to tolerate outpatient therapy, or failure to improve; reassess within 72 hours .
- Exam trap: Do not wait for NAAT results before indicated empiric therapy.

تشخيص PID لا تنتظر نتائج STI غالبًا سريري؛ إذا توفرت معايير العلاج، لكن استبعد ectopic pregnancy وtorsion أولًا.

Topic 2: Urologic Emergencies — Infected Stone & Torsion

- Core concept: An infected obstructed kidney with sepsis or anuria is a urologic emergency: start antibiotics and resuscitation, obtain cultures, and urgently decompress by ureteral stent or nephrostomy; delay definitive stone removal until infection resolves.
- High-yield fact: Testicular torsion is time-critical: sudden unilateral scrotal pain, nausea, high-riding/transverse testis, or strong clinical suspicion warrants immediate urologic/surgical review.
- High-yield fact: Ultrasound must not delay exploration when suspicion is high.
- Management priority: Assess for sepsis/anuria and obtain cultures; begin antibiotics and arrange urgent decompression.
- High-yield fact: Ureteral stent and nephrostomy are accepted drainage options; definitive stone treatment follows infection control.
- High-yield fact: Suspected testicular torsion requires immediate urologic/surgical review and exploration when indicated.
- High-yield fact: Analgesia, NPO, and transfer/escalation should not delay definitive care.

في infected obstructed stone فكّر بـ urgent المشكلة ليست الألم فقط بل عدوى خلف انسداد؛ decompression وفي torsion الوقت مهم. مع علاج الإنتان.

Topic 3: Gynecologic Emergencies — Ovarian Torsion & Heavy Bleeding

- Core concept: Adnexal torsion presents classically with sudden intermittent unilateral abdominal/pelvic pain and nausea/vomiting; no clinical or imaging criteria reliably confirm it preoperatively, and preserved Doppler flow does not exclude torsion.

• High-yield fact: Acute heavy uterine bleeding: first assess hypovolemia and hemodynamic instability, stabilize, exclude pregnancy and nonuterine bleeding, then evaluate causes and use medical or surgical management according to stability, contraindications, and fertility goals.

• Exam trap: Do not delay surgery for a reassuring Doppler when clinical suspicion remains high.

طبيعي التشخيص؛ استمر بالتقييم السريري، خصوصًا مع ألم حوض Doppler قد لا ينفي في adnexal torsion مفاجئ.

II. Obstetric Emergencies & Maternal Care

Topic 4: Complications of Pregnancy in the ED

• Management priority: Assess maternal stability first; pregnancy does not change the need for prompt treatment of life-threatening illness.

• Management priority: Establish gestational age, symptoms, bleeding, pain, fetal movement and relevant obstetric history, then involve obstetrics early.

• High-yield fact: Early pregnancy with pain or bleeding requires pregnancy-location assessment and follow-up until ectopic pregnancy is excluded; rupture can cause internal hemorrhage and shock.

• High-yield fact: Severe-range hypertension, neurologic symptoms, epigastric/right-upper- quadrant pain or seizure raises concern for preeclampsia/eclampsia and requires urgent stabilization and obstetric management.

• High-yield fact: For trauma, resuscitate the mother, assess for hemorrhage/abruption and coordinate fetal assessment according to gestational age and local protocol.

• High-yield fact: Choose imaging and medication by maternal risk-benefit, not by withholding necessary care.

• High-yield fact: ACOG obstetric triage identifies vaginal bleeding, severe abdominal pain, seizure and major trauma as urgent presentations requiring immediate assessment.

• Exam trap: Do not use a single hCG value or an empty early ultrasound to label a pregnancy nonviable or give methotrexate without an appropriate diagnostic pathway.

• Exam trap: Do not attribute severe headache, visual symptoms, epigastric pain or seizure to normal pregnancy changes.

الحامل غير المستقرة تُنقذ أولًا. في نزف الحمل المبكر لا تعتبر الحمل داخل الرحم مؤكدًا قبل تحديد موقعه واستبعاد ectopic.

Topic 5: Medical Emergencies during Pregnancy

• Clinical Pearl: Sheehan syndrome is postpartum hypopituitarism; the supplied source identifies it as a complication occurring after delivery.

Sheehan syndrome هو postpartum hypopituitarism، أي قصور النخامى الذي يحدث بعد الولادة.

• Management priority: Treat maternal illness promptly while considering gestational age and fetal effects.

• Management priority: Assess maternal oxygenation, circulation, glucose, neurologic status, pregnancy stage and fetal concerns; involve obstetrics early for unstable or time-sensitive presentations.

• High-yield fact: Asthma, sepsis, pulmonary embolism, acute coronary syndromes, severe hypertension/eclampsia and metabolic emergencies require condition-specific treatment rather than therapeutic delay.

• Number/threshold: Severe-range blood pressure (systolic ≥160 or diastolic ≥110 mmHg) with persistent elevation or end-organ features is urgent.

• High-yield fact: For cardiac arrest with uterine fundus at/above the umbilicus, provide high-quality CPR with continuous manual left uterine displacement, prioritize airway, place IV/IO access above the diaphragm, activate the team and prepare resuscitative delivery from recognition, aiming to complete by 5 minutes if ROSC is not achieved.

• High-yield fact: AHA 2025 pregnancy-arrest algorithm: continuous manual left uterine displacement when fundus is at/above the umbilicus; prepare resuscitative delivery at arrest recognition and target completion by 5 minutes if no ROSC.

• High-yield fact: Maternal resuscitation is the primary fetal intervention; use standard defibrillation/CPR when indicated and remove fetal monitors during arrest care.

• High-yield fact: Persistent severe-range BP or neurologic/epigastric symptoms can indicate severe preeclampsia; urgently stabilize and involve obstetrics.

• High-yield fact: Normal pregnancy can alter symptoms and physiology; use disease-specific pathways for asthma, VTE, ACS, sepsis and metabolic illness.

المرض الحاد أثناء الحمل لا يبرر تأخير الإنعاش. اختر فحصًا وعلاجًا يناسب حالة الأم وعمر الحمل، واستدعِ obstetrics مبكرًا.

Topic 6: Drug Therapy in Pregnancy

• Core concept: Make medication decisions by balancing maternal benefit, fetal risk, gestational timing, dose, route, duration and available alternatives.

• High-yield fact: Untreated maternal disease can itself harm both patients.

• High-yield fact: Reconcile prescribed, over-the-counter, herbal and recreational exposures; identify urgent toxicity or withdrawal risk; consult a current drug-information source, obstetric/pharmacy expertise and local protocol.

• High-yield fact: Modern FDA labeling uses narrative risk summaries and evidence rather than the former A/B/C/D/X letter categories.

• Management priority: Avoid starting or stopping an essential chronic medicine abruptly without an individualized plan.

• High-yield fact: In emergencies, treat the mother with effective evidence-based care and document the risk-benefit reasoning.

• High-yield fact: FDA PLLR replaced the old pregnancy letter categories with narrative risk summaries, clinical considerations and available data; do not quote A/B/C/D/X as current labeling.

• High-yield fact: Assess gestational timing, dose, route, duration, maternal condition, risks of untreated disease and safer alternatives when reviewing medication during pregnancy.

• Exam trap: Do not equate ‘natural’ or OTC products with proven pregnancy safety.

• Number/threshold: The AQRAB source diagram labels organogenesis at 3-8 gestational weeks as the period of highest teratogenic vulnerability; this is a timing schematic, not a drug-specific safety assessment. [AQRAB PDF p. 329; original visual panel]

المرحلة المبكرة للـorganogenesis أكثر حساسية بحسب الرسم الأصلي، لكن تقييم الدواء يبقى خاصًا بالمادة والجرعة والتوقيت.

الدواء في الحمل يحتاج تقييم indication والجرعة والتوقيت وتأثيره على الأم والجنين؛ ليس كل علاج ممنوعًا تلقائيًا.

Topic 7: Labor & Emergency Delivery

• Core concept: First determine whether labor is established, the stage, gestational age, maternal stability, fetal status, presentation and likelihood of imminent birth.

• Management priority: Arrange obstetric transfer when safe; if crowning or birth is imminent, prepare to deliver where the patient is, summon neonatal help, protect privacy and use clean equipment.

• High-yield fact: Support controlled delivery of the head and body, assess for immediate maternal hemorrhage and newborn transition, and be ready for neonatal resuscitation.

• Number/threshold: ACOG defines active labor beginning at 6 cm; apply current obstetric definitions and local pathways for labor arrest rather than diagnosing dystocia early.

• Management priority: Escalate immediately for malpresentation, cord prolapse, shoulder dystocia, maternal instability or nonreassuring fetal status.

• High-yield fact: ACOG 2024: active labor begins at 6 cm; active-phase arrest requires no cervical change at ≥6 cm with ruptured membranes despite adequate duration of contractions/augmentation.

• High-yield fact: If birth is imminent, do not attempt unsafe transfer; call obstetric and neonatal teams and prepare the delivery space and resuscitation equipment.

• High-yield fact: After birth, assess maternal bleeding and uterine tone while evaluating newborn breathing, tone and heart rate; anticipate neonatal resuscitation if transition is poor.

• High-yield fact: Cord prolapse, malpresentation, shoulder dystocia, maternal instability and nonreassuring fetal status require immediate escalation.

• Number/threshold: The AQRAB fetal-monitoring panel gives a normal fetal heart-rate baseline of 110-160 beats/min and asks the examiner to assess variability, accelerations and decelerations. [AQRAB PDF p. 331; original visual panel]

• Key discriminator: The pictured Category III fetal tracing is abnormal and includes absent variability with recurrent late decelerations or a sinusoidal pattern; the panel calls for prompt evaluation. [AQRAB PDF p. 331; original visual panel]

Domain 10 — Medical Emergencies in Pregnancy

Pass 5 | Integrated source-derived clinical refinements

Management context: The AQRAB pregnancy infographic describes acute-asthma exacerbation management using inhaled beta2-agonists, short-course steroids and magnesium sulfate when indicated; its illustrated maternal oxygen-saturation goal is at least 95%. This is the cited figure pathway, not independent verification of a universal target.

الصورة تركز على علاج النوبة وعدم إهمال أكسجة الأم؛ قيمة الأكسجة مذكورة ضمن مسار المصدر وليست حكمًا عامًا خارج سياق الحالة.

Differential / testing pathway: The same AQRAB panel calls pregnancy hypercoagulable and identifies leg swelling, chest pain, dyspnea and hypoxia as DVT/PE triggers; its diagnostic examples include CTPA and V/Q scanning in the setting specified in the panel. The figure alone does not select a universal imaging sequence.

الحمل يرفع الاشتباه بالجلطات عند وجود الأعراض المناسبة؛ اختيار التصوير يعتمد على السياق ولا يُستنتج من الرسم وحده.

Placed with Medical Emergencies During Pregnancy; individual-source clinical-pathway context retained. No drug dose added.

- Management priority: For shoulder dystocia, the source image outlines the HELPERR sequence: call for Help, consider Episiotomy, McRoberts Legs position, suprapubic Pressure, internal Enter maneuvers, Remove posterior arm and Roll to an alternative position. [AQRAB PDF p. 331; original visual panel]
- Differential point: The source-image framework for postpartum hemorrhage lists the 4 Ts: Tone (uterine atony), Tissue (retained placenta), Trauma (lacerations/uterine rupture), and Thrombin (coagulopathy). [AQRAB PDF p. 331; original visual panel]
- Management priority: The AQRAB image lists uterine massage, uterotonics, source assessment and early escalation for postpartum hemorrhage; it specifically places tranexamic acid within 3 hours in its pathway. [AQRAB PDF p. 331; original visual panel]

في مخاض الطوارئ افحص baseline والـvariability، وتذكّر HELPERR عند shoulder dystocia. وفي نزف ما بعد الولادة صنّف السبب إلى 4 Ts قبل الاكتفاء بتكرار العلاج.

في الطوارئ أثناء الولادة: حدد المرحلة، واستدعِ المساعدة، واستعد لنزف ما بعد الولادة ومشكلات الوليد.

Topic 8: Trauma in Pregnancy

- Management priority: Use standard trauma priorities and stabilize the mother first; maternal resuscitation provides the best fetal resuscitation.
- High-yield fact: Screen for pregnancy when relevant, estimate gestational age and consider concealed hemorrhage, placental abruption, uterine injury and fetomaternal hemorrhage.
- Management priority: Use indicated imaging when it changes care; do not withhold necessary diagnostic imaging solely because of pregnancy.
- High-yield fact: After the primary survey and maternal stabilization, involve obstetrics and assess the fetus according to viability and local resources.
- Number/threshold: The 2025 EAST guideline conditionally recommends at least 4–6 hours of formal observation for trauma in viable pregnancy and fetal assessment after rapid maternal evaluation; its recommendations reflect limited evidence and should be applied with local protocols.
- High-yield fact: In maternal arrest with fundus at/above the umbilicus, follow current AHA resuscitative-delivery algorithm.
- High-yield fact: EAST 2025: maternal resuscitation is central; fetal assessment follows the rapid maternal primary survey.
- High-yield fact: For trauma with a viable pregnancy, EAST conditionally recommends a formal 4–6-hour observation period; evidence is limited and extend evaluation for concerning findings per local pathway.
- Exam trap: Do not withhold medically indicated imaging because of pregnancy; balance actual radiation risk against the danger of missed maternal injury.
- Clinical Pearl: The source-image seat-belt guidance places the lap belt below the gravid uterus across the hips and the shoulder belt between the breasts toward the side of the uterus. [AQRAB PDF p. 333; original visual panel]
- Number/threshold: The source image gives a protocol example of Rh(D) immune globulin 300 micrograms IM for a previously unsensitized Rh-negative pregnant patient after significant trauma; retain the indication and use gestational/local protocol context rather than treating this as a universal dose. [AQRAB PDF p. 333; original visual panel]

في الرضوض، الحزام السفلي يكون تحت الرحم. وجرعة Rh(D) هنا مذكورة في لوحة المصدر كمثال مشروط بنوع الأم والتحسس وسياق البروتوكول.

في trauma during pregnancy، لا تستبعد إنعاش الأم هو الأساس. ورتّب abruption بسبب غياب نزف واضح، حسب عمر الحمل fetal monitoring.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 9: Placental abruption clinical clues

- Key discriminator: Placental abruption may present with sudden severe abdominal pain, a rigid tender uterus and vaginal bleeding.
- Clinical trigger: Clinical concern for abruption after maternal trauma requires urgent obstetric assessment and fetal monitoring according to gestational age.

ابحث عن ألم ونزف وتوتر الرحم وتدهور الأم/الجنين؛ لا تنتظر العلامات الكلاسيكية كاملة في abruption.

Topic 10: Postpartum hemorrhage: uterine atony and thresholds

- Key discriminator: Uterine atony is described as the most common cause of primary postpartum hemorrhage.
- Number/threshold: One definition in CLEAN V3 uses cumulative postpartum blood loss of at least 1,000 mL or bleeding with signs of hypovolemia within 24 hours.

عالج الصدمة والـ uterine atony في postpartum bleeding وفق السياق، ولا تحصر القرار بحدّ رقمي واحد.

Topic 11: Shoulder dystocia and associated injury

- High-yield fact: Shoulder dystocia is associated in the source with brachial plexus injury, clavicular fracture, fetal hypoxia and postpartum hemorrhage.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 12: Uterine rupture and prior uterine surgery

- Red flag: Prior cesarean surgery is a clinical risk clue for uterine rupture, an emergency with maternal and neonatal morbidity.

تذكّر أن shoulder dystocia و uterine rupture طوارئ وقتية، والتصعيد السريع أساسي.

Topic 13: Anti-D and sensitizing event timing

- Number/threshold: CLEAN V3 cites administering anti-D immunoglobulin within 72 hours after a potentially sensitizing event.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 14: Preeclampsia and postpartum risk window

- High-yield fact: Preeclampsia may develop with new hypertension after 20 weeks of gestation and can involve headache or visual symptoms.
- Number/threshold: CLEAN V3 describes eclampsia risk extending into the postpartum period, including an example up to eight weeks.

الصداع واضطراب الرؤية أو الاختلاج في الحمل مع ارتفاع الضغط علامات خطورة تستدعي تقييمًا عاجلًا.

Topic 15: Uterine rupture and bedside ultrasound

- Clinical Pearl: Intraperitoneal free fluid in a pregnant patient with suspected uterine rupture may be seen on a right-upper-quadrant POCUS view in the cited case.

تذكّر أن shoulder dystocia و uterine rupture طوارئ وقتية، والتصعيد السريع أساسي.

Obstetrics & Gynecology | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

Intrauterine pregnancy on transvaginal ultrasonography

• High-yield fact: A yolk sac or fetal pole within an intrauterine gestational sac is the definitive intrauterine-pregnancy finding described in the source.

وجود yolk sac أو fetal pole داخل كيس حمل رحمي في TVUS هو العلامة المذكورة في المصدر لتثبيت الحمل داخل الرحم.

Painless bleeding and placenta previa

• High-yield fact: Placenta previa should be considered when painless vaginal bleeding occurs in the second half of pregnancy; this clue is explicitly stated in the source.

نزف مهبلي غير مؤلم في النصف الثاني من الحمل يثير احتمال placenta previa كما ورد بالمصدر.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 211

72 hours	Number/threshold: Hospitalize for pregnancy, severe illness, abscess, inability to tolerate outpatient therapy, or failure to improve; reassess within 72 hours . Exam trap: Do not wait for NAAT results before indicated empiric therapy.
≥110 mmHg	metabolic emergencies require condition-specific treatment rather than therapeutic delay. Number/threshold: Severe-range blood pressure (systolic ≥160 or diastolic ≥110 mmHg) with persistent elevation or end-organ features is urgent.
5 minutes	continuous manual left uterine displacement, prioritize airway, place IV/IO access above the diaphragm, activate the team and prepare resuscitative delivery from recognition, aiming to complete by 5 minutes if ROSC is not achieved. High-yield fact: AHA 2025 pregnancy-arrest algorithm: continuous manual left uterine displacement when fundus is
5 minutes	High-yield fact: AHA 2025 pregnancy-arrest algorithm: continuous manual left uterine displacement when fundus is at/above the umbilicus; prepare resuscitative delivery at arrest recognition and target completion by 5 minutes if no ROSC.
6 cm	newborn transition, and be ready for neonatal resuscitation. Number/threshold: ACOG defines active labor beginning at 6 cm; apply current obstetric definitions and local pathways for labor arrest rather than diagnosing dystocia early.
6 cm, ≥6 cm	or nonreassuring fetal status. High-yield fact: ACOG 2024: active labor begins at 6 cm; active-phase arrest requires no cervical change at ≥6 cm with ruptured membranes despite adequate duration of contractions/augmentation.
4–6 hours	to viability and local resources. Number/threshold: The 2025 EAST guideline conditionally recommends at least 4–6 hours of formal observation for trauma in viable pregnancy and fetal assessment after rapid maternal evaluation; its recommendations reflect limited
1,000 mL	Key discriminator: Uterine atony is described as the most common cause of primary postpartum hemorrhage. Number/threshold: One definition in CLEAN V3 uses cumulative postpartum blood loss of at least 1,000 mL or bleeding with signs of hypovolemia within 24 hours.
24 hours	Number/threshold: One definition in CLEAN V3 uses cumulative postpartum blood loss of at least 1,000 mL or bleeding with signs of hypovolemia within 24 hours. في حال الاشتباه بنزف ما بعد الولادة، يجب تقييم سريع وعلاج الصدمة في أول ساعة, postpartum bleeding uterine atony .
72 hours	Number/threshold: CLEAN V3 cites administering anti-D immunoglobulin within 72 hours after a potentially sensitizing event.
20 weeks	High-yield fact: Preeclampsia may develop with new hypertension after 20 weeks of gestation and can involve headache or visual symptoms.
500-1,000 mL, 1,000 mL	PPH threshold Blood loss 500-1,000 mL after Cumulative loss >=1,000 mL Keep definition and setting vaginal delivery or hypovolemic signs within 24 attached; do not collapse

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 211

72 hours	01. CLEAN V3 cites administering anti-D immunoglobulin within 72 hours after a potentially sensitizing event. 02. CLEAN V3 describes eclampsia risk extending into the postpartum period, including an example up to eight weeks.
≥110 mmHg	02. CLEAN V3 describes eclampsia risk extending into the postpartum period, including an example up to eight weeks. 03. Severe-range blood pressure (systolic ≥160 or diastolic ≥110 mmHg) with persistent elevation or end-organ features is urgent.
1,000 mL	features is urgent. 04. One definition in CLEAN V3 uses cumulative postpartum blood loss of at least 1,000 mL or bleeding with signs of hypovolemia within 24 hours.
24 hours	04. One definition in CLEAN V3 uses cumulative postpartum blood loss of at least 1,000 mL or bleeding with signs of hypovolemia within 24 hours. 05. Hospitalize for pregnancy, severe illness, abscess, inability to tolerate outpatient therapy, or failure to improve;
72 hours	05. Hospitalize for pregnancy, severe illness, abscess, inability to tolerate outpatient therapy, or failure to improve; reassess within 72 hours . 06. ACOG defines active labor beginning at 6 cm; apply current obstetric definitions and local pathways for labor arrest
6 cm	reassess within 72 hours . 06. ACOG defines active labor beginning at 6 cm; apply current obstetric definitions and local pathways for labor arrest rather than diagnosing dystocia early.
4–6 hours	obstetrics early for unstable or time-sensitive presentations. 30. The 2025 EAST guideline conditionally recommends at least 4–6 hours of formal observation for trauma in viable pregnancy and fetal assessment after rapid maternal evaluation
20 weeks	39. Ureteral stent and nephrostomy are accepted drainage options; definitive stone treatment follows infection control. 40. Preeclampsia may develop with new hypertension after 20 weeks of gestation and can involve headache or visual symptoms.
500–1,000 mL, 1,000 mL	CLEAN V3: CLEAN V3 includes the 500–1,000 mL minor PPH convention and also >=1,000 mL / hypovolemic signs definition.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 211

Zika infection - pregnancy exposure [#074]

Key discriminator: Zika is transmitted primarily by Aedes mosquitoes and is relevant to pregnancy exposure assessment.

ينتقل Zika أساسًا بواسطة بعوض Aedes؛ يجب أخذ تاريخ التعرض بالحسبان أثناء الحمل.

PASS 9 — Source-backed clinical recovery | Domain 10

Obstetrics & Gynecology

dosing or medical-correctness certification.

Ovarian torsion risk factor

Key discriminator: CLEAN V3 identifies a mobile ovarian mass, including a cyst, as a major risk factor for ovarian torsion.

وجود كتلة مبيضية متحركة يزيد الاشتباه بـ ovarian torsion في سياق الألم الحاد.

Lactational mastitis

Source-specific management: The CLEAN V3 lactational-mastitis source example recommends continuing breastfeeding from both breasts, provided the case matches that pathway.

في سياق mastitis المرتبط بالرضاعة، يذكر المصدر استمرار الرضاعة بدل إيقافها آليًا.

Endometritis and retained tissue

Differential point: CLEAN V3 describes retained products after vaginal birth as a possible contributor to postpartum endometritis, in addition to cesarean-associated risk.

بعد الولادة الطبيعية، وجود retained products قد يكون من أسباب endometritis بحسب المصدر.

High-Yield Comparison Tables

Early pregnancy emergency pattern — Ectopic pregnancy / Other early pregnancy pain/bleeding

Feature	Ectopic pregnancy	Other early-pregnancy pain	Exam point
Pregnancy location	Not established as intrauterine; rupture causes hemorrhage	Location may still be uncertain	Positive test alone does not prove intrauterine pregnancy
Unstable presentation	Resuscitate and obtain urgent gynecology input	Clinical severity still drives care	Do not wait for one hCG or an inconclusive ultrasound

Postpartum bleeding terminology — One supplied definition / Another supplied definition

Feature	Definition 1	Definition 2	Exam point
PPH threshold	Blood loss 500-1,000 mL after vaginal delivery	Cumulative loss >=1,000 mL or hypovolemic signs within 24 h	Keep definition and setting attached; do not collapse thresholds
Management	Treat maternal instability and suspected atony	Treat maternal instability and suspected atony	Bleeding physiology outranks a cutoff

Rapid Recall: Must-Know Facts

01. CLEAN V3 cites administering anti-D immunoglobulin within 72 hours after a potentially sensitizing event.

02. CLEAN V3 describes eclampsia risk extending into the postpartum period, including an example up to eight weeks.

03. Severe-range blood pressure (systolic ≥160 or diastolic ≥110 mmHg) with persistent elevation or end-organ features is urgent.

04. One definition in CLEAN V3 uses cumulative postpartum blood loss of at least 1,000 mL or bleeding with signs of hypovolemia within 24 hours.

05. Hospitalize for pregnancy, severe illness, abscess, inability to tolerate outpatient therapy, or failure to improve; reassess within 72 hours .

06. ACOG defines active labor beginning at 6 cm; apply current obstetric definitions and local pathways for labor arrest rather than diagnosing dystocia early.

07. Do not wait for NAAT results before indicated empiric therapy.

08. Do not equate ‘natural’ or OTC products with proven pregnancy safety.

09. Do not delay empiric treatment for STI test results when criteria are met.

10. Do not delay surgery for a reassuring Doppler when clinical suspicion remains high.

11. Treat maternal illness promptly while considering gestational age and fetal effects.

12. Uterine atony is described as the most common cause of primary postpartum hemorrhage.

13. Obtain gonorrhea/chlamydia testing and assess for complications such as tubo-ovarian abscess.

14. Assess for sepsis/anuria and obtain cultures; begin antibiotics and arrange urgent decompression.

15. Avoid starting or stopping an essential chronic medicine abruptly without an individualized plan.

16. Evaluate pregnancy and urgent mimics including ectopic pregnancy, appendicitis, and ovarian torsion.

17. Do not attribute severe headache, visual symptoms, epigastric pain or seizure to normal pregnancy changes.

18. Placental abruption may present with sudden severe abdominal pain, a rigid tender uterus and vaginal bleeding.

19. Assess maternal stability first; pregnancy does not change the need for prompt treatment of life-threatening illness.

20. Use indicated imaging when it changes care; do not withhold necessary diagnostic imaging solely because of pregnancy.

21. Prior cesarean surgery is a clinical risk clue for uterine rupture, an emergency with maternal and neonatal morbidity.

22. Use standard trauma priorities and stabilize the mother first; maternal resuscitation provides the best fetal resuscitation.

23. Escalate immediately for malpresentation, cord prolapse, shoulder dystocia, maternal instability or nonreassuring fetal status.

24. Establish gestational age, symptoms, bleeding, pain, fetal movement and relevant obstetric history, then involve obstetrics early.

25. Clinical concern for abruption after maternal trauma requires urgent obstetric assessment and fetal monitoring according to gestational age.

26. Do not withhold medically indicated imaging because of pregnancy; balance actual radiation risk against the danger of missed maternal injury.

27. Do not use a single hCG value or an empty early ultrasound to label a pregnancy nonviable or give methotrexate without an appropriate diagnostic pathway.

28. Arrange obstetric transfer when safe; if crowning or birth is imminent, prepare to deliver where the patient is, summon neonatal help, protect privacy and use clean equipment.

29. Assess maternal oxygenation, circulation, glucose, neurologic status, pregnancy stage and fetal concerns; involve obstetrics early for unstable or time-sensitive presentations.

30. The 2025 EAST guideline conditionally recommends at least 4–6 hours of formal observation for trauma in viable pregnancy and fetal assessment after rapid maternal evaluation

31. Untreated maternal disease can itself harm both patients.

32. Analgesia, NPO, and transfer/escalation should not delay definitive care.

33. Choose imaging and medication by maternal risk-benefit, not by withholding necessary care.

34. Suspected testicular torsion requires immediate urologic/surgical review and exploration when indicated.

35. EAST 2025: maternal resuscitation is central; fetal assessment follows the rapid maternal primary survey.

36. In maternal arrest with fundus at/above the umbilicus, follow current AHA resuscitative-delivery algorithm.

37. In emergencies, treat the mother with effective evidence-based care and document the risk-benefit reasoning.

38. Modern FDA labeling uses narrative risk summaries and evidence rather than the former A/B/C/D/X letter categories.

39. Ureteral stent and nephrostomy are accepted drainage options; definitive stone treatment follows infection control.

40. Preeclampsia may develop with new hypertension after 20 weeks of gestation and can involve headache or visual symptoms.

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

Postpartum hemorrhage definitions

- CLEAN V3: CLEAN V3 includes the 500–1,000 mL minor PPH convention and also >=1,000 mL / hypovolemic signs definition.
- AQRAB: AQRAB focuses on clinically significant bleeding, maternal stability, and local pathway rather than one standalone threshold.
- Nature of difference: Different thresholds and practical definition context; retain both source wordings.
- Classification: Definition/context - not automatically direct conflict

Final clinical topics: 15. Retained integrated English bullet points: 79. Rapid Recall: 40 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 10; AQRAB-only 69; confirmed both-source 0. These are point-by-

Source-screening notes: 37 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Audit Pass 3: Supplemental image-derived facts added from original AQRAB PDF panels, with page-number evidence embedded in each added point. This targeted visual pass is not a full medical validation or proof of complete two-source coverage.

Obstetrics & Gynecology | Pass 8 | Explicit CLEAN V3 source locations

Untreated pelvic inflammatory disease — later ectopic risk

• High-yield fact: CLEAN V3 links untreated pelvic inflammatory disease, fallopian-tube scarring, and increased future ectopic-pregnancy risk.

إهمال PID قد يترك تندبًا بالأنابيب ويرتبط لاحقًا بـEctopic Pregnancy.

Postmenopausal vaginal bleeding — assessment

• High-yield fact: CLEAN V3 states that postmenopausal vaginal bleeding requires a full evaluation rather than assuming a benign atrophic cause.

حتى لو كانت Atrophy سببًا محتملًا، النزف بعد انقطاع الطمث يحتاج تقييمًا كما ورد في المصدر.

High-flow priapism — arterio-cavernosal fistula

• High-yield fact: CLEAN V3 associates high-flow priapism with a fistula between the cavernosal artery and corpus cavernosum and describes cavernosal blood-gas testing as a diagnostic discriminator.

في High-flow Priapism يشير المصدر إلى Arterio-cavernosal fistula، ويذكر Cavernosal Blood Gas ضمن التقييم.

Testicular malignancy — painless mass

• High-yield fact: CLEAN V3 describes a painless unilateral testicular mass as a presentation suggestive of testicular germ-cell malignancy.

وجود كتلة خصية أحادية غير مؤلمة يستدعي التفكير في ورم خصوي وفق النص.

Obstetrics & Gynecology

Supplementary Clinical Points

Topic: Ectopic pregnancy: implantation location

Anatomic discriminator: CLEAN V3 identifies the fallopian tube, especially its ampullary region, as the most common location for ectopic implantation. Its attached 95% figure is not transferred as an ampullary-specific proportion because the extracted sentence does not separate tubal from ampullary denominators.

الـampulla من أكثر مواقع ectopic pregnancy شيوعًا؛ لم أنقل نسبة 95% كأنها تخص ampulla وحدها.

Source evidence: C03290: DOCX paragraph 3290

EDIC / Emergency & Critical Care

11 — Renal, Electrolyte, Metabolic & Endocrine Emergencies

I. Renal & Electrolyte Emergencies

Topic 1: Acute Kidney Injury — Causes, Complications & Urine Output

- Core concept: AKI is an acute change in kidney function identified by creatinine trend and/or urine output; compare with baseline and assess time course.
- Management priority: Evaluate volume/perfusion, sepsis, obstruction, nephrotoxins, and intrinsic renal causes.
- Management priority: Check electrolytes, acid- base status, urinalysis, medication exposures, and bladder/renal imaging when obstruction is possible.
- Management priority: Treat the cause, optimize perfusion without fluid overload, stop or adjust nephrotoxic/renally cleared drugs, and monitor urine output and biochemical trajectory.
- Management priority: Escalate for refractory hyperkalemia/acidosis, pulmonary edema, uremic complications, or persistent oliguria.
- High-yield fact: Establish baseline and trend creatinine; monitor urine output and hemodynamics.
- Management priority: Review volume/perfusion, sepsis, obstruction, nephrotoxins, urinalysis, electrolytes, and acid-base status.
- Management priority: Use bladder scan/renal imaging when obstruction is plausible.
- High-yield fact: Urgent renal replacement assessment for refractory hyperkalemia/acidosis, pulmonary edema, or uremic complications.

في AKI و راقب السوائل. انسداد، أو كلوية، أذية ترويّة، نقص أولًا: السبب حدّد Kوالـ urine output ولا تعتمد على creatinine وحده.

Topic 2: Hyperkalemia & Dialysis-Related Emergencies

- Clinical trigger: Peritonitis is a potentially serious complication in patients receiving peritoneal dialysis.
- Number/threshold: The supplied PD-peritonitis example describes a peritoneal-fluid WBC count of 100 cells/µL with more than 50% neutrophils; this is source-specific wording.

في مريض peritoneal dialysis يجب الانتباه لاحتمال peritonitis. الرقم المذكور جاء ضمن مثال المصدر ولا يُعمم خارج سياقه.

- Core concept: Hyperkalemia can cause lethal conduction abnormalities; obtain ECG promptly in severe elevation or clinical concern, but a normal ECG does not exclude danger.
- High-yield fact: Confirm an unexpected result when pseudohyperkalemia is plausible without delaying treatment for instability or ECG changes.
- Management priority: Stabilize the myocardium when indicated, shift potassium intracellularly, and arrange potassium removal; these actions have different purposes.
- Management priority: Review renal failure/dialysis status, medications, acid-base state, and ongoing potassium release.
- High-yield fact: Recheck potassium and glucose after temporizing therapy; involve nephrology early for refractory/severe cases or missed dialysis.
- Management priority: Obtain ECG and continuous monitoring for severe or symptomatic hyperkalemia.

- Management priority: Treat ECG toxicity promptly; intracellular shift is temporary, so plan definitive potassium removal.
- Management priority: Check for pseudohyperkalemia when appropriate, without delaying care in unstable patients.
- High-yield fact: After insulin-based treatment for hyperkalemia, monitor glucose and repeat potassium; involve renal services for refractory disease or missed dialysis.

Hyperkalemia افحص رقم. خطر كهربائي قبل أن يكون مجرد ECG، ثم حرّك K إلى ثبّت القلب عند اللزوم، داخل الخلايا وعالج إخراجه.

Topic 3: Electrolyte Emergencies — ECG Before the Lab

- Core concept: Electrolyte emergencies can trigger dysrhythmia, seizure or weakness before the laboratory result is confirmed.
- Management priority: Assess ABCs, obtain ECG and monitored access when clinically indicated, and interpret potassium, sodium, calcium and magnesium with renal function, medications and acid–base status.
- High-yield fact: A normal ECG does not exclude dangerous hyperkalemia.
- High-yield fact: If a result may be spurious, repeat it promptly while treating instability or toxic ECG features without waiting.
- Management priority: Stabilize the myocardium, shift potassium intracellularly and arrange definitive removal when indicated; reassess for rebound and treatment-related hypoglycemia.
- Management priority: Obtain and repeat ECGs during treatment when potassium is significantly abnormal; use continuous monitoring for severe cases. With ECG toxicity, distinguish myocardial stabilization, intracellular potassium shift and definitive potassium removal.
- Management priority: Check glucose after insulin-based potassium shifting and recheck potassium because effects may rebound.
- Management priority: Assess renal failure, dialysis access, medication causes, acid–base disturbance and other concurrent electrolyte abnormalities.

ECG التغيرات على electrolyte disturbance قد تسبق أو لا تطابق الرقم المخبري في أولًا عالج الخطر القلبي؛ عند اللزوم.

II. Acid–Base, Glycemic & Muscle Emergencies

Topic 4: Acid–Base I — Anion Gap & Compensation

- Management priority: Use a reproducible sequence: verify the sample and clinical context, identify acidemia/alkalemia from pH, determine whether PaCO2 or HCO3− best explains the direction, then assess expected compensation.
- High-yield fact: In metabolic acidosis calculate the anion gap using the local laboratory convention and consider albumin; compare measured respiratory compensation with Winter’s formula.
- High-yield fact: A mismatch suggests an additional primary disorder.
- High-yield fact: A near-normal pH can conceal mixed disorders.
- Management priority: Treat the cause and immediate physiologic threat rather than the laboratory label alone.
- High-yield fact: Confirm specimen type, timing and clinical context; venous and arterial values are not interchangeable for every question.
- High-yield fact: Read pH, identify the primary respiratory/metabolic direction, then test whether compensation is appropriate.
- High-yield fact: For metabolic acidosis calculate anion gap with the laboratory reference range; consider albumin and delta-gap context.
- High-yield fact: Compare measured PaCO2 with expected compensation; investigate a mixed disorder when outside the expected range.

acid–base احسب anion gap في مختلطًا اضطرابًا يكشف غير المتوقع التعويض. وراجع التعويض المتوقع.

Topic 5: Acid–Base II — Respiratory & Mixed Disorders

- Core concept: Respiratory acidosis reflects inadequate alveolar ventilation; respiratory alkalosis reflects increased ventilation, but both require a search for the cause rather than a label.
- Management priority: Assess oxygenation, ventilation, work of breathing, mental status, medications and acute cardiopulmonary illness.
- High-yield fact: Compensation varies with duration and renal response.
- High-yield fact: When measured pH/PaCO2/HCO3− do not fit the expected acute or chronic pattern, consider a mixed disorder, sampling/timing issues or evolving illness.
- Management priority: Treat ventilation failure or the precipitating disease and repeat gases when the clinical course changes.
- Management priority: Assess airway, respiratory effort, oxygenation, ventilation, consciousness, medications and precipitating cardiopulmonary Distinguish acute from chronic respiratory disturbance using the clinical timeline and expected renal compensation; do not infer chronicity from one gas alone.
- High-yield fact: If PaCO2, HCO3− and pH do not fit expected compensation, assess for a mixed metabolic process and verify sampling/timing.
- Management priority: Treat the cause and ventilatory failure; repeat blood gas testing when it will change management or the patient deteriorates.
- Exam trap: Do not assume anxiety is the cause of respiratory alkalosis before evaluating hypoxemia, sepsis, pulmonary embolism, pregnancy or toxicity.

اضطرابات التنفس و CO2في، راقب pH إلى قد يشير قراءتين غير متوافقتين وجود ؛والتعويض mixed disorder.

Topic 6: Diabetic Emergencies — DKA, HHS & Hypoglycemia

- Core concept: DKA requires diabetes/hyperglycemia, significant ketonemia and metabolic acidosis; glucose may be below traditional thresholds, including euglycemic DKA.
- High-yield fact: HHS is severe hyperglycemia and hyperosmolality without severe ketonemia or acidosis; overlap occurs.
- Management priority: Assess volume, mental status, potassium, ketones, acid–base state and precipitant.
- High-yield fact: In DKA, potassium may fall rapidly during treatment, so check it before and during insulin.
- High-yield fact: Hypoglycemia is immediately dangerous: treat promptly, recheck glucose and investigate recurrence or the precipitating medication/illness.
- Management priority: Use the 2024 multisociety consensus criteria; do not exclude DKA because glucose is below 200 mg /dL when diabetes, ketones and acidosis criteria are met.
- Management priority: Assess DKA/HHS overlap, osmolality, mental status, volume status, potassium, renal function and precipitating illness.
- High-yield fact: In DKA, check potassium before insulin and monitor replacement during treatment because insulin can rapidly lower serum potassium. In hypoglycemia, restore glucose promptly and recheck for recurrence.
- Exam trap: Do not start or continue insulin without considering severe hypokalemia and the local protocol.

DKA قد يحدث مع glucose راقب. غير مرتفع جدًا ketones وacidosis وK قبل insulin أما .وخلاله HHS
فـhyperosmolality بارزان والجفاف.

Topic 7: Rhabdomyolysis — Muscle Injury & Renal Risk

- Core concept: Rhabdomyolysis releases CK, myoglobin and intracellular electrolytes after muscle injury from crush, prolonged immobilization, exertion, seizure, heat, ischemia, infection or drugs.
- Management priority: Assess CK trend, renal function, potassium, calcium, phosphate, urine output and compartment syndrome risk.
- High-yield fact: Give goal-directed isotonic crystalloid when appropriate and reassess perfusion and congestion.
- Management priority: Avoid routine bicarbonate or mannitol solely to prevent AKI; use renal replacement therapy for standard indications, not CK elevation alone.
- Exam trap: Do not continue escalating fluid in an oliguric, overloaded patient without reassessing renal/cardiac status.

Rhabdomyolysis ارتفاع ليس CK راقب ؛فقط renal function وelectrolytes وurine output مع علاج المسبب.

III. Endocrine Crises

Topic 8: Thyroid Storm & Adrenal Crisis

- Core concept: Thyroid storm is a clinical diagnosis of severe thyrotoxicosis with systemic decompensation; do not wait for thyroid tests to begin urgent specialist-led treatment.
- Management priority: Stabilize, treat the precipitant and use coordinated therapy in the correct sequence.
- High-yield fact: Adrenal crisis may cause shock, vomiting, abdominal symptoms, hyponatremia, hypoglycemia and—especially in primary adrenal failure —hyperkalemia.
- High-yield fact: Draw cortisol/ACTH first only if this causes no delay; give emergency glucocorticoid and isotonic fluid promptly, correct glucose and monitor electrolytes/hemodynamics.
- High-yield fact: Recognize thyroid storm clinically from thyrotoxicosis plus systemic decompensation; treat the trigger and coordinate ICU/endocrinology care.
- High-yield fact: Sequence storm therapy according to current consensus; use beta-blockade cautiously when heart failure or hemodynamic instability is possible.
- High-yield fact: For suspected adrenal crisis, do not delay parenteral hydrocortisone for confirmatory tests; draw cortisol/ACTH first only if immediately feasible.
- High-yield fact: Give isotonic fluid and monitor glucose, sodium, potassium, blood pressure and response; consider steroid dependence and primary versus secondary disease.

Thyroid storm وadrenal crisis طوارئ سريرية؛ لا تنتظر كل الاختبارات إذا كانت الدورة الدموية والأعضاء مهددة.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 9: DKA: phosphate loss and euglycemia

- High-yield fact: DKA can produce hypophosphatemia through osmotic diuresis and urinary phosphate losses.
- Clinical Pearl: SGLT2 inhibitors can maintain relatively normal glucose while ketogenesis continues in euglycemic DKA.

علاج DKA يحتاج مراقبة K وacid–base، ولا يكفي متابعة glucose وحده.

Topic 10: HHS: disproportionate water depletion

- Key discriminator: The greater water depletion of HHS can produce higher serum sodium than a comparable DKA presentation.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

Topic 11: Dialysis disequilibrium syndrome

- Core concept: Dialysis disequilibrium syndrome is described as neurologic deterioration related to rapid urea clearance during hemodialysis.
- Red flag: Dialysis disequilibrium can cause headache, nausea, disorientation and, in severe cases, seizure or coma.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

Topic 12: Dialysis and severe potassium disturbances

- Management priority: The supplied end-stage kidney disease example combines IV calcium for severe hyperkalemia with urgent hemodialysis.

Renal, Electrolyte, Metabolic & Endocrine Emergencies | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

Amiodarone iodine load

• High-yield fact: The source describes amiodarone as approximately 37% iodine by weight; a 200-mg tablet contains approximately 75 mg iodine in its type-I AIT discussion.

الرقم هنا يشرح iodine load في سياق Amiodarone-induced thyrotoxicosis، وليس جرعة علاجية مقترحة.

Hypoparathyroidism biochemical pattern

• High-yield fact: Parathyroid hormone deficiency is associated with hypocalcemia and hyperphosphatemia in the supplied source.

نقص PTH يرتبط في النص بانخفاض calcium وارتفاع phosphate، وهي نقطة تمييز أساسية.

PASS 9 — Source-backed clinical recovery | Domain 11

Renal, Electrolyte, Metabolic & Endocrine Emergencies

dosing or medical-correctness certification.

Urine chloride in metabolic alkalosis

Number/threshold: In the CLEAN V3 described metabolic-alkalosis classification, a urine chloride below 20 mmol/L supports a chloride-responsive cause.

Urine chloride أقل من 20 mmol/L في تصنيف المصدر يشير إلى نمط chloride-responsive.

Primary hyperthyroidism lab pattern

Key discriminator: CLEAN V3 describes primary hyperthyroidism with decreased TSH and elevated free T4.

النمط المذكور في primary hyperthyroidism هو TSH منخفض وFT4 مرتفع.

وخطوة التقييم التالية، مع مراعاة سياق المصدر Red flags اربط هذه النقاط بالمظهر السريري الفعلي وحدّد.

Topic 13: Adrenal insufficiency: sodium and potassium

- High-yield fact: Mineralocorticoid deficiency in adrenal insufficiency can cause hyponatremia and hyperkalemia.

ربط renal injury تفويت المضاعفات يمنع الدموية والحالة والبوتاسيوم بالإدرار.

Topic 14: Renal tubular acid–base patterns

- Key discriminator: Type 4 renal tubular acidosis is described with a normal-anion-gap metabolic acidosis.

ربط renal injury تفويت المضاعفات يمنع الدموية والحالة والبوتاسيوم بالإدرار.

High-Yield Comparison Tables

Hyperglycemic crises — DKA / HHS

Feature	DKA	HHS	Exam point
Core process	Ketones plus metabolic acidosis	Marked hyperglycemia and hyperosmolality without severe ketonemia/acidosis	Overlapping physiology is possible
Diagnostic trap	May occur with relatively low glucose	Dehydration and high osmolality prominent	Glucose alone is not diagnostic
Treatment caution	Check potassium before and during insulin	Assess hydration, osmolality and potassium	Do not ignore potassium or mental status

Potassium emergency — Hyperkalemia / Hypokalemia

Feature	Hyperkalemia	Hypokalemia	Exam point
Threat	ECG/conduction instability	Arrhythmia and weakness risk	Integrate ECG with clinical situation
Principle	Stabilize heart and shift/remove potassium when indicated	Replace deficit with monitoring	Do not extrapolate one regimen to all causes

Rapid Recall: Must-Know Facts

01. Use bladder scan/renal imaging when obstruction is plausible.
02. Obtain ECG and continuous monitoring for severe or symptomatic hyperkalemia.
03. Assess volume, mental status, potassium, ketones, acid–base state and precipitant.
04. Stabilize, treat the precipitant and use coordinated therapy in the correct sequence.
05. Type 4 renal tubular acidosis is described with a normal-anion-gap metabolic acidosis.
06. Treat the cause and immediate physiologic threat rather than the laboratory label alone.
07. Evaluate volume/perfusion, sepsis, obstruction, nephrotoxins, and intrinsic renal causes.
08. Check for pseudohyperkalemia when appropriate, without delaying care in unstable patients.
09. Do not start or continue insulin without considering severe hypokalemia and the local protocol.
10. Review renal failure/dialysis status, medications, acid-base state, and ongoing potassium release.
11. Treat ECG toxicity promptly; intracellular shift is temporary, so plan definitive potassium removal.
12. The greater water depletion of HHS can produce higher serum sodium than a comparable DKA presentation.
13. Check glucose after insulin-based potassium shifting and recheck potassium because effects may rebound.
14. Treat ventilation failure or the precipitating disease and repeat gases when the clinical course changes.
15. Dialysis disequilibrium can cause headache, nausea, disorientation and, in severe cases, seizure or coma.
16. Assess CK trend, renal function, potassium, calcium, phosphate, urine output and compartment syndrome risk.

17. Escalate for refractory hyperkalemia/acidosis, pulmonary edema, uremic complications, or persistent oliguria.
18. Do not continue escalating fluid in an oliguric, overloaded patient without reassessing renal/cardiac status.
19. Assess oxygenation, ventilation, work of breathing, mental status, medications and acute cardiopulmonary illness.
20. The supplied end-stage kidney disease example combines IV calcium for severe hyperkalemia with urgent hemodialysis.
21. Assess DKA/HHS overlap, osmolality, mental status, volume status, potassium, renal function and precipitating illness.
22. Treat the cause and ventilatory failure; repeat blood gas testing when it will change management or the patient deteriorates.
23. Assess renal failure, dialysis access, medication causes, acid–base disturbance and other concurrent electrolyte abnormalities.
24. Do not assume anxiety is the cause of respiratory alkalosis before evaluating hypoxemia, sepsis, pulmonary embolism, pregnancy or toxicity.
25. Avoid routine bicarbonate or mannitol solely to prevent AKI; use renal replacement therapy for standard indications, not CK elevation alone.
26. Stabilize the myocardium when indicated, shift potassium intracellularly, and arrange potassium removal; these actions have different purposes.
27. Use the 2024 multisociety consensus criteria; do not exclude DKA because glucose is below 200 mg /dL when diabetes, ketones and acidosis criteria are met.
28. Treat the cause, optimize perfusion without fluid overload, stop or adjust nephrotoxic/renally cleared drugs, and monitor urine output and biochemical trajectory.
29. Assess ABCs, obtain ECG and monitored access when clinically indicated, and interpret potassium, sodium, calcium and magnesium with renal function, medications and acid–base status.
30. A near-normal pH can conceal mixed disorders.
31. A mismatch suggests an additional primary disorder.
32. A normal ECG does not exclude dangerous hyperkalemia.
33. Compensation varies with duration and renal response.
34. Establish baseline and trend creatinine; monitor urine output and hemodynamics.
35. DKA can produce hypophosphatemia through osmotic diuresis and urinary phosphate losses.
36. In DKA, potassium may fall rapidly during treatment, so check it before and during insulin.
37. Mineralocorticoid deficiency in adrenal insufficiency can cause hyponatremia and hyperkalemia.
38. Give goal-directed isotonic crystalloid when appropriate and reassess perfusion and congestion.
39. HHS is severe hyperglycemia and hyperosmolality without severe ketonemia or acidosis; overlap occurs.
40. SGLT2 inhibitors can maintain relatively normal glucose while ketogenesis continues in euglycemic DKA.

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

DKA and HHS classification

- CLEAN V3: CLEAN V3 contains scenario-specific numeric and metabolic thresholds.
- AQRAB: AQRAB describes the multidimensional criteria including ketonemia, acidosis, hyperosmolality and euglycemic DKA.
- Nature of difference: Different framing/diagnostic generation and patient context; do not equate a single glucose cutoff to all cases.
- Classification: Protocol context

Final clinical topics: 14. Retained integrated English bullet points: 74. Rapid Recall: 40 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 8; AQRAB-only 66; confirmed both-source 0. These are point-by-

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 224

100 cells/µL	Clinical trigger: Peritonitis is a potentially serious complication in patients receiving peritoneal dialysis. Number/threshold: The supplied PD-peritonitis example describes a peritoneal-fluid WBC count of 100 cells/µL with more than 50% neutrophils; this is source-specific wording.
50%	Number/threshold: The supplied PD-peritonitis example describes a peritoneal-fluid WBC count of 100 cells/µL with more than 50% neutrophils; this is source-specific wording.
37%	High-yield fact: The source describes amiodarone as approximately 37% iodine by weight; a 200-mg tablet contains approximately 75 mg iodine in its type-I AIT discussion.
75 mg	High-yield fact: The source describes amiodarone as approximately 37% iodine by weight; a 200-mg tablet contains approximately 75 mg iodine in its type-I AIT discussion. thyrotoxicosis سياق في iodine load يشرح هنا الرقم .مقترحة عالجية جرعة وليس ،Amiodarone-induced
200 mg	actions have different purposes. 27. Use the 2024 multisociety consensus criteria; do not exclude DKA because glucose is below 200 mg /dL when diabetes, ketones and acidosis criteria are met.

Domain 11 | Renal, Electrolyte, Metabolic & Enc Emergencies

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 224

Refeeding syndrome - electrolyte shifts [#087]

High-yield fact: Refeeding after prolonged malnutrition can cause severe electrolyte shifts and cardiac dysfunction.

إعادة التغذية بعد سوء تغذية طويل قد تسبب تبدلات شديدة في الشوارد واختلال وظيفة القلب.

Rhabdomyolysis - acute kidney injury and electrolytes [#089]

Key discriminator: Early hypocalcemia can occur in rhabdomyolysis as calcium enters injured muscle.

قد يحدث نقص كالسيوم مبكر في Rhabdomyolysis نتيجة دخول الكالسيوم إلى العضلات المصابة.

Source-screening notes: 30 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Renal, Electrolyte, Metabolic & Endocrine Emergencies | Pass 8 | Explicit CLEAN V3 source locations

Chronic glucocorticoid use — adrenal suppression

• High-yield fact: The source identifies chronic corticosteroid therapy as a common cause of suppression of the adrenal hormonal axis.

استخدام Corticosteroids المزمن قد يسبب Suppression للـAdrenal axis كما يذكر المصدر.

Renal, Electrolyte, Metabolic & Endocrine Emergencies

Supplementary Clinical Points

Topic: Thyroid nodule with new hoarseness

Red flag: CLEAN V3 highlights a firm, nonmobile solitary thyroid nodule with new hoarseness as a concerning presentation requiring evaluation for possible thyroid malignancy; its specific cancer subtype in the vignette is not diagnostic without further evidence.

عقدة درقية قاسية وثابتة مع بحة جديدة ترفع الاشتباه؛ لا تجزم بنوع الورم من هذه العلامات وحدها.

Source evidence: C03556: DOCX paragraph 3556

Topic: Hypocalcemia, phosphate and intracranial calcification

Pathophysiology: CLEAN V3 describes low calcium with relative phosphate excess and calcium-phosphate deposition in soft tissues, including basal-ganglia calcification in the cited endocrine disorder.

قد يترافق ارتفاع phosphate النسبي مع ترسبات calcium-phosphate حتى مع انخفاض calcium بالدم بحسب المثال.

Source evidence: C03566: DOCX paragraph 3566

Topic: Mineralocorticoid excess: biochemical pattern

Clinical trigger: CLEAN V3 describes hypertension, hypokalemia and metabolic alkalosis with relatively high urine chloride as a pattern that raises concern for mineralocorticoid excess such as primary hyperaldosteronism.

ارتفاع الضغط مع انخفاض K وmetabolic alkalosis يوجه في المصدر إلى mineralocorticoid excess.

Source evidence: C04733: DOCX paragraph 4733

Topic: Peritoneal dialysis and peritonitis consequences

Clinical Pearl: CLEAN V3 describes peritonitis as a major peritoneal-dialysis complication that may lead to catheter loss or transition to hemodialysis; the mentioned intraperitoneal antibiotic combination was not copied as a universal regimen.

التهاب الصفاق عند مرضى PD قد يؤدي لخسارة catheter أو الحاجة إلى HD؛ نظام المضاد يتبع سياق الحالة.

Source evidence: C03540: DOCX paragraph 3540

Renal, Electrolyte, Metabolic & Endocrine Emergencies. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 226

Hyponatremia - neurologic manifestations [#078]

Key discriminator: Neurologic manifestations of hyponatremia depend strongly on how quickly sodium falls, not simply on the final number.

تتأثر الأعراض العصبية في نقص الصوديوم بسرعة انخفاضه، لا بالرقم النهائي وحده.

Myxedema coma - clinical recognition [#083]

High-yield fact: Myxedema coma is severe decompensated hypothyroidism and may present with altered consciousness and hypothermia.

Myxedema coma هو قصور درقي شديد متدهور قد يترافق مع انخفاض الحرارة واضطراب الوعي.

Hypercalcemia - clinical manifestations [#084]

Key discriminator: Hypercalcemia may manifest as stones, bone pain, GI symptoms and neuropsychiatric changes.

قد يتظاهر فرط الكالسيوم بحصيات وألم عظمي وأعراض هضمية وتغيرات عصبية نفسية.

EDIC / Emergency & Critical Care

12 — Hematology & Oncology

I. Blood Transfusion & Anemia

Topic 1: Blood Transfusion — Indications & Acute Reactions

- Core concept: Transfusion decisions depend on active bleeding, hemodynamics, symptoms, comorbidity and the clinical context; a hemoglobin threshold is not a substitute for resuscitation or individualized judgment.
- Number/threshold: For most hemodynamically stable hospitalized adults, AABB 2023 supports considering red-cell transfusion below 7 g/dL, with selected thresholds around 7.5–8 g/dL in specified surgical/cardiovascular populations.
- High-yield fact: If a reaction is suspected, stop the transfusion, assess ABC/vitals, maintain IV access with compatible fluid per local policy, recheck patient/unit identity, notify the blood bank and investigate hemolysis, sepsis, allergy or TACO/TRALI.
- High-yield fact: In hemorrhage or shock, prioritize clinical resuscitation and bleeding control; do not apply stable-patient thresholds mechanically.
- High-yield fact: For most stable hospitalized adults, consider RBC transfusion below 7 g/dL; individualize for symptoms, comorbidity and procedure-specific guidance.
- High-yield fact: Classify the suspected transfusion reaction as hemolytic, septic, allergic/anaphylactic, TACO or TRALI, and arrange investigation through the transfusion service.
- Exam trap: Do not restart a suspected transfusion reaction merely because symptoms improve before evaluation. Acute dyspnea during transfusion requires prompt consideration of both TACO and TRALI.

عند transfusion reaction قيّم فورًا، أوقف الوحدة ABC، وأبلغ بنك الدم قبل تحقّق من هوية الدم والمريض. أي إعادة تشغيل.

Topic 2: Anemia — Indices, Causes & Sickle Cell Emergencies

- Core concept: Classify anemia by acuity, hemodynamic effect, MCV, reticulocyte response and smear while looking for bleeding, hemolysis, marrow failure, nutritional deficiency, renal disease and chronic inflammation.
- High-yield fact: In sickle cell disease, acute pain is an emergency; rapidly treat pain and screen for acute chest syndrome, stroke, sepsis, aplastic crisis and splenic sequestration.
- High-yield fact: New fever, chest symptoms/hypoxia, neurologic deficit, pallor with sudden hemoglobin fall or enlarged spleen needs urgent escalation.
- High-yield fact: Transfusion is indication-specific and not routine for uncomplicated pain; involve hematology and follow local SCD protocol.
- Management priority: Assess stability and bleeding first; classify anemia using MCV, reticulocyte response, smear and targeted tests.
- Management priority: Treat acute sickle pain promptly under the patient's plan/local protocol and reassess frequently.
- Management priority: Escalate chest pain, fever, hypoxia, neurologic deficit, sudden anemia/reticulocytopenia or splenic enlargement urgently.
- Exam trap: Do not transfuse uncomplicated vaso-occlusive pain routinely; choose simple/exchange transfusion based on complication and specialist protocol.

في anemia حدّد النزف والـhemolysis وreticulocytes. وفي sickle cell انتبه للحمى وأعراض الصدر والعجز العصبي.

II. White Cells, Hemostasis & Cancer

Topic 3: White Cell Disorders — Interpret the Differential

- Core concept: Interpret the absolute count and differential with symptoms, medications, infection risk, smear and prior trends.
- High-yield fact: Neutrophilia and leukopenia are patterns, not diagnoses; consider infection, inflammation, marrow disease, drugs and physiologic stress.
- High-yield fact: Fever with severe neutropenia is a time-critical infection pathway.
- High-yield fact: Blasts, cytopenias, rapidly rising counts or respiratory/neurologic symptoms with suspected leukostasis require urgent hematology involvement.
- High-yield fact: Leukostasis risk depends on cell type and clinical physiology; there is no single safe universal WBC cutoff.

عدد WBC افحص differential والـneutrophils وحده لا يفسّر الحالة؛ والمظهر السريري، ولا تنسَ العدوى حتى مع leukopenia.

Topic 4: Hemostasis — TTP, ITP, HIT & Hemophilia

- Core concept: First classify the problem: bleeding, thrombosis, platelet disorder, coagulation-factor deficiency or microangiopathic hemolysis.
- High-yield fact: Thrombocytopenia plus schistocytes/hemolysis or neurologic/cardiac injury raises urgent concern for TTP; collect ADAMTS13 samples before plasma therapy when feasible, but do not delay urgent specialist-led treatment when suspicion is high.
- High-yield fact: Suspected HIT is prothrombotic: use 4Ts assessment, stop heparin when indicated and follow a non-heparin anticoagulation pathway.
- High-yield fact: Severe ITP bleeding and hemophilia-related critical-site bleeding require urgent hematology-directed hemostatic management.
- Exam trap: Do not reflexively transfuse red cells or give large fluid volumes in suspected leukostasis without specialist input.

التفريق بين TTP وITP وHIT وDIC يعتمد على الصفائح والـhemolysis والتخثر والسياق؛ لا تعامل كل thrombocytopenia بطريقة واحدة.

Topic 5: Oncologic Emergencies — Febrile Neutropenia & More

- Number/threshold: In the supplied leukemia material, hyperleukocytosis associated with leukostasis is described at WBC >100,000/µL, especially with AML.
- High-yield fact: The supplied Burkitt lymphoma description notes a MYC proto-oncogene translocation involving chromosome 8.

احفظ ارتباط AML بـhyperleukocytosis/leukostasis وارتباط Burkitt lymphoma بـMYC كما ورد في المصدر، على chromosome 8.

- Core concept: Cancer-related emergencies may arise from malignancy or its treatment.
- High-yield fact: Fever during neutropenia is time-critical: assess stability, obtain cultures and key tests promptly, and start empiric antibiotics rapidly per the local pathway.
- High-yield fact: Look for tumor lysis syndrome (TLS): rising urate, potassium, phosphate and creatinine with falling calcium can cause arrhythmia, seizure and acute kidney injury.
- High-yield fact: Also consider leukostasis, hypercalcemia, spinal cord compression, neutropenic enterocolitis and treatment toxicity.
- High-yield fact: Early oncology/hematology involvement and monitored reassessment are essential.
- High-yield fact: Recognize febrile neutropenia as an emergency; risk-stratify, obtain cultures and begin empiric antimicrobial therapy promptly under local protocol.

- Management priority: Assess tumor lysis syndrome using trends in potassium, phosphate, calcium, urate, creatinine, fluid balance and ECG; involve hematology. Treat associated electrolyte emergencies, seizures, shock and acute kidney injury through the relevant emergency pathways.
- High-yield fact: Consider cord compression, leukostasis, hypercalcemia, neutropenic enterocolitis and drug toxicity alongside infection.
- Exam trap: Do not wait for hypotension or culture results before starting indicated febrile-neutropenia antibiotics.

Febrile neutropenia طارئ زمني؛ لا تنتظر إثبات العامل الجرثومي قبل بدء التقييم والعلاج المناسب ضمن البروتوكول.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 6: Transfusion yield and adult hemoglobin increment

- Number/threshold: CLEAN V3 states that one unit of packed red cells increases hemoglobin by about 1 g/dL in an adult example.

Red flags اربط هذه النقاط بالمظهر السريري الفعلي وحدّد، وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 7: ITP treatment distinction

- Management priority: CLEAN V3 identifies corticosteroids as an initial therapy for adult immune thrombocytopenic purpura.

Red flags اربط هذه النقاط بالمظهر السريري الفعلي وحدّد، وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 8: Suspected thrombotic microangiopathy

- Clinical trigger: Small-vessel platelet-rich thrombi with thrombocytopenia and microangiopathic hemolytic anemia suggest a thrombotic microangiopathy requiring urgent hematology assessment.

Red flags اربط هذه النقاط بالمظهر السريري الفعلي وحدّد، وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 9: CML and the Philadelphia chromosome

- High-yield fact: The Philadelphia chromosome is associated with chronic myelogenous leukemia in the supplied review.

Red flags اربط هذه النقاط بالمظهر السريري الفعلي وحدّد، وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 10: Massive bleeding component balance

- High-yield fact: CLEAN V3 includes balanced packed red-cell, plasma and platelet components in a 1:1:1 major-hemorrhage pathway example.

Red flags اربط هذه النقاط بالمظهر السريري الفعلي وحدّد، وخطوة التقييم التالية، مع مراعاة سياق المصدر.

High-Yield Comparison Tables

Transfusion pulmonary reactions — TRALI / TACO

Feature	TRALI	TACO	Exam point

Hematology & Oncology | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

Tumour lysis and phosphate

• High-yield fact: Tumour lysis syndrome is cited as a cause of severe hyperphosphatemia, not hypophosphatemia.

في النص الأصلي TLS يرتبط بارتفاع phosphate وليس نقصه؛ احفظ اتجاه التغير.

PASS 9 — Source-backed clinical recovery | Domain 12

Hematology & Oncology

dosing or medical-correctness certification.

Type 2 von Willebrand disease

Exam trap: CLEAN V3 states that desmopressin (DDAVP) response varies with the type 2 VWD subtype (2A, 2B, 2M or 2N); do not assume one response across subtypes.

فعالية DDAVP في type 2 VWD تختلف بين الأنماط؛ لا تعاملها كحالة واحدة.

G6PD-associated hemolysis

Clinical Pearl: CLEAN V3 identifies nitrofurantoin among medication exposures that may trigger oxidative hemolysis in patients with G6PD deficiency, with Heinz-body formation described in the source.

يحذر المصدر من hemolysis التأكسدي مع nitrofurantoin عند G6PD deficiency.

Tumor lysis and calcium

Management caution: AQRAB advises against reflex treatment of asymptomatic hypocalcemia associated with tumor lysis before considering calcium-phosphate precipitation and specialist guidance.

في TLS لا تُصحح hypocalcemia عديمة الأعراض تلقائيًا دون تقييم ترسّب calcium-phosphate.

Mechanism	Noncardiogenic permeability edema	Hydrostatic/circulatory overload	Volume status and clinical findings discriminate
Response	Stop transfusion and provide supportive care	Stop transfusion and assess overload	Notify blood bank and investigate

Thrombocytopenia patterns — ITP / TTP

Feature	ITP	TTP	Exam point
Key concern	Isolated immune platelet destruction pattern	Microangiopathic hemolysis with organ injury risk	Smear and hemolysis features matter
Management direction	Different condition-specific treatment	Urgent specialty pathway	Do not treat every low platelet count alike

Rapid Recall: Must-Know Facts

01. CLEAN V3 states that one unit of packed red cells increases hemoglobin by about 1 g/dL in an adult example.

02. Treat acute sickle pain promptly under the patient’s plan/local protocol and reassess frequently.

03. CLEAN V3 identifies corticosteroids as an initial therapy for adult immune thrombocytopenic purpura.

04. Do not wait for hypotension or culture results before starting indicated febrile-neutropenia antibiotics.

05. Assess stability and bleeding first; classify anemia using MCV, reticulocyte response, smear and targeted tests.

06. Escalate chest pain, fever, hypoxia, neurologic deficit, sudden anemia/reticulocytopenia or splenic enlargement urgently.

07. Do not reflexively transfuse red cells or give large fluid volumes in suspected leukostasis without specialist input. ping partial.

08. Do not transfuse uncomplicated vaso-occlusive pain routinely; choose simple/exchange transfusion based on complication and specialist protocol.

09. Small-vessel platelet-rich thrombi with thrombocytopenia and microangiopathic hemolytic anemia suggest a thrombotic microangiopathy requiring urgent hematology assessment.

10. For most hemodynamically stable hospitalized adults, AABB 2023 supports considering red-cell transfusion below 7 g/dL, with selected thresholds around 7.5–8 g/dL in specified surgical/cardiovascular populations.

11. Fever with severe neutropenia is a time-critical infection pathway.

12. Cancer-related emergencies may arise from malignancy or its treatment.

13. Early oncology/hematology involvement and monitored reassessment are essential.

14. The Philadelphia chromosome is associated with chronic myelogenous leukemia in the supplied review.

15. Leukostasis risk depends on cell type and clinical physiology; there is no single safe universal WBC cutoff.

16. Severe ITP bleeding and hemophilia-related critical-site bleeding need immediate hematology-directed hemostatic

17. Interpret the absolute count and differential with symptoms, medications, infection risk, smear and prior trends.

18. Also consider leukostasis, hypercalcemia, spinal cord compression, neutropenic enterocolitis and treatment toxicity.

19. CLEAN V3 includes balanced packed red-cell, plasma and platelet components in a 1:1:1 major-hemorrhage pathway example.

20. Classify likely reaction (hemolytic, septic, allergic/anaphylactic, TACO/TRALI) and send indicated samples/unit per local

21. Transfusion is indication-specific and not routine for uncomplicated pain; involve hematology and follow local SCD protocol.

22. First classify the problem: bleeding, thrombosis, platelet disorder, coagulation-factor deficiency or microangiopathic hemolysis.

23. Suspected HIT is prothrombotic: use 4Ts assessment, stop heparin when indicated and follow a non-heparin anticoagulation pathway.

24. In hemorrhage or shock, prioritize clinical resuscitation and bleeding control; do not apply stable-patient thresholds mechanically.

25. New fever, chest symptoms/hypoxia, neurologic deficit, pallor with sudden hemoglobin fall or enlarged spleen needs urgent escalation.

26. Neutrophilia and leukopenia are patterns, not diagnoses; consider infection, inflammation, marrow disease, drugs and physiologic stress.

27. Blasts, cytopenias, rapidly rising counts or respiratory/neurologic symptoms with suspected leukostasis require urgent hematology involvement.

28. For most stable hospitalized adults, consider RBC transfusion below 7 g/dL; individualize for symptoms, comorbidity and procedure-specific guidance.

29. Recognize febrile neutropenia as an emergency; risk-stratify, obtain cultures and begin empiric antimicrobial therapy promptly under local protocol.

30. Look for tumor lysis syndrome (TLS): rising urate, potassium, phosphate and creatinine with falling calcium can cause arrhythmia, seizure and acute kidney injury.

31. In sickle cell disease, acute pain is an emergency; rapidly treat pain and screen for acute chest syndrome, stroke, sepsis, aplastic crisis and splenic sequestration.

32. Never restart a suspected reaction simply because symptoms improve before evaluation

33. Assess TLS risk and trends in potassium, phosphate, calcium, urate, creatinine, fluid balance and ECG

34. Transfusion decisions depend on active bleeding, hemodynamics, symptoms, comorbidity and the clinical context

35. If a reaction is suspected, stop the transfusion, assess ABC/vitals, maintain IV access with compatible fluid per local policy, recheck patient/unit identity, notify the blood bank and investigate hemolysis, sepsis, alle

36. Thrombocytopenia plus schistocytes/hemolysis or neurologic/cardiac injury raises urgent concern for TTP

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

Transfusion decision thresholds

- CLEAN V3: CLEAN V3 presents examples of restrictive thresholds in selected adult settings.
- AQRAB: AQRAB cites <7 g/dL in most stable adults, and 7.5–8 g/dL for selected cardiovascular/surgical populations.
- Nature of difference: Subpopulation-dependent thresholds; preserve indication and comorbidity.
- Classification: Population/context

Final clinical topics: 10. Retained integrated English bullet points: 39. Rapid Recall: 36 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 5; AQRAB-only 34; confirmed both-source 0. These are point-by-

Source-screening notes: 15 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Hematology & Oncology | Pass 8 | Explicit CLEAN V3 source locations

Hypercalcemia of malignancy — clinical trigger

• High-yield fact: In a patient with malignancy, CLEAN V3 highlights nausea, lethargy and altered mental state as clinical features that should prompt consideration of hypercalcemia.

عند مريض السرطان مع غثيان وخمول وتغير الوعي، يذكر المصدر Hypercalcemia كاحتمال مهم.

Acute leukemia in children — source pattern

• High-yield fact: The source states that acute leukemias predominate in pediatric leukemia presentations, whereas chronic forms are uncommon in children.

في الأطفال يصف المصدر Acute Leukemia بأنها أكثر شيوعًا من الأشكال المزمنة.

Hematology & Oncology

Supplementary Clinical Points

Topic: Tumor lysis versus differentiation syndrome

Key discriminator: In the CLEAN V3 APL discussion, tumor lysis syndrome is associated with hyperkalemia, hyperuricemia and hyperphosphatemia; the source explicitly distinguishes tumor lysis syndrome from differentiation syndrome.

في APL لا تخلط TLS مع Differentiation Syndrome؛ نمط TLS المذكور يرتبط بارتفاع K وuric acid وphosphate.

Source evidence: C03739: DOCX paragraph 3739

Topic: Ongoing coagulation activation: address inciting cause

Management priority: CLEAN V3 states that correction of the inciting illness is needed to limit continuing pathological coagulation-cascade activation; an isolated vignette answer does not supply a complete standalone treatment protocol.

علاج العامل الذي يفعّل coagulation cascade أساسي؛ النص الأصلي لا يقدم بروتوكولًا كاملًا بذاته.

Source evidence: C03787: DOCX paragraph 3787

Topic: Macrophage activation syndrome: platelet trend

Exam clue: AQRAB identifies a precipitous platelet-count decline as an important clue in macrophage activation syndrome; the clue is not sufficient alone for diagnosis.

الهبوط السريع بالصفائح قد يكون علامة إنذار لـMacrophage Activation Syndrome، لكنه ليس معيارًا منفردًا.

Source evidence: A407-32: AQRAB PDF p.407, block 32

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 235

7 g/dL, 7.5–8 g/dL	Number/threshold: For most hemodynamically stable hospitalized adults, AABB 2023 supports considering red-cell transfusion below 7 g/dL, with selected thresholds around 7.5–8 g/dL in specified surgical/cardiovascular populations. High-yield fact: If a reaction is suspected, stop the transfusion, assess ABC/vitals, maintain IV access with compatible
7 g/dL	patient thresholds mechanically. High-yield fact: For most stable hospitalized adults, consider RBC transfusion below 7 g/dL; individualize for symptoms, comorbidity and procedure-specific guidance.
1 g/dL	Number/threshold: CLEAN V3 states that one unit of packed red cells increases hemoglobin by about 1 g/dL in an adult example.
1:1:1	High-yield fact: CLEAN V3 includes balanced packed red-cell, plasma and platelet components in a 1:1:1 major- hemorrhage pathway example.
1 g/dL	01. CLEAN V3 states that one unit of packed red cells increases hemoglobin by about 1 g/dL in an adult example. 02. Treat acute sickle pain promptly under the patient’s plan/local protocol and reassess frequently.
7.5–8 g/dL	10. For most hemodynamically stable hospitalized adults, AABB 2023 supports considering red-cell transfusion below 7 g/dL, with selected thresholds around 7.5–8 g/dL in specified surgical/cardiovascular populations. 11. Fever with severe neutropenia is a time-critical infection pathway.
<7 g/dL, 7.5–8 g/dL	CLEAN V3: CLEAN V3 presents examples of restrictive thresholds in selected adult settings. AQRAB: AQRAB cites <7 g/dL in most stable adults, and 7.5–8 g/dL for selected cardiovascular/surgical populations. Nature of difference: Subpopulation-dependent thresholds; preserve indication and comorbidity.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 235

Thrombotic thrombocytopenic purpura (TTP) [#090, 091]

Key discriminator: TTP may present with microangiopathic hemolysis and thrombocytopenia, sometimes with neurologic or renal findings; the classic pentad need not be complete.

في TTP، يكفي أن يثير اجتماع الانحلال الدموي الميكرووعائي ونقص الصفائح الشك؛ لا تنتظر اكتمال الخماسية، وقد تظهر أعراض عصبية أو كلوية.

EDIC / Emergency & Critical Care

13 — Musculoskeletal & Rheumatology

I. Acute Joint, Tendon & Spine Presentations

Topic 1: Acute Arthritis — Septic Joint versus Crystals

- Core concept: A hot, painful, swollen joint may be septic even without fever or markedly abnormal blood tests.
- High-yield fact: When native-joint septic arthritis is suspected, perform urgent arthrocentesis and send synovial fluid for culture/organism identification, cell count/differential and crystals.
- High-yield fact: In a stable patient, obtain synovial and blood cultures before empiric antibiotics; in sepsis or shock, start antibiotics immediately and do not delay for aspiration.
- Management priority: Obtain urgent orthopedic/infectious- disease input for drainage and antimicrobial management.
- Management priority: Maintain suspicion in an acutely hot/swollen joint; no single clinical sign or blood test rules septic arthritis in or out.
- High-yield fact: Perform prompt arthrocentesis; prioritize bacterial identification/culture, cell count with differential and crystal analysis.
- High-yield fact: In stable patients obtain joint fluid and blood cultures before antibiotics; in sepsis/shock, treat immediately.
- Management priority: Coordinate urgent drainage/source control and empiric therapy with local resistance patterns and specialist teams.
- Exam trap: Do not wait for fever or culture results before escalating a septic patient.

ونسّق synovial fluid لا ينفي العدوى؛ اسحب crystals حتى دون حمى. وجود septic المفصل الساخن قد يكون العلاج والتفريغ.

Topic 2: Tendinopathy, Bursitis & Achilles Rupture

- Core concept: Tendinopathy and bursitis usually cause localized pain related to loading or movement; assess for infection, fracture, inflammatory disease and neurovascular compromise before conservative care.
- High-yield fact: Achilles rupture often follows sudden push-off with a snap or “kicked” sensation, weak plantar flexion and an abnormal calf-squeeze (Thompson) test.
- High-yield fact: Protect the ankle in plantar flexion, limit weight bearing, and arrange prompt orthopedic assessment; ultrasound can help when the diagnosis is uncertain.
- High-yield fact: Operative and nonoperative pathways are both used, chosen with the patient and specialist.

ولا neurovascular status قد تخدعك حركة الطرف؛ افحص الوظيفة والـ tendon injury أو Achilles rupture في تستبعد التمزق بسبب وجود بعض الحركة.

Topic 3: Low Back Pain — Conservative Care or Urgent MRI?

- Core concept: Most acute low back pain is nonspecific and does not need routine imaging in a non-specialist setting.
- High-yield fact: Screen for serious causes: new bladder/bowel/sexual dysfunction, saddle sensory loss, progressive bilateral weakness, cancer, fever/immunosuppression, significant trauma, fracture risk or vascular/abdominal symptoms.

• High-yield fact: Suspected cauda equina or cord compression requires immediate pathway escalation and urgent MRI/specialist assessment.

• Exam trap: Without red flags, encourage activity and self-management, use shared decisions and reassess if symptoms persist or evolve; tailor analgesia to risks and local guidance.

• High-yield fact: Ask specifically about bladder sensation/retention, bowel and sexual function, saddle sensation, bilateral/progressive weakness and systemic risk factors.

ألم أسفل الظهر مع urinary retention أو ضعف عصبي متزايد = red flag ؛ لا تعالجه كشدّ عضلي دون تقييم السبب.

II. Rheumatologic, Bone & Pediatric Injury

Topic 4: Lupus & Vasculitis — Flare, Infection or Organ Injury?

• Core concept: In systemic lupus, new symptoms may reflect inflammatory flare, infection, medication toxicity or thrombosis (including antiphospholipid syndrome).

• Management priority: Assess organ-threatening features, vital signs, medication/immunosuppression, infection exposure, prior phenotype and objective organ findings; obtain targeted labs/imaging and involve rheumatology early.

• High-yield fact: Infection can mimic flare and immunosuppression increases infection risk.

• High-yield fact: Suspected thrombosis requires standard urgent vascular evaluation.

• Management priority: Avoid reflexively escalating immunosuppression before considering infection and other mimics; treatment depends on organ involvement and severity.

• Exam trap: A single normal neurologic exam does not override evolving red flags. “No routine imaging” applies only when serious pathology is not suspected.

في lupus أو vasculitis لا تنسب الحمى تلقائيًا إلى flareً جذريًا ويختلف العلاج التهيّج قد تشبه العدوى ؛.

Topic 5: Bone & Joint Infection — Osteomyelitis

• Core concept: A hot joint or focal bone pain with fever, bacteremia, recent surgery, injection drug use or immunosuppression can represent osteomyelitis or septic arthritis.

• High-yield fact: Examine joint range of motion, adjacent soft tissue and neurovascular status.

• Management priority: Obtain blood cultures and inflammatory markers; use MRI when deeper bone disease or complications are suspected.

• High-yield fact: Aspirate an infected joint urgently for cell count, Gram stain and culture when feasible.

• High-yield fact: In sepsis, rapidly progressive infection or neurologic compromise, do not delay antibiotics and surgical consultation for sampling.

• High-yield fact: Differentiate bone infection, septic arthritis, cellulitis, fracture and crystal disease; coexistence is possible.

• High-yield fact: In children with suspected hematogenous osteomyelitis, obtain blood cultures before antimicrobials if this does not delay treatment of an ill MRI is preferred when advanced imaging is needed for pediatric osteomyelitis; ultrasound can identify an effusion for aspiration.

• High-yield fact: In suspected septic arthritis, urgent aspiration and orthopedic/ID consultation are central; start empiric therapy promptly in an unstable patient.

• Exam trap: Do not label a joint as crystal arthritis before considering infection; crystals do not exclude concomitant septic arthritis.

Bone/joint infection قد يحتاج culture وsource control مع ألم العظم أو المفصل في افحص العدوى بجدية ؛ علامات جهازية.

Topic 6: Pediatric Musculoskeletal Injuries

• Management priority: Assess mechanism, age/development, skin, deformity, focal tenderness and neurovascular status before and after immobilization.

• High-yield fact: A child’s physis may fail before surrounding ligaments; normal-appearing radiographs do not exclude an occult physeal or nondisplaced injury when focal tenderness persists.

- High-yield fact: Displaced supracondylar fractures require urgent neurovascular assessment and timely orthopedic input because of vascular injury and compartment-syndrome risk.
- Management priority: Provide age/weight- appropriate analgesia, splint in a comfortable position, document pulses/sensation/motor function and arrange follow-up for occult injury.
- High-yield fact: When the history is inconsistent with developmental ability or injury pattern, evaluate for possible inflicted injury and medical mimics without making an unsupported accusation.
- Exam trap: Do not use a normal early x-ray to dismiss persistent focal pediatric bone tenderness.
- Exam trap: Do not apply adult fracture patterns or growth-plate assumptions to a child.
- Exam trap: Do not equate a suspicious fracture pattern with proven abuse; document facts, consider mimics and follow local safeguarding policy.

إصابات الطفل ليست كسور بالغين مصغرة؛ افحص growth plates والنمط العمري والآلية والـneurovascular status.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 7: Compartment syndrome: recognition and escalation

- Core concept: Compartment syndrome can occur in a muscle group enclosed by its own fascia, and source cases call for urgent pressure assessment or surgical evaluation.
- Management priority: The supplied high-suspicion compartment-syndrome case requires urgent operative fasciotomy.

الألم المتزايد والعلامات العصبية قد يسبقان فقدان النبض في compartment syndrome.

Topic 8: Achilles rupture: immediate stabilization

- Key discriminator: The source associates Achilles rupture with an abnormal Thompson calf-squeeze test and impaired plantar flexion.
- Management priority: CLEAN V3 describes plantar-flexion splinting, avoidance of weight bearing and orthopedic follow-up.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

Topic 9: Knee dislocation and vascular compromise

- Management priority: Neurovascular compromise with posterior knee dislocation requires urgent reduction and vascular reassessment.

ولا تنس الكسور المرافقة neurovascular status وبعدها وثّق قبل reduction.

Topic 10: Intraosseous access safety with fracture

- Exam trap: CLEAN V3 warns against intraosseous access through a fractured target bone because extravasation may worsen compartment injury.

كل إصابة عظمية تحتاج فحص distal perfusion وnerve function، قبل التثبيت وبعده.

Topic 11: Exertional muscle injury with threatened compartments

- Clinical trigger: Severe muscle pain with dark urine after intense exertion raises concern for rhabdomyolysis, with compartment syndrome among the complications.

الألم المتزايد والعلامات العصبية قد يسبقان فقدان النبض في compartment syndrome.

Musculoskeletal & Rheumatology | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

Monteggia injury association

• High-yield fact: A proximal ulnar fracture should raise suspicion for associated radial-head dislocation (Monteggia injury).

عند proximal ulna fracture ابحث عن radial-head dislocation المرافقة؛ هذه هي الفكرة المذكورة بالمصدر.

Axillary nerve in anterior shoulder dislocation

• High-yield fact: The source identifies the axillary nerve as a common associated injury in anterior glenohumeral dislocation.

في anterior shoulder dislocation انتبه لاحتمال إصابة axillary nerve كما يذكر المصدر.

Rheumatoid arthritis and atlantoaxial instability

• High-yield fact: Rheumatoid arthritis is associated with risk of atlantoaxial subluxation in the source.

وجود RA يرتبط بخطر atlantoaxial subluxation؛ لا تتعامل مع ألم الرقبة بمعزل عن هذا السياق.

PASS 9 — Source-backed clinical recovery | Domain 13

Musculoskeletal & Rheumatology

dosing or medical-correctness certification.

Rheumatoid arthritis presentation

Clinical trigger: CLEAN V3 describes symmetric inflammatory peripheral polyarthritis with morning pain or stiffness lasting more than one hour as a rheumatoid-arthritis pattern.

التيبس الصباحي لأكثر من ساعة مع التهاب مفاصل متناظر يوجه إلى النمط المذكور لـRA.

Achilles tendon rupture

Key discriminator: CLEAN V3 describes a palpable tendon gap around 2-6 cm above the calcaneal insertion and possible radiographic change at Kager triangle in the cited injury.

في إصابة Achilles ابحث عن فجوة الوتر فوق الكعب؛ ويذكر المصدر Kager triangle كقرينة تصويرية.

Topic 12: Occult Scaphoid Fracture — Source Scenario

• Clinical Pearl: In the supplied case with concern for an occult scaphoid fracture, thumb-spica immobilization and orthopedic follow-up were advised.

الـoccult scaphoid fracture استدعى في الحالة المصدرية thumb-spica splint ومتابعة Orthopedics.

High-Yield Comparison Tables

Acute monoarthritis — Septic arthritis / Crystal arthritis

Feature	Septic arthritis	Crystal arthritis	Exam point
Clinical appearance	Hot and swollen; fever may be absent	Can also be acutely hot and painful	Appearance alone cannot exclude infection
Synovial analysis	Culture/organism testing central	Crystal identification helps classification	Crystals do not rule out coexistent sepsis

Limb threat — Compartment syndrome / Acute limb ischemia

Feature	Compartment syndrome	Acute limb ischemia	Exam point
Problem	Raised compartment pressure with threatened perfusion	Arterial supply compromise	Both require rapid assessment and escalation
Common mistake	Waiting for late pulse loss	Delaying vascular evaluation	Neurovascular findings and progression guide action

Rapid Recall: Must-Know Facts

01. Do not wait for fever or culture results before escalating a septic patient.
02. Do not apply adult fracture patterns or growth-plate assumptions to a child.
03. Do not use a normal early x-ray to dismiss persistent focal pediatric bone tenderness.
04. The supplied high-suspicion compartment-syndrome case requires urgent operative fasciotomy.
05. Obtain urgent orthopedic/infectious- disease input for drainage and antimicrobial management.
06. CLEAN V3 describes plantar-flexion splinting, avoidance of weight bearing and orthopedic follow-up.
07. Neurovascular compromise with posterior knee dislocation requires urgent reduction and vascular reassessment.
08. Obtain blood cultures and inflammatory markers; use MRI when deeper bone disease or complications are suspected.
09. The source associates Achilles rupture with an abnormal Thompson calf-squeeze test and impaired plantar flexion.
10. Coordinate urgent drainage/source control and empiric therapy with local resistance patterns and specialist teams.
11. Maintain suspicion in an acutely hot/swollen joint; no single clinical sign or blood test rules septic arthritis in or out.
12. Do not label a joint as crystal arthritis before considering infection; crystals do not exclude concomitant septic arthritis.
13. Assess mechanism, age/development, skin, deformity, focal tenderness and neurovascular status before and after immobilization.
14. CLEAN V3 warns against intraosseous access through a fractured target bone because extravasation may worsen compartment injury.
15. Do not equate a suspicious fracture pattern with proven abuse; document facts, consider mimics and follow local safeguarding policy.
16. Severe muscle pain with dark urine after intense exertion raises concern for rhabdomyolysis, with compartment syndrome among the complications.
17. A single normal neurologic exam does not override evolving red flags. “No routine imaging” applies only when serious pathology is not suspected.
18. Avoid reflexively escalating immunosuppression before considering infection and other mimics; treatment depends on organ involvement and severity.
19. Provide age/weight- appropriate analgesia, splint in a comfortable position, document pulses/sensation/motor function and arrange follow-up for occult injury.
20. Without red flags, encourage activity and self-management, use shared decisions and reassess if symptoms persist or evolve; tailor analgesia to risks and local guidance.

21. Suspected thrombosis requires standard urgent vascular evaluation.
22. Infection can mimic flare and immunosuppression increases infection risk.
23. Examine joint range of motion, adjacent soft tissue and neurovascular status.
24. Aspirate an infected joint urgently for cell count, Gram stain and culture when feasible.
25. Operative and nonoperative pathways are both used, chosen with the patient and specialist.
26. A hot, painful, swollen joint may be septic even without fever or markedly abnormal blood tests.
27. Most acute low back pain is nonspecific and does not need routine imaging in a non-specialist setting.
28. In stable patients obtain joint fluid and blood cultures before antibiotics; in sepsis/shock, treat immediately.
29. Differentiate bone infection, septic arthritis, cellulitis, fracture and crystal disease; coexistence is possible.
30. Suspected cauda equina or cord compression requires immediate pathway escalation and urgent MRI/specialist assessment.
31. Perform prompt arthrocentesis; prioritize bacterial identification/culture, cell count with differential and crystal analysis.
32. In sepsis, rapidly progressive infection or neurologic compromise, do not delay antibiotics and surgical consultation for sampling.
33. In systemic lupus, new symptoms may reflect inflammatory flare, infection, medication toxicity or thrombosis (including antiphospholipid syndrome).
34. In suspected septic arthritis, urgent aspiration and orthopedic/ID consultation are central; start empiric therapy promptly in an unstable patient.
35. Compartment syndrome can occur in a muscle group enclosed by its own fascia, and source cases call for urgent pressure assessment or surgical evaluation.
36. Protect the ankle in plantar flexion, limit weight bearing, and arrange prompt orthopedic assessment; ultrasound can help when the diagnosis is uncertain.
37. Ask specifically about bladder sensation/retention, bowel and sexual function, saddle sensation, bilateral/progressive weakness and systemic risk factors.
38. A hot joint or focal bone pain with fever, bacteremia, recent surgery, injection drug use or immunosuppression can represent osteomyelitis or septic arthritis.
39. Displaced supracondylar fractures require urgent neurovascular assessment and timely orthopedic input because of vascular injury and compartment-syndrome risk.
40. A child’s physis may fail before surrounding ligaments; normal-appearing radiographs do not exclude an occult physeal or nondisplaced injury when focal tenderness persists.

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

Septic arthritis with crystals

- CLEAN V3: CLEAN V3 contains limited scenario-specific diagnostic facts.
- AQRAB: AQRAB expressly states crystals do not exclude concomitant septic arthritis.
- Nature of difference: Not opposing assertions; retain full diagnostic context.
- Classification: Clarification - not direct conflict

Final clinical topics: 11. Retained integrated English bullet points: 48. Rapid Recall: 40 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 7; AQRAB-only 41; confirmed both-source 0. These are point-by-

Source-screening notes: 37 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 243

• High-yield fact: CLEAN V3 describes uncomplicated acute torticollis as commonly self· sometimes following awkward sleeping position; it mentions gentle stretching and NSAI presentation.

• High-yield fact: CLEAN V3 identifies aspirin exposure as a possible trigger for gout att altered urate excretion in the case discussion; the original excerpt does not define a un relationship.

• High-yield fact: CLEAN V3 describes dermatomyositis as inflammatory myopathy pref affecting proximal muscles, with difficulty rising to sit from a supine position in the cite

2-6 cm	Achilles tendon rupture Key discriminator: CLEAN V3 describes a palpable tendon gap around 2-6 cm above the calcaneal insertion and possible radiographic change at Kager triangle in the cited injury.

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 243

Lisfranc injury - missed midfoot trauma [#093, 094]

High-yield fact: Lisfranc injury may be missed after midfoot trauma; pain or instability after injury warrants assessment because chronic instability can follow.

قد تُفوت إصابة Lisfranc بعد رضّ منتصف القدم؛ ألم المنطقة أو عدم ثباتها يستدعي التقييم لتجنّب عدم الاستقرار المزمن.

Fibromyalgia - noninflammatory chronic widespread pain [#096]

High-yield fact: Fibromyalgia is noninflammatory widespread pain with central pain sensitization rather than destructive arthritis.

Fibromyalgia ألم واسع غير التهابي يرتبط بتحسّس الألم المركزي، لا بالتهاب مفاصل مخرّب.

Synovial fluid - inflammatory characteristics [#097]

High-yield fact: Inflammatory joint fluid loses viscosity as hyaluronic acid properties change.

تنخفض لزوجة السائل المفصلي الالتهابي مع تغير خصائص حمض الهيالورونيك؛ يفيد ذلك في التمييز.

Ankle dislocation - skin-threat recognition [#098]

High-yield fact: Tented skin and foot deformity suggest ankle dislocation with threatened soft tissues.

تشوّه الكاحل مع شد الجلد قد يعني خلعًا يهدد النسج الرخوة؛ قيّم التروية والجلد بصورة عاجلة.

EDIC / Emergency & Critical Care

14 — Dermatology

I. Dangerous Rash & Skin Disease

Topic 1: Dermatologic Emergencies — Rash with Systemic Risk

- Core concept: A rash with fever, toxicity, mucosal involvement, skin pain, blistering, rapidly spreading purpura, hypotension or neurologic symptoms can signal a life-threatening systemic disease.
- Management priority: Assess ABC, perfusion, temperature, medication exposures, onset and progression; inspect the entire skin and mucosa, including conjunctivae and hidden areas.
- High-yield fact: Consider meningococcemia/sepsis, severe cutaneous adverse reaction (SJS/TEN), necrotizing infection, vasculitis and anaphylaxis.
- High-yield fact: Resuscitate and escalate urgently; obtain cultures/labs and specialist input without delaying empiric treatment when sepsis is suspected.
- Management priority: Assess hemodynamic stability and rapidly progressive features; examine all skin, mucosal surfaces and conjunctivae.
- Management priority: Escalate immediately for nonblanching purpura with systemic illness, mucosal erosions, blistering, skin pain or necrosis. Review new medications, infection symptoms and the time course; consider SJS/TEN, sepsis, vasculitis and necrotizing infection. Treat suspected sepsis or anaphylaxis through the relevant emergency pathway and seek specialist support.
- Exam trap: Do not mistake mucosal disease or skin pain with systemic symptoms for an uncomplicated drug rash.

يعني احتمال مرض شديد؛ افحص الدورة rash مع blistering أو skin pain أو mucosal involvement وجود الدموية والجلد كاملًا ولا تكتفِ بمضاد حساسية.

II. Dermatologic Infection & Threatened Tissue

Topic 2: Skin & Soft-Tissue Infection — Necrotizing Warning

- Core concept: Cellulitis is common, but rapidly progressive pain, systemic toxicity, edema beyond erythema, wooden induration, bullae, ecchymosis, skin anesthesia or crepitus raises concern for necrotizing infection.
- High-yield fact: Resuscitate, obtain cultures if this does not delay care, start broad empiric IV antibiotics and call surgery immediately.
- High-yield fact: Imaging may help when the diagnosis is uncertain, but it must not delay operative assessment when suspicion is high.
- Management priority: Treat suspected necrotizing fasciitis as a surgical emergency: immediate surgical consultation, broad-spectrum empiric IV coverage and resuscitation.
- Management priority: Use the examination and trajectory; CT/MRI can support but a reassuring scan or score cannot safely overrule high clinical suspicion.
- High-yield fact: Search for abscess, infected wound, perineal source, foreign body, ischemia and immunocompromise.
- High-yield fact: Reassess frequently; mark erythema and document pain, perfusion, skin changes and systemic physiology.
- Exam trap: Do not let a low LRINEC score or imaging delay surgery when clinical suspicion is high.

في necrotizing infection الألم الشديد أو التدهور السريع أهم من شكل الجلد في البداية؛ لا تؤخر surgical assessment.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 3: Erythema multiforme and HSV-associated target lesions

- Core concept: Erythema multiforme is an immune-mediated cutaneous disorder characterized by target lesions.
- High-yield fact: Herpes simplex virus is an important infectious trigger of erythema multiforme.

EM يتميّز بـ target lesions، وHSV من المحرضات المذكورة في المصدر.

Topic 4: Ramsay Hunt syndrome and herpes zoster

- Clinical trigger: Ramsay Hunt syndrome can cause ear pain, vesicles of the external ear and ipsilateral facial paralysis.
- High-yield fact: Herpes zoster produces a painful vesicular eruption in a dermatomal distribution.

Ramsay Hunt يوحي بـ facial weakness قرب الأذن مع وجود vesicles؛ اربط الطفح بموقع العصب.

Topic 5: Pemphigus vulgaris and mucosal blistering

- Key discriminator: Pemphigus vulgaris causes flaccid blisters and erosions affecting skin and mucous membranes.
- High-yield fact: Pemphigus vulgaris is associated with autoantibodies against desmoglein 3 or desmogleins 1 and 3.

في pemphigus vulgaris وجود mucosal erosions مع flaccid blisters يميّزه عن طفح بسيط.

Topic 6: Seborrheic dermatitis in infants

- Clinical trigger: Seborrheic dermatitis may cause salmon-colored greasy patches with yellow scales on the scalp, diaper area or axillae.

تدعم القشور الدهنية الصفراء على فروة الرأس في الرضيع seborrheic dermatitis.

Topic 7: Erythema nodosum and inflammatory associations

- Core concept: Erythema nodosum is a form of panniculitis involving the subcutaneous fat.
- Clinical trigger: Tender red subcutaneous nodules on the shins can represent erythema nodosum; sarcoidosis is one possible association.

عقيدات مؤلمة على الساق قد تمثل panniculitis مرتبطة؛ ابحث عن أسباب جهازية.

Topic 8: Dermatitis herpetiformis and celiac disease

- High-yield fact: Dermatitis herpetiformis is an extraintestinal manifestation associated with celiac disease.

هذا الطفح قد يكون علامة خارج الأمعاء لـ celiac disease.

Topic 9: SJS/TEN recognition and drug triggers

- Number/threshold: Toxic epidermal necrolysis involves epidermal detachment affecting more than 30% of total body surface area.

• Clinical Pearl: Several medicines, including certain antiseizure drugs, sulfonamide antibiotics and allopurinol, are described as SJS triggers in CLEAN V3.

مساحة انفصال الجلد مهمة لتصنيف TEN، لكن أعراض المخاطية والأعضاء لا تقل أهمية.

Topic 10: IgA vasculitis skin and systemic clues

• Clinical trigger: Henoch-Schönlein purpura may present in a child after an upper respiratory illness with rash, abdominal pain, joint pain and hematuria.

الطفح مع ألم بطن ومفاصل و hematuria عند طفل يثير احتمال IgA vasculitis.

Topic 11: Varicella: polymorphic vesicles

• Key discriminator: Varicella produces vesicular lesions in different stages of evolution on an erythematous base.

وجود فقاعات بأطوار مختلفة يوجّه لتشخيص varicella.

Topic 12: Stasis dermatitis and topical antibiotic pitfalls

• Common mistake: Topical antibiotic preparations may provoke allergic contact dermatitis and complicate venous stasis dermatitis or ulcers.

استخدام topical antibiotics دون حاجة قد يسبب contact dermatitis ويشوّش الصورة.

Topic 13: Erythema infectiosum and parvovirus B19

• Core concept: Erythema infectiosum, also called fifth disease, is a viral exanthem caused by parvovirus B19.

تذكّر ارتباط fifth disease بـparvovirus B19 عند تقييم viral exanthem.

Topic 14: Pellagra: dermatologic metabolic presentation

• High-yield fact: Niacin (vitamin B3) deficiency can cause the pellagra triad of dermatitis, diarrhea and dementia.

ثلاثية pellagra هي dermatitis وdiarrhea وdementia، وترتبط بـniacin deficiency.

Topic 15: Erythema migrans and early Lyme infection

• Key discriminator: An expanding erythematous ring-like lesion with central clearing can represent erythema migrans in early Lyme disease.

الطفح الحلقي المتسع بعد التعرض للقراد يدعم Lyme المبكر.

Topic 16: Bullous pemphigoid as a geriatric blistering pattern

• Clinical trigger: Bullous pemphigoid is an autoimmune blistering disorder described more often in adults older than 60 years.

المرض الفقاعي عند كبار السن يحتاج تمييزًا عن pemphigus وعن severe drug reaction.

Dermatology | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

Varicella infectious interval

• High-yield fact: The source describes chickenpox as infectious until all lesions crust, commonly around 5-7 days after rash onset.

العدوى في النص تستمر حتى تجف وتتقشر جميع الآفات، وليس فقط حتى تتحسن الحرارة.

EDIC / EMERGENCY & CRITICAL CARE | SOURCE-ONLY ADDITION | NOT AN EXTERNAL CLINICAL GUIDELINE

PASS 9 — Source-backed clinical recovery | Domain 14

Dermatology

dosing or medical-correctness certification.

Scabies treatment in adults

Source-specific management: CLEAN V3 names topical permethrin 5% cream as its first-line adult scabies treatment example; the source statement does not provide a universal application schedule.

في Scabies عند البالغين يذكر المصدر permethrin 5% موضعيًا، دون تحديد كامل لجدول التطبيق هنا.

Topic 17: Other skin clues: inflammatory breast carcinoma

• Red flag: Inflammatory carcinoma of the breast may produce diffuse breast erythema, edema and skin thickening.

قد لا يكون عدوى بسيطة؛ انتبه لاحتمال edema احمرار الثدي المنتشر مع malignancy.

High-Yield Comparison Tables

Severe drug-related skin reaction — SJS/TEN / DRESS

Feature	SJS/TEN	DRESS	Exam point
Core clue	Painful epidermal/mucosal injury or skin detachment	Drug reaction with possible systemic organ injury	Do not focus only on rash appearance
Immediate step	Stop suspected drug and support organs/skin	Stop suspected drug and assess organs	Specialty escalation and follow-up are critical

Nonblanching rash — Localized skin disease / Systemically ill patient with purpura

Feature	Localized skin disease	Systemic purpura	Exam point
Initial assessment	Skin findings and local symptoms	Shock, fever, neurologic signs, mucosal findings	Do not treat dangerous systemic rash as simple dermatitis

Rapid Recall: Must-Know Facts

01. Toxic epidermal necrolysis involves epidermal detachment affecting more than 30% of total body surface area.
02. Do not let a low LRINEC score or imaging delay surgery when clinical suspicion is high.
03. Pemphigus vulgaris causes flaccid blisters and erosions affecting skin and mucous membranes.
04. Varicella produces vesicular lesions in different stages of evolution on an erythematous base.
05. Do not mistake mucosal disease or skin pain with systemic symptoms for an uncomplicated drug rash.
06. Inflammatory carcinoma of the breast may produce diffuse breast erythema, edema and skin thickening.
07. Ramsay Hunt syndrome can cause ear pain, vesicles of the external ear and ipsilateral facial paralysis.
08. Bullous pemphigoid is an autoimmune blistering disorder described more often in adults older than 60 years.
09. Assess hemodynamic stability and rapidly progressive features; examine all skin, mucosal surfaces and conjunctivae.
10. Tender red subcutaneous nodules on the shins can represent erythema nodosum; sarcoidosis is one possible association.
11. Seborrheic dermatitis may cause salmon-colored greasy patches with yellow scales on the scalp, diaper area or axillae.
12. An expanding erythematous ring-like lesion with central clearing can represent erythema migrans in early Lyme disease.
13. Use the examination and trajectory; CT/MRI can support but a reassuring scan or score cannot safely overrule high clinical suspicion.
14. Henoch-Schönlein purpura may present in a child after an upper respiratory illness with rash, abdominal pain, joint pain and hematuria.
15. Treat suspected necrotizing fasciitis as a surgical emergency: immediate surgical consultation, broad-spectrum empiric IV coverage and resuscitation.
16. Assess ABC, perfusion, temperature, medication exposures, onset and progression; inspect the entire skin and mucosa, including conjunctivae and hidden areas.
17. Erythema nodosum is a form of panniculitis involving the subcutaneous fat.
18. Herpes simplex virus is an important infectious trigger of erythema multiforme.
19. Herpes zoster produces a painful vesicular eruption in a dermatomal distribution.
20. Dermatitis herpetiformis is an extraintestinal manifestation associated with celiac disease.
21. Erythema multiforme is an immune-mediated cutaneous disorder characterized by target lesions.
22. Erythema infectiosum, also called fifth disease, is a viral exanthem caused by parvovirus B19.
23. Search for abscess, infected wound, perineal source, foreign body, ischemia and immunocompromise.
24. Pemphigus vulgaris is associated with autoantibodies against desmoglein 3 or desmogleins 1 and 3.

25. Niacin (vitamin B3) deficiency can cause the pellagra triad of dermatitis, diarrhea and dementia.

26. Reassess frequently; mark erythema and document pain, perfusion, skin changes and systemic physiology.

27. Imaging may help when the diagnosis is uncertain, but it must not delay operative assessment when suspicion is high.

28. Resuscitate, obtain cultures if this does not delay care, start broad empiric IV antibiotics and call surgery immediately.

29. Topical antibiotic preparations may provoke allergic contact dermatitis and complicate venous stasis dermatitis or ulcers.

30. Consider meningococcemia/sepsis, severe cutaneous adverse reaction (SJS/TEN), necrotizing infection, vasculitis and anaphylaxis.

31. Resuscitate and escalate urgently; obtain cultures/labs and specialist input without delaying empiric treatment when sepsis is suspected.

32. Several medicines, including certain antiseizure drugs, sulfonamide antibiotics and allopurinol, are described as SJS triggers in CLEAN V3.

33. Cellulitis is common, but rapidly progressive pain, systemic toxicity, edema beyond erythema, wooden induration, bullae, ecchymosis, skin anesthesia or crepitus raises concern for necrotizing infection.

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

Severe cutaneous reaction labels

- CLEAN V3: CLEAN V3 contains specific disorder-specific clinical vignettes.
- AQRAB: AQRAB distinguishes SJS/TEN by detached surface and DRESS/AGEP by organ and skin presentation.
- Nature of difference: Differences in classification scope; not a verified contradiction.
- Classification: Classification context

Final clinical topics: 17. Retained integrated English bullet points: 35. Rapid Recall: 33 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 20; AQRAB-only 15; confirmed both-source 0. These are point-by-

Source-screening notes: 31 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Dermatology | Pass 8 | Explicit CLEAN V3 source locations

Mupirocin — topical antimicrobial coverage

• High-yield fact: CLEAN V3 describes mupirocin as a topical treatment with activity against gram-positive organisms, including some community-acquired MRSA strains; indication and protocol remain context-dependent.

Mupirocin علاج موضعي يذكر المصدر فعاليته تجاه Gram-positive وبعض CA-MRSA؛ الاستطباب يحدد الاستخدام.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 251

30%	Number/threshold: Toxic epidermal necrolysis involves epidermal detachment affecting more than 30% of total body surface area.
5-7 days	High-yield fact: The source describes chickenpox as infectious until all lesions crust, commonly around 5-7 days after rash onset. .الحرارة تتحسن حتى فقط وليس الآفات، جميع وتتقشر تجف حتى تستمر النص في العدوى
5%	Scabies treatment in adults Source-specific management: CLEAN V3 names topical permethrin 5% cream as its first-line adult scabies treatment example; the source statement does not provide a universal application schedule.
5 %	treatment example; the source statement does not provide a universal application schedule. .هنايالتطبيق لجدول كامل تحديد دون ا، موضعًpermethrin 5 % المصدر يذكر البالغين عند Scabies في
30%	01. Toxic epidermal necrolysis involves epidermal detachment affecting more than 30% of total body surface area. 02. Do not let a low LRINEC score or imaging delay surgery when clinical suspicion is high.
60 years	07. Ramsay Hunt syndrome can cause ear pain, vesicles of the external ear and ipsilateral facial paralysis. 08. Bullous pemphigoid is an autoimmune blistering disorder described more often in adults older than 60 years. 09. Assess hemodynamic stability and rapidly progressive features; examine all skin, mucosal surfaces and conjunctivae.

EDIC / Emergency & Critical Care

15 — Procedures, Imaging & Diagnostics

I. Focused Emergency Ultrasound

Topic 1: Emergency Ultrasound / POCUS

- Core concept: Point-of-care ultrasound (POCUS) is a focused bedside examination used to answer a specific clinical question and integrate findings with history, examination and physiology.
- Management priority: Select the appropriate application, probe and views; acquire and interpret images within the operator's training and local credentialing.
- High-yield fact: Record the indication, views obtained, findings, limitations and interpretation, and save images according to local policy.
- Exam trap: A negative or technically limited study does not exclude disease when sensitivity is insufficient or the clinical probability remains concerning; obtain formal imaging or specialist review when indicated.
- High-yield fact: During CPR, scanning must not prolong chest-compression pauses; an isolated finding such as absent cardiac motion is not by itself a stopping rule.
- High-yield fact: ACEP describes core emergency ultrasound applications including trauma, pregnancy, cardiac/hemodynamic, aorta, thoracic, biliary, urinary, DVT, soft tissue, ocular, bowel and procedural guidance.
- High-yield fact: The scan is an adjunct to clinical assessment; document the question, views, findings, limitations and interpretation.
- High-yield fact: During resuscitation, only perform focused scanning that does not prolong compression pauses; no single ultrasound finding determines termination.
- Exam trap: Do not treat a focused negative scan as a universal rule-out test.

POCUS يجيب عن سؤال محدد ويكمّل الفحص؛ نتيجة سلبية أو محدودة لا تلغي مرضًا مرجحًا، ولا تُطل توقفات CPR.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 2: Ultrasound compression for suspected DVT

- Key discriminator: Normal veins collapse with gentle transducer pressure during B-mode compression ultrasonography.
- Clinical trigger: Noncompressibility of the common femoral vein with abnormal Doppler flow supports proximal deep venous thrombosis.

الوريد الطبيعي ينضغط؛ فشل الانضغاط مع Doppler غير طبيعي يدعم DVT.

Topic 3: MRI for spinal epidural abscess

- Management priority: MRI with contrast is the preferred imaging method described for suspected spinal epidural abscess.
- Clinical Pearl: In a concerning spinal epidural abscess presentation, imaging of the whole spine may be needed because examination cannot reliably localize every involved level.

في suspected spinal epidural abscess، التصوير بالرنين مهم ولا تكفي العلامات الموضعية وحدها.

Topic 4: Whole-spine imaging for malignant cord compression

• Key discriminator: Malignant spinal cord compression can affect multiple levels; MRI restricted to one spinal segment may miss disease.

قد يصيب malignant cord compression عدة مستويات؛ MRI لمستوى واحد قد لا يكفي.

Topic 5: CT angiography for cervical arterial dissection

• Key discriminator: CT angiography can delineate the vessel lumen and wall in suspected vertebral artery dissection.

CTA يوضح مسار الوعاء وجدارَه في cervical arterial dissection.

Topic 6: Duplex imaging of femoral pseudoaneurysm

• High-yield fact: Duplex ultrasonography provides noninvasive real-time assessment for suspected femoral pseudoaneurysm.

Duplex يسمح بتقييم pseudoaneurysm دون تدخل مباشر.

Topic 7: Plain-film sign of scapholunate dissociation

• Key discriminator: The cortical ring or signet-ring sign reflects rotary subluxation of the scaphoid on an anteroposterior wrist radiograph.

علامة cortical ring في wrist radiograph تساعد على كشف scapholunate injury.

Topic 8: Musculoskeletal ultrasound and pediatric hip evaluation

• Clinical trigger: For an infant with clinical signs of developmental dysplasia of the hip, ultrasound is the imaging modality described in CLEAN V3.

الـhip ultrasound خيار موصوف عند الرضيع مع علامات DDH.

Topic 9: Ultrasound for breast abscess

• Management priority: Ultrasonography can help evaluate whether a suspected breast infection contains a drainable abscess.

السونار يساعد على التفريق بين التهاب الثدي و abscess قابل للتفريغ.

Topic 10: Ultrasound settings: understanding gain

• Core concept: Ultrasound gain controls the brightness or amplitude of returned echoes.

gain يغيّر سطوع الصورة، وليس دلالة المرض بحد ذاته.

Topic 11: Soft-tissue foreign body imaging

- Key discriminator: High-resolution sonography is particularly useful for detecting wooden soft-tissue foreign bodies.

يمكن للـultrasound كشف foreign body خشبي قد لا يظهر في فحوص أخرى.

Topic 12: Lumbar puncture anatomic landmarks

- High-yield fact: A lumbar puncture needle traverses skin, subcutaneous tissue, supraspinous and interspinous ligaments, ligamentum flavum, epidural space, dura and arachnoid.
- Clinical Pearl: The described tactile pops during lumbar puncture occur at the ligamentum flavum and dura mater.

احفظ طبقات LP ومواضع الإحساس بتجاوز ligamentum flavum وdura.

Topic 13: Focused cardiac ultrasound views

- High-yield fact: The apical four-chamber view is obtained with the probe near the cardiac apex in the cited source description.
- High-yield fact: The parasternal long-axis view is obtained at the left sternal border with a source-specific orientation marker position.

اختيار probe position والـview orientation قبل تفسير أساسي POCUS القلبي.

Topic 14: Anterior and posterior epistaxis procedures

- Management priority: Anterior epistaxis is initially managed with compression of the anterior nares; the source also describes oxymetazoline.
- Differential point: A posterior epistaxis requiring packing may use a nasal balloon catheter in the cited source.

فرّق بين anterior epistaxis و الذي يبدأ بضغط الأنفposterior packing عند الحاجة.

Topic 15: Confirming suspected CSF leak

- Key discriminator: Beta-2 transferrin testing is described as a highly specific investigation for suspected cerebrospinal fluid rhinorrhea.

يُستخدم beta-2 transferrin عند الشك في CSF rhinorrhea.

Topic 16: Diagnostic joint aspiration

- Management priority: A red, hot, swollen knee with effusion requires consideration of diagnostic arthrocentesis to assess for septic arthritis.

قد يحتاج المفصل الساخن المتورم aspiration قبل افتراض gout.

Topic 17: Gentle tick extraction

- Management priority: The source describes grasping an attached tick close to the skin with forceps and pulling upward gently.

إزالة tick غير ضرورية تتم بإمساكه قرب الجلد وسحبه بلطف لتجنب أذية.

Procedures, Imaging & Diagnostics | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

Fascia iliaca block plane

• High-yield fact: The fascia iliaca block is described as placement of local anesthetic below the fascia iliaca, which covers the iliopsoas and femoral-nerve region.

موضع الحقن المذكور هو beneath fascia iliaca؛ هذه معلومة تشريحية للإجراء وليست وصفًا كاملًا لطريقته.

EDIC / EMERGENCY & CRITICAL CARE | SOURCE-ONLY ADDITION | NOT AN EXTERNAL CLINICAL GUIDELINE

Topic 18: Simple ring entrapment without ischemia

• Key discriminator: When finger perfusion is preserved and ischemic signs are absent, the source describes less-invasive ring-removal methods before more destructive approaches.

تأكد من تروية الإصبع قبل اختيار طريقة ring removal.

High-Yield Comparison Tables

Focused ultrasound result — Positive/question answered / Negative or technically limited

Feature	Positive / answered	Negative / limited	Exam point
Meaning	Supports a targeted clinical decision	Does not universally exclude disease	Pretest probability and test limitations matter
During CPR	Focused assessment without long pauses	No cardiac motion is not a sole termination rule	Maintain compressions

Procedure and test interpretation — Static single result / Dynamic re-assessment

Feature	Static result	Dynamic reassessment	Exam point
Risk	May miss evolving disease	Integrates trend, physiology and changes	Treat the patient, not an isolated number

Rapid Recall: Must-Know Facts

01. Do not treat a focused negative scan as a universal rule-out test.
02. Normal veins collapse with gentle transducer pressure during B-mode compression ultrasonography.
03. CT angiography can delineate the vessel lumen and wall in suspected vertebral artery dissection.
04. MRI with contrast is the preferred imaging method described for suspected spinal epidural abscess.
05. High-resolution sonography is particularly useful for detecting wooden soft-tissue foreign bodies.
06. Ultrasonography can help evaluate whether a suspected breast infection contains a drainable abscess.
07. The source describes grasping an attached tick close to the skin with forceps and pulling upward gently.
08. Noncompressibility of the common femoral vein with abnormal Doppler flow supports proximal deep venous thrombosis.
09. Malignant spinal cord compression can affect multiple levels; MRI restricted to one spinal segment may miss disease.
10. Anterior epistaxis is initially managed with compression of the anterior nares; the source also describes oxymetazoline.
11. Beta-2 transferrin testing is described as a highly specific investigation for suspected cerebrospinal fluid rhinorrhea.
12. The cortical ring or signet-ring sign reflects rotary subluxation of the scaphoid on an anteroposterior wrist radiograph.
13. A red, hot, swollen knee with effusion requires consideration of diagnostic arthrocentesis to assess for septic arthritis.
14. For an infant with clinical signs of developmental dysplasia of the hip, ultrasound is the imaging modality described in CLEAN V3.
15. Select the appropriate application, probe and views; acquire and interpret images within the operator’s training and local credentialing.
16. When finger perfusion is preserved and ischemic signs are absent, the source describes less-invasive ring-removal methods before more destructive approaches.
17. Ultrasound gain controls the brightness or amplitude of returned echoes.
18. A posterior epistaxis requiring packing may use a nasal balloon catheter in the cited source.
19. The described tactile pops during lumbar puncture occur at the ligamentum flavum and dura mater.
20. Duplex ultrasonography provides noninvasive real-time assessment for suspected femoral pseudoaneurysm.
21. The apical four-chamber view is obtained with the probe near the cardiac apex in the cited source description.
22. The scan is an adjunct to clinical assessment; document the question, views, findings, limitations and interpretation.
23. The parasternal long-axis view is obtained at the left sternal border with a source-specific orientation marker position.

24. Record the indication, views obtained, findings, limitations and interpretation, and save images according to local policy.

25. During resuscitation, only perform focused scanning that does not prolong compression pauses; no single ultrasound finding determines termination.

26. During CPR, scanning must not prolong chest-compression pauses; an isolated finding such as absent cardiac motion is not by itself a stopping rule.

27. A lumbar puncture needle traverses skin, subcutaneous tissue, supraspinous and interspinous ligaments, ligamentum flavum, epidural space, dura and arachnoid.

28. In a concerning spinal epidural abscess presentation, imaging of the whole spine may be needed because examination cannot reliably localize every involved level.

29. Point-of-care ultrasound (POCUS) is a focused bedside examination used to answer a specific clinical question and integrate findings with history, examination and physiology.

30. A negative or technically limited study does not exclude disease when sensitivity is insufficient or the clinical probability remains concerning

31. ACEP describes core emergency ultrasound applications including trauma, pregnancy, cardiac/hemodynamic, aorta, thoracic, biliary, urinary, DVT, soft tissue, ocular, bowel and procedural guidance.

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

Negative focused scan

- CLEAN V3: CLEAN V3 contains multiple test-specific sensitivity and diagnostic examples.
- AQRAB: AQRAB states a negative or limited POCUS is not a universal rule-out test.
- Nature of difference: Disease and pretest-probability context; no direct contradiction identified.
- Classification: Test-context

Final clinical topics: 18. Retained integrated English bullet points: 31. Rapid Recall: 31 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 22; AQRAB-only 9; confirmed both-source 0. These are point-by-

Source-screening notes: 43 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Procedures, Imaging & Diagnostics

Supplementary Clinical Points

Topic: Posterior shoulder dislocation: orthogonal imaging

Imaging discriminator: CLEAN V3 names modified axillary-lateral and trans-scapular Y-view radiographs when assessing possible posterior glenohumeral dislocation. Its garbled follow-on sentence was not used to create another imaging rule.

قد تحتاج posterior shoulder dislocation إلى modified axillary أو Y-view؛ لم أعتمد الجزء المبتور من المصدر.

Source evidence: C04309: DOCX paragraph 4309

Topic: Thromboelastography: maximum amplitude

Key discriminator: CLEAN V3 describes thromboelastography maximum amplitude (MA) as a measure of overall clot strength reflecting platelet–fibrin interaction.

MA في TEG يعبّر عن clot strength الناتجة من تفاعل platelets مع fibrin.

Source evidence: C04726: DOCX paragraph 4726

Topic: Arterial-line waveform: underdamping artifact

Common mistake: CLEAN V3 warns that an underdamped or resonant invasive arterial-pressure trace can overestimate systolic pressure; waveform quality must be considered before trusting the apparent systolic peak.

Underdamping قد يعطي systolic pressure أعلى من الحقيقي على arterial line؛ انتبه لشكل الموجة.

Source evidence: C04752: DOCX paragraph 4752

Topic: Anterior epistaxis after failed first aid

Source-specific management: CLEAN V3 identifies silver-nitrate chemical cautery as a next-step option for a visible anterior bleeding point after initial first-aid measures fail, within the cited case context.

إذا كان مصدر النزف الأمامي مرئيًا وفشلت الإسعافات الأولية، يذكر المصدر silver nitrate كخيار موضعي.

Source evidence: C04729: DOCX paragraph 4729

Topic: Basal skull fracture: source-specific CT timing

Source-dependent threshold: In the CLEAN V3 NICE head-injury teaching pathway, a sign of basal skull fracture is linked to head CT within one hour. This is a named source protocol, not an all-jurisdiction rule.

ضمن مسار NICE المذكور، علامة كسر قاعدة الجمجمة مرتبطة بتصوير CT خلال ساعة؛ لا تُنقل دون اسم المسار.

Source evidence: C03958: DOCX paragraph 3958

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 259

Cardiac devices - pacemaker versus ICD magnet response [#101]

High-yield fact: Magnets affect pacemakers and defibrillators differently; device-specific interpretation is essential.

تختلف استجابة Pacemaker وICD للمغناطيس؛ لا تعمّم تأثير جهاز على الآخر.

EDIC / Emergency & Critical Care

16 — Emergency Systems, Ethics & Evidence

I. Safety, Abuse & Equitable Care

Topic 1: Child Abuse & Safeguarding

- Management priority: Treat immediate injuries first and ensure the child's current safety.
- Management priority: Obtain a neutral, developmentally appropriate history, preferably from the child when suitable, using open questions and recording the caregiver's words without leading prompts.
- High-yield fact: Compare the mechanism with developmental ability, timing and injury pattern; perform a complete examination and document objective findings.
- High-yield fact: Consider accidental injury, medical mimics and maltreatment together.
- Number/threshold: For suspected abusive head trauma, acute head CT is commonly the first study when urgent cranial injury is suspected; children 2 years or younger with concern for physical abuse generally need a skeletal survey and follow-up survey in about 2–3 weeks as recommended by specialist guidance.
- Management priority: Involve a child-protection team and follow local mandatory-reporting law; proof is not required to raise a concern.
- Exam trap: Do not conduct a leading or repeated forensic interview; preserve the child's exact spontaneous words and use trained specialists.
- Exam trap: Do not let a normal single radiograph end the safety assessment.
- Exam trap: Do not import another country's mandatory-reporting law; apply local jurisdictional rules.

وفق النظام safeguarding وثّق الوقائع الموضوعية وافصلها عن الاستنتاجات، وفعّل child abuse عند الشك في المعتمد.

Topic 2: Geriatric Abuse & Neglect

- Core concept: Consider mistreatment when injuries, dehydration, malnutrition, medication access, hygiene or history raise concern, while recognizing that none alone proves abuse.
- High-yield fact: Speak with the older person privately when safe, assess decision-making capacity and immediate danger, use a professional interpreter when needed, and document objective findings and the person's own words.
- Management priority: Treat urgent injury/illness and develop a safe disposition with social work, safeguarding or adult-protection teams.
- Management priority: Avoid confronting a suspected perpetrator in a way that increases danger.
- High-yield fact: Definitions, reporting duties and consent rules vary by jurisdiction; follow local law and institutional policy, and consult specialists for uncertainty.
- High-yield fact: CDC defines older-person abuse broadly to include physical, sexual, psychological, financial abuse, neglect and abandonment by a trusted person/caregiver.
- Management priority: Assess immediate safety and capacity; document specific findings and statements without judgment, and interview privately Look for patterns such as delayed care, inconsistent explanations, repeated injuries, unmet basic needs or medication/financial control; consider alternative medical explanations too.
- High-yield fact: Reporting thresholds, confidentiality and adult-protection pathways are jurisdiction-specific; consult local safeguarding/social- work teams.
- Exam trap: Do not confront a suspected abuser in a way that could increase danger after discharge.

يحتاج بيئة مقابلة آمنة، توثيقًا مستقلًا، وتفعيل إجراءات الحماية المناسبة دون افتراض السبب. Elder abuse أو neglect

Topic 3: Diversity, Cultural Humility & Care Delivery

- High-yield fact: The supplied source emphasizes the use of a trained medical interpreter when communicating with patients who face language barriers.

استخدم medical interpreter مدربًا عندما توجد language barrier بدل الاعتماد على فهم غير موثوق.

- Management priority: Provide clinically equivalent, respectful care by asking each patient about their goals, communication needs, preferred name/pronouns and relevant cultural or family context without stereotyping.
- Management priority: Use a qualified medical interpreter for consequential communication when needed; speak to the patient, use short segments, avoid family members or children as interpreters when accuracy/confidentiality may be compromised, and confirm understanding with teach-back.
- Management priority: Assess literacy, access barriers, disability accommodations, financial/social needs and the feasibility of the plan.
- High-yield fact: Cultural humility is an ongoing practice of listening and adapting, not memorizing assumptions about a group.
- High-yield fact: HHS CLAS standards support timely, understandable and respectful care and competent language assistance for patients with limited English proficiency.
- High-yield fact: For covered U.S. programs, Section 1557 language access may require meaningful access; use qualified assistance and follow the applicable law and institutional policy.
- Management priority: Use teach-back to check understanding; do not infer comprehension from nodding, English fluency in casual conversation or family translation.
- High-yield fact: Ask about the individual’s priorities and barriers; avoid stereotyping, diagnostic overshadowing and assumptions based on identity.
- Exam trap: Do not assume cultural identity predicts an individual’s beliefs, adherence or decision-making preference.

محترف عند حاجز اللغة بدل الاعتماد interpreter تعني الاستماع دون افتراضات، واستخدام Cultural humility على التخمين.

Topic 4: Human Trafficking & Exploitation

- Core concept: Consider exploitation when history, injuries, control by another person, inability to speak privately, lack of access to identification/money/phone or unsafe work/sexual conditions raise concern; no single indicator confirms trafficking.
- Management priority: Treat acute medical and mental-health needs, create a private and safe opportunity to talk, explain confidentiality and its limits, use a qualified interpreter, ask permission before sensitive questions, and let the patient choose how much to disclose.
- Management priority: Avoid repeated retelling, confrontation or promises of rescue.
- High-yield fact: Offer discreet, patient-led options and trained social-work/advocacy support; consider safe contact and discharge planning.
- High-yield fact: Mandatory reporting, adult consent and child-protection duties differ by jurisdiction, so verify local rules and immediate-danger exceptions.
- High-yield fact: The HHS/ACF PEARR approach emphasizes Privacy, Educate, Ask, Respect and Respond; screening should be trauma- informed and patient-led.
- High-yield fact: A disclosure is not required before offering resources; the goal is safety and support, not law-enforcement evidence collection.
- Management priority: Use a trained interpreter and private conversation; assess immediate danger and safe methods of follow-up before providing written materials or making contact.
- High-yield fact: Apply local mandatory-reporting rules for minors and other protected groups; adult reporting/consent requirements vary by jurisdiction.

قد لا يستطيع المريض الحديث بحرية؛ وفّر خصوصية وأمانًا وتوثيقًا موضوعيًا دون مواجهات trafficking في تزيد الخطر.

Topic 5: LGBTQIA+ Patients & Sexual/Gender Diversity

- Management priority: Provide routine, nonjudgmental emergency care without treating sexual orientation, gender identity or expression as pathology.
- High-yield fact: Ask name, pronouns and anatomy/exposure information only when clinically relevant; explain why sensitive questions are needed and protect privacy.
- High-yield fact: Base screening and differential diagnoses on organs present, surgeries, hormones, sexual practices, symptoms and individual risk—not identity labels alone.
- High-yield fact: Offer an inclusive history, obtain consent before examination, use the patient's terminology and avoid assumptions about partners or reproduction.
- High-yield fact: Recognize that discrimination and barriers can delay care; assess safety, mental-health needs and access to follow-up without presuming distress.
- High-yield fact: For transgender patients, consider hormone- or procedure-related complications when relevant and consult current specialty guidance.
- High-yield fact: Sexual orientation, gender identity and gender expression are distinct; use the patient's stated terminology and maintain Base testing on anatomy, organs present, exposure and symptoms; do not assume sexual practices or screening needs from identity alone.
- High-yield fact: For transgender and gender-diverse patients, CDC recommends inclusive terminology and comprehensive sexual-health history tailored to the patient.
- High-yield fact: Ask permission before sensitive questions/examination and avoid outing the patient in shared spaces or to companions without consent, subject to applicable law.

قدّم رعاية محترمة بحسب المشكلة الطبية الفعلية والهوية التي يذكرها المريض؛ لا تجعل الافتراضات تحجب التشخيص.

Topic 6: Social Determinants of Health

- Core concept: Social determinants are the conditions in which people are born, grow, live, work and age, and the wider forces shaping them.
- High-yield fact: In the ED, ask about barriers only when relevant to care: housing, food, transport, safety, medication access, language, disability and social support.
- Management priority: Stabilize urgent illness first, then use a warm handoff to available social work, community health, shelter, food or follow-up resources.
- Management priority: Document the patient's stated needs and agreed plan; avoid blaming patients or treating a screening score as a diagnosis.
- High-yield fact: Available services and eligibility vary by location.
- High-yield fact: Social conditions shape health and access to care; identify a specific barrier that affects today's treatment or safe disposition.
- Management priority: Use targeted screening and a practical referral or warm handoff when available; document the patient's preference and plan.
- High-yield fact: A positive social-needs screen is not itself a medical diagnosis and does not establish that a resource is locally available.
- Exam trap: Do not delay stabilization while completing social-needs screening.

SDOH تؤثر على القدرة على الالتزام والمتابعة؛ اسأل عن العوائق وأدرجها ضمن خطة الخروج الآمنة.

Topic 7: Community Violence

- Core concept: Community violence is a public-health and safety concern with immediate injury, psychological trauma and longer-term health effects.
- High-yield fact: In the ED, provide usual trauma resuscitation and injury care, assess ongoing danger and safe disposition, offer trauma- informed support, and connect interested patients with hospital-based violence intervention or community services when available.
- High-yield fact: Population-level prevention addresses interacting individual, relationship, community and societal conditions; risk factors are not deterministic and never make a victim responsible.
- Management priority: Coordinate with security or law enforcement only as clinically necessary and consistent with patient safety, consent and local law.

- Management priority: Treat immediate injury and assess whether the patient faces a credible ongoing threat before discharge.
- Management priority: Use trauma-informed, nonjudgmental communication and offer locally available support or violence-intervention services.
- High-yield fact: CDC describes prevention across social-ecological levels; risk factors are not individual predictions or proof of causation.
- Exam trap: Do not assume gang involvement, perpetrator identity or future violence from neighborhood or demographic characteristics.
- Exam trap: Do not use risk factors to blame the patient or deny care.

Community violence تحتاج تقييم الإصابة والخطر المستمر، مع توثيق موضوعي وخطة أمان مناسبة.

Topic 8: Sexual Assault

- Core concept: First address life-threatening injuries and urgent medical needs.
- High-yield fact: Offer a private, safe setting, explain choices, obtain consent separately for each examination, specimen and evidence step, and allow the patient to pause or decline without losing medical care.
- Management priority: Involve a trained sexual-assault examiner/advocate when available.
- Management priority: Assess pregnancy possibility, STI exposure, hepatitis B/HPV vaccination and HIV exposure; offer emergency contraception when indicated.
- Number/threshold: Current CDC guidance recommends case- specific HIV nPEP assessment and starting indicated nPEP as soon as possible, no later than 72 hours , for 28 days .
- High-yield fact: Forensic collection, documentation, chain of custody, reporting and follow-up must follow local protocol and the survivor’s choices.
- High-yield fact: Absence of visible injury does not exclude sexual assault; offer a consent-based, survivor-centered history and examination.
- High-yield fact: CDC STI recommendations include site- and exposure-based testing, pregnancy assessment and vaccination/prophylaxis decisions; adapt to patient and local protocol.
- High-yield fact: Forensic evidence collection and clinical treatment are related but distinct; treatment must not depend on evidence collection.

في sexual assault و قدّم العلاج العاجل أولًا، واشرح الخيارات وconsent، ولا تربط الرعاية بإجراء forensic evidence.

Topic 9: Intimate Partner Violence & Abuse

- Core concept: IPV may include physical or sexual violence, coercive control, psychological abuse and financial control.
- High-yield fact: When safe, speak with the patient alone, use a private and nonjudgmental approach, validate disclosure, assess immediate danger and urgent injury, and ask what help the patient wants.
- High-yield fact: Offer individualized safety planning and referral to an advocate or local service; avoid confronting the partner or sending unsafe materials home.
- Management priority: Document objective findings and the patient’s words.
- Management priority: Explain confidentiality and any limits before disclosure when feasible.
- High-yield fact: Reporting duties, consent rules and available resources vary by jurisdiction, age and capacity.
- Exam trap: Do not interview in front of a possible abusive partner or use a companion as interpreter for sensitive disclosure.
- Exam trap: Do not pressure the patient to leave, report or accept a referral; support informed choices unless law or imminent safety duties require action.
- Exam trap: Do not document speculation as fact or expose printed discharge material when it could increase danger.

في intimate partner violence وفّر مقابلة منفردة وآمنة واحترم القرارات والخصوصية وفق التزامات السلامة.

II. Global Care, Ethics & Medicolegal Practice

Topic 10: Global Emergency Medicine

- Core concept: Global emergency medicine considers how emergency care systems deliver timely, safe and equitable assessment and treatment across different settings and resource levels.
- High-yield fact: Begin with systematic triage and ABCDE, recognize time-sensitive illness or injury, provide interventions that are feasible and evidence-based in the local context, and arrange safe transfer or follow-up when capability is exceeded.
- High-yield fact: System strengthening includes prehospital response, emergency-unit readiness, trained teams, essential supplies, referral pathways, quality improvement and preparedness.
- High-yield fact: Adapt protocols to local epidemiology, scope of practice and resources; a resource-limited pathway should preserve essential care and dignity rather than assume a single high- resource model.
- High-yield fact: WHO Basic Emergency Care teaches a systematic approach to time-sensitive illness and injury, including trauma, breathing difficulty, shock and altered mental status.
- High-yield fact: The WHO emergency-care framework spans scene, transport, emergency-unit and early inpatient care; capacity varies by setting. Use context-appropriate resources, referral/transfer pathways and continuous quality-improvement processes.
- Exam trap: Do not assume that high-income resource availability or referral times apply in every health system.
- Exam trap: Do not confuse resource adaptation with withholding essential emergency care.
- Exam trap: Do not use a system framework as a patient-specific treatment guideline.

Global EM يعتمد على الموارد الفعلية والوبائيات المحلية؛ لا تنقل بروتوكولًا من نظام آخر بلا تكييف.

Topic 11: Humanitarian Aid in Disaster & Conflict

- Core concept: Humanitarian emergencies disrupt routine services and may produce displacement, mass injury, outbreaks and prolonged interruption of care.
- High-yield fact: Priorities include staff and patient safety, rapid needs assessment, clear triage and referral, continuity of essential services, protection of vulnerable people, coordination with local authorities and humanitarian partners, and reliable communication.
- High-yield fact: In a mass-casualty incident, use the locally adopted incident command and triage system; priorities may shift because demand exceeds available capacity.
- High-yield fact: Adapt interventions to local epidemiology, supplies and workforce, while preserving dignity, impartiality and essential care.
- High-yield fact: Humanitarian packages are planning aids and must be contextualized.
- High-yield fact: MCI triage prioritizes patients by urgency and likely benefit when demand exceeds available resources; use the locally adopted protocol.
- High-yield fact: Humanitarian health-service packages are starting points and should be adapted to local burden, delivery platforms and system capacity.
- High-yield fact: Preparedness includes roles, communication, supplies, referral, staff safety and post-event recovery.
- Exam trap: Do not apply routine one-patient-at-a-time resource assumptions during a declared MCI without using the incident plan.

في الكوارث الإنسانية احمِ العاملين والمرضى، ورتّب الفرز وتوزيع الموارد حسب الأولوية والقدرة.

Topic 12: End of Life & Palliative Emergencies

- Core concept: Palliative care addresses physical, psychological, social and spiritual suffering and can accompany disease-directed treatment.
- High-yield fact: In an emergency, assess reversible causes and immediate distress, ask the patient’s goals and what matters now, review available advance directives and identify the appropriate surrogate if capacity is absent.
- High-yield fact: A do-not-resuscitate order addresses CPR and does not automatically mean no treatment, symptom relief or evaluation of reversible problems.
- Management priority: Provide comfort, communication and a feasible plan for medications, support and follow-up.
- Exam trap: Do not apply foreign forms or surrogate hierarchies without checking local law and document validity.
- Exam trap: A DNR order does not by itself prohibit analgesia, antibiotic treatment or evaluation for reversible illness.

- Key discriminator: The documented wishes of a patient with decision-making capacity take precedence over an unsupported competing family preference.

DNR يخص CPR، ناقش goals of care ولا يعني إيقاف المسكنات أو التقييم أو كل العلاجات. والوثائق والممثل القانوني.

Topic 13: Bioethics in Emergency Medicine

- Core concept: Ethical practice in emergency medicine commonly balances respect for autonomy, beneficence, nonmaleficence and justice while protecting privacy and professional integrity.
- High-yield fact: For a capable adult, explain the condition, options, material risks and benefits, and respect an informed voluntary decision, including refusal.
- High-yield fact: Capacity is decision-specific and may fluctuate; assess understanding, appreciation, reasoning and ability to communicate a choice.
- High-yield fact: If immediate treatment is needed to prevent serious harm and the patient lacks capacity with no available authorized surrogate, emergency treatment may be ethically justified within applicable law and policy; inform the patient or surrogate when possible afterward.
- Exam trap: Do not equate disagreement or an unwise decision with incapacity.
- Exam trap: Do not use implied consent to override an explicit refusal by a patient with decision-making capacity.
- Exam trap: Do not assume local rules on minors, guardians, reporting or surrogate hierarchy are universal.

Decision-making capacity خاصة بالقرار وقد تتغير؛ وثّق الفهم والتقدير والاستدلال والاختيار، واحترم الرفض المستنير.

Topic 14: EMTALA & Medicolegal Issues

- High-yield fact: In the supplied medicolegal material, an occurrence insurance policy covers an incident according to when it occurred, regardless of when the claim was filed.
- Key discriminator: The supplied EMTALA example distinguishes a receiving hospital lacking capability or capacity when evaluating whether a transfer may be declined.

يرتبط بتاريخ وقوع الحدث، وليس occurrence policy. وcapacity وcapability يميّز النص بين EMTALA في تاريخ تقديم المطالبة.

- Core concept: EMTALA is a United States federal law, not a general international rule.
- High-yield fact: For Medicare-participating U.S. hospitals with emergency services, CMS describes duties to provide an appropriate medical screening examination, stabilizing treatment for an emergency medical condition within the hospital’s capability, or an appropriate transfer when required.
- High-yield fact: The details depend on current statute, regulations, CMS interpretive guidance and case facts.
- High-yield fact: Keep clinical documentation objective: record presentation, findings, decisions, discussions, consent/refusal, consultations and transfer arrangements contemporaneously.
- High-yield fact: In Jordan or another jurisdiction, use local law, hospital policy and qualified legal advice; do not import U.S. rules as local law.
- High-yield fact: EMTALA is U.S.-specific: appropriate medical screening, stabilizing treatment within capability, or appropriate transfer as applicable.
- High-yield fact: A medical screening examination is not merely triage; it must be appropriate to presentation and hospital capability under applicable rules.
- Management priority: Document contemporaneous facts, clinical reasoning, consent/refusal, consultation and transfer communication; follow local policy and law.
- Exam trap: Do not describe EMTALA as Jordanian or universal law.

القواعد القانونية تختلف بحسب الولاية والنظام؛ فرّق بين واجب التقييم الإسعافي وبين إجراءات النقل والتوثيق.

Topic 15: Forensic Emergency Medicine

- Core concept: Forensic emergency medicine applies careful clinical assessment and objective documentation when injuries, abuse, assault or other events may have legal relevance.
- Management priority: Obtain informed consent for examination and evidence collection where possible; explain choices, privacy and limits of confidentiality.
- High-yield fact: Record the patient's words in quotation marks when useful, distinguish reported history from observed findings and clinical interpretation, use body diagrams or photographs only with consent and policy, and preserve chain of custody when evidence is collected by trained staff.
- Exam trap: Do not infer a perpetrator, mechanism or timing beyond what findings support.
- High-yield fact: Follow applicable local law and institutional protocol and involve trained forensic specialists when needed.
- Management priority: Stabilize urgent illness or injury before evidence collection; coordinate with trained forensic examiners when available.
- High-yield fact: Separate patient statements, observed findings and clinical interpretation in the record; avoid unsupported certainty about mechanism or timing.
- High-yield fact: Evidence collection requires consent, proper handling and chain-of-custody procedures under local protocol; clinical care must not depend on it.
- Exam trap: Do not delay lifesaving treatment to preserve forensic evidence.

في forensic care: ثبّت الحالة أولًا، ثم وثّق كلمات المريض والفحص الموضوعي منفصلين عن الاستنتاجات واحترم consent.

III. Quality, EMS, Disaster & Humanitarian Response

Topic 16: Quality Improvement & Patient Safety

- Core concept: Patient safety improvement treats errors and harm as signals from a complex care system, not simply as individual failure.
- Management priority: Stabilize and disclose/manage immediate patient harm according to institutional policy, preserve relevant records, report safety events through the approved system and support affected patients and staff.
- High-yield fact: For improvement work, define a measurable problem, examine workflow and contributing conditions, test a small change, study results and adapt; monitor balancing effects and sustain successful changes.
- High-yield fact: Distinguish a hazard, near miss, no-harm incident and harmful incident using local definitions.
- High-yield fact: Reporting and review should support learning, accountability and prevention while respecting confidentiality and legal requirements.
- High-yield fact: WHO frames patient safety as coordinated system-wide action to reduce avoidable harm and build learning systems.
- High-yield fact: A practical improvement cycle defines a problem and measure, tests a change, studies results and adapts; monitor unintended effects.
- High-yield fact: Event reporting supports learning but does not replace disclosure, clinical care, formal investigation or required reporting.
- Exam trap: Do not label every adverse outcome as negligence or every error as patient harm.

في patient safety الهدف كشف أخطاء النظام وتحسين العمليات، وليس الاكتفاء بلوم الفرد. راقب النتائج بعد التدخل.

Topic 17: Patient Experience in the Emergency Department

- Core concept: A good ED experience combines timely, clinically safe care with respect, clear communication, empathy and inclusion.
- Management priority: Explain what is happening, likely waits and next steps; listen without interruption, use understandable language and a qualified interpreter when needed, check understanding and invite questions.
- High-yield fact: Protect privacy, include family or support people only with patient permission when possible, and address pain, accessibility, cultural needs and uncertainty.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 267

2 years	Number/threshold: For suspected abusive head trauma, acute head CT is commonly the first study when urgent cranial injury is suspected; children 2 years or younger with concern for physical abuse generally need a skeletal survey and follow-up survey in about 2–3 weeks as recommended by specialist guidance.
2–3 weeks	cranial injury is suspected; children 2 years or younger with concern for physical abuse generally need a skeletal survey and follow-up survey in about 2–3 weeks as recommended by specialist guidance. Management priority: Involve a child-protection team and follow local mandatory-reporting law; proof is not required
72 hours, 28 days	Number/threshold: Current CDC guidance recommends case- specific HIV nPEP assessment and starting indicated nPEP as soon as possible, no later than 72 hours , for 28 days . High-yield fact: Forensic collection, documentation, chain of custody, reporting and follow-up must follow local

- High-yield fact: Patient feedback can identify communication or process problems, but satisfaction scores alone do not establish clinical quality or safety.
- Exam trap: Never promise a specific wait time or outcome that the team cannot control.
- High-yield fact: Clear empathetic communication, understandable explanations, privacy and inclusion of patient p can improve the care experience.
- Management priority: Use a qualified interpreter for important communication when needed and verify understanding with teach-back.
- High-yield fact: Patient experience measures complement, but do not replace, clinical quality, safety and outcome measures.
- Exam trap: Do not give false certainty about waiting times, diagnosis or treatment outcomes.

Patient experience لا يعني رضا شكليًا؛ التواصل الواضح وإعادة التقييم وخطة الخروج جزء من سلامة الرعاية.

Topic 18: Physician Wellness, Stress & Impairment

- Core concept: Physician wellness is shaped by workload, scheduling, staffing, exposure to suffering, workplace safety, administrative burden, stigma and access to confidential care.
- High-yield fact: WHO classifies burnout as an occupational phenomenon related to chronic workplace stress, not a medical diagnosis.
- High-yield fact: Prevention therefore requires organizational changes as well as individual support.
- High-yield fact: Seek timely clinical help for mental or physical illness, substance-use concerns or exhaustion that affects safe practice.
- High-yield fact: If a clinician may be impaired, protect patients immediately through the established confidential peer-support, occupational-health and fitness-for-duty pathway; avoid gossip or unsupported diagnosis.
- High-yield fact: Local law and institutional policy govern reporting and practice restrictions.
- High-yield fact: WHO defines burnout as an occupational phenomenon resulting from chronic workplace stress; it is not classified as a medical condition.
- Exam trap: Do not reduce burnout prevention to self-care while ignoring staffing, schedules and workplace conditions.
- Exam trap: Do not diagnose impairment from a single report or disclose sensitive information unnecessarily.

Burnout أو impairment قد يؤثران في جودة القرار؛ اعرف مسارات الدعم والسلامة المهنية دون إهمال المرضى.

Topic 19: Emergency Medical Services & Ground Transport

- Core concept: EMS is a system linking public access, dispatch, field assessment and treatment, medical oversight, transport, receiving facilities and quality improvement.
- Management priority: Use the locally authorized triage and clinical protocols, match interventions to provider scope and resources, and communicate with medical control when required.
- High-yield fact: During transport, reassess the patient, secure equipment, continue indicated monitoring and treatment, and give a structured handover with the reason for transfer, key findings, treatment, response and outstanding risks.
- High-yield fact: EMS models, staffing, scope, dispatch and destination rules differ by jurisdiction; the WHO 2025 operational guidance is a systems reference rather than a substitute for local protocols.
- High-yield fact: WHO 2025 ambulance-system guidance covers governance, dispatch, medical oversight, protocols, equipment, medication lists, handover and infection control.
- High-yield fact: Choose a destination and transport plan based on time-sensitive needs, available capability, patient condition and local system rules.
- High-yield fact: Reassessment and structured handover are essential; transport does not replace ongoing stabilization and monitoring.
- Exam trap: Do not assume the same EMS scope, destination rules or medical-control model across jurisdictions.
- Exam trap: Do not transfer without matching crew skill, equipment and monitoring to the patient’s anticipated risks.

EMS handover ينبغي أن يوضح الحالة والتدخلات والاتجاه الزمني، مع اتصال مبكر عند الأمراض الحساسة للوقت.

Topic 20: Air Medical Transport

- Core concept: Air medical transport may help when a patient needs specialized care unavailable locally, a time-sensitive intervention is meaningfully closer by air, or geography prevents timely ground access.
- High-yield fact: Compare expected clinical benefit with activation, launch and transfer time, ground alternatives, weather, landing access, aircraft/crew capability and aviation risk.
- Management priority: Coordinate dispatch, receiving-facility acceptance, stabilization, en-route monitoring and handover.
- High-yield fact: Air transport is not automatically faster door-to-door or safer for every patient; the pilot/operator retains aviation safety authority.
- High-yield fact: Patient selection and flight operations must follow local EMS and aviation regulations and the operator’s protocols.
- High-yield fact: Joint NAEMSP/ACEP/AMPA guidance frames air-medical need around advanced care, time to definitive treatment or difficult access, alongside risk and system integration.
- High-yield fact: Compare total time to definitive care, not flight speed alone; account for activation, packaging, landing and handover.
- High-yield fact: Aviation safety assessment is independent and may override the medical mission; use operator and local regulatory procedures.
- Exam trap: Do not equate air distance or flight time with faster definitive treatment.

وازن سرعة النقل والمخاطر والمعدات والقدرة السريرية والظروف الجوية ضمن خطة air transport في الفريق.

Topic 21: Disaster Preparedness

- Core concept: Disaster readiness is a continuous cycle of mitigation, preparedness, response and recovery.
- High-yield fact: Facilities should maintain a hazard- informed plan with clear activation criteria, incident command roles, communications, staff accountability, surge space, supplies, patient tracking, triage, referral/transport, decontamination interfaces and continuity of essential services.
- High-yield fact: Train staff and exercise the plan with partner agencies; review lessons and update procedures.
- High-yield fact: When an incident overwhelms routine capacity, activate the locally adopted mass-casualty plan and its specific triage system.
- High-yield fact: Reassess priorities as patient volume, resources and incident conditions change.
- High-yield fact: Planning must fit the facility and region rather than copy a generic template unchanged.
- High-yield fact: WHO recommends facility-specific mass-casualty plans, activation criteria, role clarity, drills and coordination across the emergency unit.
- High-yield fact: MCI triage is distinct from routine triage when patient demand exceeds available resources; use the locally adopted tool.
- High-yield fact: Preparedness includes communication, workforce, space, supplies, patient flow, transfer and recovery, not triage alone.

Disaster preparedness يبدأ بـ incident command الاستجابة العشوائية، لا بالفرز والاتصال ومراجعة الجاهزية.

Topic 22: Weapons of Mass Destruction & CBRN Events

- Core concept: A suspected chemical, biological, radiological or nuclear (CBRN) event creates hazards for patients, responders and the receiving facility.
- Exam trap: Do not enter an unsafe or potentially contaminated area without the required training, command authorization and hazard-appropriate protection.
- High-yield fact: Activate the incident plan, notify hazardous-materials/public-health partners, establish safe zones and prevent secondary contamination.
- High-yield fact: Decontaminate as directed by the agent and local protocol, then assess and treat immediate threats; chemical exposure may require rapid removal from exposure and decontamination while supportive care proceeds.
- High-yield fact: Radiation monitoring, PPE, agent-specific diagnostics and antidotes are specialist- and protocol-dependent.
- High-yield fact: Notify the receiving hospital before transport and document exposure details.

• High-yield fact: Responder safety and contamination control come first; identify the hazard and use trained HAZMAT support and appropriate Prevent secondary contamination by coordinating decontamination before transport when indicated and alerting the receiving Treatment, antidotes, monitoring and PPE are agent-specific; use current specialist and local protocols rather than generic assumptions.

• Exam trap: Do not send unprotected staff into an unknown or unsafe environment.

• Exam trap: Do not transport potentially contaminated patients without coordinating the required decontamination and receiving- facility precautions.

حافظ على سلامة المستجيبين ومنع التلوث الثانوي، ثم إزالة التلوث وتقييم المصابين حسب CBRN في البروتوكول.

Topic 23: Tactical EMS & Urban Search and Rescue

• Core concept: Tactical EMS and urban search and rescue (USAR) involve care in austere or hazardous operations where access, communications, extraction and responder safety are constrained.

• High-yield fact: Medical teams should be integrated into incident planning, trained for their assigned role, equipped for the environment and coordinated with incident command, law enforcement/fire- rescue and receiving EMS.

• Exam trap: Do not enter an unsafe area outside the team’s authorization or protection.

• High-yield fact: Prioritize immediate threats to life, request technical rescue for complex extrication, and plan casualty collection, evacuation, reassessment and handover.

• High-yield fact: Scope, operational zones, reporting and clinical protocols vary; use the agency’s current local procedures and recognized training.

• High-yield fact: This overview is educational, not an operational tactics manual.

• Exam trap: Do not direct clinicians into an unsecured or uncontrolled hazard area.

• Exam trap: Do not confuse a medical summary with authority to perform tactical operations or technical rescue.

• Exam trap: Do not assume one zone model, scope or tactical protocol applies across agencies and countries.

أمان المشهد والاتصال والاستخراج أهم من دخول منطقة لا يسمح الفريق بدخولها USARو tactical EMS في.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 24: Confirmation bias and diagnostic momentum

• Common mistake: Confirmation bias is the tendency to seek or interpret information that supports a pre-existing belief.

• Clinical Pearl: Diagnostic momentum occurs when later clinicians adopt a diagnosis without critical reassessment of the evidence.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags وخطوة التقييم التالية، مع مراعاة سياق المصدر.

Topic 25: Decision-specific capacity

• High-yield fact: A diagnosis such as Down syndrome does not automatically mean that a patient lacks capacity for a specific care decision.

الـcapacity ليست صفة ثابتة لكل القرارات؛ وثّق تقييمها في الموقف الحالي.

Topic 26: Forest Plot Interpretation

• High-yield fact: On a forest plot, the square or box marks an individual study’s point estimate and its horizontal line represents the confidence interval.

• Key discriminator: A forest plot diamond represents the overall pooled effect estimate; its width represents the pooled confidence interval.

Emergency Systems, Ethics & Evidence | Only CLEAN V3 DOCX | Pass 7 evidence insertion

> These source-specific points were not evident as complete standalone propositions in the prior integrated body. Their wording is preserved with explicit original paragraph references. They do not override local policies or competing source protocols.

UK Mental Capacity Act — decision specificity

• High-yield fact: Under the UK Mental Capacity Act 2005 framework described in CLEAN V3, capacity assessment is decision-specific and not inferred from age or diagnosis alone.

هذه قاعدة ضمن UK Mental Capacity Act 2005؛ يجب عدم تعميم الإطار القانوني على كل بلد.

UK Montgomery ruling — consent context

• High-yield fact: CLEAN V3 cites Montgomery v Lanarkshire Health Board as a UK Supreme Court ruling affecting the standard for consent; it is jurisdiction-specific.

هذا مثال قانوني خاص بالمملكة المتحدة، ولا يجوز تقديمه كقاعدة قانونية عامة في جميع الأنظمة.

PASS 9 — Source-backed clinical recovery | Domain 16

Emergency Systems, Ethics & Evidence

dosing or medical-correctness certification.

Forest plot and statistical significance

Exam interpretation: In the CLEAN V3 teaching example, a confidence interval that touches or crosses the line of no effect is not statistically significant at the stated level.

في forest plot عبور أو ملامسة CI لخط no effect يعني عدم ثبوت الدلالة وفق المثال.

METHANE major incident report

Management priority: CLEAN V3 instructs the first senior responder at a potential mass-casualty incident to declare a major incident and give a structured METHANE report.

في major incident يذكر المصدر الإعلان المبكر واستخدام METHANE لتقرير منظم.

Quality improvement: system factors

Exam trap: AQRAB cautions that staff retraining alone is insufficient when equipment, workflow or system design contributed to a patient-safety problem.

تحسين السلامة يحتاج معالجة سبب الخلل في النظام وليس الاكتفاء بتدريب الأفراد.

Emergency Systems, Ethics & Evidence

Precisely source-linked additions after the priority-safety review. Not independent guideline verification.

Topic: Emergency contraception following sexual assault

Time-critical window: CLEAN V3 states that emergency contraception should be offered for consideration up to 5 days after vaginal assault; the related case selects ulipristal. Preserve consent, eligibility, safeguarding and local pathway context rather than converting a vignette into an automatic drug choice.

فترة الخمسة أيام في المصدر مرتبطة بخيارات Emergency Contraception بعد الاعتداء؛ القرار يحتاج موافقة المريضة والسياق المناسب.

> Do not merge case-specific, adult and pediatric regimens into a single dose. Source-dependent conflicts remain unresolved.

في forest plot: المربع يمثل تقدير الدراسة الواحدة مع CI، والماسة تمثل pooled estimate وعرضها يعبر عن CI للمحصلة.

High-Yield Comparison Tables

End-of-life concepts — DNR / Comfort-only plan

Feature	DNR	Comfort-only care	Exam point
Meaning	Addresses CPR	Describes broader agreed treatment goals	DNR alone does not prohibit other therapy
Communication	Check valid order and wishes	Clarify scope with capable patient/surrogate	Do not assume limits not documented

Ethical decision — Patient with capacity / Patient without capacity

Feature	Patient with capacity	Patient without capacity	Exam point
Choice	Patient informed decision or refusal	Use valid directive/authorized surrogate under local law	Capacity is decision-specific
Documentation	Record options and decision	Record authority, values and reasoning	Family preference does not override a capable patient

Rapid Recall: Must-Know Facts

01. Current CDC guidance recommends case- specific HIV nPEP assessment and starting indicated nPEP as soon as possible, no later than 72 hours , for 28 days .
02. Document objective findings and the patient’s words.
03. Do not describe EMTALA as Jordanian or universal law.
04. Do not use risk factors to blame the patient or deny care.
05. Avoid repeated retelling, confrontation or promises of rescue.
06. Do not let a normal single radiograph end the safety assessment.
07. Do not delay lifesaving treatment to preserve forensic evidence.
08. Do not equate disagreement or an unwise decision with incapacity.
09. Involve a trained sexual-assault examiner/advocate when available.
10. Do not delay stabilization while completing social-needs screening.
11. Do not send unprotected staff into an unknown or unsafe environment.
12. Treat immediate injuries first and ensure the child’s current safety.
13. Explain confidentiality and any limits before disclosure when feasible.
14. Do not direct clinicians into an unsecured or uncontrolled hazard area.
15. Do not use a system framework as a patient-specific treatment guideline.
16. Avoid confronting a suspected perpetrator in a way that increases danger.
17. Never promise a specific wait time or outcome that the team cannot control.
18. Do not equate air distance or flight time with faster definitive treatment.
19. Do not enter an unsafe area outside the team’s authorization or protection.
20. Do not confuse resource adaptation with withholding essential emergency care.
21. Do not infer a perpetrator, mechanism or timing beyond what findings support.
22. Do not label every adverse outcome as negligence or every error as patient harm.
23. Do not give false certainty about waiting times, diagnosis or treatment outcomes.
24. Do not import another country’s mandatory-reporting law; apply local jurisdictional rules.
25. Provide comfort, communication and a feasible plan for medications, support and follow-up.
26. Do not assume local rules on minors, guardians, reporting or surrogate hierarchy are universal.
27. Do not assume one zone model, scope or tactical protocol applies across agencies and countries.
28. Do not diagnose impairment from a single report or disclose sensitive information unnecessarily.
29. Do not assume the same EMS scope, destination rules or medical-control model across jurisdictions.
30. Do not confuse a medical summary with authority to perform tactical operations or technical rescue.

31. Do not assume that high-income resource availability or referral times apply in every health system.
32. Coordinate dispatch, receiving-facility acceptance, stabilization, en-route monitoring and handover.
33. Do not apply foreign forms or surrogate hierarchies without checking local law and document validity.
34. Do not use implied consent to override an explicit refusal by a patient with decision-making capacity.
35. Treat immediate injury and assess whether the patient faces a credible ongoing threat before discharge.
36. Do not document speculation as fact or expose printed discharge material when it could increase danger.
37. Do not transfer without matching crew skill, equipment and monitoring to the patient's anticipated risks.
38. Do not assume cultural identity predicts an individual's beliefs, adherence or decision-making preference.
39. Do not reduce burnout prevention to self-care while ignoring staffing, schedules and workplace conditions.
40. A DNR order does not by itself prohibit analgesia, antibiotic treatment or evaluation for reversible illness.

Conflict Resolution Register

Only verified opposing statements are classified as "Conflict — source-dependent". Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

EMTALA and local legal duties

- CLEAN V3: CLEAN V3 includes jurisdiction-specific emergency-law and ethics examples.
- AQRAB: AQRAB emphasizes valid local laws, patient capacity and appropriate documentation.
- Nature of difference: Different jurisdictions and legal frameworks; cannot generalize across systems.
- Classification: Jurisdiction-dependent

Final clinical topics: 25. Retained integrated English bullet points: 206. Rapid Recall: 40 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 3; AQRAB-only 203; confirmed both-source 0. These are point-by-

Source-screening notes: 17 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Medicine bullet removed. No missing clinical content has been guessed.

Emergency Systems, Ethics & Evidence | Pass 8 | Explicit CLEAN V3 source locations

> **Single-blind study — masking definition**
>
> • High-yield fact: CLEAN V3 defines a single-blind study as a trial in which one party, usually participants, is unaware of treatment allocation while investigators remain aware in the cited example.
>
> في Single-blind يكون طرف واحد فقط غير عالم بتوزيع العلاج، وغالبًا المشاركون حسب المثال.

Emergency Systems, Ethics & Evidence

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 274

5 days	Time-critical window: CLEAN V3 states that emergency contraception should be offered for consideration up to 5 days after vaginal assault; the related case selects ulipristal. Preserve consent, eligibility, safeguarding and local pathway context rather than converting a vignette

72 hours, 28 days	01. Current CDC guidance recommends case- specific HIV nPEP assessment and starting indicated nPEP as soon as possible, no later than 72 hours , for 28 days . 02. Document objective findings and the patient’s words.

Supplementary Clinical Points

Topic: Funnel plots: evidence interpretation

Exam point: CLEAN V3 describes the inverted-funnel shape of a funnel plot. The shape itself does not establish that a group of studies is free from bias; the source line provides no such conclusion.

Funnel plot شكله قمع مقلوب؛ لا تستنتج من هذا الوصف وحده غياب bias.

Source evidence: C04607: DOCX paragraph 4607

Topic: ROC curves and classification thresholds

Core concept: AQRAB defines a receiver-operating-characteristic (ROC) curve as a way to display how a binary classifier performs as its discrimination threshold is varied.

ROC curve يوضح كيف يتغير أداء الاختبار عند تغيير cutoff، بدل الاعتماد على نقطة واحدة.

Source evidence: A435-35: AQRAB PDF p.435, block 35

Topic: Informed refusal and documentation

Management priority: CLEAN V3 emphasizes documenting a capacity assessment and discussion of the options, benefits and risks when a patient declines recommended treatment, including serious consequences where relevant.

عند رفض العلاج، دوّن تقييم capacity والنقاش حول الفوائد والمخاطر؛ مجرد عبارة «رفض» لا تكفي في سياق المصدر.

Source evidence: C04629: DOCX paragraph 4629

Emergency Systems, Ethics & Evidence. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

EDIC / Emergency & Critical Care

17 — Mixed Presentations & Clinical Principles

I. Psychiatric & Behavioral Presentations

Topic 1: Psychosis — Medical Mimics & Antipsychotic Safety

- Core concept: New or acute psychosis in the ED requires safety assessment and a search for delirium, intoxication/withdrawal, medication effects, neurologic illness and metabolic/infectious causes.
- Management priority: Establish onset, baseline, substances, prescribed drugs, vital signs, attention and neurologic findings.
- Management priority: Use targeted investigations guided by history and examination rather than assuming a primary psychiatric disorder.
- High-yield fact: De-escalation and a low-stimulation setting are first-line for agitation when safe; if medication is required, consider cause, prior response, interactions, QT risk and local rapid-tranquillisation protocol.
- High-yield fact: Recognize dystonia, akathisia, neuroleptic malignant syndrome and serotonin toxicity.
- Management priority: Assess immediate risk to patient/others, capacity, vital signs, glucose when indicated, attention and neurologic status.
- High-yield fact: Evaluate medical causes of acute psychosis, including delirium, intoxication/withdrawal, infection, metabolic disturbance, head injury, seizures and medication effects. Use targeted tests guided by the presentation and coordinate psychiatric assessment after medical stabilization.
- High-yield fact: If antipsychotic medication is needed, consider QT/cardiac risk, interactions and adverse effects, and follow the local monitoring protocol.

ابدأ بنفي الأسباب العضوية والدوائية، وقيّم الخطر والقدرة على اتخاذ القرار قبل اختصار acute psychosis في الحالة كمرض نفسي.

Topic 2: Mood Disorders — Depression, Mania & Differential

- Core concept: ED assessment of severe depression or mania begins with safety: suicidal intent, self-harm, psychosis, inability to care for self and danger to others.
- Management priority: Assess sleep, energy, speech, thought speed, impulsivity, substances, prescribed drugs and prior episodes.
- High-yield fact: Exclude medical mimics including intoxication, withdrawal, thyroid/metabolic disease, infection and neurologic illness as indicated.
- High-yield fact: Suspected mania or severe depression with danger warrants urgent specialist mental-health assessment.
- Management priority: Avoid starting or changing long-term mood stabilizers as an ED-only decision without specialist plan and required monitoring.
- Management priority: Assess suicide/self-harm risk, psychosis, self-neglect, vulnerability and risk to others; create a safe environment.
- High-yield fact: Consider substance/medication effects and medical mimics, including thyroid/metabolic and neurologic causes.
- High-yield fact: Urgently refer suspected mania or severe depression with danger to specialist mental-health services.
- Management priority: Review pregnancy, comorbidity and medication history; follow local crisis pathway and involve supports when appropriate.

شدة الأعراض تحدّد ؛ medical mimics والـsuicidality والـsleep change ابحث عن depression أو mania في مستوى التدخل.

Topic 3: Anxiety & Panic — Recognize Medical Mimics

- Core concept: A panic attack can cause abrupt intense fear with palpitations, dyspnea, chest discomfort, dizziness and paresthesia, but first consider time-critical physical causes based on presentation: acute coronary syndrome, arrhythmia, pulmonary embolism, asthma, hypoglycemia, intoxication/withdrawal and pregnancy-related emergencies.
- Management priority: Obtain focused history, examination and only the investigations needed to exclude acute physical problems.
- High-yield fact: Once stable, explain symptoms, reduce stimulation, use calm reassurance and arrange follow-up; do not diagnose panic disorder from a single ED episode.
- Management priority: Use symptom onset, risk factors, vital signs and examination to identify dangerous cardiopulmonary, metabolic or toxic NICE recommends the minimum investigations necessary to exclude acute physical problems for a panic attack presenting After stabilization, provide clear explanation, reassurance and follow-up for ongoing anxiety/panic disorder assessment.
- High-yield fact: Reassess if symptoms persist, change, recur with exertion or include syncope, hypoxia or abnormal examination.
- Exam trap: Do not order an indiscriminate test battery; select investigations based on clinical risk.

لا يُشخّص من تشابه الأعراض وحده؛ استبعد الأسباب الخطرة للصدر والتنفس والقلب بحسب الحالة Panic.

Topic 4: Somatic Symptoms, Factitious Disorder & Malingering

- Core concept: Persistent physical symptoms are real and deserve respectful evaluation even when a clear disease is not found.
- High-yield fact: In the ED, first assess acute threats and review prior workup without reflexive repetition; look for positive functional findings, distress and impairment.
- High-yield fact: Factitious disorder involves intentional symptom falsification to assume the patient role; malingering is intentional symptom production for external gain.
- High-yield fact: Neither label should be inferred from inconsistent tests, frequent visits or social context alone.
- Management priority: Document objective findings and maintain a therapeutic, nonjudgmental approach.
- High-yield fact: Evaluate new red flags and time-sensitive disease before attributing symptoms to a functional or psychiatric cause.
- Management priority: Review records and prior testing; avoid duplicate low-yield investigations while explaining the reasoning.
- High-yield fact: Distinguish symptom experience from intent: symptoms can be genuine without an identified structural disease.
- Management priority: Use neutral language, objective documentation, clear follow-up and consistent team communication.

بحسب الدافع؛ لا تُصدر أحكامًا دون أدلة، وواصل تقييم المشكلات الطبية malingering يختلف عن Factitious الواقعية.

Topic 5: Suicide Risk — Emergency Assessment & Safety

- Core concept: After self-harm or suicidal presentation, treat physical injury/poisoning and assess immediate safety, distress, intent, plan, access to means, past behavior, supports and safeguarding needs.
- Management priority: Arrange psychosocial assessment by a mental-health professional as soon as possible, concurrently with physical care when feasible.
- High-yield fact: NICE advises against using risk scales or low/medium/high labels to predict suicide or decide discharge.
- High-yield fact: Develop an individualized safety plan, address access to lethal means, involve supports with consent, and make a clear follow-up and escalation plan.

• High-yield fact: Ask directly about thoughts, intent, plan, access to means, prior attempts, supports and reasons for living.

• High-yield fact: Arrange psychosocial assessment promptly; do not use risk scores or global low/medium/high categories to predict suicide or decide discharge.

يحتاج تقييم suicide risk أسئلة مباشرة عن النية والخطة والوسائل والدعم؛ سلامة المريض أهم من الاكتفاء بتشخيصه.

Topic 6: The Combative & Difficult Patient

• Core concept: Severe agitation is a high-risk presentation and may reflect hypoxia, hypoglycemia, hyperthermia, intoxication/withdrawal, neurologic disease, infection, trauma, psychiatric illness or a mixed cause.

• High-yield fact: Protect the patient, staff and bystanders with adequate personnel, a low-stimulation environment and verbal de-escalation when feasible.

• High-yield fact: Rapidly assess vital signs, glucose, temperature, oxygenation, trauma and toxidrome while maintaining a safe approach.

• Management priority: Use oral medication if safe and feasible; if severe agitation prevents care, choose parenteral medication using current protocol, patient factors and local resources, with continuous airway, oxygenation, cardiac and temperature monitoring.

• High-yield fact: Restraint is a last-resort safety intervention, applied by trained staff with frequent reassessment and prompt release when safe.

• High-yield fact: ACEP’s 2024 adult policy reviews heterogeneous evidence and does not establish one universally superior regimen.

• High-yield fact: ACEP 2024 addresses adult severe agitation; studies are heterogeneous and the policy does not identify one universally superior parenteral regimen.

• High-yield fact: Verbal de-escalation is first-line when feasible; oral/sublingual medication can be considered if safe and practical.

• Management priority: Treat agitation as a possible medical emergency: check oxygenation, glucose, temperature, trauma, intoxication/withdrawal and neurologic causes.

يسبقان افتراض السبب intoxication أو hypoglycemia أو hypoxia أمان الجميع وتقييم، في agitated patient النفسي.

II. Geriatric & Special-Population Emergencies

Topic 7: Care of the Older Adult in the ED

• Management priority: Assess the older adult’s baseline function, cognition, mobility, medications, goals of care, social supports and physiologic reserve.

• High-yield fact: New confusion may be hypoactive delirium from infection, hypoxia, metabolic disease, medication toxicity, stroke or ACS; establish baseline with a caregiver when possible and search for reversible causes.

• High-yield fact: A low-energy fall can cause major injury, and normal-range vital signs may mask shock.

• Management priority: Review anticoagulants, sedatives, anticholinergics and polypharmacy; use age-appropriate pain control and minimize deliriogenic exposure.

• Management priority: Check for occult injury, frailty, pressure injury and elder mistreatment.

• High-yield fact: Plan disposition around function, safe mobility, medication access and clear follow-up, not diagnosis alone.

• Exam trap: Do not dismiss acute confusion as dementia; establish baseline and evaluate time-sensitive causes including infection, metabolic disturbance, hypoxia, stroke and ACS.

• High-yield fact: Older adults can have major trauma after low-energy mechanisms and may not show expected tachycardia or hypotension; reassess trajectory and perfusion.

• High-yield fact: Medication reconciliation should identify anticoagulants, sedatives, anticholinergic burden, interactions and recent changes.

• Diagnostic criteria: The source-image Confusion Assessment Method (CAM) requires acute/fluctuating change (1) plus inattention (2), with either disorganized thinking (3) or altered level of consciousness (4). [AQRAB PDF p. 335; original visual panel]

• Differential point: The illustrated geriatric assessment distinguishes basic ADLs (bathing, dressing, toileting, transferring, continence, feeding) from IADLs (medication management, finances, shopping, cooking and transport). [AQRAB PDF p. 335; original visual panel]

في delirium تذكّر CAM: 1 + 2 + (3 أو 4). وخلال تقييم القدرة الوظيفية فرّق بين ADLs الأساسية وIADLs الأكثر تعقيدًا.

بدل اعتبار delirium والأدوية و baseline function المسن قد يظهر المرض الخطير بصورة غير نمطية. راقب التغيّر «شيخوخة».

Topic 8: Geriatric Trauma

• Management priority: Use an age-aware trauma assessment: frailty, baseline function, cognition, anticoagulants/antiplatelets, comorbidity and goals of care all affect risk.

• High-yield fact: A low-energy fall can cause intracranial, cervical-spine, rib, pelvic or long-bone injury; tachycardia may be blunted and hypotension may be late.

• Management priority: Obtain collateral history and medication list, assess head strike and loss of consciousness, perform repeated neurologic/perfusion checks, and use imaging based on injury risk rather than reassuring mechanism alone.

• High-yield fact: Identify anticoagulant exposure and active bleeding, then reverse only according to the specific drug, injury, severity and current local protocol.

• High-yield fact: Prevent hypothermia, delirium, pressure injury and functional decline; plan mobility and safe disposition early.

• High-yield fact: ACS geriatric trauma guidance emphasizes age-specific criteria; normal-range vital signs and low-energy mechanism do not safely exclude major injury.

• High-yield fact: Anticoagulant or antiplatelet exposure changes the bleeding assessment; establish the exact agent, last dose, renal function and injury before choosing reversal.

• High-yield fact: Consider occult head, neck, chest, pelvic and long-bone injury after falls; repeat examination and use imaging when clinical risk supports it.

• High-yield fact: Preserve baseline function: use hearing/vision aids, prevent hypothermia and delirium, mobilize safely and involve caregivers/interdisciplinary supports.

• Imaging sign: The image illustrates an epidural hematoma as a lens-shaped/biconvex collection and a subdural hematoma as a crescent-shaped collection on head CT; imaging must still be interpreted in clinical context. [AQRAB PDF p. 337; original visual panel]

• Clinical Pearl: The geriatric-trauma infographic highlights significant pulmonary complications after rib fractures, including pneumonia and respiratory failure, and supports careful analgesia and pulmonary hygiene. [AQRAB PDF p. 337; original visual panel]

في كبار السن لا تعتمد على آلية رض بسيطة لنفي إصابة خطيرة؛ في CT انتبه للفارق بين EDH العدسي وSDH الهلالي، ولعواقب كسور الأضلاع.

في geriatric trauma قد تكون الإصابة شديدة بآلية بسيطة؛ لا تُطمئن لوجود vital signs أولية مقبولة.

Topic 9: Geriatric Drug Therapy

• Core concept: Medication harm in older adults is shaped by renal/hepatic clearance, altered sensitivity, frailty, cognition, nutrition and polypharmacy—not chronological age alone.

• High-yield fact: Reconcile prescribed, OTC and herbal medicines, recent changes, adherence and the patient’s actual use.

• High-yield fact: For suspected toxicity, identify dose, timing, formulation, indication, interactions, renal function and toxidrome; support airway/circulation and consult poison control/pharmacy when appropriate.

• Management priority: Use the AGS Beers Criteria as a screening aid for potentially inappropriate medicines, not as an absolute prohibition or a substitute for individualized decisions.

• Management priority: Start cautiously when clinically appropriate, monitor response and adverse effects, and consider deprescribing or alternatives at transitions of care.

• Management priority: Review medication indication, dose, last use, renal function, interactions and recent changes; several modest risks can combine into delirium, falls or hypotension.

• High-yield fact: AGS Beers Criteria identify medicines that are potentially inappropriate in many older adults or require caution; apply patient- specific judgment.

• High-yield fact: Anticholinergic and sedative burden can worsen confusion, urinary retention, falls and respiratory depression.

• High-yield fact: During transitions, reconcile and communicate medication holds, changes, restart plans and monitoring clearly.

يغيّران خطر الأدوية عند كبار السن؛ راجع القائمة كاملة والتداخلات عند تقييم Polypharmacy وrenal function أي تدهور.

Topic 10: The Immunocompromised Patient

• Core concept: Recognize that immunosuppression may blunt fever, leukocytosis and local inflammatory signs.

• Management priority: Establish the immune defect and timeline: neutropenia/cancer therapy, transplant, steroids/biologics, HIV, asplenia or primary immune disorder; review prophylaxis, devices, prior organisms and recent antimicrobials.

• Management priority: Assess sepsis and organ dysfunction early, obtain appropriate cultures without delaying treatment, and use the relevant institutional empiric pathway with infectious-disease/oncology input.

• High-yield fact: Febrile neutropenia is time-critical; ASCO/IDSA recommends initial empiric antibacterial therapy within 1 hour of triage for patients with cancer and febrile neutropenia.

• High-yield fact: Consider opportunistic, viral, fungal and routine bacterial infection plus medication toxicity, malignancy and inflammatory mimics.

• High-yield fact: Febrile neutropenia is an emergency; ASCO/IDSA recommends initial empiric antibacterial therapy within 1 hour of triage for the defined cancer population.

• High-yield fact: The 2018 ASCO/IDSA outpatient guideline applies to adults treated for malignancy; do not generalize its outpatient eligibility to unstable patients or other immune defects.

• High-yield fact: Immunosuppression can mute classic signs; a normal temperature or white-cell count does not exclude serious infection.

• High-yield fact: Tailor investigation and empiric therapy to immune defect, depth/duration of neutropenia, transplant timeline, prior cultures, prophylaxis and local resistance.

• Number/threshold: The AQRAB immune-status illustration associates CD4 counts below 200 cells/μL with Pneumocystis jirovecii, below 100 with Toxoplasma gondii and below 50 with Mycobacterium avium complex; these are source-image risk cues, not stand-alone diagnoses. [AQRAB original PDF p. 343, visual panel]

• Diagnostic criteria: The AQRAB neutropenic-fever infographic uses a single temperature of at least 38.3°C or at least 38.0°C sustained for 1 hour together with ANC below 500 cells/μL or expected to fall below 500; interpret as the figure's stated definition. [AQRAB original PDF p. 343, visual panel]

• Clinical Pearl: The source-image examples show bilateral ground-glass lung opacity with Pneumocystis infection, ring-enhancing brain lesions with toxoplasmosis, and pulmonary nodules/halo signs with invasive aspergillosis; appearances are suggestive, not pathogen-specific proof. [AQRAB original PDF p. 343, visual panel]

ربط CD4 بالـopportunistic infection يساعد ترتيب الـdifferential، لكن يلزم تقييم الأعراض والفحوص. معيار neutropenic fever هنا مُقتبس من صورة المصدر.

قد لا يطور المريض immunocompromised علامات عدوى كلاسيكية؛ انخفاض العتبة للتقييم مهم عند التدهور.

Topic 11: The Solid Organ Transplant Patient

• Core concept: An ED presentation after kidney, liver, heart, lung or intestinal transplant can reflect infection, rejection, medication toxicity/interactions, vascular or surgical complications, malignancy or unrelated emergency disease.

• High-yield fact: Identify the graft, transplant date, baseline function, immunosuppressants, prophylaxis, recent rejection/infection, last doses and transplant-center contact.

• High-yield fact: Fever, hypotension, dyspnea, reduced urine output, graft-area pain, diarrhea or new organ dysfunction warrants prompt evaluation; symptoms can be muted.

• Management priority: Stabilize first, obtain graft-appropriate tests and cultures, check drug levels/renal function when relevant, and contact the transplant team early.

Domain 17 — Geriatric Safety & Medication Review

Pass 5 | Additional source-image facts for clinical discrimination

Red flag / elder abuse: The AQRAB geriatric-abuse image highlights a history inconsistent with the observed injury, delay in seeking care, patterned bruises, injury at different stages of healing, unexplained burns and genital/anal injuries as reasons to assess for potential abuse.

تعدد الإصابات وتناقض القصة أو التأخر في طلب الرعاية علامات تستدعي تقييمًا دقيقًا؛ ولا تثبت الإساءة وحدها.

Red flag / neglect: The image separately identifies poor hygiene, dehydration or malnutrition, untreated medical conditions, an unsafe environment and pressure ulcers as possible neglect clues; assess other explanations rather than treating any single clue as proof.

تميّز بين الإساءة والإهمال، وقيّم النظافة والتغذية والعلاج والبيئة المنزلية ضمن الصورة السريرية كاملة.

Documentation and jurisdiction: The pictured pathway advises objective injury descriptions, direct quotations, and photography with consent. Its mandatory-reporting reference specifically describes U.S. state-law Adult Protective Services requirements, not a universal rule for all jurisdictions.

وثّق الوقائع باحتراف، واستأذن للتصوير. إجراءات الإبلاغ المذكورة أمريكية خاصة بالولاية، ولا تُعمم تلقائيًا على بلد آخر.

Medication risk framework: AQRAB uses five or more medications as its illustrated polypharmacy definition and links high-risk drug classes in older adults (sedatives, anticholinergics, long-acting opioids and NSAIDs) to adverse effects such as falls, delirium, bleeding and kidney injury. This is a source-image risk inventory, not an individualized prescribing rule.

عند كبار السن افحص تعدد الأدوية، خصوصًا المهدئات ومضادات الكولين والأفيونات الطويلة وNSAIDs بسبب مخاطر التداخلات والسقوط والاختلاطات.

Medication reconciliation: The image presents a four-part transition-of-care pathway: obtain a complete medication history, compare it with prior lists and confirm indications, identify and resolve discrepancies, then educate the patient or caregiver and communicate the plan.

عند انتقال الرعاية لا يكفي نسخ قائمة الأدوية؛ يجب مطابقتها، تصحيح اختلافاتها، ثم توضيح الخطة للمريض أو المرافق.

These source-page recoveries belong to the older-adult/geriatrics topics. Jurisdiction and patient-specific safety qualifications were preserved.

- Exam trap: Do not independently stop immunosuppression or give empiric anti- rejection therapy without specialist direction unless a resuscitation protocol requires immediate action.
- High-yield fact: Transplant recipients have overlapping risks: infection, rejection, drug toxicity/interactions, graft dysfunction and ordinary ED emergencies.
- High-yield fact: The organ and time since transplant shape likely complications; obtain transplant date, baseline function, immunosuppression/prophylaxis and recent specialist contact.
- High-yield fact: Fever or new organ symptoms warrant prompt assessment and early transplant-team communication; immunosuppression may blunt inflammatory findings.
- Management priority: Check medication interactions and organ function; avoid unilateral immunosuppressant interruption or anti-rejection treatment without specialist coordination.
- Clinical trigger: The transplant infographic stratifies complications by time: under 1 month emphasizes surgical/nosocomial problems; months 1-6 emphasize opportunistic infection and acute rejection; beyond 6 months it highlights community-acquired infection, chronic rejection and malignancy. [AQRAB PDF p. 345; original visual panel]
- High-yield fact: The image lists calcineurin inhibitors (tacrolimus/cyclosporine) with nephrotoxicity, neurotoxicity, hypertension and hyperkalemia; antimetabolites (mycophenolate/azathioprine) with marrow suppression and gastrointestinal intolerance. [AQRAB PDF p. 345; original visual panel]
- Key discriminator: The graft-specific panel distinguishes kidney obstruction/vascular complications, liver hepatic-artery or biliary complications, heart rejection/arrhythmias/vasculopathy, and lung rejection/infection/airway complications. [AQRAB PDF p. 345; original visual panel]

في transplant حدّد العضو والوقت منذ الزراعة والأدوية. المشاكل المبكرة قد تكون جراحية، وفي الأشهر التالية تزيد احتمالات opportunistic infection، لكن لا تختصر التشخيص في رفض العضو فقط.

time since transplantation وorgan لديه مخاطر عدوى ورفض وتداخلات دوائية؛ راجع transplant المريض بعد والأدوية.

Topic 12: The Patient with Obesity

- Management priority: Use the patient’s actual physiology, body habitus, comorbidities and functional needs rather than assumptions based on appearance or BMI alone.
- High-yield fact: Obesity can reduce functional residual capacity, increase airway obstruction risk and shorten safe apnea time; plan ramped/head-elevated positioning, preoxygenation, appropriate equipment and backup airway strategies.
- Management priority: Assess for obesity hypoventilation, obstructive sleep apnea, cardiometabolic disease, thromboembolism and technically difficult examination/imaging.
- High-yield fact: Drug dosing is drug- and context-specific: confirm the correct weight scalar (total, ideal, adjusted or lean body weight), maximum dose, renal function and local reference; avoid blanket dose reductions or total-body-weight calculations.
- High-yield fact: Obesity can reduce FRC and oxygen reserve; head-elevated/ramped positioning and optimized preoxygenation improve airway preparation.
- High-yield fact: Medication weight scalar varies by drug; verify total/ideal/adjusted/lean body weight, maximum dose and institutional Anticipate obstructive sleep apnea, obesity hypoventilation, difficult mask ventilation, equipment limits and imaging challenges.
- Management priority: Use respectful language, ask permission before discussing weight and address the acute complaint without diagnostic overshadowing.
- Exam trap: Do not assume a single scalar (actual or ideal body weight) is correct for every medication.
- Exam trap: Do not position a high-risk airway flat without considering ramping/head elevation and rapid desaturation risk.

في obesity والـ airway قد تتغيّر المراقبة والجرعات ضمن السياق للمعدات عملية خطة ضع والإجراءات؛.

Topic 13: The Geriatric Emergency Department

- Core concept: Older adults may present with atypical illness, multimorbidity, polypharmacy, sensory impairment, frailty and reduced physiologic reserve.
- High-yield fact: Ask about baseline cognition, function, mobility, falls, medications, goals and support; assess delirium and reversible causes, pain, injury and medication harm.

CLINICAL NUMBERS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 283

1 hour	High-yield fact: Febrile neutropenia is time-critical; ASCO/IDSA recommends initial empiric antibacterial therapy within 1 hour of triage for patients with cancer and febrile neutropenia. High-yield fact: Consider opportunistic, viral, fungal and routine bacterial infection plus medication toxicity,
1 hour	High-yield fact: Febrile neutropenia is an emergency; ASCO/IDSA recommends initial empiric antibacterial therapy within 1 hour of triage for the defined cancer population. High-yield fact: The 2018 ASCO/IDSA outpatient guideline applies to adults treated for malignancy; do not generalize
200 cells/µL	timeline, prior cultures, prophylaxis and local resistance. Number/threshold: The AQRAB immune-status illustration associates CD4 counts below 200 cells/µL with Pneumocystis jirovecii, below 100 with Toxoplasma gondii and below 50 with Mycobacterium avium complex;
1 hour, 500 cells/µL	Diagnostic criteria: The AQRAB neutropenic-fever infographic uses a single temperature of at least 38.3°C or at least 38.0°C sustained for 1 hour together with ANC below 500 cells/µL or expected to fall below 500; interpret as the figure's stated definition. [AQRAB original PDF p. 343, visual panel]
1 month	interruption or anti-rejection treatment without specialist coordination. Clinical trigger: The transplant infographic stratifies complications by time: under 1 month emphasizes surgical/nosocomial problems; months 1-6 emphasize opportunistic infection and acute rejection; beyond 6 months it
6 months	Clinical trigger: The transplant infographic stratifies complications by time: under 1 month emphasizes surgical/nosocomial problems; months 1-6 emphasize opportunistic infection and acute rejection; beyond 6 months it highlights community-acquired infection, chronic rejection and malignancy. [AQRAB PDF p. 345; original visual panel]
35/min	Marked bradycardia in anorexia nervosa Red flag: The CLEAN V3 example identifies a heart rate of 35/min in anorexia nervosa as severe bradycardia requiring hospital assessment; this is a case threshold, not a general rule.

- Management priority: Use hearing/vision aids, clear communication and collateral history with patient permission when possible.
- High-yield fact: Base disposition on acute risk, function, supports, patient p and feasible follow-up, not age alone.
- High-yield fact: Geriatric ED models use coordinated staffing, education, equipment, policies, follow-up and quality improvement; local resources determine implementation.
- High-yield fact: A new change in attention or cognition suggests delirium until assessed; compare with baseline and seek reversible causes.
- High-yield fact: Medication review, falls/function, sensory support, goals and safe disposition are central geriatric ED domains.
- High-yield fact: ACEP geriatric ED guidance is a systems template for staffing, equipment, education, policies, follow-up and performance improvement.
- Exam trap: Do not label acute confusion as dementia without establishing baseline and evaluating delirium causes.

يحتاج تقييم الإدراك والوظيفة والسقوط والأدوية والدعم الاجتماعي مع خطة خروج آمنة Geriatric ED.

Additional High-Yield Clinical Discriminators from CLEAN V3

Topic 14: Malingering and external incentive

- Key discriminator: Malingering is the intentional production of symptoms for external or secondary gain.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

Topic 15: Obesity and chest wall mechanics

- High-yield fact: The supplied physiologic review states that obesity decreases chest-wall compliance.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

Topic 16: Safe verbal de-escalation in agitation

- Management priority: The source recommends calm, non-threatening verbal de-escalation as the first approach in an agitated patient.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

Topic 17: Restraints and patient/staff safety

- Exam trap: The source describes physical restraints as a last resort after attempted de-escalation and medication when needed for immediate safety.

اربط هذه النقاط بالمظهر السريري الفعلي وحدّد Red flags مع مراعاة سياق المصدر، وخطوة التقييم التالية.

High-Yield Comparison Tables

Cognitive presentation — Delirium / Dementia

Feature	Delirium	Dementia	Exam point
Typical presentation	Acute/fluctuating attention disturbance	Usually more chronic cognitive course	Look for underlying acute illness
Emergency priority	Investigate reversible causes and safety	Establish baseline function and acute change	Do not assume new confusion is age alone

PASS 9 — Source-backed clinical recovery | Domain 17

Mixed Presentations & Clinical Principles

dosing or medical-correctness certification.

Marked bradycardia in anorexia nervosa

Red flag: The CLEAN V3 example identifies a heart rate of 35/min in anorexia nervosa as severe bradycardia requiring hospital assessment; this is a case threshold, not a general rule.

بطء القلب الشديد مع anorexia nervosa إنذار؛ رقم 35/min وارد ضمن حالة المصدر.

Rifampicin-tacrolimus interaction

Key discriminator: CLEAN V3 describes rifampicin induction of CYP3A4/CYP3A5 and P-glycoprotein, potentially reducing tacrolimus exposure and raising rejection risk in transplant recipients.

Rifampicin قد يخفض tacrolimus بسبب enzyme induction، ما يزيد القلق من rejection.

Avulsed primary tooth

Exam trap: CLEAN V3 warns that an avulsed primary tooth must not be replanted because of potential damage to the developing permanent tooth.

السن اللبني المقلوع لا يعاد زرعه وفق المصدر حفاظًا على السن الدائم النامي.

Uncomplicated traumatic tympanic membrane perforation

Source-specific follow-up: The CLEAN V3 small central perforation case advises water precautions and ENT referral if it persists at six weeks; the timeframe is linked to that case.

في ثقب الطبلة الصغير غير المعقد يحذّر المصدر من الماء ويذكر إحالة ENT إذا استمر بعد 6 أسابيع.

Intentional symptom production — Factitious disorder / Malingering

Feature	Factitious disorder	Malingering	Exam point
Motivation	Assumption of sick role without obvious external reward	External incentive/secondary gain	Avoid labeling without adequate evidence

Rapid Recall: Must-Know Facts

01. Check for occult injury, frailty, pressure injury and elder mistreatment.
02. Document objective findings and maintain a therapeutic, nonjudgmental approach.
03. Malingering is the intentional production of symptoms for external or secondary gain.
04. Do not order an indiscriminate test battery; select investigations based on clinical risk.
05. Do not assume a single scalar (actual or ideal body weight) is correct for every medication.
06. Use neutral language, objective documentation, clear follow-up and consistent team communication.
07. Do not label acute confusion as dementia without establishing baseline and evaluating delirium causes.
08. Establish onset, baseline, substances, prescribed drugs, vital signs, attention and neurologic findings.
09. Assess sleep, energy, speech, thought speed, impulsivity, substances, prescribed drugs and prior episodes.
10. Obtain focused history, examination and only the investigations needed to exclude acute physical problems.
11. Review records and prior testing; avoid duplicate low-yield investigations while explaining the reasoning.
12. Use hearing/vision aids, clear communication and collateral history with patient permission when possible.
13. The source recommends calm, non-threatening verbal de-escalation as the first approach in an agitated patient.
14. Do not position a high-risk airway flat without considering ramping/head elevation and rapid desaturation risk.
15. Use targeted investigations guided by history and examination rather than assuming a primary psychiatric disorder.
16. Assess suicide/self-harm risk, psychosis, self-neglect, vulnerability and risk to others; create a safe environment.
17. Assess immediate risk to patient/others, capacity, vital signs, glucose when indicated, attention and neurologic status.
18. Review pregnancy, comorbidity and medication history; follow local crisis pathway and involve supports when appropriate.
19. Avoid starting or changing long-term mood stabilizers as an ED-only decision without specialist plan and required monitoring.
20. Use respectful language, ask permission before discussing weight and address the acute complaint without diagnostic overshadowing.
21. Arrange psychosocial assessment by a mental-health professional as soon as possible, concurrently with physical care when feasible.
22. Assess the older adult’s baseline function, cognition, mobility, medications, goals of care, social supports and physiologic reserve.
23. The source describes physical restraints as a last resort after attempted de-escalation and medication when needed for immediate safety.
24. Review anticoagulants, sedatives, anticholinergics and polypharmacy; use age-appropriate pain control and minimize deliriogenic exposure.
25. Use the patient’s actual physiology, body habitus, comorbidities and functional needs rather than assumptions based on appearance or BMI alone.
26. Treat agitation as a possible medical emergency: check oxygenation, glucose, temperature, trauma, intoxication/withdrawal and neurologic causes.
27. Stabilize first, obtain graft-appropriate tests and cultures, check drug levels/renal function when relevant, and contact the transplant team early.
28. Assess for obesity hypoventilation, obstructive sleep apnea, cardiometabolic disease, thromboembolism and technically difficult examination/imaging.
29. Start cautiously when clinically appropriate, monitor response and adverse effects, and consider deprescribing or alternatives at transitions of care.
30. Use an age-aware trauma assessment: frailty, baseline function, cognition, anticoagulants/antiplatelets, comorbidity and goals of care all affect risk.
31. Check medication interactions and organ function; avoid unilateral immunosuppressant interruption or anti-rejection treatment without specialist coordination.
32. Use the AGS Beers Criteria as a screening aid for potentially inappropriate medicines, not as an absolute prohibition or a substitute for individualized decisions.

33. Review medication indication, dose, last use, renal function, interactions and recent changes; several modest risks can combine into delirium, falls or hypotension.

34. Do not independently stop immunosuppression or give empiric anti- rejection therapy without specialist direction unless a resuscitation protocol requires immediate action.

35. Assess sepsis and organ dysfunction early, obtain appropriate cultures without delaying treatment, and use the relevant institutional empiric pathway with infectious-disease/oncology input.

36. The supplied physiologic review states that obesity decreases chest-wall compliance.

37. Recognize dystonia, akathisia, neuroleptic malignant syndrome and serotonin toxicity.

38. A low-energy fall can cause major injury, and normal-range vital signs may mask shock.

39. Recognize that immunosuppression may blunt fever, leukocytosis and local inflammatory signs.

40. Neither label should be inferred from inconsistent tests, frequent visits or social context alone.

Conflict Resolution Register

Only verified opposing statements are classified as “Conflict — source-dependent”. Different patient populations, legal jurisdictions, diagnostic criteria or guideline generations are documented as contextual differences, not silently harmonized.

No confirmed direct source-to-source clinical contradiction was established in the selected, fully contextualized points for this domain. This is not a claim that all unseen or image-only details have been exhaustively reconciled.

Psychiatric versus medical cause of behavior

- CLEAN V3: CLEAN V3 includes exam findings in individual case contexts.
- AQRAB: AQRAB stresses examination for medical mimics and immediate safety.
- Nature of difference: Different contexts; no direct contradiction confirmed.
- Classification: Differential context

Final clinical topics: 17. Retained integrated English bullet points: 121. Rapid Recall: 40 derivative points. Comparison tables: 2.

Editorial provenance of retained points: CLEAN V3-only 4; AQRAB-only 117; confirmed both-source 0. These are point-by-

Source-screening notes: 29 CLEAN V3 source bullets were withheld from automatic inclusion because they were incomplete, overly case-bound or failed the conservative screening criteria; withholding does not prove every one is medically incorrect. 1 contextual difference(s) are documented above; 0 direct inter-file clinical conflicts were confirmed within the selected contextualized points.

No external medical information was added. No claim of independent guideline verification. Source text contains historical or jurisdiction-dependent thresholds; use their full context. AQRAB PDF image-only diagrams or infographics may require separate source-to-image verification before claiming absolutely exhaustive extraction.

Audit Pass 3: Supplemental image-derived facts added from original AQRAB PDF panels, with page-number evidence embedded in each added point. This targeted visual pass is not a full medical validation or proof of complete two-source coverage.

addendum count is not part of the original Merge Quality Note editorial tally. No external clinical source was consulted.

Mixed Presentations & Clinical Principles. All entries below preserve statements already appearing in the integrated study text. Numbers are kept with the clinical context given in the two supplied source documents, without adding outside guidelines.

Primary evidence: original CLEAN V3 DOCX and AQRAB v18 PDF. Registered status: source-limited, not a verified clinical guideline.

> Administrative closure of a discrepancy means its type and safe editorial disposition are recorded. It does **not** mean the clinical dispute was resolved, or that all unknown source conflicts were discovered. Only an actually opposed same-population claim is classified as a direct inter-file conflict.

Direct inter-file recommendation conflict (1)

X01 — Routine P2Y12 pretreatment in NSTEMI [Domain 02]

Source statement A: CLEAN V3 DOCX ¶583: "In addition to aspirin, clopidogrel should be given to patients with NSTEMI in the emergency department."

Source statement B: AQRAB PDF pp.30,35: selection/timing depends on PCI/invasive strategy and bleeding/CABG risk; not one pretreatment policy for every patient.

Classification: Conflict — source-dependent

Editorial disposition: Retain both source wordings; do not prescribe a universal pretreatment rule.

Definition versus action window (1)

X02 — Status epilepticus: 30-minute historical definition vs 5-minute action threshold [Domain 05]

Source statement A: CLEAN V3 DOCX ¶1793: >=5-minute action/diagnostic criterion; ¶1796: historical >=30-minute generalized convulsive definition.

Source statement B: AQRAB PDF p.140: treat >=5-minute convulsion or recurrent seizures without recovery; do not wait 30 minutes.

Classification: Context difference — not a direct therapeutic conflict

Editorial disposition: Distinguish older definition from early intervention trigger; do not silently discard either source wording.

Internal first-source discrepancies (6)

I01 — Adult IM adrenaline dose examples in anaphylaxis [Domain 01]

Source statement A: CLEAN V3 source-internal conflict register: 0.5 mg IM (1 mg/mL) in multiple passages.

Source statement B: CLEAN V3 same register: 0.3 mg IM (1 mg/mL) in PEER-derived examples.

Classification: Source-dependent internal protocol variation

Editorial disposition: Keep as unresolved cited dose conventions; do not choose one.

I02 — Pediatric IM adrenaline repeat interval [Domain 01]

Source statement A: CLEAN V3 register: repeat after 3-5 or 5 minutes.

Source statement B: CLEAN V3 register: 5-10 minutes in another cited passage.

Classification: Source-dependent internal protocol variation

Editorial disposition: Preserve interval by named pathway; no single universal time.

I03 — Observation after anaphylaxis [Domain 01]

Source statement A: CLEAN V3 register: 2, 6 or >12 hours according to risk.

Source statement B: CLEAN V3 register: 6-12 hours or longer in selected higher-risk adults.

Classification: Source-dependent internal framework variation

Editorial disposition: Preserve start clock and risk criteria rather than picking a number.

I04 — Refractory VF dual sequential defibrillation [Domain 01]

Source statement A: CLEAN V3 ¶256 and global register: one passage reports early-study improvement.

Source statement B: CLEAN V3 same register: another advises against DSED outside research.

Classification: Conflict — source-dependent (within file 1)

Editorial disposition: Differentiate vector-change technique; do not merge incompatible recommendations.

I05 — Early neonatal sepsis time window [Domain 07]

Source statement A: CLEAN V3 ¶2454: early-onset window around first 72 h.

Source statement B: CLEAN V3 ¶2465: first 7 days / 72 h convention also appears.

Classification: Definition variation inside file 1

Editorial disposition: Preserve original named framework; not a new between-file contradiction.

I06 — Traumatic hemothorax output criterion [Domain 03]

Source statement A: CLEAN V3 Domain 3 register: 200 mL/hour for 3 hours example.

Source statement B: CLEAN V3 same passage: 250 mL/hour for 3 hours answer fragment.

Classification: Internal numeric-context tension

Editorial disposition: Neither value should become a universal standalone decision rule.

Answer-stem or extraction reversals (1)

I07 — Hydroxyethyl starch / glutamine / ESBL reversal stems [Domain 07]

Source statement A: CLEAN V3 ¶2441, ¶2476, ¶2712: short answer-stems suggesting actions or negations.

Source statement B: Same paragraphs: explanations indicate different recommendation scope or the reverse answer.

Classification: Extraction/answer-stem inversion — quarantine

Editorial disposition: Do not transfer stem as advice; preserve full question/explanation for later review.

Source corruption / exclusion (2)

I08 — ADH and collecting-duct permeability [Domain 11]

Source statement A: CLEAN V3 ¶3671: "collecting duct becomes impermeable to water in the presence of ADH".

Source statement B: Same paragraph: ADH/V2 receptor mechanism is provided but without resolving the first claim.

Classification: Incomplete/internally suspect source — not included

Editorial disposition: Quarantine, do not correct from memory or external medicine.

I09 — Delirium versus delirium [Domain 17]

Source statement A: CLEAN V3 ¶4727: acute delirium contrasted with "slower onset of delirium".

Source statement B: No coherent opposite entity stated in the extracted source sentence.

Classification: Extraction-corrupted — not included

Editorial disposition: Quarantine, do not guess the missing comparison term.

Named protocol differences (5)

P01 — Adult symptomatic-bradycardia atropine [Domain 02]

Source statement A: AQRAB PDF p.17: RCUK initial 500 mcg IV, repeat within its pathway.

Source statement B: AQRAB PDF p.18: AHA initial 1 mg IV, repeat within its pathway.

Classification: Two protocols within same file — context retained

Editorial disposition: Always name pathway; do not mark a direct inter-file conflict.

P02 — Adenosine dose sequence [Domain 02]

Source statement A: AQRAB PDF p.17: RCUK sequence 6/12/18 mg.

Source statement B: AQRAB PDF p.12: AHA example 6/12 mg.

Classification: Different named pathway — context retained

Editorial disposition: Keep distinct monitoring/eligibility context.

P05 — COPD oral corticosteroid regimens [Domain 03]

Source statement A: CLEAN V3 and AQRAB examples: prednisone equivalent 40 mg for 5 days.

Source statement B: AQRAB domain register: NICE prednisolone 30 mg for 5 days.

Classification: Different source-era pathways — not direct inter-file conflict

Editorial disposition: Do not collapse into one dosing recommendation.

P07 — Adult CAP with positive respiratory virus [Domain 03]

Source statement A: AQRAB: specified ATS 2025 populations receive antibiotics in source-described setting.

Source statement B: AQRAB: describes IDSA nonendorsement of that specific recommendation.

Classification: Guideline disagreement reported within file 2

Editorial disposition: Do not turn an internal source report into a between-file discrepancy.

P09 — Postpartum hemorrhage labels [Domain 10]

Source statement A: CLEAN V3: 500-1000 mL minor PPH convention and >=1000 mL/hypovolemia definition.

Source statement B: AQRAB: physiologic bleeding severity/maternal stability emphasized.

Classification: Different definitions and clinical emphasis

Editorial disposition: No one numeric rule substituted for physiology.

Different definitions (1)

P03 — Septic shock definitions [Domain 01]

Source statement A: CLEAN V3: hypotension after 30 mL/kg or lactate >=4 pathway statement.

Source statement B: AQRAB PDF p.203: Sepsis-3 vasopressor MAP>=65 and lactate >2 despite adequate fluids.

Classification: Different definitional systems — not a direct source conflict

Editorial disposition: Do not substitute one named definition for another.

Different contextual thresholds (1)

P04 — Shock index warning levels [Domain 01]

Source statement A: CLEAN V3 ¶342 and ¶425: >0.7 for occult shock warning example.

Source statement B: AQRAB PDF p.213: approximately >=0.9 contextual concerning level.

Classification: Different risk-warning thresholds — unresolved context

Editorial disposition: Retain both as non-diagnostic contextual examples.

Patient or selection differences (2)

P06 — Stable primary spontaneous pneumothorax observation [Domain 03]

Source statement A: CLEAN V3: size-related management examples; >20% fragment without full case.

Source statement B: AQRAB: selected stable minimally symptomatic adult PSP conservative pathway independent of size.

Classification: Selection/case context incomplete — cannot prove direct conflict

Editorial disposition: Preserve clinical qualifiers; no universal rule.

P08 — Naloxone titration example [Domain 08]

Source statement A: CLEAN V3: 0.4 mg then 0.8 mg at about 60 s in an example.

Source statement B: AQRAB PDF p.236: 0.2-2 mg parenteral or 2-4 mg IN; repeat 2-3 minutes in another example.

Classification: Compatible ranges/contexts — not verified direct conflict

Editorial disposition: Maintain route, titration, and respiratory endpoint.

Jurisdiction-dependent wording (1)

P10 — Emergency-law duties and capacity [Domain 16]

Source statement A: CLEAN V3: UK-focused Mental Capacity Act / Montgomery and other legal source examples.

Source statement B: AQRAB: regional EMTALA, consent, and local law material.

Classification: Different jurisdictional legal frameworks — not a clinical conflict

Editorial disposition: Retain source-specific jurisdiction; avoid universal legal claims.

Conflict count refers to the curated register only. Previously unrecognized conflicts may exist in unadjudicated original source items.

Content-to-Source Closure Audit — Pass 7

Source-limited editorial reconciliation of 2 uploaded source files against the integrated 17-Domain clinical body (excluding prior audit appendices).

> **Current decision: NOT VERIFIED FINAL.** All 8,606 extracted items have been placed in a documented review/disposition screen, but the great majority have not received individual human semantic signoff. No automatic similarity score is equated to confirmed clinical coverage.

1. Full extracted-source inventory

Unit	Count	Interpretation
CLEAN V3 clinical bullets	2,397	Paragraph-anchored medical points extracted from DOCX
AQRAB extractable blocks	6,209	Text/PDF-layout segments; may include metadata and repeated content
All source-screen rows	8,606	Every row has source ID, original text, candidate destination and Pass 7 status

2. Disposition screen — explicit status for every source row

Status	Rows
Case Specific Antibiotic Example Held	1
Drug Context Held	1
Drug Name Missing Held	1
Exact Duplicate Source Item Screen	83
Exact Wording Present In Pass6 Body	387
High Lexical Candidate Not Medically Verified	456
Jurisdictional Legal Claim Held	1
Manual Source Verified Recovered In Pass7	12
Moderate Lexical Candidate Needs Adjudication	1,875
Navigation Bibliography Nonclinical Screen	92
No Proven Source To Output Match	5,690
Patient Specific Drug Example Held	1
Protocol Dependent Drug Example Held	2
Source Corruption Quarantined	1
Source Dependent Threshold Held	1
Source Fragment Not Safe To Reconstruct	1
Source Internal Inconsistency Quarantined	1

All lexical and metadata classifications are triage labels only. Even exact wording in the integrated text is not proof that numeric conditions, exceptions, or appropriate patient populations were independently clinically validated.

3. Human-reviewed examples and source recovery

12 source-contained points with missing or incomplete independent representation were restored into their clinical domains, preserving CLEAN V3 paragraph IDs and bilingual concise explanations. **11 other examined points** were held or quarantined for case/protocol/legal context, incomplete text or internal contradictions. No extrapolated medical content was inserted.

Domain	Recovery source IDs
10 - Obstetrics & Gynecology	C03341, C03320
11 - Renal, Electrolyte, Metabolic & Endocrine Emergencies	C03600, C03555
12 - Hematology & Oncology	C03738
13 - Musculoskeletal & Rheumatology	C03887, C03872, C04032
14 - Dermatology	C04161
15 - Procedures, Imaging & Diagnostics	C04742
16 - Emergency Systems, Ethics & Evidence	C04641, C04660

4. Global Conflict Resolution Register

21 known issues from prior local and global registers were classified and given an editorial disposition. Of these, **1** is retained as a direct source-to-source disagreement (NSTEMI pretreatment), while the rest are historical definition/action differences, internal source discrepancies, or population/guideline/jurisdiction contexts. None is silently clinically decided.

5. Remaining release blockers

To substantiate VERIFIED FINAL, every clinically significant point still needs a traceable human decision (included / meaningfully merged / relocated / valid exclusion / unresolved) after the full source and destination are viewed. Unreviewed AQRAB figures in text-rich as well as sparse-text pages must be checked. Numeric/therapeutic context must be reviewed independently of similarity and formatting. This pass does not certify a zero-missing-content result.

6. Supplied evidence and limits

Source files: CLEAN V3 DOCX and AQRAB v18 PDF. Working destination: Pass 6 integrated 17-Domain body, 224 PDF pages before prior audit appendices. The consolidated Conflict Register includes both original source document references and corresponding AQRAB page locations when available. No internet or outside medical guidance was consulted. No prescription endorsement is implied.

Audit evidence: EDIC_Pass7_Source_Disposition_8606_Rows.csv; EDIC_Pass7_Manually_Adjudicated_Source_Items.csv; EDIC_Pass7_Global_Conflict_Dispositions.csv

EDIC Integrated Reference | Audit Pass 9

Targeted source-to-output review of 4,359 previously unproven matches

Sources restricted to CLEAN V3 (DOCX) and AQRAB v18 (PDF). Target: Audit Pass 8 working reference. No web or external medical knowledge used.

1. Result and verification boundary

All 4,359 rows received a source-aware triage category and retained original text, source identifiers and closest Pass 8 destination candidates. 34 selected independent statements were reviewed against their source text and added in the relevant Domains. This does not certify semantic equivalence for every remaining record or medical correctness of source recommendations.

2. Trace-status counts

Status	Rows
Fragment Requires Page Join	1487
Priority 3 General Or Context	1212
Priority 2 Clinical Paraphrase Review	571
Exam Stem Or Clue Not Standalone	433
Priority 1 Clinical Safety	297
Patient Specific Context Required	118
Priority 2 Numeric Context	113
Priority 1 Numeric Or Therapeutic	48
Admin Or Reference Text Screen	46
Source Point Reviewed And Added Pass9	34

3. Source recovery by clinical domain

Domain	Clinical area	New points
1	Resuscitation, Airway & Shock	2
2	Cardiovascular Emergencies	1
3	Respiratory & Thoracic Emergencies	2
4	Gastrointestinal & Abdominopelvic Emergencies	1
5	Neurology & Ophthalmology	4
6	Trauma, Burns & Environmental Emergencies	2
8	Toxicology & Pharmacology	2
9	Pediatrics & Neonatology	2
10	Obstetrics & Gynecology	3
11	Renal, Electrolyte, Metabolic & Endocrine Emergencies	2
12	Hematology & Oncology	3
13	Musculoskeletal & Rheumatology	2
14	Dermatology	1
16	Emergency Systems, Ethics & Evidence	3
17	Mixed Presentations & Clinical Principles	4

4. Interpretation of unresolved records

A low similarity score is not evidence of a missing medical fact. AQRAB extraction contains broken lines, questions, diagrams, references and repeated editorial text. A complete case-linked statement can still require the original question and answer before it can be reused safely. Every unresolved row remains visible in the CSV, without being silently called missing, duplicated or clinically verified.

5. High-risk work that remains

- Review all priority-1 numerical and safety-sensitive statements with indication, route, age, timing, source version and negation.
- Rejoin AQRAB source fragments to their complete PDF page/clinical teaching panel before adjudication.
- For candidate paraphrases, inspect original context and destination topic rather than relying on lexical scores.
- Continue resolving conflicts only as source-dependent; never silently select an external guideline.

6. Deliverables and provenance

This package contains a 4,359-row individual routing ledger, a source-backed recovery ledger, the updated 17-Domain PDF and this QA report. Domain additions preserve CLEAN V3 DOCX paragraph identifiers or original AQRAB PDF page numbers. The working study reference must not be labelled VERIFIED FINAL on the basis of this audit pass.

EDIC / EMERGENCY & CRITICAL CARE — SOURCE-LIMITED EDITORIAL REVIEW

Audit Pass 11: Medication, Numeric & Incomplete Fragment Closure

> What “closed” means: All 48 specifically flagged numeric/therapeutic source records now have a definitive editorial decision: included with preserved context, already represented, retained only as a case observation, or excluded because the two supplied files do not support a safe stand-alone statement. This is NOT independent guideline verification or a certificate for all medication doses in the 276-page master.

Final disposition of 48 source records

48/48 have an explicit outcome and rationale; no record is labeled “pending”. Exclusion does not imply a medical judgment that a dose is wrong; it means the supplied source text cannot support a complete instruction.

Record and location	Final source-limited disposition	What the sources establish
A081-15 AQRAB PDF p.81, block 15	ALREADY IN REFERENCE	Adult severe-asthma source example appears in Domain 3; keep clinical severity and reassessment context.
A145-03 AQRAB PDF p.145, block 3	ALREADY IN REFERENCE	MS relapse >24 h, exclude infection/pseudorelapse, and corneal protection covered in Domain 5.

A152-00 AQRAB PDF p.152, block 0	ALREADY IN REFERENCE	Suspected encephalitis: prompt IV aciclovir and source-described ideal early window appear in Domain 5.
A204-13 AQRAB PDF p.204, block 13		AQRAB older-adult sepsis MAP 60–65 mmHg conditional range is explicitly source-qualified in Pass 10.
A236-34 AQRAB PDF p.236, block 34		The AQRAB p.236 pH 7.50–7.55 and Na 155 limits are TCA/sodium-channel-blocker bicarbonate-table targets, not salicylate targets.
A262-09 AQRAB PDF p.262, block 9		EXTRIP lithium >4 with renal impairment; >5 suggestions; neurologic/cardiac criteria appear in Pass 10 with specialist context.
A314-03 AQRAB PDF p.314, block 3	ALREADY IN REFERENCE	Pediatric urinary criteria and torsion/obstruction already source-qualified; guideline scope separately stated.
A416-38 AQRAB PDF p.416, block 38		AQRAB pediatric status case: 40 mg/kg over 5 min; pediatric explanatory range 30–60 mg/kg max 3 g, separate from adult regimen.
A433-39 AQRAB PDF p.433, block 39	CLOSED NONTHERAPEUTIC CONTEXT	30 minutes means time until end of shift in a handoff case, not seizure treatment time.

Record and location	Final source-limited disposition	What the sources establish
A445-25 AQRAB PDF p.445, block 25	CLOSED PATIENT VIGNETTE	Onset 2 hours earlier is one patient’s stroke presentation, not a universal cutoff.
C01163 DOCX paragraph 1163	CLOSED CASE EXAMPLE	500 mL warmed saline and group O blood occur in a hemorrhage vignette; another 1000 mL fluid example belongs to a different case; neither is a universal prescription.
C01169 DOCX paragraph 1169	ALREADY IN REFERENCE	Immediate traumatic chest output >1500 mL is present in Domain 6.
C01256 DOCX paragraph 1256	ALREADY IN REFERENCE	Terlipressin IV regimen from BSG-cited variceal bleeding passage retained with that specific source framework.
C01572 DOCX paragraph 1572	RESTORED SOURCE QUALIFIER	CLEAN V3 describes delayed onset >6 hours after wild-mushroom ingestion; preserved as source-defined red flag.
C01695 DOCX paragraph 1695	ALREADY IN REFERENCE	195/105 mmHg is from a non-thrombolysis stroke vignette; not a generic IV-thrombolysis target.
C01708 DOCX paragraph 1708	CLOSED EXCLUDED AMBIGUOUS	Sentence collides previous hemorrhagic stroke and age >75 without coherent eligibility criteria. Incomplete in source — not included as treatment rule.

C01717 DOCX paragraph 1717	CLOSED PATIENT VIGNETTE	CSF tube counts are one SAH patient example, not general diagnostic thresholds.
C01722 DOCX paragraph 1722	CLOSED EXAM CLAIM ONLY	AQRAB p.405 supplies an answer to a thrombolysis contraindication MCQ involving GI bleed within 21 days; insufficient as universal current eligibility algorithm. Audit-only exam claim, not advice.
C01778 DOCX paragraph 1778	RESTORED SOURCE SPECIFIC CASE	CLEAN V3 gives IV levetiracetam 30 mg/kg over 5 minutes after failed first-line benzodiazepines in an unspecified case; separated from pediatric 40 and adult 60 mg/kg examples.
C01780 DOCX paragraph 1780	CLOSED CASE EXAMPLE	Fosphenytoin 20 phenytoin-equivalents/kg is tied to a prolonged-seizure vignette. Without infusion/rate/ contraindication context, excluded from stand-alone dosing table.
C01789 DOCX paragraph 1789		First unprovoked adult seizure recurrence up to 45% within 2 years included as source-specific estimate.
C01793 DOCX paragraph 1793	ALREADY IN REFERENCE	Convulsive status treatment trigger >=5 min is included; distinguish from historical 30-min definition.
C01794 DOCX paragraph 1794	RESTORED SOURCE SPECIFIC CASE	CLEAN V3 selects repeat lorazepam 4 mg IV for one ongoing-status case; AQRAB p.140 describes selected adult 4 mg IV over 2 min, with monitoring and pathway-guided repeat. Not a universal repeat dose.

Record and location	Final source-limited disposition	What the sources establish
C01796 DOCX paragraph 1796	CLOSED HISTORICAL VIGNETTE	Phenytoin 1000 mg over 20 min is a patient example; 30-minute convulsive status is historical definition, not the 5-min emergency-action threshold.
C01802 DOCX paragraph 1802	CLOSED SOURCE OBSERVATION	Nine-fold postpartum seizure comparison is a source statement without defined comparator; keep as an observation, not independent prognostic rate.
C01806 DOCX paragraph 1806	ALREADY IN REFERENCE	Febrile seizure pattern and lack of universal minimum fever threshold already covered.
C02195 DOCX paragraph 2195	ALREADY IN REFERENCE	NICE-cited CT within 1 hour for initial GCS <=12 is present.
C02357 DOCX paragraph 2357	CLOSED DUPLICATE	Same GCS/head-CT teaching point as C02195; one canonical assertion only.
C02461 DOCX paragraph 2461	CLOSED EXCLUDED COLLISION	Coarctation and sepsis statements merged by extraction; no reliable conclusion about "most common" neonatal shock cause. Incomplete in source — not included.

C02467 DOCX paragraph 2467	CLOSED EXCLUDED DOSE	Isolated hydrocortisone 3 mg/kg lacks indication/timing/ route, and is conflated with Phoenix sepsis criteria. Incomplete in source — not included as dose.
C02468 DOCX paragraph 2468	CLOSED CASE EXAMPLE	Additional 1 L lactated Ringer bolus belongs to one case; adult septic-shock definition handled separately.
C02626 DOCX paragraph 2626	CLOSED EXCLUDED DOSE	CLEAN V3 lists penicillin VK 50 mg/kg/day QID for 10 days in GAS pharyngitis, but no maximum dose or comprehensive patient caveats appear in either source. Not carried as a clinical dosing directive.
C02646 DOCX paragraph 2646	RESTORED INDICATION SEPARATION	CLEAN V3 parenteral PID example: ceftriaxone 1 g IV + doxycycline 100 mg PO/IV + metronidazole 500 mg PO/IV; AQRAB p.133 supplies separate eligible nonpregnant outpatient regimen (ceftriaxone 500 mg IM once then doxycycline/metronidazole 100/500 mg PO BID x14d). Not interchangeable.
C02703 DOCX paragraph 2703	RESTORED INDICATION SEPARATION	CLEAN V3 cites azithromycin 500 mg daily for 3 days as an adult pertussis treatment example; AQRAB p.401 and CLEAN V3 para 1040 cite 500 mg day1 then 250 mg days2–5 for adult postexposure prophylaxis. Different indications; do not infer equivalence.
C02876 DOCX paragraph 2876	ALREADY IN REFERENCE	Benztropine 2 mg IM example for acute drug-induced dystonia is present.
C03168 DOCX paragraph 3168	CLOSED CASE EXAMPLE	48–72 h pediatric dental follow-up is tied to one tender, immobile tooth case; not a blanket dental-trauma rule.

Record and location	Final source-limited disposition	What the sources establish
C03173 DOCX paragraph 3173		SCIWORA younger-than-8 years teaching statement and pediatric trauma referral are in Pass 10.
C03179 DOCX paragraph 3179	ALREADY IN REFERENCE	Source describes pediatric nasal compression 5–10 min, not medication dose.
C03283 DOCX paragraph 3283	ALREADY IN REFERENCE	Placental-abruption clinical pattern included; 34 weeks is a specific patient context.
C03297 DOCX paragraph 3297	ALREADY IN REFERENCE	Postpartum hemorrhage source-dependent >500 vs >=1000 mL/hypoperfusion definitions preserved separately.
C03307 DOCX paragraph 3307	CLOSED DEFINITIONAL DUPLICATE	>500-mL vaginal delivery convention is source-specific and already compared with the >=1000-mL/hypoperfusion convention.
C03382 DOCX paragraph 3382		Emergency contraception up to five days after vaginal assault source example retained with consent and eligibility context.

C03607 DOCX paragraph 3607	RESTORED SOURCE OBSERVATION	CLEAN V3 cites minimum 6-hour observation after symptom resolution when two IM adrenaline doses were needed. Source-dependent pathway; not a blanket observation duration.
C03608 DOCX paragraph 3608	ALREADY IN REFERENCE	Adult adrenaline 500 micrograms IM in source-dependent anaphylaxis pathways is already present.
C04374 DOCX paragraph 4374	CLOSED EXCLUDED CORRUPTED	Mismatched D-dimer 500 mg/L, (<0.5), troponin, urgent surgery, symptoms. Incomplete/extraction-corrupted — not included.
C04394 DOCX paragraph 4394	CLOSED CONFLICT REGISTER	Direct contradictory claims on early post-arrest MRI/ biomarkers within 48h. Both recorded in conflict register, neither entered as settled recommendation.
C04454 DOCX paragraph 4454		Pediatric retropharyngeal abscess 3–5-year source pattern and case-specific airway plan included.
C04458 DOCX paragraph 4458	ALREADY IN REFERENCE	Seizure >5 minutes is the active treatment trigger; isolated lorazepam drug name without dose is not separate dosing claim.

Source disagreements and constraints

Azithromycin: adult pertussis treatment (CLEAN V3 para 2703) and adult contact postexposure prophylaxis (CLEAN V3 para 1040; AQRAB PDF p.401) are different indications, not an automatically reconciled prescription.

PID: source parenteral regimen (CLEAN V3 para 2646) and eligible outpatient regimen (AQRAB PDF p.133) must not be equated.

Antiepileptics: do not collapse a CLEAN V3 seizure-case levetiracetam 30 mg/kg, an AQRAB selected adult 60 mg/kg, and an AQRAB pediatric case 40 mg/kg into one universal regimen.

Unusable source excerpts: isolated hydrocortisone 3 mg/kg, garbled D-dimer unit/cutoff text, contradictory early post-arrest prognostication, and stitched neonatal shock statements are explicitly excluded rather than silently guessed.

Remaining project scope

This pass closes the listed 48 source-context records editorially. It does not assert semantic validation of the other 297 high-priority source units, the separate 802-unit comparison set, or all medication statements elsewhere in the original 17 domains. Those claims require source-by-source adjudication; declaring them complete without reading them would be misleading.

Evidence files: EDIC_Pass11_48_Numeric_Therapy_Final_Dispositions.csv and the two user-supplied original sources.

Mixed Presentations & Clinical Principles

Supplementary Clinical Points

Topic: Consciousness: two components

Core concept: CLEAN V3 distinguishes arousal from awareness as the two components of consciousness in its teaching definition.

الوعي ليس مجرد فتح العين؛ المصدر يفرّق بين arousal وawareness.

Source evidence: C04739: DOCX paragraph 4739

Topic: Delirium Triage Screen: source-specific test performance

Source-reported number: AQRAB reports a 98% sensitivity and negative likelihood ratio of 0.04 for the Delirium Triage Screen (DTS) in its cited teaching note. These are source-reported figures, not independently validated or universal across populations.

هذه أرقام مذكورة في AQRAB لأداء DTS؛ لا تُعمم على كل بيئة أو مريض بلا معرفة سياق الدراسة.

Source evidence: A407-03: AQRAB PDF p.407, block 3

ADDITIONAL CLINICAL PEARLS · TOPIC CONTEXT

Integrated near original chapter content · Source topic page 301

Mania - core clinical features [#103]

Key discriminator: Mania features sustained mood elevation or irritability with increased energy and reduced need for sleep.

من خصائص Mania ارتفاع المزاج أو تهيّجه المستمر، زيادة الطاقة، وقلة الحاجة للنوم.

Supplement IDs 103 | PreTest Emergency Medicine 5e (2021) | PDF p. 600 | Question 536

Fabricated or induced illness - safeguarding [#104]

High-yield fact: Fabricated or induced illness by a caregiver is a safeguarding concern, not simply a diagnostic curiosity.

افتعال المرض أو إحداثه بواسطة مقدم الرعاية قضية حماية طفل وتوثيق، وليس مجرد فضول تشخيصي.

Supplement IDs 104 | 1200 by Subjects | PDF p. 773 | Question 8

Global Conflict Resolution Register — Pass 7